

1 Investigation of Halide Perovskite Materials, Thermal
2 Sensing Concepts, and Radiation-Tolerant FPGA
3 Instrumentation for Advanced Radiation Monitoring

4 Dissertation

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Abstract

17 Radiation monitoring systems for nuclear environments require coordinated ad-
 18 vances across functional materials, detection devices, and radiation-tolerant electronic
 19 instrumentation. This dissertation investigates halide perovskite materials, fiber-optic
 20 thermal sensors, and FPGA-based digital electronics as complementary technologies
 21 for advanced radiation detection and monitoring, organized around a materials-to-
 22 systems hierarchy.

23 The first research thrust characterizes the radiation response of photovoltaic-
 24 architecture $\text{Cs}_{0.1}\text{FA}_{0.9}\text{PbI}_3$ perovskite solar cells under X-ray, gamma, and neutron
 25 irradiation. Under ^{137}Cs gamma irradiation at $44.16 \text{ rad hr}^{-1}$, an induced current
 26 density of 2.19 nA cm^{-2} was measured with a confirmed step response upon source
 27 insertion and withdrawal. Extended ^{60}Co irradiation at $355.61 \text{ rad hr}^{-1}$ yielded
 28 3.56 nA cm^{-2} , demonstrating that the current-mode detection approach scales across
 29 nearly an order of magnitude in dose rate. A linear degradation rate of approximately
 30 0.74 nA rad^{-1} was observed during six-day continuous irradiation, attributable to cu-
 31 mulative defect creation and substrate darkening.

32 The second thrust develops perovskite semiconductor radiation detectors for direct-
 33 conversion operation. A two-dimensional perovskite X-ray detector incorporating

5F-PEAI surface passivation was characterized, addressing the voltage-induced instability problem caused by ion migration under applied bias. A MAPbI₃-hBN composite detector was demonstrated for X-ray detection, and a solution-grown FAPbBr₃ single crystal achieved proof-of-concept gamma spectroscopy. For fast neutron detection, the hydrogen-rich MHyPbCl₃ perovskite was synthesized and tested at the Ohio State University Research Reactor Fast Beam Facility, exhibiting a resistivity of $4.43 \times 10^{11} \Omega \text{ cm}$ and an X-ray sensitivity of $61.74 \mu\text{C Gy}_{\text{air}}^{-1} \text{ cm}^{-2}$. Distinct radiation-induced pulses were recorded from the reactor fast neutron beam, and gamma discrimination analysis using beam filtering confirmed that the observed signals were predominantly attributable to neutron interactions.

The third thrust investigates thermal radiation sensing through simulation and experimental testing of an Optical Fiber-Based Gamma Thermometer (OFBGT) and a proposed Optical Fiber-Based Neutron Dosimeter (OFBND). Analytical thermal models predicted temperature differentials of 27–30 °C for the OFBGT and 155–159 °C for the OFBND. Experimental testing of the OFBGT yielded a measured differential of approximately 17 °C, with the discrepancy attributed to the omission of radiative heat transfer in the original simulations.

The fourth thrust addresses radiation-tolerant FPGA-based digital instrumentation through a Flip-Flop Shift Register (FFSR) with dual modular redundancy and a fixed-point thermal control benchmark, both tested under fast neutron irradiation. During two-day testing of five Spartan-7 FPGAs, configuration memory single-event upsets were observed at moderate fluence levels ($\sim 10^9 \text{ n/cm}^2$), while higher fluences ($\sim 10^{11} \text{ n/cm}^2$) produced latch-up events, a multi-SEU cluster, and measurable total ionizing dose degradation. These results demonstrate that configuration memory

58 scrubbing and defense-in-depth strategies are essential for SRAM-based FPGAs in
59 sustained radiation environments.

60 Collectively, this work establishes that no single technology addresses all radiation
61 monitoring requirements and that an integrated architecture combining perovskite
62 detectors, thermal sensors, and radiation-tolerant digital electronics is necessary for
63 comprehensive nuclear environment monitoring.

⁶⁴ To my Lord and Savior Jesus Christ for without I could do no things. Also to my
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Publications

Research Publications

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Fields of Study

104 Major Field: Nuclear Engineering

105

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Chapter 1: Introduction

647 Nuclear power occupies a singular position among low-carbon energy sources: it is
648 the only technology that delivers firm, dispatchable, high-density electricity without
649 direct greenhouse-gas emissions at a scale already proven across decades of opera-
650 tion. Yet the environments that make nuclear power possible, intense neutron and
651 gamma fields, high temperatures, and chemically aggressive primary coolants (e.g.,
652 liquid sodium, molten fluoride salts, and high-pressure water), are precisely the en-
653 vironments that destroy the instruments needed to monitor and control it. Reliable
654 radiation monitoring is not incidental to safe reactor operation; it is the primary
655 mechanism by which operators verify that every defense-in-depth barrier to fission-
656 product release — fuel cladding, reactor pressure boundary, and containment — re-
657 mains intact. As the existing fleet ages and a generation of advanced and small
658 modular reactors (SMRs) moves from concept to construction, the instrumentation
659 technologies that have served the industry for half a century are confronting hard
660 limits in both technical performance and economic viability. This dissertation inves-
661 tigates two complementary approaches to resolving that tension: the development
662 of perovskite semiconductors as radiation-responsive detector materials, and the im-
663 plementation of radiation-tolerant digital monitoring systems built on commercially
664 available field-programmable gate arrays (FPGAs). Taken together, these efforts span

the full stack from detector material to deployable system and aim to demonstrate that next-generation nuclear instrumentation can be both technically capable and represent a meaningful improvement in economic viability over current approaches.

1.1 Nuclear Power and the Global Decarbonization Context

The scale of the decarbonization challenge facing the global electricity system makes any realistic pathway to net-zero heavily dependent on nuclear energy. At the COP28 climate summit in December 2023, twenty-two nations signed a declaration to triple global nuclear generating capacity from approximately 375 GWe to more than 1,125 GWe by 2050; that coalition has since grown to thirty-three countries [99]. The International Energy Agency’s net-zero pathway requires nuclear capacity to reach approximately 1,079 GWe by 2050, demanding that annual investment in the sector roughly double from its current level of \$65–75 billion to \$120–150 billion per year by 2030 [44]. The International Atomic Energy Agency (IAEA) has revised its projections upward for five consecutive years, with its 2025 high-case scenario projecting 992 GWe by 2050 [43].

The existing global fleet comprises approximately 408 operating reactors in thirty-one countries, with a combined net capacity that reached a record 369 GWe in 2024 [88]. Nuclear electricity generation also set an all-time high of 2,667 TWh in 2024, narrowly surpassing the 2006 record, though nuclear’s share of global electricity has fallen to approximately 9% as variable renewables have expanded at a far faster pace [88]. The fleet’s average age of 32.4 years — with 141 units exceeding 41 years of operation — makes life extension, instrumentation modernization, and new construction simultaneously urgent priorities.

688 Development of advanced reactors is more active and better funded than at any
689 point in the last several decades. In the United States, TerraPower’s 345 MW(e)
690 Natrium sodium-cooled fast reactor received its NRC construction permit in March
691 2026, the first commercial non-light-water reactor approval in over forty years [83].
692 Kairos Power’s Hermes fluoride-salt-cooled test reactor began nuclear safety-related
693 construction in 2025 [47], and X-energy’s Xe-100 pebble-bed high-temperature gas-
694 cooled reactor is under development for an industrial host site with over \$1.2 billion in
695 DOE cost-share [113]. Among light-water SMRs, GE-Hitachi’s BWRX-300 is under
696 construction at Ontario’s Darlington site and has received a \$400 million DOE award
697 for deployment at TVA’s Clinch River site [108]. Each of these programs introduces
698 novel coolant chemistries, fuel geometries, and multi-module control architectures
699 that existing monitoring technologies were never designed to address.

700 1.2 The Regulatory Foundation for Radiation Monitoring

701 Continuous radiation monitoring across neutron flux, gamma dose, primary coolant
702 activity, and effluent pathways is mandated by the regulatory foundations of nuclear
703 operation, not merely encouraged by best practice. The NRC’s General Design Cri-
704 teria (GDC), codified in 10 CFR 50 Appendix A, establish this requirement at the
705 highest level of the licensing framework. GDC 13 requires instrumentation capable
706 of monitoring all variables that affect the reactor core, coolant pressure boundary,
707 and containment integrity; GDC 64 mandates monitoring of the containment atmo-
708 sphere, recirculation spaces, and effluent pathways for radioactivity; and GDC 19
709 requires that control room operators maintain safe-shutdown capability with ade-
710 quate instrumentation even under accident conditions [102]. The IAEA mirrors these

711 requirements through the Safety Requirements for Nuclear Power Plants (SSR-2/1)
712 and provides detailed I&C system guidance in SSG-39 [42, 41].

713 The technical scope of these monitoring requirements is exceptionally broad. Neu-
714 tron flux measurement alone spans approximately twelve decades — from a few counts
715 per second during refueling to approximately $10^{15} \text{ n cm}^{-2} \text{ s}^{-1}$ at full power — requiring
716 three overlapping detector subsystems. Source-range monitors based on proportional
717 counters or fission chambers operate in pulse-counting mode across the lowest six
718 to seven decades during startup and shutdown. Intermediate-range monitors em-
719 ploy compensated ionization chambers in Campbell mode across approximately eight
720 to ten decades. Power-range monitors using uncompensated boron-lined ionization
721 chambers cover the upper two decades and are calibrated against secondary-side heat
722 balance. Each of these subsystems must meet independent safety-channel redundancy
723 requirements, typically two to four independent channels per range [72].

724 Beyond neutron flux, plants deploy area gamma monitors throughout contain-
725 ment, auxiliary buildings, fuel handling areas, and spent fuel pools to enforce occu-
726 pational dose limits under 10 CFR 20. On-line gamma spectroscopy of the primary
727 coolant monitors fission-product signatures that indicate fuel-cladding breaches. Con-
728 tainment atmosphere monitors classified as Category 1 under Regulatory Guide 1.97,
729 Revision 5, must survive combined temperature, pressure, radiation, and chemical-
730 spray environments for the full duration of a design-basis accident [101]. Stack and
731 liquid effluent monitors verify compliance with 10 CFR 50 Appendix I dose objec-
732 tives, while the Radiological Environmental Monitoring Program tracks off-site air,
733 water, and vegetation samples under RG 4.1.

The Fukushima Daiichi accident of March 2011 provided a severe demonstration of the consequences when monitoring fails. The tsunami rendered 23 of 24 on-site radiation monitoring stations nonfunctional. The subsequent loss of all AC power and battery depletion eliminated reactor vessel pressure, temperature, and water-level indication, forcing operators to manage three simultaneous core-melt progressions with essentially no real-time process data [63, 64]. Spent fuel pool instrumentation was unavailable for nearly two weeks. The NRC’s post-Fukushima response, including Order EA-12-051 requiring backup-powered spent fuel pool level instrumentation, the FLEX diverse-coping strategy described in NEI 12-06, and the addition of Type F severe-accident management variables to RG 1.97 Rev. 5, underscored that reliable monitoring is not a compliance formality but the operational foundation of accident management.

1.3 Limitations of Current Radiation Detector Technologies

The detector materials deployed across the nuclear industry each carry critical limitations that constrain their use in existing plants and threaten to preclude their adoption in advanced reactor environments. High-purity germanium (HPGe) remains the gold standard for gamma spectroscopy, achieving better than 0.2% full-width at half-maximum energy resolution at 662 keV. However, its narrow bandgap of 0.67 eV mandates cryogenic operation below 120 K. Liquid-nitrogen dewars are impractical for unattended or remote monitoring; mechanical coolers add substantial cost, complexity, and microphonic noise that degrades resolution at low energies. HPGe is therefore confined almost exclusively to laboratory and assay applications rather than continuous in-plant monitoring [22].

757 Silicon detectors dominate charged-particle detection and, in research settings, are
758 used for fast-neutron spectrometry via proton-recoil interactions [67], but suffer irre-
759 versible displacement damage above approximately 10^{14} – 10^{15} n_{eq} cm^{−2} fluence, with
760 charge-collection efficiency falling toward 50% — a threshold easily surpassed in the
761 lifetime of a power-reactor instrument [67]. Silicon’s low atomic number ($Z = 14$)
762 also provides poor photoelectric stopping power for gamma-ray energies above a few
763 hundred keV, making it poorly suited to the mixed neutron-gamma fields present
764 throughout a reactor plant.

765 Scintillators constitute the most widely deployed class of gamma-detection instru-
766 ments and are valued for their ability to produce measurable light from a wide range of
767 ionizing radiations. Sodium iodide thallium-doped [NaI(Tl)] crystals, the workhorse
768 material, achieve approximately 6–7% energy resolution at 662 keV but are highly
769 hygroscopic, mechanically fragile, and must be coupled to photomultiplier tubes that
770 are sensitive to magnetic fields, fragile under shock and vibration, and require high-
771 voltage supplies of 800–1200 V [57]. Advanced scintillators such as LaBr₃(Ce) achieve
772 approximately 3% resolution but contain intrinsic radioactivity from ¹³⁸La that cre-
773 ates a flat background continuum complicating low-level measurements. Organic
774 scintillators, including plastic scintillators and stilbene crystals, are well established
775 for fast neutron detection through proton-recoil interactions, but their low effective
776 atomic number renders them essentially transparent to gamma rays, requiring either
777 active discrimination electronics or companion gamma-shielding that adds mass, cost,
778 and complexity.

779 Cadmium zinc telluride (CZT) represents the most mature room-temperature
780 semiconductor alternative, offering energy resolutions approaching 0.7% at 662 keV

781 in pixelated geometries with full single-carrier-transport correction. However, CZT
782 crystal growth requires extremely precise control of stoichiometry and vapor-phase
783 conditions during directional solidification: high cadmium vapor pressure promotes
784 vacancy defects, tellurium inclusions limit boule diameter, and critical shear stresses
785 of only 0.3 MPa make the material mechanically fragile during processing [109]. Pro-
786 duction costs remain high, supply is constrained, and the situation deteriorated sig-
787 nificantly in 2025 when the People’s Republic of China imposed export controls on
788 germanium, gallium, and related critical materials including cadmium telluride com-
789 pounds.

790 For thermal neutron detection, the ^3He tube has been the standard technology
791 since the 1950s by virtue of its high neutron absorption cross-section (5330 barns at
792 0.025 eV), excellent gamma discrimination, and compatibility with conventional pulse-
793 counting electronics. The supply of ^3He — produced exclusively as a decay product of
794 tritium from the nuclear-weapons stockpile — was recognized as critically constrained
795 following the post-September 11 deployment of approximately 1,300 radiation portal
796 monitors for border-security applications. Annual demand reached roughly 70,000 L
797 against a production rate of only approximately 8,000 L yr $^{-1}$, driving prices from
798 approximately \$100 L $^{-1}$ to over \$2,000 L $^{-1}$ and prompting DOE to begin rationing
799 allocation in 2009 [19]. The episode exposed the fragility of depending on a single non-
800 renewable detection medium and has accelerated development of alternative neutron
801 detectors using ^{10}B -lined tubes and $^6\text{LiF}:\text{ZnS}$ scintillator screens.

802 Halide perovskites have emerged as a compelling alternative class of radiation-
803 responsive materials. Their high effective atomic number — CsPbBr_3 has $Z_{\text{eff}} =$
804 65.9, compared to 50.2 for CZT — provides excellent photoelectric stopping power

805 for gamma radiation [28]. Single-crystal MAPbBr₃ and CsPbBr₃ devices, operat-
 806 ing at room temperature, have demonstrated mobility-lifetime products of up to
 807 $8 \times 10^{-3} \text{ cm}^2 \text{ V}^{-1}$ and energy resolutions approaching 1.4% at 662 keV [36], approach-
 808 ing CZT performance without cryogenic cooling or Bridgman growth. A distinc-
 809 tive self-healing mechanism — in which the soft perovskite lattice allows radiation-
 810 displaced ions to migrate back toward equilibrium positions at room temperature
 811 — has been confirmed under proton irradiation to $10^{13} \text{ p cm}^{-2}$ and extended gamma
 812 exposures [27]. The methylhydrazinium lead chloride (MHyPbCl₃) composition inves-
 813 tigated in this work introduces a high-hydrogen-density 3D perovskite framework with
 814 a hydrogen density of $3.79 \times 10^{22} \text{ cm}^{-3}$ comparable to organic scintillators, enabling
 815 direct fast-neutron conversion through proton recoil without requiring a separate
 816 moderator or conversion layer [77].

817 Significant challenges remain. Ion migration under applied electric fields and ele-
 818 vated humidity drives hysteresis in current-voltage characteristics and accelerates de-
 819 vice failure at the biases required for thick-crystal direct-conversion operation. Long-
 820 term stability data at power-reactor fluence levels (10^{14} – $10^{15} \text{ n cm}^{-2}$) remain absent
 821 from the literature. Lead toxicity presents regulatory and environmental considera-
 822 tions for deployed instruments. These challenges motivate the stabilization strategies
 823 — fluorinated surface passivation, hexagonal boron nitride composites, and vacuum
 824 operation — investigated in the present work.

825 **1.4 The Economics of Nuclear Instrumentation**

826 Cost overruns in nuclear construction have become the defining obstacle to ex-
 827 pansion in liberalized electricity markets. The two AP1000 units completed at Plant

828 Vogtle in Georgia cost approximately \$35 billion against an original estimate of \$14
829 billion [6]. France’s Flamanville 3 EPR reached €23.7 billion — seven times its orig-
830 inal budget — after seventeen years of construction. The UK’s Hinkley Point C has
831 escalated to an estimated £35 billion in 2015 prices against an initial projection of
832 approximately £18 billion [21]. New nuclear levelized cost of electricity in the United
833 States ranges from approximately \$110 MWh⁻¹ (EIA) to \$180 MWh⁻¹ (Lazard), com-
834 pared to approximately \$55 MWh⁻¹ for utility-scale solar and \$40 MWh⁻¹ for onshore
835 wind. The existing U.S. fleet operates far more competitively — NEI reports a 2023
836 average total generating cost of \$31.76 MWh⁻¹, the lowest on record [71] — but twelve
837 commercial reactors have closed since 2013 due to competitive pressure, and emer-
838 gency support programs including a \$6 billion Civil Nuclear Credit allocation have
839 been required to prevent further premature retirements [68].

840 Instrumentation and control systems, while representing an estimated 5–10% of
841 total plant capital cost, exert a disproportionate influence on both construction time-
842 lines and long-term operating expenses. A typical large light-water reactor contains
843 approximately 10,000 sensors, 5,000 km of I&C cabling, and hundreds of electronic
844 cabinets. Nuclear-grade quality assurance requirements under 10 CFR 50 Appendix B
845 impose a cost premium that pervades the entire supply chain: components manufac-
846 tured to identical functional specifications have been documented at price ratios of
847 4:1 or greater between nuclear-qualified and commercial-grade procurement, driven by
848 QA documentation requirements, environmental qualification testing, and the severe
849 scarcity of nuclear-qualified suppliers [4]. Dominion Energy’s comprehensive digital
850 I&C modernization across four units at Surry and North Anna is budgeted at \$5.45
851 billion as part of subsequent license renewal [82]. The DOE Light Water Reactor

852 Sustainability (LWRS) Program has explicitly cautioned that “wide-scale moderniza-
853 tion of legacy I&C systems could prove to be cost-prohibitive unless implemented to
854 enable significant business innovation.”

855 The cost of radiation-hardened digital electronics amplifies this problem at the
856 component level. Purpose-built radiation-hardened FPGAs — the Microchip RTG4,
857 AMD/Xilinx XQRKU060, and their equivalents — carry price tags of tens of thou-
858 sands to hundreds of thousands of dollars per device, a premium of $10\times$ to $100\times$
859 over commercial equivalents that cost \$10–500 [13]. The radiation-hardened FPGA
860 market, valued at \$465.5 million in 2024, serves primarily space and defense appli-
861 cations where unit volumes are small and alternatives do not exist. These devices
862 trail commercial FPGAs by two to three process generations: the Microchip RTG4
863 uses a 65 nm flash-based process while commercial devices operate at 3–5 nm, a gap
864 representing multiple orders of magnitude in logic density, performance, and power
865 efficiency [60, 3]. Long lead times of 6–18 months, ITAR export restrictions, and the
866 absence of advanced intellectual property cores compound the challenge. For SMRs
867 targeting a levelized cost of electricity of \$60–90 MWh⁻¹, this cost structure is unten-
868 able: I&C requirements do not scale down proportionally with reactor output, making
869 instrumentation a higher fraction of total cost in smaller plants. The DOE’s GEM-
870 INA program has set an aspirational target of reducing advanced reactor operations
871 and maintenance cost from \$23 MWh⁻¹ to \$2 MWh⁻¹ through autonomous operations
872 and advanced diagnostics — a goal that cannot be achieved without fundamentally
873 cheaper and more capable sensing and processing hardware [100].

874 The alternative to purpose-built rad-hard electronics is software-defined radiation
875 tolerance using commercial off-the-shelf FPGAs with designer-inserted mitigation.

876 Triple modular redundancy (TMR) triplicates critical logic with majority voting to
 877 correct single-event upsets in real time; configuration memory scrubbing periodically
 878 corrects bit-flip errors in SRAM-based FPGAs before they propagate; and partial
 879 dynamic reconfiguration enables real-time repair of corrupted fabric regions. CERN's
 880 two-year study of Kintex-7 FPGAs under mixed-field radiation found no permanent
 881 operational failures to 300 krad total ionizing dose with TMR and scrubbing [26].
 882 Montana State University's RadSat program demonstrated COTS FPGA-based ra-
 883 diation tolerance at "a cost $100\times$ lower than existing radiation-hardened computers."
 884 A TMR-with-spares architecture for nuclear reactor protection system logic has been
 885 validated under cyclotron irradiation [29]. These results suggest that COTS FP-
 886 GAs with mature mitigation strategies can meet nuclear monitoring requirements
 887 at a fraction of the cost of purpose-built rad-hard devices, provided the mitigation
 888 architecture is co-designed with the application from the outset.

889 **1.5 Research Objectives and Dissertation Organization**

890 The research presented in this dissertation is organized around three complemen-
 891 tary objectives aligned with the materials-to-systems hierarchy described above.

892 The first objective is to characterize the radiation response of halide perovskite
 893 materials across a range of compositions, device architectures, and radiation types
 894 relevant to nuclear monitoring. This encompasses gamma-photon response from Cs-
 895 137 and Co-60 sources, X-ray sensitivity as a function of applied bias, fast-neutron re-
 896 sponse, and the material instability mechanisms — ion migration, humidity-dependent
 897 breakdown, and charge trapping — that currently limit perovskite detector lifetimes.

898 The second objective is to develop and demonstrate perovskite-based radiation
899 detection devices optimized for specific applications in the nuclear fuel cycle: high-
900 energy photon monitoring for post-detonation forensics, direct-conversion X-ray de-
901 tection, and direct fast-neutron conversion using the hydrogen-rich MHyPbCl_3 com-
902 position. Each device is characterized through electrical transport measurements,
903 Monte Carlo simulation, and experimental irradiation at a research reactor facility.

904 The third objective is to implement and validate a radiation-tolerant FPGA-based
905 digital monitoring system suitable for nuclear plant instrumentation applications.
906 This includes design and testing of a fault-tolerant flip-flop scrubbing register (FFSR)
907 architecture, integration of TMR voter logic, and validation of the combined system
908 against a digital plant simulation environment representing nuclear I&C functions.

909 The remainder of this dissertation is organized as follows. Chapter 2 provides
910 the perovskite-focused background necessary for the materials and device investiga-
911 tions, encompassing perovskite crystal structure, radiation interaction physics, ion
912 migration mechanisms, and the current state of perovskite-based radiation detectors.
913 Chapter 3 provides the electronics background, covering radiation effects on digital de-
914 vices, single-event effects in SRAM-based FPGAs, and the taxonomy of fault-tolerant
915 design approaches including triple modular redundancy and configuration scrubbing.
916 Chapter 4 presents the characterization of $\text{Cs}_{0.1}\text{FA}_{0.9}\text{PbI}_3$ perovskite solar cells un-
917 der extended gamma irradiation. Chapter 5 addresses the voltage-induced instability
918 problem in two-dimensional perovskite detectors and the 5F-PEAI surface passivation
919 strategy developed to overcome it. Chapter 6 describes the MAPbI_3 -hBN compos-
920 ite detector and its characterization for X-ray detection. Chapter 7 presents the

921 MHyPbCl₃ fast neutron detector, from crystal synthesis through beam-facility test-
922 ing and gamma discrimination analysis. Chapter 8 describes the radiation-tolerant
923 FPGA monitoring system, including the FFSR design, TMR integration, and digital
924 plant simulation validation. Chapter 9 provides overall conclusions and recommen-
925 dations for future work.

Chapter 2: Background: Halide Perovskite Semiconductors for Radiation Detection

The research reported in Chapters 4–?? rests on three intersecting bodies of knowledge: the crystal chemistry and electronic structure of halide perovskites, the physics governing how ionizing radiation deposits energy in matter, and the device physics that determines how efficiently that deposited energy is converted into a measurable electrical signal. This chapter develops each body of knowledge in sufficient depth to motivate the experimental choices made in later chapters and to place the results in the context of the broader literature on semiconductor radiation detectors. The FPGA-based systems work is treated separately in Chapter 3.

2.1 The Perovskite Crystal Structure and Material Families

2.1.1 The ABX_3 Archetype

The term “perovskite” derives from the mineral calcium titanate ($CaTiO_3$), named for the Russian mineralogist Lev von Perovski and first synthesized in the nineteenth century [106]. The defining structural motif is a three-dimensional network of corner-sharing BX_6 octahedra in which a larger A-site cation occupies the 12-coordinate cuboctahedral void at the unit-cell body-center (in the ideal cubic setting, space group $Pm\bar{3}m$). The versatility of this archetype arises from the wide range of chemistries

that can be accommodated without destroying the framework: in the oxide family, B-sites range from Ti^{4+} to Pb^{4+} , while in the halide family addressed here, B is almost exclusively Pb^{2+} (or Sn^{2+}) and X is a monovalent halide (I^- , Br^- , or Cl^-).

The stability of a given ABX_3 composition is conveniently predicted by the Goldschmidt tolerance factor,

$$t = \frac{r_A + r_X}{\sqrt{2}(r_B + r_X)}, \quad (2.1)$$

where r_i denotes the ionic radius of species i [33]. Perovskite formation is generally favored for $0.80 \leq t \leq 1.05$; values close to unity favor the ideal cubic structure, while smaller values drive cooperative octahedral tilting that lowers the symmetry to tetragonal or orthorhombic phases. Kieslich et al. extended this framework to hybrid organic-inorganic systems by defining effective radii for polyatomic organic cations from the root-mean-square displacement of their hydrogen envelope [49], enabling quantitative phase stability predictions for the methylammonium and formamidinium compounds central to this work.

A complementary criterion is the octahedral factor $\mu = r_B/r_X$, which must satisfy $0.414 \leq \mu \leq 0.732$ for the B cation to fit stably inside the halide octahedron. Pb^{2+} ($r = 1.19 \text{ \AA}$) paired with I^- ($r = 2.20 \text{ \AA}$) gives $\mu = 0.54$, comfortably within the stable range and consistent with the broad tunability of the Pb-halide framework.

2.1.2 Three-Dimensional Structures

Methylammonium lead iodide (MAPbI_3). The most extensively studied hybrid halide perovskite, MAPbI_3 (methylammonium = CH_3NH_3^+ , MA^+), adopts a cubic structure above 327 K, a tetragonal structure between 162 K and 327 K, and an orthorhombic structure below 162 K [7]. At room temperature, the tetragonal phase

(space group $I4/mcm$) is therefore the operative structure for detector applications. MAPbI₃ has a direct bandgap of approximately 1.55 eV and a high optical absorption coefficient ($> 10^4 \text{ cm}^{-1}$ above the bandgap), along with the excellent charge transport described in Section 2.4.2. Large single crystals with centimeter-scale dimensions can be grown from solution by inverse temperature crystallization [87], making bulk detector fabrication feasible. The principal weakness of MAPbI₃ for detector applications is the light A-site cation: the low atomic masses of C, N, and H contribute minimally to gamma-ray stopping power, and the modest hydrogen density in the cubic CH₃NH₃⁺ cation limits fast-neutron response. The MAPbI₃-hBN composite detector described in Chapter ?? addresses the neutron sensitivity limitation by incorporating high- n boron-containing nanoflakes within a MAPbI₃ matrix.

Cesium lead bromide (CsPbBr₃). Replacing the organic MA⁺ cation with the monoatomic Cs⁺ ($r = 1.67 \text{ \AA}$) yields an all-inorganic perovskite with significantly improved thermal and photochemical stability [95]. CsPbBr₃ undergoes orthorhombic-to-tetragonal ($\sim 88 \text{ }^\circ\text{C}$) and tetragonal-to-cubic ($\sim 130 \text{ }^\circ\text{C}$) transitions, remaining in the orthorhombic phase at room temperature. Its bandgap of approximately 2.3 eV is wide enough to suppress intrinsic carrier excitation at 300 K, yielding resistivities exceeding $10^9 \text{ } \Omega \text{ cm}$ in single crystals [36] — a prerequisite for the low dark-current operation required in radiation counting. Most critically for gamma detection, the presence of both Cs ($Z = 55$) and Pb ($Z = 82$) gives CsPbBr₃ a calculated effective atomic number of approximately 65.9, far exceeding that of conventional detector materials such as silicon ($Z_{\text{eff}} = 14$) and approaching that of cadmium zinc telluride ($Z_{\text{eff}} \approx 49.1$) [28]. This high Z_{eff} makes CsPbBr₃ among the most promising semiconductor candidates for room-temperature gamma spectroscopy.

990 **Mixed cation $\text{Cs}_{0.1}\text{FA}_{0.9}\text{PbI}_3$.** Pure formamidinium lead iodide (FAPbI_3 , FA^+
991 $= \text{CH}(\text{NH}_2)_2^+$) has a narrower bandgap of approximately 1.48 eV and superior thermal
992 stability relative to MAPbI_3 , but the desired black α -phase is metastable at room tem-
993 perature, spontaneously converting to the photoinactive yellow δ -phase [105]. Partial
994 substitution of FA^+ with Cs^+ (typically 5–15 mol%) suppresses this phase transition
995 by reducing the tolerance factor toward unity, entropy-stabilizing the cubic α -phase at
996 ambient conditions [116]. The composition $\text{Cs}_{0.1}\text{FA}_{0.9}\text{PbI}_3$ investigated in Chapter 4
997 exploits this stabilization while retaining the narrow bandgap and large absorption
998 cross-section of FAPbI_3 , enabling its use as a photon-responsive active layer in a
999 photovoltaic architecture repurposed as a radiation monitor.

1000 2.1.3 Two-Dimensional Ruddlesden-Popper Structures

1001 The three-dimensional perovskite framework can be sliced along specific crystal-
1002 lographic planes and the resulting slabs separated by layers of large organic spacer
1003 cations to form two-dimensional (2D) or quasi-2D structures. In the Ruddlesden-
1004 Popper series the general formula is $(R)_2(\text{MA})_{n-1}\text{Pb}_n\text{X}_{3n+1}$, where R is a large or-
1005 ganic cation that cannot fit the A-site of the 3D structure, n is the number of corner-
1006 sharing PbX_6 octahedral layers between adjacent organic spacer bilayers, and the
1007 ratio of inorganic to organic content increases with n [98]. In the limit $n \rightarrow \infty$ the
1008 three-dimensional structure is recovered.

1009 For the present work, the relevant composition is the $n = 3$ member $\text{PEA}_2\text{MA}_2\text{Pb}_3\text{I}_{10}$,
1010 where PEA^+ denotes the phenylethylammonium cation ($\text{C}_6\text{H}_5(\text{CH}_2)_2\text{NH}_3^+$). The al-
1011 ternating organic-inorganic layering creates a natural quantum-well electrostatic land-
1012 scape in which charge is preferentially confined within the higher-dielectric inorganic

1013 slabs. From a detector perspective, this layered architecture confers two important
 1014 advantages over its 3D counterpart: the bulky PEA^+ layers act as physical barriers
 1015 to halide vacancy migration, substantially suppressing the ion-transport instabilities
 1016 detailed in Section 2.6; and the organic layers provide intrinsic passivation of the inor-
 1017 ganic slab surfaces, reducing trap-state densities at the heterojunction interfaces [18].
 1018 The price paid is a modest reduction in charge-carrier mobility relative to three-
 1019 dimensional analogs, since carriers must traverse the organic-inorganic boundaries.
 1020 Chapter ?? investigates whether surface-passivation strategies can recover sufficient
 1021 transport quality to enable practical detector operation while retaining the stability
 1022 advantages of the 2D architecture.

1023 **2.1.4 Methylhydrazinium Lead Chloride (MHyPbCl_3)**

1024 Methylhydrazinium lead chloride, MHyPbCl_3 ($\text{MHy}^+ = \text{CH}_3\text{NH}_2\text{NH}_2^+$), is a rela-
 1025 tively recently reported member of the halide perovskite family in which the conven-
 1026 tional monovalent cation is replaced by the methylhydrazinium ion [70]. This compo-
 1027 sition is of particular interest for fast-neutron detection because of its exceptionally
 1028 high hydrogen content: the $\text{CH}_3\text{NH}_2\text{NH}_2^+$ cation carries five hydrogen atoms, and the
 1029 resulting crystallographic hydrogen density is approximately $3.79 \times 10^{22} \text{ H cm}^{-3}$ [77].
 1030 This exceeds the hydrogen density of many purpose-built neutron moderator and
 1031 detector materials. Fast neutrons (energies above $\sim 1 \text{ MeV}$) interact primarily with
 1032 hydrogen nuclei via elastic scattering, transferring up to 100% of their kinetic energy
 1033 to the recoil proton in a head-on collision; those recoil protons then ionize the de-
 1034 tector medium in the same manner as a directly ionizing particle. The combination
 1035 of high hydrogen density (for neutron moderation and proton production), moderate

bandgap (~ 3.0 eV, appropriate for room-temperature operation), and the semiconducting perovskite framework (for direct electronic readout) makes MHyPbCl₃ a compelling candidate for a compact direct-conversion fast-neutron detector. Chapter ?? presents the synthesis, characterization, and fast-neutron response measurements for this material.

2.2 Radiation Detection Fundamentals

2.2.1 Mechanisms of Ionizing Radiation Detection

A radiation detector converts the energy deposited by an ionizing particle or photon into a measurable quantity — in semiconductor detectors, this quantity is an electrical charge pulse whose amplitude is proportional to the deposited energy. The process proceeds in three stages: (1) the primary ionizing event, in which the incident radiation transfers energy to the detector medium, producing a spatial distribution of electron-hole pairs along the particle track or at the absorption site; (2) charge transport, in which the liberated carriers drift under the applied bias field toward the collection electrodes; and (3) signal induction and readout, in which the moving charges induce a current pulse on the electrodes that is amplified and digitized [51].

The mean energy required to create one electron-hole pair in a semiconductor is given by Klein’s empirical rule,

$$W \approx 2.67 E_g + 0.87 \text{ eV}, \quad (2.2)$$

where E_g is the bandgap energy [50]. For MAPbI₃ ($E_g \approx 1.55$ eV), this gives $W \approx 5.0$ eV per pair; for CsPbBr₃ ($E_g \approx 2.3$ eV), $W \approx 7.0$ eV per pair. These values compare favorably with conventional semiconductor detectors: silicon ($W = 3.6$ eV) and germanium ($W = 2.9$ eV) produce more pairs per unit energy, but those materials

1058 require cryogenic cooling or high-purity growth technologies incompatible with cost-
 1059 effective deployment. The higher W of perovskites is partially offset by their far
 1060 greater stopping power per unit thickness (Section 2.4), so the total charge produced
 1061 by a given photon energy can be comparable to or exceed that in a thinner silicon
 1062 detector despite the higher pair creation energy.

1063 2.2.2 Energy Resolution and Pulse Height Analysis

1064 The ability to resolve closely spaced photon energies — and thereby identify spe-
 1065 cific radionuclides — is quantified by the energy resolution R , conventionally ex-
 1066 pressed as the fractional full-width at half-maximum (FWHM) of a spectral peak:

$$R = \frac{\Delta E_{\text{FWHM}}}{E_0}, \quad (2.3)$$

1067 where E_0 is the incident photon energy. In an ideal semiconductor detector in which
 1068 Poisson statistics govern pair production, the minimum achievable resolution is set
 1069 by the Fano limit:

$$\Delta E_{\text{FWHM}}^{\text{Fano}} = 2.35 \sqrt{F \cdot W \cdot E_0}, \quad (2.4)$$

1070 where F is the Fano factor ($F \approx 0.1$ – 0.2 for most semiconductors) [51]. In practice,
 1071 the measured resolution is degraded by incomplete charge collection due to carrier
 1072 trapping, electronic noise from the amplifier chain, and dielectric polarization from
 1073 ion migration (the last factor being specific to halide perovskites and discussed in Sec-
 1074 tion 2.6). The benchmark figure commonly cited in the perovskite detector literature
 1075 is the 1.4% resolution at 662 keV (^{137}Cs) achieved by He et al. for a CsPbBr_3 single
 1076 crystal coupled to a custom application-specific integrated circuit (ASIC) [36], which
 1077 at the time of publication was the best room-temperature resolution demonstrated
 1078 by any non-CZT semiconductor detector.

Energy-resolved radiation measurements are performed by connecting the detector preamplifier output to a pulse-height analyzer (PHA), which bins each charge pulse by its amplitude and accumulates a histogram — the pulse-height spectrum — whose peak positions encode incident photon energies and whose areas are proportional to the detected activity. The shaping time of the amplifier is a critical parameter: it must be long enough to integrate the slow hole component of the charge pulse (hole mobilities in halide perovskites are typically $5\text{--}10\times$ lower than electron mobilities) but short enough to avoid excessive pileup at practical count rates.

2.2.3 X-Ray Detector Sensitivity

For applications in medical and industrial X-ray imaging, the relevant figure of merit is not energy resolution but sensitivity S , defined as the charge collected per unit area per unit X-ray exposure:

$$S = \frac{\Delta Q}{A \cdot \Delta X} \quad [\mu\text{C Gy}^{-1} \text{ cm}^{-2}], \quad (2.5)$$

where ΔX is the absorbed dose rate. Sensitivity is controlled by the X-ray absorption efficiency (determined by the stopping power and detector thickness), the charge collection efficiency (determined by $\mu\tau$ products and applied field), and the dark current level (which sets the minimum detectable signal). Perovskite thick-film detectors fabricated from polycrystalline MAPbI_3 or MAPbBr_3 have demonstrated sensitivities exceeding $4,500 \mu\text{C Gy}^{-1} \text{ cm}^{-2}$ under 8 keV X-rays, more than 10^4 times higher than conventional amorphous selenium panels [104]. The high atomic number of the Pb sublattice is the primary driver of this extraordinary sensitivity at soft X-ray energies, where the photoelectric cross-section scales roughly as $Z^{4.5}$.

1100 2.2.4 Neutron Detection and Discrimination

1101 Neutron detection in semiconductor materials is indirect: neutrons carry no charge
1102 and interact negligibly with electrons, so they must first undergo a nuclear reaction
1103 or elastic scattering event that produces a charged secondary particle capable of ion-
1104 izing the detector. Thermal neutrons (energies below ~ 0.5 eV) are most efficiently
1105 captured via exothermic reactions such as $^{10}\text{B}(\text{n},\alpha)^7\text{Li}$ and $^6\text{Li}(\text{n},\alpha)^3\text{H}$. Fast neutrons
1106 above ~ 1 MeV interact principally through elastic scattering from hydrogen nuclei,
1107 producing recoil protons with a continuous energy spectrum up to the full neutron
1108 energy. The detection efficiency for fast neutrons is proportional to the hydrogen
1109 areal density in the active medium, making hydrogen-rich materials intrinsically ad-
1110 vantageous. The measured ^1H density in MHyPbCl_3 of $3.79 \times 10^{22} \text{ cm}^{-3}$ exceeds
1111 that of common polymer moderator materials and places this perovskite in a unique
1112 position as a combined moderating and detecting medium for fast neutrons [77].

1113 Discrimination between neutron and gamma signals is accomplished by exploiting
1114 differences in the ionization density (linear energy transfer, LET) of the respective
1115 secondary particles. Recoil protons from fast-neutron elastic scattering and alpha
1116 particles from thermal capture reactions deposit energy far more densely than the
1117 Compton electrons produced by gamma photons of similar incident energy. In systems
1118 with sufficient time resolution, pulse-shape discrimination (PSD) can separate the fast
1119 and slow scintillation components that arise from these different ionization densities;
1120 in solid-state detectors, differences in carrier extraction dynamics under the high local
1121 ionization density of heavy charged particles can be exploited in a similar manner [51].

1122 **2.3 Wide Bandgap Semiconductors for Room-Temperature De-** 1123 **tection**

1124 **2.3.1 The Room-Temperature Constraint**

1125 The operating requirement most often overlooked in academic detector compar-
1126 isons — yet most consequential for deployment in nuclear plants, field instruments,
1127 and space systems — is that the detector must function reliably at room temperature
1128 without active cooling. This single constraint eliminates two of the highest-performing
1129 semiconductor detector technologies: germanium (HPGe) must be operated below
1130 120 K to suppress thermally generated leakage current, requiring liquid-nitrogen de-
1131 wars or mechanical cryocoolers that add cost, mass, and failure modes incompatible
1132 with unattended monitoring [51]. Silicon drift detectors, while operable at modest
1133 Peltier cooling (-20 to -40 °C), still suffer from displacement damage under high
1134 neutron fluences that degrade their spectroscopic performance irreversibly.

1135 The root cause of the temperature sensitivity in narrow-gap semiconductors is the
1136 intrinsic carrier concentration n_i , which increases exponentially with temperature as:

$$n_i = \sqrt{N_c N_v} \exp\left(-\frac{E_g}{2kT}\right), \quad (2.6)$$

1137 where N_c and N_v are the effective density of states in the conduction and valence
1138 bands, respectively, and k is Boltzmann’s constant [96]. At room temperature ($T =$
1139 300 K), germanium ($E_g = 0.67$ eV) has $n_i \approx 2.4 \times 10^{13}$ cm $^{-3}$, producing a leak-
1140 age current density many orders of magnitude larger than the signal currents from
1141 typical radiation fields. Silicon ($E_g = 1.12$ eV) fares considerably better with $n_i \approx$

1142 $1.5 \times 10^{10} \text{ cm}^{-3}$, but displacement damage from neutron irradiation introduces com-
 1143 pensating defects that effectively raise the intrinsic carrier density and degrade the
 1144 depletion voltage over time.

1145 For a semiconductor to operate as a room-temperature detector without active
 1146 cooling or impractically high reverse bias, n_i must be sufficiently low that the thermal
 1147 generation current is smaller than the smallest radiation-induced signal of interest. A
 1148 widely cited rule of thumb requires $n_i \lesssim 10^7 \text{ cm}^{-3}$, which for silicon-like band struc-
 1149 ture translates to a minimum bandgap of approximately 1.5 eV [74]. This threshold
 1150 defines the wide bandgap semiconductor regime relevant to room-temperature radia-
 1151 tion detection.

1152 **2.3.2 Figures of Merit for Wide Bandgap Detector Materials**

1153 The suitability of a wide bandgap semiconductor for radiation detection is char-
 1154 acterized by several interrelated figures of merit beyond the bandgap itself:

1155 **Resistivity** (ρ , $\Omega \text{ cm}$). High bulk resistivity minimizes the dark current $I_{\text{dark}} =$
 1156 $V/R = VA/(L\rho)$ that flows under the applied bias field and sets the noise floor.
 1157 For spectroscopic applications, resistivities above $10^9 \Omega \text{ cm}$ are typically required; for
 1158 imaging applications where high flux dominates shot noise, values above $10^7 \Omega \text{ cm}$ may
 1159 be acceptable. Wide bandgap materials achieve high resistivity by a combination of
 1160 large E_g (reducing n_i) and the absence of shallow donor or acceptor impurities in the
 1161 bandgap. CsPbBr₃ single crystals have been measured at $\rho > 10^{10} \Omega \text{ cm}$ at room
 1162 temperature [36], exceeding the resistivity of commercial CdZnTe by approximately
 1163 one order of magnitude.

1164 **Mobility-lifetime product** ($\mu\tau$, $\text{cm}^2 \text{V}^{-1}$). As derived in Section 2.4.2, $\mu\tau$ di-
1165 rectly governs the mean free path of carriers between generation and trapping events.
1166 Wide bandgap semiconductors generally have lower carrier mobilities than narrow-
1167 gap materials (because heavier effective masses accompany larger bandgaps in many
1168 crystal systems), but the lifetime can be extraordinarily long in materials with low
1169 defect densities, partially compensating. The target for spectroscopic detectors is
1170 $\mu_e\tau_e \gtrsim 10^{-3} \text{ cm}^2 \text{V}^{-1}$; for hole collection, the equivalent target is often relaxed to
1171 $10^{-4} \text{ cm}^2 \text{V}^{-1}$ when single-polarity collection schemes are employed [56].

1172 **Effective atomic number** (Z_{eff}). For gamma and X-ray detection, Z_{eff} deter-
1173 mines the photoelectric stopping power, which scales as $Z^{4.5}$. Wide bandgap semicon-
1174 ductors based on heavy elements (Pb, Bi, Hg, Cd, Te) combine the thermal benefits
1175 of a large bandgap with high stopping power in a way that narrow-gap elemental
1176 semiconductors cannot achieve. Silicon ($Z = 14$) requires centimeters of active thick-
1177 ness to stop 662 keV photons, while a few millimeters of CsPbBr_3 ($Z_{\text{eff}} = 65.9$) or
1178 CdZnTe ($Z_{\text{eff}} = 49.1$) provide equivalent absorption [62].

1179 **Pair creation energy** (W , eV). As noted in Eq. 2.2, W is approximately pro-
1180 portional to E_g . Wide bandgap materials therefore produce fewer electron-hole pairs
1181 per unit of deposited energy than narrow-gap ones, increasing the statistical (Fano)
1182 contribution to the energy resolution. This is the principal penalty paid for room-
1183 temperature operability: it must be offset by large $\mu\tau$ products that ensure complete
1184 charge collection, since any additional variance from trapping further degrades the
1185 resolution beyond the Fano limit.

1186 2.3.3 The Wide Bandgap Material Landscape

1187 Table 2.1 contextualizes the halide perovskites investigated in this dissertation
1188 within the wider landscape of wide bandgap semiconductor detector materials.

Table 2.1: Selected properties of wide bandgap semiconductor detector materials at 300 K. CZT = cadmium zinc telluride $\text{Cd}_{0.9}\text{Zn}_{0.1}\text{Te}$. Data compiled from [74, 36, 28, 51].

Material	E_g (eV)	Z_{eff}	ρ ($\Omega\text{ cm}$)	$\mu_e\tau_e$ ($\text{cm}^2\text{ V}^{-1}$)	W (eV)	Best R_{662}
Si	1.12	14	$> 10^4$	> 1	3.6	$< 0.1\%$ (cryo)
CdTe	1.50	49.5	10^9	3×10^{-3}	4.4	1.7%
CZT	1.57	49.1	10^{10}	10^{-2}	4.6	0.5%
GaAs	1.42	31.6	10^7	10^{-5}	4.2	$\sim 2\%$
4H-SiC	3.26	14	$> 10^{12}$	10^{-6}	8.0	$\sim 1\%^*$
HgI ₂	2.13	62.2	10^{13}	10^{-4}	4.2	0.5%
CsPbBr ₃	2.30	65.9	$> 10^{10}$	1.3×10^{-2}	7.0	1.4%
MAPbI ₃	1.55	51.4	10^8	8×10^{-3}	5.0	$\sim 6\%$

*For alpha particles; gamma spectroscopy performance is limited by the low Z_{eff} .

1189 Several observations emerge from this comparison. CdZnTe has long occupied
1190 the upper tier of room-temperature gamma spectrometers, combining a near-ideal
1191 bandgap for thermal leakage suppression, high Z_{eff} , and a $\mu_e\tau_e$ product sufficient for
1192 millimeter-to-centimeter-scale crystals — but its dominance rests entirely on the ex-
1193 traordinary effort invested in Bridgman crystal growth and post-growth processing
1194 that has accumulated over three decades. HgI₂ offers the highest Z_{eff} and resistiv-
1195 ity of the conventional materials but suffers from mechanical fragility and a phase
1196 transition at 127 °C that has limited commercial uptake. Silicon carbide (4H-SiC)

1197 is exceptional in its radiation hardness and thermal tolerance but its low Z_{eff} con-
 1198 fines it to charged-particle and neutron applications where stopping power from the
 1199 heavy-element sublattice is not required [74].

1200 CsPbBr₃ occupies a remarkably favorable position in this landscape. Its Z_{eff} of
 1201 65.9 is the highest of any material in common laboratory study, its $\mu_e\tau_e$ is competitive
 1202 with CdZnTe despite far simpler solution-based crystal growth, and its resistivity
 1203 exceeds that of CdZnTe by an order of magnitude. The 1.4% energy resolution
 1204 at 662 keV achieved by He et al. is already within a factor of three of the best
 1205 CdZnTe single-element detectors, and it was achieved after only a few years of focused
 1206 development [36]. The remaining gap is attributable primarily to the ion-migration
 1207 polarization effect (Section 2.6), which introduces a time-dependent component to
 1208 charge collection efficiency not present in purely electronic semiconductors.

1209 **2.3.4 Why Bandgap Engineering Matters: The Perovskite Ad-** 1210 **vantage**

1211 The halide perovskite family is unique among wide bandgap detector candidates in
 1212 that the bandgap is continuously tunable over the range 1.2–3.5 eV by compositional
 1213 substitution at all three crystallographic sites, without changing the crystal structure
 1214 or growth method. Replacing iodide with bromide shifts the bandgap from 1.55 eV
 1215 (MAPbI₃) to 2.3 eV (MAPbBr₃); substituting Pb²⁺ with Sn²⁺ narrows the gap to
 1216 ~ 1.2 eV; replacing MA⁺ with the larger FA⁺ cation narrows the gap marginally
 1217 to ~ 1.48 eV [95]. This tunability enables the optimization of three competing re-
 1218 quirements simultaneously: (1) E_g large enough to suppress thermal leakage at the
 1219 maximum operating temperature; (2) E_g small enough to keep W — and therefore

1220 the Fano noise — acceptably low; and (3) Z_{eff} high enough for efficient photon stop-
1221 ping in the energy range of interest. No other material family offers all three degrees
1222 of freedom within a single, solution-processable crystal system, which is the central
1223 motivation for investigating multiple perovskite compositions across the chapters that
1224 follow.

1225 **2.4 Radiation Interaction with High- Z Semiconductors**

1226 **2.4.1 Photon Interactions**

1227 Three mechanisms govern the interaction of gamma and X-ray photons with mat-
1228 ter: the photoelectric effect, Compton scattering, and pair production. Each domi-
1229 nates over a different energy range and scales differently with the atomic number of
1230 the absorber, making the choice of detector material central to the detection efficiency
1231 for a given source energy.

1232 The *photoelectric effect* is dominant below approximately 100 keV in high- Z ma-
1233 terials. An incident photon is completely absorbed by an inner-shell electron, which
1234 is ejected with kinetic energy $E_e = h\nu - E_b$, where E_b is the binding energy of
1235 that shell. The photoelectric cross-section scales strongly with atomic number and
1236 inversely with photon energy as $\sigma_{\text{pe}} \propto Z^{4.5}/E^{3.5}$, so high- Z semiconductors provide
1237 orders-of-magnitude higher photoelectric absorption than silicon or germanium for the
1238 same detector volume. The complete energy deposition of the photoelectric process
1239 produces the full-energy peak in the pulse-height spectrum, making it the principal
1240 mechanism for photon spectroscopy.

1241 *Compton scattering* dominates for photon energies between approximately 100 keV
1242 and several MeV in heavy materials. An incident photon transfers a fraction of its

energy to a loosely bound or free electron, with the scattered photon carrying the remainder. Because only partial energy is deposited, Compton events populate a continuum below the full-energy peak rather than contributing to it. The Compton cross-section is proportional to Z , so high- Z absorbers still offer a substantial geometric advantage, but the spectroscopic information is contained only in the photopeak fraction.

Pair production becomes significant above the threshold energy of 1.022 MeV and dominates at multi-MeV photon energies. The photon is converted into an electron-positron pair in the Coulomb field of the nucleus; the 511-keV annihilation photons from the subsequent positron annihilation appear as characteristic escape peaks in the spectrum. The pair-production cross-section scales as Z^2 .

The combined effect of these mechanisms is captured by the mass attenuation coefficient μ/ρ , tabulated as a function of energy for all elements by the National Institute of Standards and Technology (NIST XCOM database). For a compound material, the effective atomic number is calculated as:

$$Z_{\text{eff}} = \left(\sum_i w_i Z_i^n \right)^{1/n}, \quad (2.7)$$

where w_i is the weight fraction of element i and the exponent $n \approx 3.5$ is chosen to match the photoelectric cross-section scaling [62]. Applying this formula to CsPbBr₃ (Cs: $Z = 55$, Pb: $Z = 82$, Br: $Z = 35$; stoichiometric weight fractions of 0.229, 0.449, and 0.322, respectively) yields $Z_{\text{eff}} \approx 65.9$, substantially higher than silicon ($Z_{\text{eff}} = 14$), germanium (32), or even CdZnTe ($Z_{\text{eff}} \approx 49.1$) [28].

2.4.2 Charge Transport and the Hecht Equation

Even if radiation deposits charge in the bulk of a detector, that charge must be transported to the collection electrodes without recombination or trapping if a proportional signal is to be measured. The quality of charge transport is characterized by the mobility-lifetime product $\mu\tau$ for electrons and holes separately. Carrier *mobility* μ ($\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$) quantifies the drift velocity per unit electric field; carrier *lifetime* τ (s) is the mean time before a carrier is irreversibly trapped at a deep defect and removed from the circuit. The product $\mu\tau$ has units of $\text{cm}^2 \text{V}^{-1}$ and represents the mean distance a carrier drifts per unit of applied field before being trapped; a large $\mu\tau$ indicates that carriers can traverse a thick crystal without significant charge loss.

The fractional charge collection for a detector of thickness L under applied field E can be derived rigorously for a planar single-carrier case. For a general double-carrier detector with the interaction site at depth x from the cathode, the Hecht equation gives:

$$Q(x) = Q_0 \left[\frac{\mu_e \tau_e E}{L} (1 - e^{-(L-x)/\mu_e \tau_e E}) + \frac{\mu_h \tau_h E}{L} (1 - e^{-x/\mu_h \tau_h E}) \right], \quad (2.8)$$

where $Q_0 = eN_0$ is the charge generated by the initial ionizing event [37]. For uniform irradiation, averaging Eq. 5.1 over x from 0 to L gives the mean charge collection efficiency (CCE), which approaches unity only when both $\mu_e \tau_e E/L \gg 1$ and $\mu_h \tau_h E/L \gg 1$. In practice this requires either very high applied fields, very thin detectors, or very large $\mu\tau$ products.

Halide perovskites have demonstrated $\mu\tau$ products competitive with those of mature compound semiconductors. Single-crystal MAPbI₃ exhibits $\mu_e \tau_e \approx 8 \times 10^{-3} \text{ cm}^2 \text{V}^{-1}$ and $\mu_h \tau_h \approx 3 \times 10^{-3} \text{ cm}^2 \text{V}^{-1}$ [87], values that in a 1-mm-thick crystal at a modest field

1285 of 100 V cm^{-1} yield a mean electron drift length of $\sim 8 \text{ mm}$ — far exceeding the crys-
1286 tal thickness. CsPbBr₃ single crystals have achieved $\mu_e \tau_e$ up to $1.34 \times 10^{-2} \text{ cm}^2 \text{ V}^{-1}$,
1287 supporting the 1.4% energy resolution reported by He et al. [36]. These values are
1288 within a factor of two to three of CdZnTe ($\mu_e \tau_e \sim 10^{-2} \text{ cm}^2 \text{ V}^{-1}$), a material that
1289 has required decades of crystal growth refinement to achieve its current performance
1290 levels.

1291 The hole transport is typically the limiting factor in halide perovskites, as in most
1292 compound semiconductors. The hole $\mu\tau$ in both MAPbI₃ and CsPbBr₃ is typically
1293 $5\text{--}10\times$ lower than the electron value, leading to a characteristic single-polarity col-
1294 lection signature in which events occurring near the cathode (from which electrons
1295 must traverse the full crystal thickness) are preferentially detected. Electronic tech-
1296 niques such as single-carrier collection (using a Frisch collar or virtual Frisch ring)
1297 and waveform analysis can mitigate poor hole collection [56], and perovskite detectors
1298 are amenable to these approaches.

1299 **2.5 Schottky Diode Contacts in Semiconductor Detectors**

1300 **2.5.1 Metal-Semiconductor Junction Physics**

1301 When a metal is brought into contact with a semiconductor, charge redistributes
1302 at the interface until the Fermi levels align, creating a built-in potential difference
1303 and a depletion region in the semiconductor. For a metal in contact with an n-type
1304 semiconductor, the Schottky barrier height is:

$$\phi_B = \phi_m - \chi_s, \quad (2.9)$$

1305 where ϕ_m is the metal work function and χ_s is the electron affinity of the semicon-
 1306 ductor [96]. The depletion region extends a distance W into the semiconductor:

$$W = \sqrt{\frac{2\varepsilon\varepsilon_0(V_{\text{bi}} - V)}{eN_D}}, \quad (2.10)$$

1307 where ε is the static dielectric constant, V_{bi} is the built-in voltage, V is the applied
 1308 reverse bias, and N_D is the donor concentration. The rectifying current-voltage rela-
 1309 tionship of the Schottky diode follows the thermionic emission model:

$$J = J_0 (e^{qV/nkT} - 1), \quad J_0 = A^{**}T^2 e^{-q\phi_B/kT}, \quad (2.11)$$

1310 where A^{**} is the effective Richardson constant and n is the ideality factor [96].

1311 In contrast, an *ohmic* contact is formed when the metal work function is matched
 1312 to the Fermi level of the semiconductor such that no barrier to majority-carrier flow
 1313 exists. Ohmic contacts allow free exchange of majority carriers with the external
 1314 circuit and are necessary on the collection side of the detector to avoid field distortion
 1315 near the electrode.

1316 2.5.2 Application to Halide Perovskite Detectors

1317 The asymmetric contact architecture — one ohmic and one Schottky electrode —
 1318 has become the standard configuration for halide perovskite radiation detectors be-
 1319 cause it simultaneously achieves low dark current and efficient charge collection [103].
 1320 Under reverse bias, the Schottky junction is depleted, suppressing majority-carrier
 1321 injection from the blocking contact; minority carriers generated by ionizing radiation
 1322 within the depletion region are swept by the electric field toward the appropriate elec-
 1323 trode. Because the dark current is thermally limited by the Schottky barrier rather
 1324 than set by the bulk resistivity, even moderate barrier heights of 0.5–0.8 eV can reduce

1325 the dark current density by several orders of magnitude relative to the ohmic-contact
1326 geometry.

1327 For MAPbI₃-based detectors, gold (Au, $\phi_m = 5.1$ eV) forms a near-ohmic contact
1328 on the p-type surface, while chromium (Cr, $\phi_m = 4.5$ eV), gallium, or indium contacts
1329 on the opposite face exhibit Schottky character [103]. For CsPbBr₃, the typical
1330 asymmetric pair is a carbon paste or Au ohmic contact on one face and a Ga or
1331 indium-gallium eutectic Schottky contact on the other. The net result is a device
1332 structure functionally analogous to a *p-i-n* junction in which the intrinsic region (the
1333 semi-insulating bulk crystal) contains the active detector volume, and the contact
1334 asymmetry enforces unipolar drift under bias [36].

1335 For the 2D perovskite devices studied in Chapter ??, the layered crystal architec-
1336 ture introduces an additional consideration: charge transport perpendicular to the
1337 organic spacer layers (the out-of-plane direction, which is the relevant direction for
1338 planar detector geometry) is more resistive than in-plane transport, and the contact
1339 interface with the topmost organic spacer layer must be carefully engineered to avoid a
1340 secondary barrier that would impede collection. Surface passivation using Lewis-base
1341 molecules that coordinate to surface Pb²⁺ sites can simultaneously reduce trap-state
1342 density and tune the contact work function alignment.

1343 2.6 Ion Migration in Halide Perovskites

1344 2.6.1 Origins and Mechanisms

1345 The same structural softness that enables solution-processable crystal growth from
1346 precursor salts also makes halide perovskites susceptible to ionic transport under ap-
1347 plied fields — a property that distinguishes them fundamentally from conventional

covalent or ionic semiconductors such as silicon or CdZnTe. Density functional theory calculations and activation energy measurements converge on halide vacancy migration as the dominant ionic conduction mechanism, with activation energies of approximately 0.1–0.6 eV depending on the halide and A-site cation [24]. For comparison, the thermal energy at room temperature ($kT \approx 26$ meV) is sufficient to drive significant vacancy hopping across these barriers on detector-relevant timescales, particularly under the electric fields of 10–1,000 V cm⁻¹ routinely applied to semiconductor detectors.

The migration pathway for iodide vacancies in MAPbI₃ proceeds via nearest-neighbor hopping through the faces of the PbI₆ octahedra, with a computed activation barrier of approximately 0.58 eV [24]. Bromide vacancies in CsPbBr₃ have a somewhat higher barrier of ~ 0.74 eV, consistent with the greater thermal stability of that compound. Chloride vacancies, which are the operative mobile species in MHyPbCl₃, have not been as extensively studied computationally, but experimental activation energies inferred from temperature-dependent dark current measurements suggest similar or slightly higher barriers than for iodide.

2.6.2 Consequences for Detector Performance

Ion migration manifests in detector operation through three interrelated phenomena. First, under sustained bias, mobile ions drift to the electrodes and accumulate, establishing an internal dipole that *screens* the applied field and reduces the effective field within the crystal bulk. This polarization effect depresses charge collection efficiency over time and produces the characteristic dark current spike followed by slow decay observed in the current transient of a freshly biased perovskite crystal [36].

1371 Second, the accumulation of ions at the electrode interfaces modifies the contact bar-
1372 rier height, causing the rectification ratio and dark current to evolve continuously
1373 during operation — a direct threat to the baseline stability required in nuclear in-
1374 strumentation. Third, if the bias polarity is reversed (as during AC-coupled readout
1375 or impedance spectroscopy measurements), the accumulated ion profiles transiently
1376 drive the internal field in the opposite direction to the intended collection field, pro-
1377 ducing severe measurement artifacts.

1378 These effects are not merely theoretical concerns: in a systematic study of CsPbBr₃
1379 single-crystal detectors, Pan et al. demonstrated that the charge collection efficiency
1380 under a fixed field of 200 V cm⁻¹ dropped by approximately 30% over the first 100 sec-
1381 onds of bias application before reaching a new quasi-equilibrium dominated by ionic
1382 rather than electronic currents [76]. The voltage-induced degradation was partially
1383 reversible upon removal of bias, with recovery occurring on timescales of minutes to
1384 hours — consistent with thermal relaxation of the accumulated ion profiles [117].

1385 2.6.3 Mitigation Strategies

1386 Several strategies have been demonstrated to suppress or circumvent ion-migration
1387 instability in perovskite detectors. *2D spacer layers* (the Ruddlesden-Popper architec-
1388 ture) interpose wide-gap organic bilayers at regular intervals through the crystal thick-
1389 ness, acting as electrostatic barriers to halide vacancy drift while preserving charge
1390 transport via tunneling or thermally activated hopping [98]. *Surface passivation* with
1391 Lewis-base ligands such as phenethylammonium iodide (PEAI), methylammonium
1392 chloride, or thiophene-2-carboxylic acid coordinates unsaturated surface Pb²⁺ sites
1393 that would otherwise serve as ion-migration nucleation points and electronic trap

states [18]. *All-inorganic compositions* such as CsPbBr₃ exhibit significantly slower ion drift than their hybrid organic-inorganic analogs because the rigid Cs⁺ sublattice provides fewer low-energy migration pathways. Finally, *low-temperature operation* reduces the ionic conductivity exponentially (Arrhenius behavior) and has been used to virtually eliminate polarization artifacts in experimental detector characterizations, though at the cost of the room-temperature operability that is one of perovskite’s primary advantages [36]. Chapter ?? investigates the combination of 2D architecture with chemical surface passivation as a room-temperature solution to the voltage-induced instability problem.

2.7 Prior Art in Perovskite Radiation Detectors

2.7.1 Early Single-Crystal Gamma Detectors

The first demonstration of halide perovskite single crystals as room-temperature radiation detectors appeared in a pair of landmark papers in 2016. Wei et al. grew large MAPbI₃ single crystals by inverse temperature crystallization and contacted them in an asymmetric Au/perovskite/Cr geometry; at 662 keV they observed a resolved ¹³⁷Cs photopeak with an energy resolution of approximately 6.8% [103]. While this was inferior to the best CdZnTe devices, the demonstration that a solution-processed hybrid perovskite could rival polycrystalline CdZnTe — at a fraction of the material cost and with far simpler growth conditions — generated immediate and sustained interest. Concurrently, Yakunin et al. reported sensitivity values exceeding 10³ μC Gy⁻¹ cm⁻² for MAPbI₃ thick-film X-ray detectors, establishing the perovskite platform for medical and industrial X-ray imaging [115].

1416 **2.7.2 CsPbBr₃ and the Resolution Record**

1417 The shift from hybrid to all-inorganic CsPbBr₃ was motivated by the thermal
1418 and photo-instability of MAPbI₃, which loses iodide at temperatures above $\sim 85^\circ\text{C}$
1419 and degrades rapidly in ambient humidity. Pan et al. demonstrated in 2019 that
1420 centimeter-scale CsPbBr₃ single crystals grown by the Bridgman method exhibit a
1421 resistivity exceeding $10^9\ \Omega\text{cm}$, $\mu_e\tau_e$ products above $10^{-2}\ \text{cm}^2\text{V}^{-1}$, and a 662 keV en-
1422 ergy resolution of approximately 3.9% [76]. The subsequent integration of CsPbBr₃
1423 detectors with custom-designed ASICs — rather than laboratory charge-sensitive am-
1424 plifiers — addressed the electronic noise limitation and enabled the record 1.4% res-
1425 olution at 662 keV reported by He et al. in 2021 [36], subsequently approached and
1426 extended by other groups employing refined crystal growth and improved contact
1427 engineering [30].

1428 **2.7.3 X-Ray Detection and Imaging**

1429 Polycrystalline thick films of MAPbI₃, MAPbBr₃, and CsPbI₂Br have been investi-
1430 gated extensively as direct-conversion X-ray detector arrays for applications in digital
1431 radiography and computed tomography. The high atomic number of the Pb sublattice
1432 confers exceptional sensitivity at soft X-ray energies (below $\sim 30\ \text{keV}$), where the
1433 photoelectric cross-section makes lead-bearing compounds intrinsically more absorb-
1434 ing than the amorphous selenium currently deployed in flat-panel detectors. Wei et
1435 al. reported a sensitivity of $4,500\ \mu\text{C Gy}^{-1}\text{cm}^{-2}$ for a 2-mm-thick solution-processed
1436 MAPbI₃ film under an 8 keV X-ray beam [104], and subsequent optimization of film
1437 deposition and contact passivation has pushed sensitivities above $10^4\ \mu\text{C Gy}^{-1}\text{cm}^{-2}$

1438 in several laboratories. The principal challenges remaining for commercial deploy-
1439 ment are long-term operational stability (particularly under the high photon fluences
1440 of medical imaging), the toxicity of lead-bearing materials in a clinical environment,
1441 and the uniformity of thick-film deposition over large detector areas.

1442 **2.7.4 Neutron Detection**

1443 Neutron detection by perovskite materials has been demonstrated via two com-
1444plementary approaches: neutron-capture via the incorporation of ^{10}B or ^6LiI into the
1445 perovskite matrix, and direct fast-neutron detection via hydrogen-recoil in hydrogen-
1446 rich perovskites. The composite approach was demonstrated in this group’s own prior
1447 work: MAPbI_3 crystals incorporating hexagonal boron nitride (hBN) nanoflakes were
1448 shown to exhibit a measurable response to thermal neutrons, with the neutron signal
1449 distinguished from the ambient gamma background by pulse-shape analysis [77]. The
1450 MHyPbCl_3 results reported in Chapter ?? represent the first systematic investigation
1451 of a perovskite designed from the outset for high hydrogen content, aiming to max-
1452imize the fast-neutron elastic-scatter cross-section in a single-phase semiconducting
1453 medium. The availability of a structurally well-characterized perovskite with both
1454 high H density and a bandgap suitable for room-temperature electronic detection
1455 opens a path toward compact direct-conversion fast-neutron detectors that do not
1456 require external moderators or ^3He -filled proportional tubes.

1457 2.7.5 Mixed-Cation and 2D Detector Architectures

1458 The instability of single-phase 3D perovskites under sustained bias has driven
1459 interest in compositionally engineered architectures designed to suppress ion migra-
1460 tion while retaining the favorable electronic properties of the perovskite host. Two-
1461 dimensional Ruddlesden-Popper compounds were first applied to radiation detection
1462 by Xu et al., who demonstrated that $\text{PEA}_2\text{MA}_3\text{Pb}_4\text{I}_{13}$ ($n = 4$) single crystals exhib-
1463 ited suppressed polarization current compared to 3D MAPbI_3 under identical bias
1464 conditions, attributed to the PEA^+ barriers impeding halide vacancy drift [114]. The
1465 compositions studied in Chapter ?? extend this concept to the $n = 3$ compound
1466 $\text{PEA}_2\text{MA}_2\text{Pb}_3\text{I}_{10}$, where the additional organic layer density provides even stronger
1467 suppression of ion migration, and investigates the interaction between the 2D ar-
1468 chitecture and chemical surface passivation as a dual strategy for achieving stable,
1469 low-dark-current detector operation.

1470 2.8 Summary

1471 This chapter has established the conceptual framework necessary to interpret the
1472 experimental results reported in Chapters 4 through ??. The ABX_3 halide perovskite
1473 framework accommodates a diverse family of compositions spanning 3D, quasi-2D,
1474 and layered architectures, with A-site chemistry tunable from inorganic Cs^+ through
1475 hybrid organic-inorganic MA^+ and FA^+ to the hydrogen-rich methylhydrazinium
1476 cation of MHyPbCl_3 . The physics of semiconductor radiation detectors connects
1477 material properties — bandgap, effective atomic number, $\mu\tau$ product — to measur-
1478 able figures of merit such as energy resolution, sensitivity, and neutron detection
1479 efficiency via the Hecht equation and Klein’s pair-creation rule. The Schottky diode

1480 contact architecture suppresses dark current to levels compatible with radiation count-
1481 ing without sacrificing charge collection, provided that ion migration is controlled by
1482 compositional engineering or surface passivation. The problem of voltage-induced
1483 instability driven by halide vacancy migration is identified as the central unsolved
1484 challenge for room-temperature perovskite detector operation, motivating the 2D and
1485 passivation strategies pursued in the experimental chapters that follow.

Chapter 3: Background: Radiation Effects on Digital Electronics and FPGA-Based Instrumentation

The deployment of field-programmable gate arrays (FPGAs) in nuclear instrumentation and control (I&C) systems represents a convergence of two historically separate engineering disciplines: the radiation effects community, which has characterized and mitigated the response of microelectronics to ionizing radiation primarily in the context of space and defense applications, and the nuclear industry, which is confronting the simultaneous pressures of fleet aging, digital modernization, and cost reduction described in Chapter ???. This chapter provides the technical background required to understand the FPGA-based monitoring system developed in Chapter ??. It begins with the radiation environments of concern, then develops the physical mechanisms by which ionizing particles damage or disturb microelectronic devices, surveys the taxonomy of single-event effects (SEEs) with emphasis on those most relevant to SRAM-based FPGAs, describes the fault-tolerance architectures used to mitigate those effects, and concludes with the testing methodologies used to characterize and qualify devices for radiation environments.

1502 3.1 Radiation Environments for Digital Electronics

1503 Digital electronics encounter ionizing radiation in three principal deployment con-
1504 texts relevant to this work: the near-Earth space environment, the nuclear reactor
1505 plant environment, and the terrestrial cosmic-ray environment at ground level. Each
1506 environment differs in particle species, energy spectra, and dose rates, and each im-
1507 poses distinct qualification requirements.

1508 3.1.1 The Space Radiation Environment

1509 The space environment exposes electronics to three primary sources of energetic
1510 particles. *Trapped protons and electrons* in the Van Allen belts (primarily 1–10 MeV
1511 protons in the inner belt, up to several hundred MeV in the South Atlantic Anomaly)
1512 produce continuous total ionizing dose accumulation and displacement damage that
1513 degrades device performance over mission lifetime [78]. *Galactic cosmic rays* (GCRs)
1514 are fully ionized nuclei spanning the entire periodic table, with energies from tens of
1515 MeV/nucleon to above 10^{20} eV; they are isotropic, difficult to shield, and responsible
1516 for the bulk of single-event effects at high altitude [118]. *Solar energetic particles*
1517 (SEPs), produced during coronal mass ejections and solar flares, deliver intense but
1518 episodic fluences of protons (up to hundreds of MeV) and heavy ions that can cause
1519 acute upsets far exceeding the GCR background [25].

1520 The cumulative radiation dose received by a spacecraft depends strongly on or-
1521 bit, shielding, and mission duration. Low-Earth orbit (LEO) satellites at Interna-
1522 tional Space Station altitude (~ 400 km, 51.6° inclination) accumulate roughly 5–
1523 20 krad(Si)/year behind 100 mil aluminum shielding, while geostationary (GEO) and
1524 medium-Earth orbit (MEO) satellites may see 30–100 krad(Si)/year due to prolonged

1525 residence in the Van Allen belts [89]. Qualification standards such as MIL-STD-883
1526 and ESA ESCC 22900 require demonstration of margin over these mission doses be-
1527 fore flight hardware can be approved.

1528 **3.1.2 The Nuclear Plant Radiation Environment**

1529 Within an operating nuclear power plant, electronics installed in or near the re-
1530 actor building are exposed to a mixed neutron-gamma field whose intensity depends
1531 strongly on location. In-core instrumentation must withstand integrated fast-neutron
1532 fluences of 10^{17} – 10^{19} n/cm² and total gamma doses of 10^6 – 10^8 rad over the reactor
1533 lifetime. Ex-core instrumentation in the primary containment sees lower but still sig-
1534 nificant fluences of 10^{13} – 10^{16} n/cm² and gamma doses of 10^4 – 10^6 rad [40]. Safety I&C
1535 systems in the auxiliary building and control room operate at much lower dose rates
1536 (typically well below 1 rad/year) but must maintain functionality under design-basis
1537 accident conditions that could spike the local dose rate by several orders of magnitude
1538 for hours to days [66].

1539 The neutron fluence in reactor environments produces displacement damage in
1540 semiconductor lattices that is qualitatively distinct from ionizing dose effects: ener-
1541 getic neutrons displace silicon atoms from lattice sites, creating vacancy-interstitial
1542 pairs and vacancy clusters that act as generation-recombination centers, increase
1543 leakage current, reduce minority carrier lifetime, and degrade transistor gain [59].
1544 SRAM-based FPGAs, whose configuration memory cells rely on stable stored charge,
1545 are particularly sensitive to neutron-induced displacement damage in the pass tran-
1546 sistors that gate cell access.

1547 **3.1.3 The Terrestrial Cosmic-Ray Environment**

1548 At sea level, the secondary cosmic-ray flux consists primarily of muons, with a
1549 smaller but device-relevant component of high-energy neutrons (“spallation” neutrons)
1550 produced by hadronic showers in the upper atmosphere. The terrestrial fast-neutron
1551 flux at sea level is approximately 1.3×10^{-2} n/cm²/s for neutrons above 10 MeV,
1552 increasing by roughly a factor of three per kilometer of altitude [69, 118]. Although
1553 this flux is orders of magnitude lower than in accelerator or reactor environments, it
1554 is ubiquitous and non-negligible for high-density memories operating in safety-critical
1555 ground applications. The JEDEC standard JESD89A quantifies the terrestrial soft-
1556 error rate environment and defines the reference neutron spectrum at sea level against
1557 which device qualification measurements are normalized [45].

1558 **3.2 Total Ionizing Dose Effects**

1559 Total ionizing dose (TID) refers to the cumulative energy deposited in a material
1560 by ionizing radiation — principally energetic electrons, protons, and gamma photons
1561 — integrated over the operating lifetime of the device. In metal-oxide-semiconductor
1562 (MOS) structures, which form the transistor basis of all modern digital FPGAs, the
1563 primary site of TID damage is the gate oxide and the field oxide used to isolate
1564 adjacent transistors.

1565 **3.2.1 Charge Trapping in Gate Oxides**

1566 When ionizing radiation passes through a SiO₂ gate oxide, it generates electron-
1567 hole pairs with a yield proportional to the local electric field and the oxide thickness.
1568 Electrons, which have high mobility in SiO₂ ($\mu_e \approx 20$ cm²/V·s), sweep out of the

oxide rapidly under the applied gate field on nanosecond timescales. Holes, with a mobility orders of magnitude lower, undergo a dispersive hopping transport toward the Si/SiO₂ interface, where a fraction become trapped at pre-existing or radiation-induced oxide trap sites [73]. The resulting positive oxide trapped charge shifts the transistor threshold voltage V_T in the negative direction (for n-channel MOSFETs), eventually causing the device to conduct when it should be off — a failure mode known as threshold-voltage rolloff [32].

In parallel, radiation generates interface trap states at the Si/SiO₂ boundary by breaking Si-H bonds. These interface traps contribute both a fixed charge and a frequency-dependent recombination current that degrades transistor transconductance, increases $1/f$ noise, and reduces drive current. Unlike oxide trapped charge, which anneals partially at room temperature, interface traps are thermally stable on device-lifetime timescales [73].

3.2.2 TID Effects in Modern FPGAs

The susceptibility of a CMOS process to TID is strongly dependent on gate oxide thickness: thinner oxides, which dominate in sub-micron nodes, trap fewer holes because the thin-oxide regime favors rapid recombination before hole transport can occur. A 65-nm process FPGA may tolerate 300–500 krad(Si) before functional failure, compared to 30–100 krad(Si) for older 1- μ m process devices [89]. However, the isolation structures (shallow trench isolation, STI) in modern processes are relatively thick and remain a TID-sensitive element, causing leakage current between adjacent transistors at doses that do not yet affect the gate oxides.

For the nuclear instrumentation context, the TID tolerance of commercial SRAM FPGAs in the control room or auxiliary building is generally adequate without special hardening, as cumulative doses over a plant lifetime at those locations are well below the few-krad level [4]. The primary concern for FPGAs in nuclear plants is therefore not TID but the stochastic single-event effects described in the next section, which occur at arbitrarily low dose rates and cannot be mitigated by simply limiting exposure time.

3.3 Single-Event Effects: Physical Mechanisms

3.3.1 Ion Track Formation and Charge Collection

A single-event effect (SEE) is a functional or physical disturbance in a microelectronic device caused by the passage of a single energetic particle. The mechanism begins with the energy deposition of the incident particle along its track through the semiconductor. Charged particles — heavy ions, protons, and the recoil nuclei from neutron elastic scattering or nuclear reactions — lose energy primarily through Coulomb interactions with the semiconductor electrons, creating a dense column of electron-hole plasma along the track.

The key parameter governing SEE sensitivity is the linear energy transfer (LET), defined as the energy deposited per unit path length:

$$\text{LET} = -\frac{1}{\rho} \frac{dE}{dx} \quad [\text{MeV} \cdot \text{cm}^2/\text{mg}], \quad (3.1)$$

where ρ is the material density. For a silicon target, an LET of 1 MeV·cm²/mg deposits approximately 1.6×10^{-13} J per micron of path length, producing roughly 10⁴ electron-hole pairs per micron. The LET of a given ion depends on its atomic number Z , mass A , and kinetic energy; heavy ions near the middle of the periodic

1613 table (e.g., Fe, Kr, Xe, Bi) with energies of tens of MeV/nucleon achieve the highest
 1614 LETs, up to $\sim 100 \text{ MeV}\cdot\text{cm}^2/\text{mg}$ in silicon [78].

1615 Following the primary ionization event, the excess carriers are collected at the
 1616 nearest reverse-biased junction by two mechanisms illustrated in Figs. 3.1 and 3.2.
 1617 *Drift collection* occurs because the high carrier density temporarily collapses the
 1618 depletion region and extends the electric field deep into the substrate — a phe-
 1619 nomenon known as *funneling* — sweeping charge toward the junction on picosecond
 1620 timescales [23]. *Diffusion collection* then operates on slower nanosecond timescales
 1621 as carriers generated outside the funnel diffuse through the neutral substrate and are
 1622 eventually collected. The total collected charge Q_{coll} is determined by the integral of
 1623 the current transient at the junction and must exceed the critical charge Q_{crit} of the
 1624 struck node to cause a SEE.

1625 3.3.2 The SEE Sensitivity Figure of Merit: Cross-Section

1626 The probability that a particle causes a SEE is characterized by the device SEE
 1627 cross-section σ , defined as the number of events observed per unit incident particle
 1628 fluence:

$$\sigma(L) = \frac{N_{\text{events}}}{\Phi} \quad [\text{cm}^2/\text{device or cm}^2/\text{bit}] , \quad (3.2)$$

1629 where Φ is the fluence in particles/cm². The cross-section is measured as a function
 1630 of LET or proton energy and typically exhibits a threshold below which no events
 1631 are observed and a plateau at high LET where every ion strike within the sensitive
 1632 volume causes an event. The threshold LET (LET_{th}) and the saturation cross-section
 1633 (σ_{sat}) together characterize the inherent hardness of the device [80].

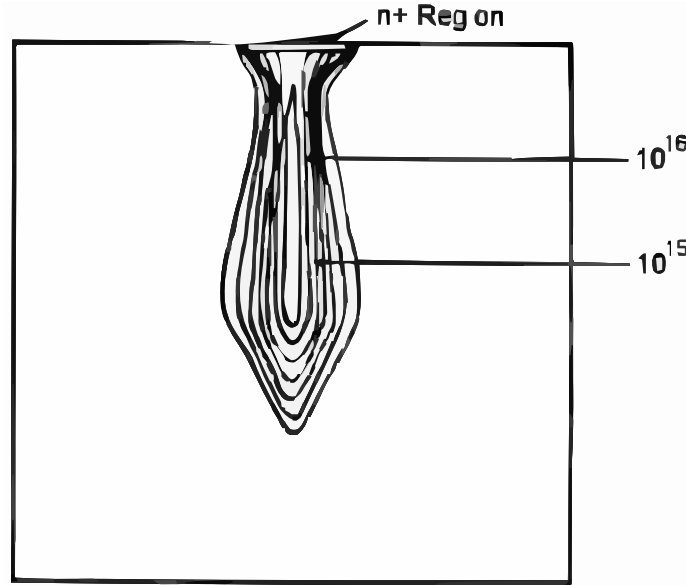


Figure 3.1: Simulated electron concentration during the funneling phase following a heavy-ion strike in a reverse-biased n^+/p junction. The dense electron-hole plasma temporarily extends the high-field depletion region deep into the substrate, collecting charge from a volume far larger than the geometric depletion region.

1634 The on-orbit SEE rate is computed by convolving the measured cross-section
 1635 curve with the environment's differential LET spectrum, a calculation codified in
 1636 tools such as CREME96 and OMERE [79]. For a device with $\sigma_{\text{sat}} = 10^{-6} \text{ cm}^2/\text{bit}$
 1637 and $\text{LET}_{\text{th}} = 3 \text{ MeV}\cdot\text{cm}^2/\text{mg}$ in a LEO orbit, the predicted upset rate is typically
 1638 10^{-5} – 10^{-3} upsets/bit/day, meaning a 16-Mbit configuration memory will experience
 1639 roughly 10–1,000 upsets per day without mitigation.

1640 3.4 Single-Event Effects Taxonomy

1641 SEEs span a range of severity from fully recoverable bit flips to permanent device
 1642 destruction. Fig. 3.3 presents the standard SEE taxonomy tree, which organizes
 1643 effects by their recoverability and the circuit element affected.

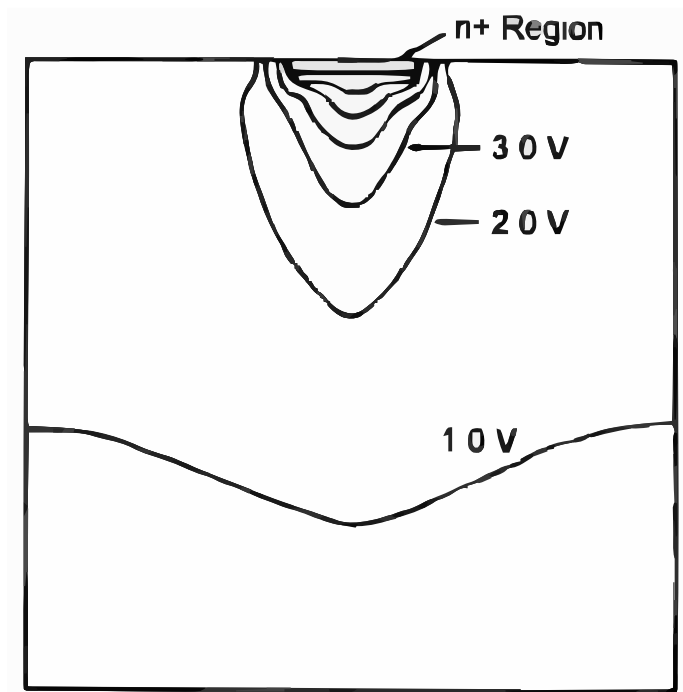


Figure 3.2: Electrostatic potential distribution during funneling, showing the field distortion caused by the ion track. The funnel depth scales with LET and substrate doping, reaching several micrometers for high-LET heavy ions in lightly doped substrates [23].

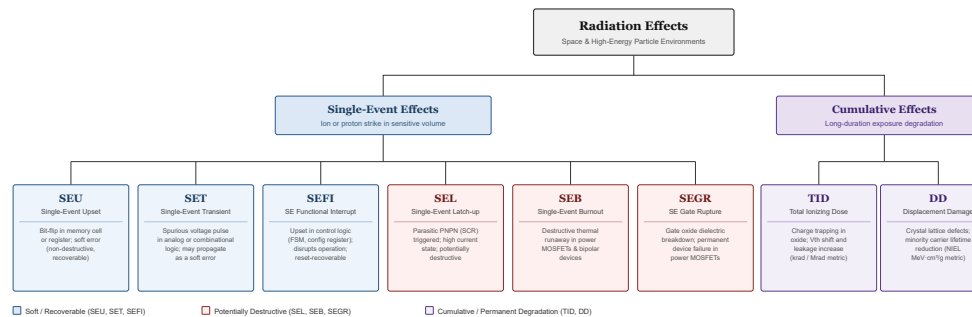


Figure 3.3: Taxonomy of single-event effects in microelectronic devices, organized by recoverability (soft vs. hard) and affected circuit element. Effects in bold are those most critical for SRAM-based FPGA operation in nuclear instrumentation applications.

1644 **3.4.1 Single-Event Upset (SEU)**

1645 A single-event upset (SEU) is a non-destructive, recoverable change in the logic
1646 state of a bistable storage element — a flip-flop, SRAM cell, or register — caused by
1647 charge collection from a single particle strike. The struck node’s voltage is transiently
1648 driven past the switching threshold of the latch, causing it to adopt the opposite
1649 state [23]. In an SRAM-based FPGA, the configuration memory that encodes the
1650 user’s design is implemented as an array of SRAM cells; an SEU in this array changes
1651 the programmed function of a logic element, routing multiplexer, or interconnect
1652 switch, potentially altering the circuit’s behavior in ways that are not immediately
1653 apparent without continuous monitoring [85].

1654 The critical charge for SEU in a modern 28-nm SRAM cell is on the order of
1655 1–10 fC, achievable by the recoil nucleus from a proton nuclear interaction or by
1656 a moderate-LET heavy ion [8]. As process nodes have scaled, Q_{crit} has decreased
1657 roughly proportionally with feature size, partially offset by reductions in node capac-
1658 itance and improvements in cell drive strength.

1659 **3.4.2 Multiple Cell Upset (MCU)**

1660 A multiple cell upset (MCU) occurs when a single particle deposits sufficient
1661 charge to simultaneously flip two or more adjacent memory cells. MCUs are of par-
1662 ticular concern because they can defeat single-bit error correction codes (SECDED)
1663 that are effective against isolated SEUs. The probability of MCU scales with LET
1664 and is strongly geometry-dependent: cells laid out in adjacent rows and columns of
1665 the SRAM array are co-vulnerable, while cells interleaved across different physical
1666 locations are not [14]. In modern planar CMOS processes, MCU events can involve

1667 tens of cells from a single high-LET heavy-ion strike, while proton-induced MCUs
 1668 typically involve two to four cells from the recoil of a proton nuclear reaction [8].

1669 3.4.3 Single-Event Transient (SET)

1670 A single-event transient (SET) is a voltage glitch on a combinational logic node
 1671 caused by charge collection at a circuit junction. Unlike an SEU, which corrupts a
 1672 storage element, an SET propagates through combinational logic and may or may not
 1673 be captured by a downstream flip-flop, depending on the timing relationship between
 1674 the transient and the clock edge. Fig. 3.4 illustrates the generation of an SET at an
 1675 inverter output, and Fig. 3.5 shows the conditions under which an SET propagating
 1676 through a logic cone is captured by a flip-flop.

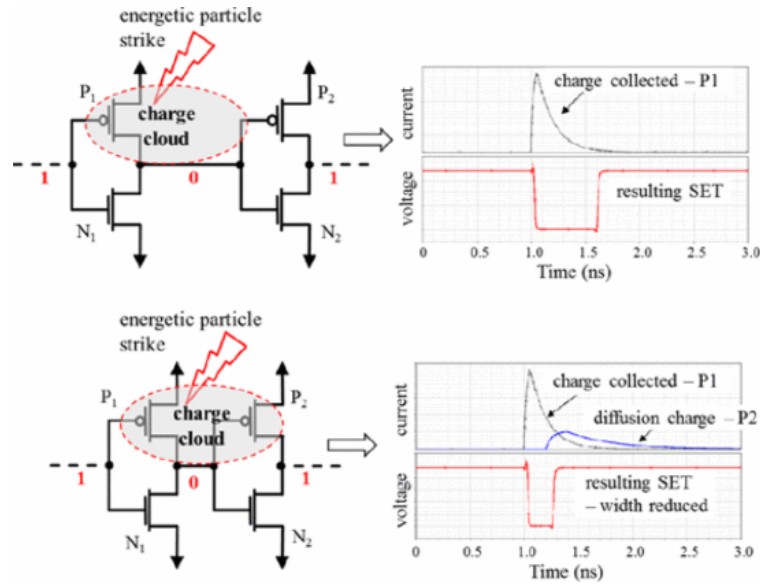


Figure 3.4: Generation of a single-event transient (SET) at a CMOS inverter output. The ion track deposits charge at a reverse-biased drain junction, driving the output voltage transiently toward the supply rail. Pulse width is determined by the collected charge and the drive strength of adjacent transistors [31].

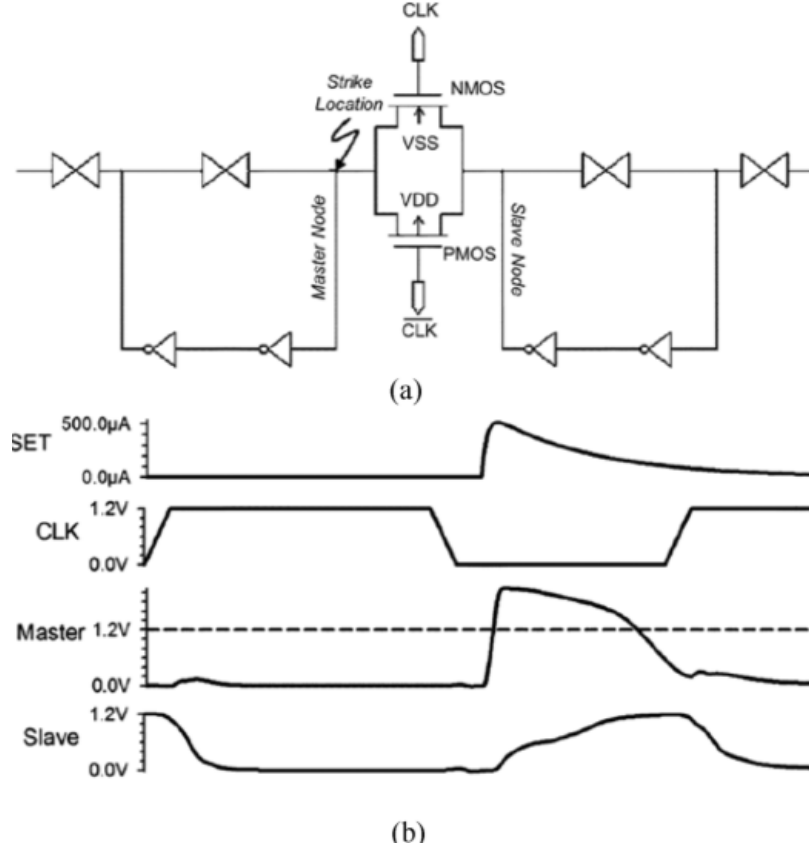


Figure 3.5: SET propagation and capture by a flip-flop. The SET (shaded pulse) traverses the combinational logic cone and arrives at the flip-flop's data input. It is captured and causes an SEU only if it overlaps with the flip-flop's setup-and-hold window around the clock edge [31].

1677 In high-speed digital logic, SET pulse widths of tens to hundreds of picoseconds are
 1678 comparable to clock periods, making SET-induced SEUs a dominant error mechanism
 1679 in modern FPGAs operating above 100 MHz [31]. At the lower clock rates typical
 1680 of nuclear I&C applications, SET capture probability is reduced, but long transients
 1681 from high-LET events remain a concern.

1682 **3.4.4 Single-Event Functional Interrupt (SEFI)**

1683 A single-event functional interrupt (SEFI) is an SEU in a control register, finite
1684 state machine, or configuration memory that causes a device to enter an erroneous
1685 operating mode — resetting itself, locking up, or outputting invalid data — that
1686 requires a system-level reset to clear. In FPGAs, SEFIs in the configuration con-
1687 troller or global control registers can cause a partial or complete loss of the user’s
1688 programmed logic, a far more serious consequence than a single flipped bit in the
1689 data path [85]. SEFI rates are typically much lower than SEU rates per bit (be-
1690 cause control registers represent a small fraction of total bits), but their operational
1691 consequence is disproportionately severe.

1692 **3.4.5 Single-Event Latchup (SEL)**

1693 Single-event latchup (SEL) is a potentially destructive condition in which a par-
1694 ticle strike triggers the parasitic p-n-p-n thyristor inherent in bulk CMOS processes,
1695 causing a low-impedance path to form between the supply rails. If not cleared by
1696 power cycling within milliseconds, the resulting current can permanently destroy the
1697 device through thermal runaway [90]. SEL susceptibility is strongly suppressed in
1698 silicon-on-insulator (SOI) processes (which lack the bulk substrate needed for the
1699 parasitic thyristor) and by epitaxial substrates with heavily doped buried layers.
1700 Modern commercial SRAM FPGAs fabricated in bulk CMOS are SEL-susceptible
1701 and require current-limiting power supply protection and power-cycling circuitry in
1702 space and nuclear applications.

1703 3.4.6 Single-Event Burnout (SEB) and Gate Rupture (SEGR)

1704 Single-event burnout (SEB) and single-event gate rupture (SEGR) are catas-
 1705 trophic destructive effects confined primarily to power MOSFETs and bipolar tran-
 1706 sistors operating at high drain-source voltages. SEB occurs when a heavy-ion strike
 1707 in the epitaxial drain region of an OFF-state power MOSFET triggers a parasitic
 1708 n-p-n transistor, causing sustained high-current conduction that thermally destroys
 1709 the device [90]. SEGR results from the momentary field enhancement in the gate
 1710 oxide caused by the ion-track plasma, which can exceed the oxide breakdown field
 1711 and permanently rupture the gate dielectric. Figs. 3.6, 3.7, and 3.8 illustrate the
 1712 cross-sectional geometry and onset conditions for these effects.

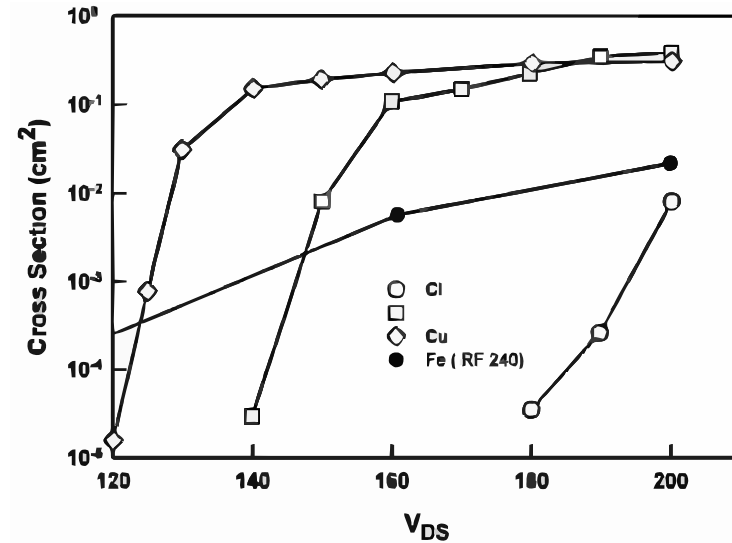


Figure 3.6: Cross-sectional schematic of single-event burnout (SEB) in an n-channel power MOSFET. The heavy-ion track (dashed) deposits charge in the epitaxial drain that forward-biases the parasitic n-p-n transistor base-emitter junction, initiating regenerative feedback and destructive current [90].

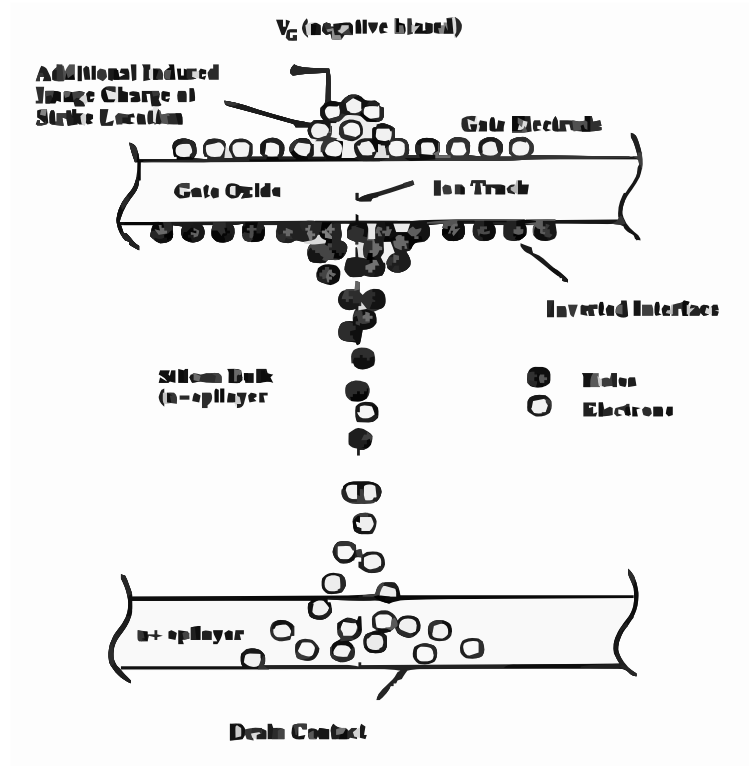


Figure 3.7: Single-event gate rupture (SEGR) mechanism. The transient increase in gate-oxide field caused by the ion-track plasma can locally exceed the intrinsic breakdown field of the SiO_2 , permanently rupturing the gate dielectric and rendering the transistor non-functional [90].

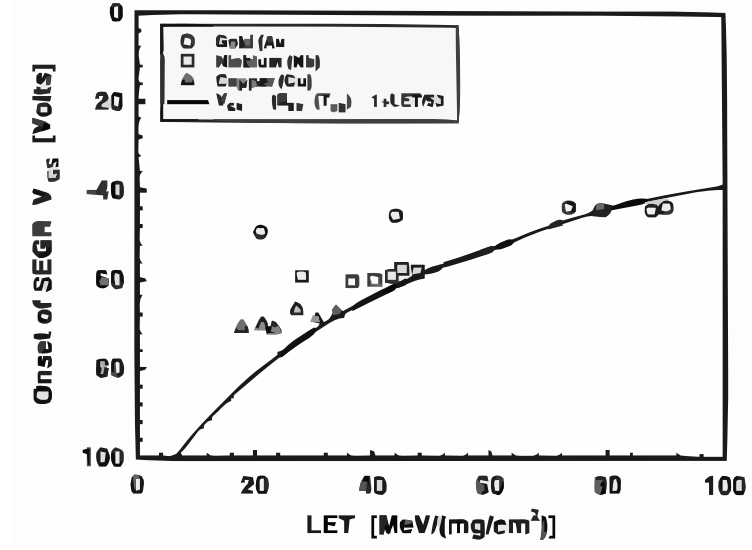


Figure 3.8: Measured SEGR onset V_{GS} as a function of heavy-ion LET for several gate-metal species. The onset voltage decreases with increasing LET, imposing a derating requirement on operating voltage for power devices in heavy-ion environments [90].

SEB and SEGR are not directly applicable to logic-level SRAM FPGAs, which operate at supply voltages far below the breakdown thresholds of their internal power transistors. They are, however, relevant to the power supply circuitry and gate drivers that support an FPGA-based system, and must be considered at the system level in any radiation-hardened design.

3.5 FPGA Architecture and Radiation Vulnerability

3.5.1 SRAM-Based FPGA Architecture

Modern high-density FPGAs are organized as a two-dimensional array of configurable logic blocks (CLBs), interconnected by a programmable routing fabric and surrounded by dedicated I/O, clock management, block RAM, DSP, and hard-IP

peripherals. The entire programmable function of the device is encoded in a configuration memory bitstream stored in an on-chip SRAM array, which for large devices may contain 50–500 million configuration bits [52]. The configuration SRAM is loaded from external non-volatile storage (typically QSPI flash or EEPROM) on power-up and retained in SRAM for the duration of operation.

This architecture has a fundamental radiation vulnerability: every bit of the configuration SRAM is continuously susceptible to SEU throughout device operation. An SEU in the configuration memory can alter the function of a lookup table (LUT), change the routing of a signal, enable or disable a flip-flop reset, or modify the address decode of a block RAM — depending on which configuration bit is struck. Unlike an SEU in a user-data register, which affects only the current computation, a configuration memory SEU silently corrupts the device’s programmed behavior until it is detected and corrected [48].

The sensitivity of different FPGA configuration bit types to functional corruption varies substantially. Bits that control LUT contents are always functionally significant; bits that control unused routing resources may have no effect. Studies of Xilinx Virtex and Kintex FPGAs have found that roughly 15–30% of configuration bits are “essential” — their upset causes a detectable output error — with the remainder corresponding to unused or redundant resources [85, 53]. This essential bit fraction is critical for predicting the functional upset rate from a measured raw upset rate.

3.5.2 Radiation Sensitivity Across FPGA Generations

Quinn et al. surveyed the radiation reliability concerns of Xilinx FPGAs from the Virtex through Virtex-5 generations, finding that each new process node brought

1746 reduced Q_{crit} (worsening raw SEU sensitivity per bit) but also reduced bit count
 1747 per logic function (improving functional upset rate per implemented operation) [85].
 1748 SRAM cell cross-sections measured at particle accelerators for 90-nm-class devices
 1749 (Virtex-4/5) are typically in the range of 10^{-14} – 10^{-13} cm²/bit at high LET, corre-
 1750 sponding to functional upset rates of 10^{-3} – 10^{-2} upsets per device per day in a rep-
 1751 resentative LEO orbit. Rakhman et al. characterized the SRAM configuration cells
 1752 of a Xilinx Kintex-7 FPGA under both heavy-ion and proton irradiation, finding a
 1753 saturation cross-section of approximately 5×10^{-14} cm²/bit and a proton cross-section
 1754 of $\sim 10^{-15}$ cm²/bit at 200 MeV [86].

1755 For nuclear plant environments, the neutron-induced upset rate at control-room
 1756 locations (fast-neutron flux $< 10^3$ n/cm²/s) is orders of magnitude lower than in
 1757 space and may be further reduced by modest concrete shielding. However, design-
 1758 basis accident scenarios that expose I&C systems to elevated fluences over hours to
 1759 days require that the system either be immune to configuration upsets or be capable
 1760 of detecting and correcting them within the plant’s safety response time.

1761 **3.6 Fault-Tolerant Mitigation Techniques**

1762 **3.6.1 Triple Modular Redundancy**

1763 The foundational technique for SEU mitigation in digital systems is triple mod-
 1764 ular redundancy (TMR), originally proposed by Lyons and Vanderkulk in 1962 for
 1765 improving the reliability of vacuum-tube computers [58]. In TMR, the protected
 1766 function is replicated three times, and a majority voter selects the output of the two
 1767 (or three) agreeing copies. A single SEU in any one replica is masked because the two
 1768 healthy copies outvote the corrupted one, and the voter output remains correct. The

1769 technique requires the voter itself to be robust, either because it is implemented in
 1770 radiation-hard technology or because the voter is a simple gate whose critical charge
 1771 greatly exceeds that of the storage elements it protects.

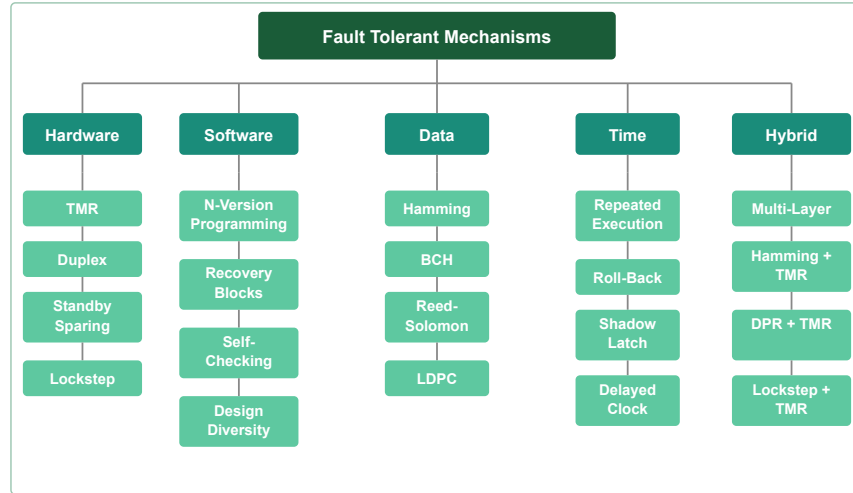


Figure 3.9: Overview of fault-tolerant mechanisms for SRAM-based FPGAs, ranging from single-level flip-flop TMR to full-system architectural redundancy. Each level adds correction capability at the cost of increased resource utilization [48].

1772 In FPGA implementations, TMR is applied at multiple granularities. *Full TMR*
 1773 triplicates every logic element, flip-flop, and routing resource in the design, provid-
 1774 ing maximum fault coverage at the cost of roughly $3\times$ resource usage and power.
 1775 *Partial TMR* applies triplication only to the most safety-critical paths, reducing
 1776 overhead while protecting against the most consequential upsets [84]. *Register-file*
 1777 *TMR* protects only the storage elements (flip-flops and block RAMs) while leaving
 1778 combinational logic unprotected, trading reduced area for acceptance of transient
 1779 SET-induced errors that are not captured [92]. Fig. 3.9 illustrates these trade-offs,
 1780 and Fig. 3.10 shows the voter logic in detail.

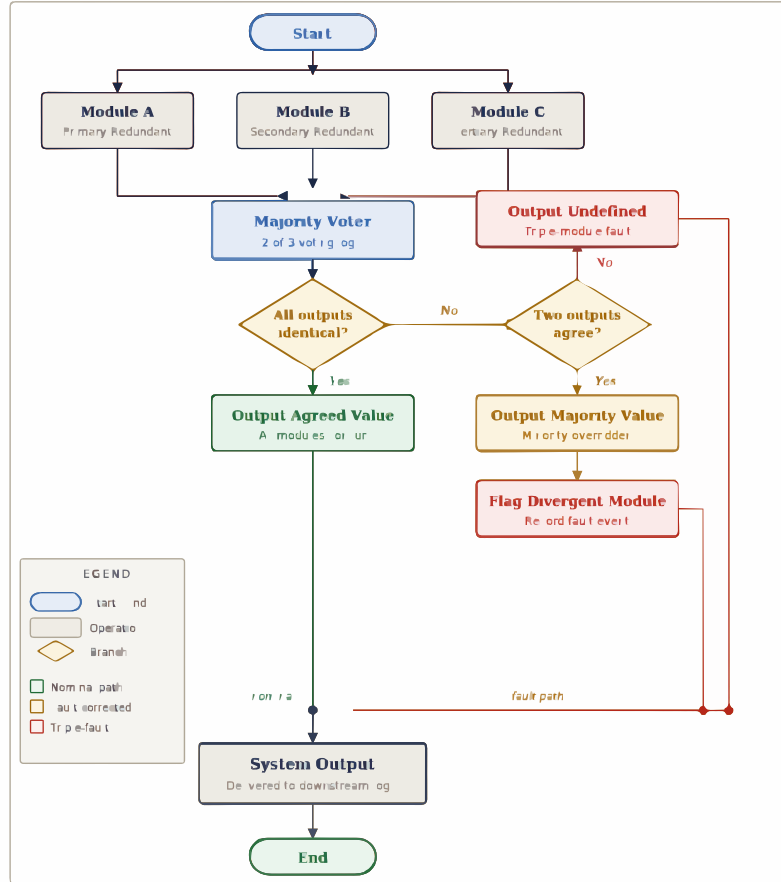


Figure 3.10: Voter logic flowchart for a TMR system. Each output bit from the three replicas is majority-voted independently; disagreement among replicas can be flagged as a fault indicator without interrupting correct output generation [58, 46].

1781 The TMR approach is well-established in space applications and has been demon-
1782 strated to reduce functional upset rates by approximately two orders of magnitude
1783 in Xilinx Virtex-class FPGAs at representative GCR fluences [110]. For nuclear I&C
1784 applications, Sciber et al. demonstrated a TMR-protected FPGA-based safety system
1785 for a research reactor that maintained correct output through simulated configuration
1786 upsets at a rate $500\times$ the expected plant environment level [29].

1787 3.6.2 Configuration Scrubbing

1788 TMR corrects the voter output but does not repair the configuration memory of
1789 the corrupted replica. After a sufficient number of upsets accumulate, two replicas
1790 may be simultaneously corrupted, defeating majority voting. Configuration scrubbing
1791 addresses this by periodically refreshing the FPGA's configuration memory from a
1792 known-good reference stored in non-volatile memory [9].

1793 In *external scrubbing*, an external processor reads the current configuration bit-
1794 stream from the FPGA, compares it bit-by-bit with the golden reference, and writes
1795 back corrected frames where differences are detected. This approach can detect and
1796 correct errors at rates determined by the scrub period, which must be short com-
1797 pared to the mean time between double upsets in the TMR replicas [38]. For a device
1798 with 10^7 configuration bits, a bit-upset rate of 10^{-3} upsets/bit/day, and a required
1799 double-upset probability below 10^{-6} per mission, the required scrub period is on the
1800 order of seconds — achievable with standard scrubbing controllers operating at JTAG
1801 bitstream rates [111].

1802 *Partial reconfiguration scrubbing* uses the FPGA vendor’s partial reconfiguration
1803 infrastructure to overwrite individual configuration frames without resetting the en-
1804 tire device, allowing scrubbing to proceed concurrently with normal device opera-
1805 tion [84]. Xilinx Virtex/UltraScale and Intel Stratix devices both support partial
1806 reconfiguration, and open-source scrubbing controllers (e.g., the Scrub Engine devel-
1807 oped at Brigham Young University) implement this approach [111]. Herrera-Alzu
1808 and López-Vallejo systematically characterized the scrubber performance for Xilinx
1809 Virtex-5 FPGAs, demonstrating that scrub periods below 50 ms reduced the effective
1810 functional upset rate below 10^{-6} upsets/day at LEO conditions [39].

1811 **3.6.3 Combined TMR-Scrubbing and Hybrid Approaches**

1812 For the highest reliability requirements, TMR and scrubbing are deployed to-
1813 gether: TMR provides instant correction of single upsets, while scrubbing prevents
1814 the accumulation of uncorrected errors that would eventually defeat TMR [48]. The
1815 combination reduces the effective uncorrectable upset rate to the probability of two
1816 upsets occurring within a single scrub period before the first is corrected — typically
1817 several orders of magnitude lower than either technique alone.

1818 Hybrid approaches extend these concepts to the system level. Stoddard et al.
1819 demonstrated a configuration in which an FPGA running a spacecraft attitude con-
1820 trol algorithm was protected by TMR at the register level, partial-reconfiguration
1821 scrubbing at the configuration level, and watchdog-timer reset at the system level,
1822 achieving an estimated mean time between failures of $> 10^5$ hours in a GEO radia-
1823 tion environment [94]. The architecture developed in Chapter ?? for nuclear plant
1824 monitoring adopts a similar layered approach, with TMR protecting the safety logic,

1825 periodic configuration scrubbing maintaining the configuration integrity, and a redun-
1826 dant processor monitoring the voter-disagreement signal to detect any uncorrected
1827 upset before it propagates to a safety output.

1828 **3.6.4 Error Detection and Correction in Block RAMs**

1829 Block RAMs (BRAMs) in FPGAs store application data — not configuration bits
1830 — and are equally susceptible to SEUs. Xilinx and Intel FPGAs provide optional
1831 built-in single-error correction, double-error detection (SECDED) Hamming codes for
1832 BRAM data, which correct single-bit upsets transparently and flag double-bit upsets
1833 as uncorrectable [61]. For nuclear applications where the BRAM stores measurement
1834 data or plant state variables, SECDED with periodic scrubbing provides an acceptable
1835 level of protection, while dual-port BRAM with cross-checking between two separate
1836 memory arrays provides an additional layer of protection against MCU events that
1837 span multiple bits within the same Hamming word.

1838 **3.7 COTS FPGAs in Nuclear and Radiation-Critical Applica-** 1839 **tions**

1840 **3.7.1 The Cost-Performance Trade-off**

1841 Purpose-built radiation-hardened FPGAs — such as the Microchip RTG4 (130-nm
1842 flash-based, inherently SEU-immune configuration) and the AMD/Xilinx XQRKU060
1843 (16-nm FinFET, rad-hard by design) — provide guaranteed TID tolerance exceeding
1844 300 krad(Si) and quantified SEL immunity, but at unit prices of \$50,000–\$500,000 and
1845 with logic densities one to two generations behind commercial equivalents [60, 3]. For
1846 nuclear plant I&C modernization programs that may require hundreds of processing
1847 nodes, this cost premium is prohibitive.

1848 Commercial off-the-shelf (COTS) FPGAs from the Xilinx 7-series, UltraScale, and
1849 Intel Cyclone/Arria families offer 50–100× lower cost per gate, higher logic density,
1850 and access to modern soft-IP ecosystems, at the cost of no radiation-hardness guar-
1851 antee from the manufacturer [13]. The engineering question is therefore not whether
1852 COTS FPGAs are radiation-tolerant, but whether their radiation response — com-
1853 bined with appropriate mitigation — can be characterized to a level of assurance
1854 sufficient for the safety classification of the application.

1855 **3.7.2 Demonstrated COTS FPGA Performance in Radiation** 1856 **Environments**

1857 Several accelerator and nuclear programs have demonstrated that COTS FPGAs
1858 with TMR and scrubbing can meet stringent reliability requirements at radiation
1859 levels consistent with nuclear plant I&C. Lee et al. irradiated a Xilinx Kintex-7
1860 FPGA (28-nm) with protons at TRIUMF and with neutrons at Los Alamos Na-
1861 tional Laboratory’s Weapons Neutron Research facility, measuring a total accumu-
1862 lated dose of 300 krad(Si) with no functional failure, and recording zero configuration
1863 upsets attributable to TID at that dose level [53]. Wirthlin et al. demonstrated a
1864 Leon3 soft-core processor implemented on a Kintex-7 FPGA that maintained cor-
1865 rect program execution through 3×10^{10} n/cm² of 14-MeV neutron irradiation with
1866 TMR+scrubbing active, corresponding to a ground-level exposure equivalent of sev-
1867 eral decades of control-room operation [112].

1868 The CERN high-energy physics community has deployed COTS FPGAs exten-
1869 sively in the LHC detector electronics, where annual doses in some locations reach tens
1870 of krad and the SEE environment from secondary shower particles is comparable to
1871 LEO. Systematic characterization programs at CERN show that 28-nm Kintex-7 and

1872 UltraScale devices exhibit no destructive effects (SEL, SEB, SEGR) at bias voltages
1873 within datasheet ratings at LET up to 62 MeV·cm²/mg, and that TMR+scrubbing
1874 reduces functional upset rates below the 10⁻⁹/hour target imposed by LHC safety
1875 systems [26].

1876 Pignol reviewed the use of COTS components in safety-critical space avionics
1877 and concluded that the combination of architectural mitigation (TMR, watchdog),
1878 physical mitigation (shielding, latchup current limiters), and comprehensive testing
1879 is sufficient to achieve failure-in-time rates competitive with purpose-built rad-hard
1880 parts, at substantially lower cost [81]. The same conclusion drives the DOE GEMINA
1881 program, which targets AI/ML-enabled COTS digital systems for advanced reactor
1882 I&C [100], and the ANS-recommended roadmap for I&C modernization at the existing
1883 LWR fleet [4].

1884 **3.7.3 Regulatory Pathway for COTS FPGAs in Nuclear Plants**

1885 The NRC’s regulatory framework for digital I&C in nuclear plants (10 CFR 50
1886 Appendix A, GDC 21–25; Regulatory Guide 1.152; NUREG/CR-6263) does not pro-
1887 hibit COTS devices but requires that the licensee demonstrate that the device meets
1888 the qualification requirements for its safety classification, including radiation qualifi-
1889 cation per IEEE Std 7-4.3.2 and post-accident monitoring per RG 1.97 [66]. The key
1890 qualification documentation for an FPGA-based safety function includes: (1) a ra-
1891 diation characterization data package demonstrating TID tolerance, SEL immunity,
1892 and SEU cross-section measurement; (2) a functional upset rate prediction for the
1893 plant-specific radiation environment; (3) a mitigation architecture description show-
1894 ing that the functional upset rate with TMR+scrubbing meets the specified single

1895 failure criterion; and (4) a test program demonstrating correct operation under the
 1896 combined effects of irradiation and functional loading [40, 65].

1897 3.8 Testing Methodologies for FPGA Radiation Qualification

1898 The qualification of FPGAs for radiation environments requires experimental mea-
 1899 surement of both TID and SEE parameters, since analytical prediction from process
 1900 parameters alone is not sufficiently accurate for safety-relevant applications. Fig. 3.11
 1901 illustrates the principal testing approaches and their relationship to the environment
 1902 being simulated.

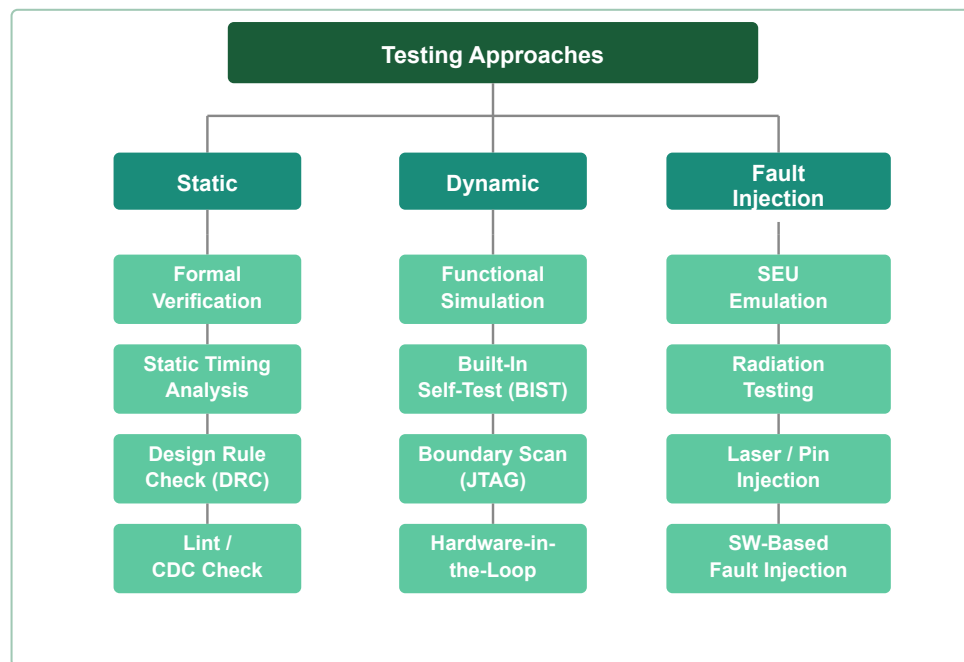


Figure 3.11: Summary of testing approaches for radiation hardness assurance of digital electronics, organized by radiation source, simulated environment, and measured effect. Each approach has characteristic advantages and limitations regarding representativeness, cost, and achievable fluence [89, 20].

1903 3.8.1 Heavy-Ion Testing

1904 Heavy-ion irradiation at particle accelerators is the primary technique for mea-
1905 suring SEE cross-sections as a function of LET and for identifying the SEL, SEB,
1906 and SEGR thresholds of a device. Facilities commonly used for FPGA radiation test-
1907 ing include the Brookhaven National Laboratory Tandem Van de Graaff (LET range
1908 1–100 MeV·cm²/mg), the Lawrence Berkeley National Laboratory 88-Inch Cyclotron
1909 (LET up to 85 MeV·cm²/mg), the Texas A&M K500 Cyclotron, and the GANIL
1910 facility in Caen, France [65].

1911 During a heavy-ion test, the device under test (DUT) is mounted in a vacuum
1912 chamber, biased and clocked at normal operating conditions, and exposed to a uniform
1913 beam of a selected heavy ion species at controlled LET and fluence. For FPGA
1914 testing, two complementary readout strategies are used: *static testing*, in which the
1915 device is irradiated with configuration memory pre-loaded and then readback to count
1916 configuration bit flips, and *dynamic testing*, in which the FPGA executes a test design
1917 that monitors its own outputs for errors in real time [85]. Dynamic testing captures
1918 both configuration upsets and user-register upsets but is limited in the depth of
1919 design validation per test run; static testing provides complete configuration memory
1920 coverage but misses upsets in the user logic flip-flops.

1921 3.8.2 Proton Testing

1922 Proton irradiation at energies of 50–200 MeV provides a practical and cost-effective
1923 alternative to heavy-ion testing for devices whose SEE cross-section vs. LET curve
1924 is required through the proton-relevant range, and for devices intended for use in
1925 proton-dominated environments (trapped proton belts, low-altitude orbits). Protons

1926 cause SEEs primarily through nuclear elastic scattering (which produces silicon or
1927 oxygen recoil nuclei with typical LET of 5–15 MeV·cm²/mg) and through non-elastic
1928 nuclear reactions (which produce heavy recoil fragments with higher LET) [80]. The
1929 resulting SEU cross-section is lower than the heavy-ion saturation cross-section by
1930 one to two orders of magnitude, but the high proton fluences available at accelerators
1931 like TRIUMF (Vancouver), IUCF (Bloomington), and the Paul Scherrer Institute
1932 (Villigen) allow statistically meaningful measurements [89].

1933 Proton testing also accumulates significant TID during the SEE test (at 100 MeV,
1934 10¹¹ p/cm² corresponds to ~1 krad(Si)), which must be accounted for in test planning
1935 to avoid confusing TID degradation with SEE events.

1936 3.8.3 Neutron Testing

1937 Neutron testing is the most directly relevant modality for nuclear plant appli-
1938 cations and for ground-level qualification [93]. The upset mechanisms for neutrons
1939 above ~10 MeV are similar to protons — nuclear elastic scattering producing silicon
1940 recoil nuclei plus non-elastic spallation reactions producing heavier fragments — but
1941 the neutron’s lack of charge means it produces no ionization along its own track and
1942 the DUT need not be operated in vacuum. This simplifies the experimental setup
1943 considerably.

1944 Several neutron irradiation facilities are available for FPGA testing. The Los
1945 Alamos Neutron Science Center (LANSCE) ICE House provides a quasi-monoenergetic
1946 neutron spectrum peaked at 800 MeV that is often used as a reference for atmospheric
1947 cosmic-ray testing [112]. The Indiana University Cyclotron Facility (IUCF) produced

1948 monoenergetic 200-MeV neutrons, now primarily replaced by the Triangle Universi-
1949 ties Nuclear Laboratory (TUNL) and the nTOF facility at CERN for intermediate
1950 energies. For reactor neutron environments, fission reactor irradiation facilities such
1951 as the MIT reactor’s fission converter beam [35] and the Ohio State University Re-
1952 search Reactor (OSURR) provide thermal-to-fast neutron spectra representative of
1953 ex-core I&C locations.

1954 The standard approach for neutron SEE testing of FPGAs follows the JEDEC
1955 JESD89A protocol: the device is irradiated with a known neutron fluence at the
1956 test facility’s reference spectrum, and the measured upset count is scaled to the
1957 terrestrial sea-level equivalent fluence using the spectral overlap integral [45]. Stirk et
1958 al. compared three neutron testing approaches (monoenergetic, white-spectrum, and
1959 reactor) for a complex system-on-chip and found that the accelerator white-spectrum
1960 test at LANSCE gave the closest agreement with reactor exposure results, while
1961 monoenergetic beam tests required more careful spectral correction [93].

1962 **3.8.4 Total Ionizing Dose Testing**

1963 TID testing is performed by exposing the device to a controlled dose from a ^{60}Co
1964 gamma source or an X-ray irradiator at a specified dose rate, with the device biased
1965 in a representative operating condition. The JEDEC/MIL-STD-883 Method 1019
1966 protocol specifies the irradiation conditions, readout intervals, and post-irradiation
1967 anneal procedure required for military and space qualification [89].

1968 A critical subtlety in TID testing is dose-rate dependence: many MOS devices
1969 exhibit enhanced low-dose-rate sensitivity (ELDRS), in which the damage at low dose
1970 rates ($<0.1 \text{ rad(Si)/s}$) exceeds that at high dose rates ($1\text{--}100 \text{ rad(Si)/s}$) for the same

total dose [32]. ELDRS occurs in bipolar transistors and certain MOS linear circuits but is less pronounced in digital CMOS; it is nonetheless a qualification concern for reactor I&C systems that accumulate dose over decades at rates of millirad per hour rather than the rad-per-second rates used in accelerated laboratory tests. The MIL-STD-883 Method 1019 elevated-temperature anneal procedure is designed to provide a conservative bound on ELDRS without requiring multi-year low-dose-rate exposures [91].

3.8.5 Ground-Level and Accelerated System Testing

For applications where the primary concern is the terrestrial cosmic-ray neutron environment, system-level testing at the actual deployment site or at a representative high-altitude location provides the most directly applicable qualification data [20]. However, at sea-level neutron flux ($\sim 10^{-2}$ n/cm²/s above 10 MeV), accumulating 10^{10} n/cm² — a fluence corresponding to several decades of ground-level exposure — would require $\sim 3 \times 10^{10}$ seconds, making direct ground-level testing impractical for qualification purposes. Accelerated testing at mountain-top observatories (e.g., Mauna Kea, 5 km altitude, flux $\sim 10\times$ sea level) or accelerator facilities bridging this gap.

Adell et al. and Artola et al. developed simulation frameworks using GEANT4 to predict ground-level neutron upset rates from device-level cross-section measurements, validating the approach against measured field failure data from commercial servers [1, 5]. The GEANT4 Monte Carlo toolkit [2], which models hadronic and

1992 electromagnetic interactions from eV to TeV energies, has become the standard sim-
1993 ulation infrastructure for predicting SEE rates in complex radiation environments
1994 including both atmospheric and reactor neutron spectra.

1995 **3.9 Summary**

1996 This chapter has established the physical and engineering framework for the
1997 FPGA-based nuclear instrumentation system developed in Chapter ???. The three
1998 radiation environments of concern — space, nuclear plant, and terrestrial cosmic-ray
1999 — each impose distinct TID and SEE demands on digital electronics. TID damage in
2000 MOS gate oxides accumulates throughout device operation and degrades transistor
2001 performance at doses that, for control-room I&C in nuclear plants, remain well within
2002 the tolerance of modern CMOS processes. The more consequential threat is stochastic
2003 single-event effects, which occur at arbitrarily low dose rates: SEUs corrupt FPGA
2004 configuration memory silently and persistently, SEFIs can interrupt device operation,
2005 and SELs can destroy devices if power cycling is not promptly applied.

2006 Triple modular redundancy, applied at the register and routing level within the
2007 FPGA fabric, corrects single upsets instantaneously without interrupting the safety
2008 function; periodic configuration scrubbing prevents upset accumulation that would
2009 eventually defeat TMR. The combination of these two techniques reduces the un-
2010 correctable upset probability to levels that can be demonstrated to meet the single-
2011 failure criterion of nuclear safety standards. COTS FPGAs from the 28-nm and 16-nm
2012 Kintex-7 and UltraScale families have been qualified for radiation environments sub-
2013 stantially more demanding than nuclear plant I&C through heavy-ion, proton, and
2014 neutron testing at national accelerator facilities, and the regulatory pathway for their

2015 use in safety-related applications is established in IAEA SSG-39 and NRC Regulatory
2016 Guide 1.152. The following chapter implements and experimentally validates these
2017 principles in a digital plant monitoring system built on a COTS Kintex-7 FPGA.

Chapter 4: $\text{Cs}_{0.1}\text{FA}_{0.9}\text{PbI}_3$ Perovskite Solar Cells as Ionizing Radiation Monitors

The preceding chapter established that halide perovskites combine the electronic properties requisite for radiation detection — high effective atomic number, adequate charge-carrier mobility-lifetime products, and tunable bandgap — with solution-processable fabrication routes inaccessible to conventional compound semiconductors. The work described in this chapter exploits a distinct but complementary strategy: rather than deploying a thick single crystal as a direct charge-collection detector, it repurposes a photovoltaic-architecture thin-film device as a *current-mode radiation monitor*. The rationale is twofold. First, modern perovskite solar cells have reached certified power-conversion efficiencies exceeding 27% [54], implying that the absorber layers have been optimized over a decade of intense materials research for the very properties — long carrier diffusion lengths, low trap-state densities, and efficient carrier extraction — that also govern sensitivity to ionizing radiation. Second, photovoltaic modules are inherently scalable in area, enabling the construction of sensor arrays whose geometric aperture is far larger than what is practical with single-crystal bulk detectors, and whose short-circuit current provides a direct analog output suitable for interfacing with standard nuclear instrumentation.

2036 The $\text{Cs}_{0.1}\text{FA}_{0.9}\text{PbI}_3$ composition is specifically chosen because partial substitution
 2037 of cesium for formamidinium suppresses the spontaneous $\alpha \rightarrow \delta$ phase transition of
 2038 pure FAPbI_3 while preserving its 1.48 eV bandgap and superior charge transport rela-
 2039 tive to MAPbI_3 [116]. The *pin* diode device structure (Section 4.1) further suppresses
 2040 dark current relative to a simple ohmic-contact geometry, and its inverted polarity
 2041 places the transparent electrode on the illuminated face — an arrangement that si-
 2042 multaneously enables photovoltaic operation under sunlight and direct exposure of
 2043 the perovskite absorber to externally incident radiation.

2044 This chapter reports visible-light temporal response measurements (Section 4.2),
 2045 X-ray response under pulsed and continuous illumination (Section 4.3), gamma-ray
 2046 response during continuous ^{137}Cs and ^{60}Co irradiation including a six-day and sixteen-
 2047 day stability study (Section 4.4), and steady-state neutron response measurements
 2048 performed at the Ohio State University Nuclear Reactor Laboratory (Section 4.5).

2049 4.1 Device Fabrication and Photovoltaic Performance

2050 Devices were fabricated by Prof. Jinsong Huang’s group at the University of North
 2051 Carolina at Chapel Hill using a standard inverted (*pin*) architecture developed for
 2052 high-efficiency perovskite photovoltaics. The composition of the absorber layer is
 2053 $\text{Cs}_{0.1}\text{FA}(\text{CH}_3\text{NH}_2)_{0.9}\text{PbI}_3$, where the 10 mol% Cs^+ substitution entropy-stabilizes the
 2054 cubic α -phase at ambient conditions as described in Section 2.1. Device fabrica-
 2055 tion proceeded by sequential solution deposition of the hole transport layer, the per-
 2056 ovskite precursor solution, the electron transport layer, and the top metal contact
 2057 onto glass/ITO substrates. The layer stack, illustrated in Figure 4.1, is ITO / hole
 2058 transport layer / $\text{Cs}_{0.1}\text{FA}_{0.9}\text{PbI}_3$ / electron transport layer / Ag.

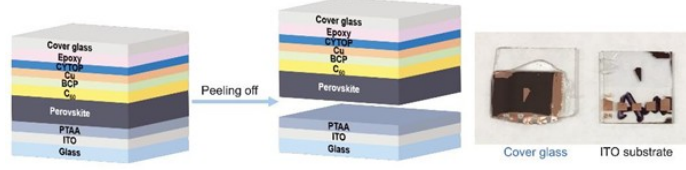


Figure 4.1: Cross-sectional schematic of the *pin* perovskite device structure used throughout this chapter. The ITO/hole-transport-layer (HTL) contact serves as the illuminated face for both photovoltaic and radiation-monitoring operation; the Ag back contact completes the $\text{Cs}_{0.1}\text{FA}_{0.9}\text{PbI}_3$ absorber sandwich.

Two device geometries were characterized: a small-area cell ($\sim 8 \text{ mm}^2$ active area) for high-throughput screening and an encapsulated module (63.7 cm^2 aperture area, four cells wired in series) intended for radiation monitoring applications. Representative current density–voltage (J–V) curves measured under 1-sun AM1.5G illumination are shown in Figure 4.2.

The small-cell PCE of 21.3% is consistent with champion efficiencies reported for the $\text{Cs}_{0.1}\text{FA}_{0.9}\text{PbI}_3$ composition in the contemporary literature [55], confirming that the devices received for radiation testing represent the current state of the art in perovskite photovoltaic fabrication. The module efficiency of 16.9% reflects the geometric fill-factor penalty inherent in a series-connected configuration and is comparable to certified module efficiencies from the same composition reported by independent groups [54].

4.2 Temporal Photoresponse Characterization

Before proceeding to ionizing radiation measurements, the time-domain photoresponse of the devices was characterized under visible light to establish the inherent

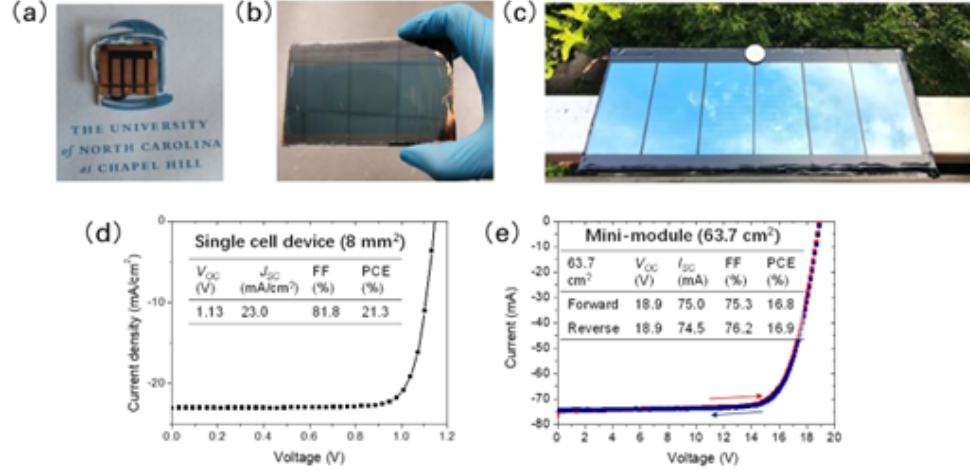


Figure 4.2: Photovoltaic performance of the $\text{Cs}_{0.1}\text{FA}_{0.9}\text{PbI}_3$ devices used in this study. (a) J-V curve for the 8 mm² small cell (PCE = 21.3%, V_{OC} = 1.13 V, J_{SC} = 23.0 mA cm⁻², FF = 81.8%). (b) J-V curve for the 63.7 cm² module (PCE = 16.9%, V_{OC} = 18.9 V, J_{SC} = 74.5 mA cm⁻², FF = 76.2%). All measurements performed at UNC Chapel Hill under AM1.5G illumination prior to shipment.

bandwidth of the detector and to identify any device-physics limitations on pulse-
following ability. Three complementary measurement modalities were employed: a
DC chopped-beam experiment to measure the turn-on and turn-off time constants, a
pulsed-laser experiment to probe the high-frequency limit, and an AC-coupled grid-
simulation experiment to assess the feasibility of power-grid synchronization using
the solar module.

4.2.1 DC Transient Response

The DC transient measurement used a 532 nm continuous-wave laser attenuated to
approximately 0.45 sun equivalent irradiance and modulated by a mechanical optical
chopper. Short-circuit current transients were acquired on a digital oscilloscope and
analyzed to extract the 10–90% rise time and the 90–10% fall time.

2085 The 8 mm² cell exhibited a rise time of approximately 33 ms, while the 28.6 cm²
 2086 module required approximately 60 ms to reach 80% of its peak short-circuit current
 2087 after the chopper opened. Both values are substantially longer than the RC time
 2088 constants computed from the device capacitance and load resistance, confirming that
 2089 the observed response time is limited by the chopper blade transit speed rather than
 2090 by any intrinsic device physics. This conclusion is consistent with the known sub-
 2091 microsecond free-carrier lifetimes in high-quality Cs-FA perovskite films [34].

2092 **4.2.2 Pulsed Laser Response**

2093 To access the true high-frequency bandwidth of the devices, a pulsed diode laser
 2094 (Thorlabs NPL41B, 405 nm central wavelength) was used in place of the chopped CW
 2095 source. The laser was operated at a repetition rate of 1 kHz with a pulse width of
 2096 20 ns (variable 6–38 ns range, up to 10 MHz maximum repetition rate). Photoresponse
 2097 pulses were amplified by a transimpedance amplifier with 10⁵ V A⁻¹ gain and 1 MHz
 2098 bandwidth before acquisition on a Keysight MSO-X-3104T oscilloscope.

2099 Two series-parallel module assemblies, denoted Cell A and Cell B, each consist-
 2100 ing of four Cs_{0.1}FA_{0.9}PbI₃ cells mounted on a printed breadboard, were character-
 2101 ized. Cell A exhibited an output pulse rise time of approximately 84 ns under these
 2102 conditions; Cell B produced a comparable waveform. The measured rise time is
 2103 dominated by the 1 MHz bandwidth of the transimpedance amplifier rather than
 2104 the device itself: the −3 dB amplifier bandwidth corresponds to a rise-time limit of
 2105 0.35/10⁶ Hz ≈ 350 ns, and the shorter observed value reflects the partial bandwidth
 2106 contribution of the oscilloscope front-end and cable parasitics. The intrinsic device

2107 response is therefore faster than the approximately 84 ns system-limited value, consis-
2108 tent with the sub-microsecond carrier lifetimes expected from the $\mu\tau$ products typical
2109 of $\text{Cs}_{0.1}\text{FA}_{0.9}\text{PbI}_3$ thin films reported in Section 2.4.2.

2110 4.2.3 AC Grid Simulation

2111 To evaluate whether the solar module could track the periodic voltage waveform
2112 produced by a single-phase power inverter — a potential application for verifying
2113 grid integrity or synchronizing a radiation monitor to a power-cycle reference —
2114 the four-cell module was connected to the input of a commercial single-phase DC-
2115 to-AC microinverter under indoor fluorescent illumination. The inverter consumed
2116 approximately 4 W at its rated load, but the module could deliver only ~ 1 W under
2117 the available illumination, leaving the inverter operating in a deeply underloaded
2118 regime. No distinct 50/60 Hz modulation of the module output current was discernible
2119 above the noise floor in this configuration. The negative result is attributed entirely
2120 to the power deficit rather than to any bandwidth limitation of the perovskite device:
2121 achieving an observable AC response would require either a larger-aperture module,
2122 a light concentrator, or a calibrated illumination source at close to 1-sun intensity.

2123 4.3 X-Ray Response

2124 X-ray characterization was performed using an Amptek Mini-X miniature X-ray
2125 tube with a gold anode, operated at tube voltages up to 50 kV. For continuous-
2126 wave measurements, the module was placed in the beam path and J-V characteristics
2127 were recorded as a function of X-ray tube voltage and current. Figure 4.3 shows the
2128 resulting I-V curves; the short-circuit current I_{SC} increases monotonically with tube
2129 voltage, consistent with the increasing photon fluence and mean photon energy.

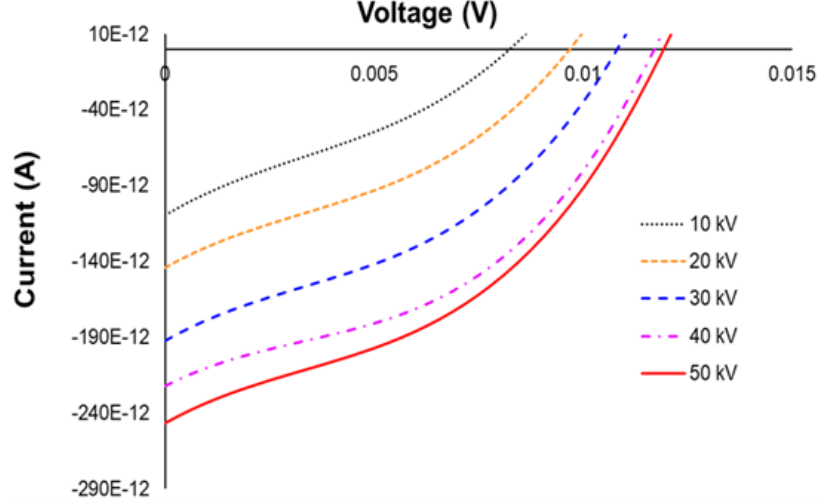


Figure 4.3: Current-voltage characteristics of the $\text{Cs}_{0.1}\text{FA}_{0.9}\text{PbI}_3$ module under Amptek Mini-X X-ray tube illumination at tube voltages from 10 kV to 50 kV. Inset: I_{SC} at 50 kV as a function of irradiation time, showing the characteristic slow decline associated with ion migration under the built-in field [97].

2130 The time-resolved I_{SC} at 50 kV, shown in the inset of Figure 4.3, exhibits the
 2131 characteristic slow-decline behavior previously reported for perovskite photovoltaic
 2132 devices under sustained illumination in the presence of a built-in electric field. The
 2133 decline is attributed to ion migration: mobile halide vacancies drift under the de-
 2134 vice's built-in field, accumulating at the charge-transport layer interfaces and partially
 2135 screening the collection field [97, 24].

2136 The absolute I_{SC} induced by the X-ray source is substantially smaller than that
 2137 observed in the silicon reference device operated under the same conditions, owing to
 2138 the much smaller active area of the perovskite module relative to the silicon photo-
 2139 diode used for comparison ($\sim 64 \text{ cm}^2$ vs. several hundred cm^2). On a per-unit-area
 2140 basis, the perovskite compares favorably due to its higher photoelectric absorption
 2141 cross section at the relevant photon energies.

For pulsed X-ray measurements, the Amptek Mini-X beam was modulated by a Thorlabs MC2000 optical chopper equipped with a 10-slot blade (760 μm stainless steel, slot width matched to achieve pulse frequencies from 10 Hz to 1 kHz). The pulsed response confirmed that the device faithfully follows the beam modulation at frequencies up to the chopper bandwidth, consistent with the pulsed laser measurements in Section 4.2.2.

4.4 Gamma-Ray Response

4.4.1 Measurement Setup and Dose Rate Calibration

Gamma-ray irradiation experiments were performed at the Ohio State University Nuclear Reactor Laboratory (OSU NRL) using two independent sources: a benchtop ^{137}Cs irradiator (six source rods, $E_\gamma = 662$ keV) and a ^{60}Co irradiator submerged in the reactor pool ($E_\gamma = 1.17$ and 1.33 MeV). Because the dose rate calibration values provided with each irradiator are given in units of roentgen per hour for air, conversion to the absorbed dose rate in the perovskite film was performed using the relation

$$\dot{D}_{\text{perovskite}} = 0.877 \times \frac{(\mu_{\text{en}}/\rho)_{\text{perovskite}}}{(\mu_{\text{en}}/\rho)_{\text{Si}}} \times \dot{X}, \quad (4.1)$$

where $\dot{X}$ is the exposure rate in roentgen per hour, 0.877 rad/R is the roentgen-to-rad conversion factor for soft tissue [51], and the ratio of mass energy-absorption coefficients (μ_{en}/ρ) was evaluated at the mean photon energy of each source using NIST XCOM tabulations [10]. This procedure yielded dose rates of 44.16 rad hr $^{-1}$ in the perovskite layer for the ^{137}Cs irradiator and 355.61 rad hr $^{-1}$ for the ^{60}Co source.

2162 4.4.2 Cesium-137 Irradiation

2163 The $\text{Cs}_{0.1}\text{FA}_{0.9}\text{PbI}_3$ module was inserted into and withdrawn from the ^{137}Cs irra-
 2164 diator while the short-circuit current was monitored in real time. Figure 4.4 shows a
 2165 time-series of I_{SC} recorded during a representative insert/withdraw cycle. A clear and
 2166 reproducible step change in I_{SC} is observed upon insertion, with the current increasing
 2167 by an amount proportional to the gamma-induced electron-hole pair generation rate
 2168 within the absorber. The induced current density at $44.16 \text{ rad hr}^{-1}$ was measured as
 2169 2.19 nA cm^{-2} , which represents the net gamma-to-current conversion sensitivity of
 2170 the device under the prevailing irradiation geometry.

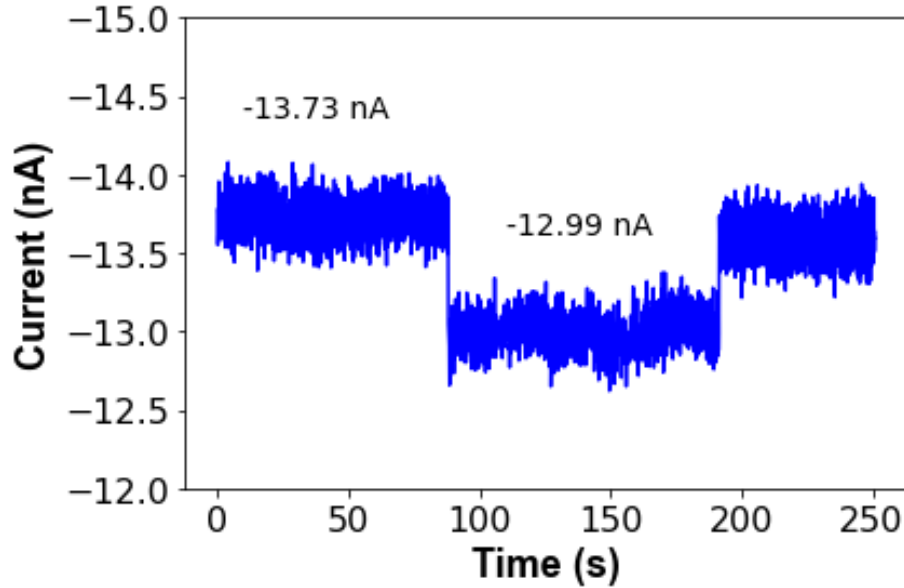


Figure 4.4: Short-circuit current I_{SC} of the $\text{Cs}_{0.1}\text{FA}_{0.9}\text{PbI}_3$ module as a function of time during insertion into and withdrawal from the ^{137}Cs irradiator (dose rate = $44.16 \text{ rad hr}^{-1}$ in perovskite). The step changes in I_{SC} provide an unambiguous current-mode detection signal.

2171 To assess the long-term stability of the perovskite module as a radiation monitor,
 2172 the device was subjected to continuous ^{137}Cs irradiation for six days without removal
 2173 from the source. The accumulated absorbed dose over the six-day period was com-
 2174 puted by integrating the dose rate over the irradiation time. Figure 4.5 shows the
 2175 evolution of I_{SC} as a function of cumulative absorbed dose. A linear decrease in I_{SC}
 2176 with dose is observed, with a slope of $-0.74 \text{ nA rad}^{-1}$, corresponding to a relative
 2177 degradation rate of approximately $[I_{\text{SC}}(0) - I_{\text{SC}}(D)] / I_{\text{SC}}(0)$ per unit absorbed dose.

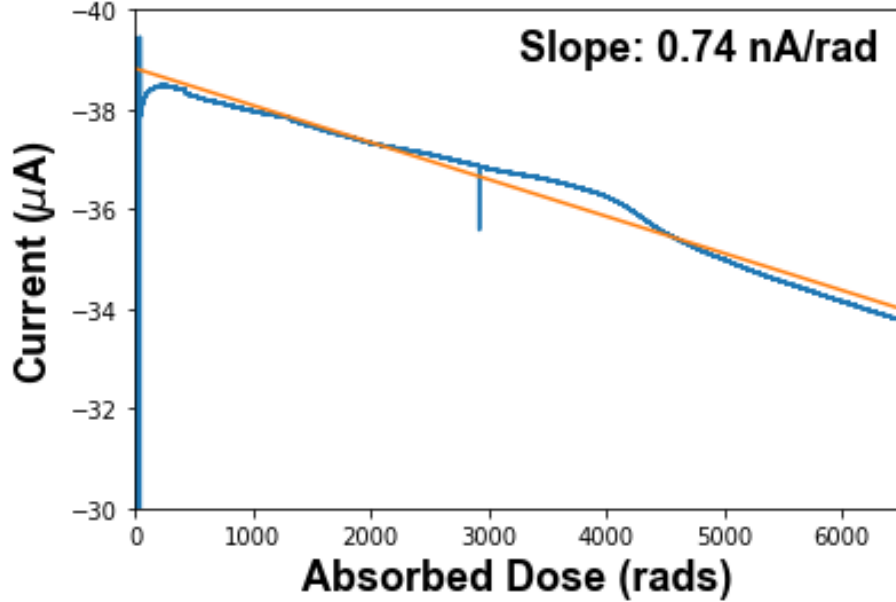


Figure 4.5: Long-term evolution of I_{SC} under continuous ^{137}Cs irradiation ($44.16 \text{ rad hr}^{-1}$) over six days. The linear decline in I_{SC} with accumulated dose is attributed to a combination of radiation-induced defect creation in the perovskite absorber and progressive reduction in the optical transmittance of the glass superstrate [16].

2178 The observed linear decline is consistent with the creation of electrically active
 2179 defects — primarily iodide Frenkel pairs — proportional to the absorbed dose, as

2180 has been documented for ^{60}Co irradiation of MAPbI_3 single crystals at doses up to
2181 1000 krad [16]. An additional contribution to the I_{SC} decline comes from cumula-
2182 tive darkening of the borosilicate glass superstrate, which reduces the visible-light
2183 transmittance and hence the photovoltaic current even in the absence of absorber
2184 degradation; this contribution has been documented separately [17].

2185 4.4.3 Cobalt-60 Irradiation

2186 The module was subsequently irradiated in the OSU NRL reactor pool using the
2187 underwater ^{60}Co irradiator at a dose rate of $355.61 \text{ rad hr}^{-1}$ in the perovskite layer —
2188 approximately eight times higher than the ^{137}Cs rate. The measurement geometry was
2189 otherwise identical: I_{SC} was recorded continuously throughout a 16-day irradiation.
2190 The induced current density at $355.61 \text{ rad hr}^{-1}$ was 3.56 nA cm^{-2} .

2191 Figure 4.6 shows the I_{SC} time series for the ^{60}Co irradiation. As with ^{137}Cs , a
2192 downward linear trend in the baseline I_{SC} is observed over the 16-day period, consis-
2193 tent with progressive defect accumulation proportional to absorbed dose.

2194 These results demonstrate that arrays of calibrated perovskite modules could, in
2195 principle, be deployed in a spatial configuration around a radiation source to trian-
2196 gulate the detonation location or measure the spatial dose distribution by comparing
2197 the I_{SC} responses of individual array elements. The linear dose–current relationship
2198 established here provides the calibration function necessary for such an application.

2199 4.5 Neutron Response

2200 Neutron response measurements were performed at the OSU Nuclear Reactor
2201 Laboratory using the facility’s Fast Beam Facility (FBF) beamport. The experimental
2202 geometry placed the perovskite module alternately outside the beam (background),

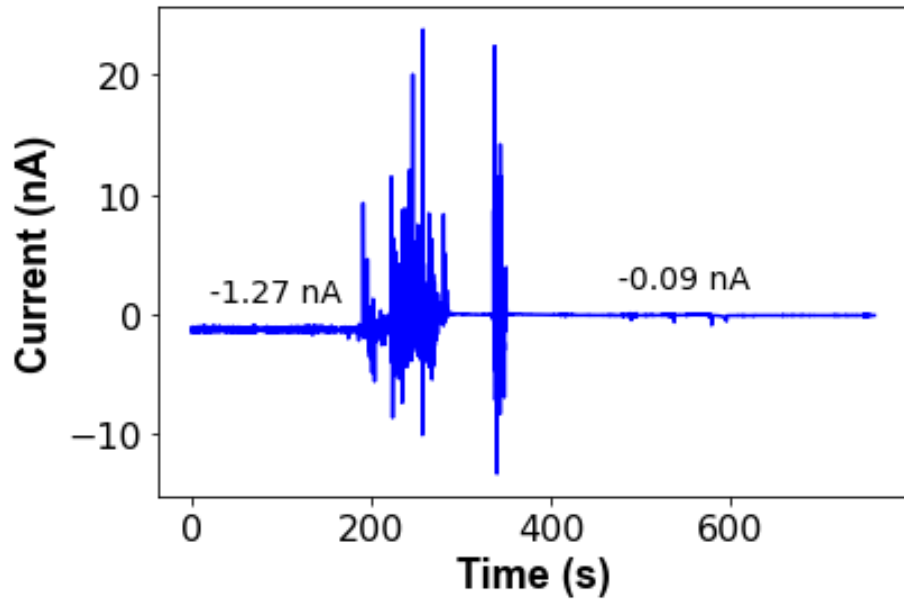


Figure 4.6: Short-circuit current I_{SC} of the module during 16-day continuous ^{60}Co irradiation (dose rate = $355.61 \text{ rad hr}^{-1}$ in perovskite). The downward trend mirrors that observed for ^{137}Cs (Figure 4.5) and scales approximately with the higher absorbed dose rate.

in the beam with the gamma-suppression filter closed, and in the beam with the filter open, enabling separation of the neutron-induced and gamma-induced current components.

4.5.1 Steady-State Reactor Measurements

Table 4.1 summarizes the short-circuit current I_{SC} measured under each condition, referenced to reactor power levels of 0 kW_{th} (reactor shutdown) and 450 kW_{th} (operating power).

Table 4.1: Steady-state short-circuit current of the Cs_{0.1}FA_{0.9}PbI₃ module at OSU NRL FBF. Uncertainties are $\pm 1\sigma$ over the measurement interval. “Gamma filter closed” and “gamma filter open” refer to the polyethylene/lead shielding assembly that preferentially attenuates the reactor gamma flux while transmitting fast neutrons.

Condition	Reactor Power	I_{SC} (pA)
Background (reactor off)	0 kW _{th}	51.29 ± 6.60
Background (reactor on)	450 kW _{th}	57.50 ± 7.39
Gamma filter closed	450 kW _{th}	57.62 ± 7.93
Gamma filter open	450 kW _{th}	59.77 ± 8.62

The data in Table 4.1 reveal a slight but statistically marginal increase in I_{SC} upon opening the gamma filter (from 57.62 ± 7.93 pA to 59.77 ± 8.62 pA). The difference of ~ 2.2 pA represents a $\sim 0.26\sigma$ effect relative to the combined quadrature uncertainty, which is not statistically significant at the 95% confidence level. The result is therefore classified as inconclusive: a neutron-induced contribution cannot be ruled out, but the signal-to-noise ratio of a single 8 mm²–63.7 cm² module is insufficient to resolve it above the fluctuating gamma and ambient backgrounds present at the FBF port.

2217 These results motivate the use of a larger-area array for future neutron sensing
2218 experiments. Scaling the active area by a factor of ten to twenty, combined with the
2219 differential signal extraction method described by Panaccione et al. [77], is expected
2220 to raise the neutron-to-background ratio above the detection threshold.

2221 4.5.2 Chopped-Beam Transient Measurements

2222 In an attempt to resolve a time-modulated neutron signal against the steady-state
2223 gamma background, the reactor beam was mechanically chopped at frequencies be-
2224 tween 1 Hz and 100 Hz using a borated polyethylene blade. Synchronous detection
2225 (lock-in analysis) of the I_{SC} transient was implemented in software. No neutron-
2226 induced current component was detected above the noise floor under any chopping
2227 frequency in this configuration, consistent with the steady-state analysis: the signal
2228 magnitude is simply too small relative to the electronic noise of the current mea-
2229 surement system at the flux levels available from the 450 kW_{th} reactor at the FBF
2230 port.

2231 4.6 Summary

2232 This chapter has demonstrated that photovoltaic-architecture Cs_{0.1}FA_{0.9}PbI₃ per-
2233 ovskite solar cells can serve as sensitive current-mode monitors for a broad spectrum
2234 of ionizing radiation. The key results are as follows.

2235 Temporally, the device bandwidth is limited by the measurement electronics rather
2236 than the intrinsic perovskite photoresponse: rise times of 33–84 ns were recorded
2237 depending on the amplifier configuration, and the actual carrier lifetime in the film
2238 is substantially shorter than the measured system rise time.

2239 Under X-ray irradiation (Amptek Mini-X, up to 50 kV), the module produces a
2240 measurable short-circuit current proportional to the beam intensity, with the char-
2241 acteristic slow I_{SC} decline associated with ion-migration field screening under the
2242 device's built-in field [97].

2243 Under ^{137}Cs gamma irradiation at $44.16 \text{ rad hr}^{-1}$, an induced current density of
2244 2.19 nA cm^{-2} was measured, and a distinct step response was observed upon inser-
2245 tion and withdrawal. Long-term (six-day) continuous irradiation revealed a linear
2246 decline of $\sim 0.74 \text{ nA rad}^{-1}$ attributable to cumulative defect creation and glass dark-
2247 ening [16, 17]. The ^{60}Co measurement at $355.61 \text{ rad hr}^{-1}$ yielded 3.56 nA cm^{-2} and
2248 an equivalent linear degradation trend over sixteen days, confirming the scalability
2249 of the current-mode detection approach across a dose-rate range spanning nearly an
2250 order of magnitude.

2251 Neutron response at the OSU NRL FBF was statistically inconclusive for a single
2252 module at 450 kW_{th} ; the signal-to-noise analysis indicates that scaling the detector
2253 array area by a factor of ten to twenty is necessary to resolve the neutron-induced
2254 current above the combined gamma and electronic noise backgrounds.

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Chapter 5: Perovskite Semiconductor Radiation Detectors

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The preceding chapter established that photovoltaic-architecture perovskite thin films can generate measurable current-mode signals in response to X-ray, gamma, and neutron irradiation, exploiting the device’s built-in electric field to sweep radiation-generated carriers to the contacts without any externally applied bias. The present chapter pursues a fundamentally different detection philosophy: deploying halide perovskites as direct-conversion semiconductor radiation detectors in which a large externally applied reverse bias sweeps charge carriers generated by individual radiation interaction events across the device and into a charge-sensitive readout chain. This mode of operation, familiar from conventional compound semiconductor detectors such as CdZnTe (CZT) and CdTe, offers the possibility of pulse-by-pulse energy spectroscopy rather than the integrated current monitoring that characterises the photovoltaic geometry of Chapter 4.

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Halide perovskites present a compelling case for direct-conversion detector development. Their heavy-metal cation sublattice — dominated by lead in the systems studied here — provides a high effective atomic number, giving large photoelectric cross sections at gamma and X-ray energies that are difficult or impossible to achieve with silicon or organic semiconductors. Some methylammonium-based and methylhydrazinium-based compositions additionally incorporate a high density

2274 of hydrogen atoms, enabling elastic proton-recoil reactions with fast neutrons that
2275 allow direct-conversion neutron detection without requiring a separate conversion
2276 layer. Solution-phase crystal growth methods developed over the past decade for
2277 photovoltaic research have produced centimetre-scale single crystals with carrier $\mu\tau$
2278 products approaching or exceeding those of polycrystalline CZT, a property critical
2279 for efficient charge collection across the millimetre-scale thicknesses required to stop
2280 MeV-energy photons and neutrons.

2281 The experimental program described in this chapter spans three radiation types
2282 and is organised by material system and detection objective. Section 5.4 treats poly-
2283 crystalline and thin-film X-ray detection, covering both the MAPbI₃-hBN device
2284 characterised in this work and the 2D perovskite (PEA₂MA₂Pb₃I₁₀) detector with
2285 5F-PEAI surface treatment reported by Tsai et al., in which the present author served
2286 as co-investigator [?]. Section 5.5 describes single-crystal bromide perovskite gamma-
2287 ray spectrometers, presenting the CsPbBr₃ and FAPbBr₃ devices and the fabrication
2288 challenges encountered with MAPbBr₃. Section 5.6 reports the novel organic-metal
2289 halide perovskite MHyPbCl₃ (methylhydrazinium lead trichloride), synthesised and
2290 tested as a fast neutron detector using the Fast Beam Facility (FBF) at the Ohio
2291 State University Research Reactor. Throughout all three application domains, a re-
2292 curring materials-physics challenge — the migration of halide ions under the large
2293 electric fields required for efficient charge collection — dominated the device perfor-
2294 mance and ultimately set the practical limits on applied bias, detector lifetime, and
2295 energy resolution. Section 5.7 draws together the ion migration observations from all
2296 three detector systems, discusses the mitigation strategies evaluated in this work, and
2297 identifies directions for future research.

2298 5.1 Perovskite Semiconductor Detector Principles

2299 A direct-conversion radiation detector converts the energy deposited by an ion-
2300 ising particle or photon directly into an electrical signal, without an intermediate
2301 scintillation step. When a gamma ray, X-ray, or charged particle deposits energy E
2302 in a semiconductor detector of bandgap E_g , approximately E/W electron-hole pairs
2303 are created, where $W \approx 2.8 E_g$ is the mean ionisation energy per pair [51]. The car-
2304 riers are swept by an applied electric field to opposite contacts, where they induce
2305 a charge pulse on the readout electronics. For spectroscopic operation, this pulse
2306 must be faithfully proportional to the deposited energy — a condition that requires
2307 complete charge collection and minimal trapping of carriers en route to the contacts.

2308 The probability that a carrier generated at depth x in the crystal (of thickness d)
2309 reaches the collecting contact is described by the Hecht equation,

$$\eta = \frac{\mu\tau E}{d} \left[1 - \exp\left(-\frac{d}{\mu\tau E}\right) \right], \quad (5.1)$$

2310 where $E = V/d$ is the applied electric field, V is the applied voltage, and $\mu\tau$ is the
2311 mobility-lifetime product of the carrier species (electrons or holes) that dominates
2312 collection [?]. Equation (5.1) shows that achieving high collection efficiency $\eta \rightarrow 1$
2313 requires either a large $\mu\tau$ product or a large applied field, or both. For a 1-mm-
2314 thick crystal with $\mu\tau = 10^{-3} \text{ cm}^2 \text{ V}^{-1}$, for instance, complete collection requires fields
2315 of order $10 \text{ V } \mu\text{m}^{-1}$, i.e. applied voltages of order 1 kV. This requirement for large
2316 applied voltages is in direct tension with the ion-migration instability described in
2317 Section 5.7.

2318 Two additional material requirements govern detector performance. First, the
 2319 dark resistivity ρ must be sufficiently high that the leakage current $I_{\text{dark}} = VA/(\rho d)$
 2320 (where A is the contact area) produces an electronic noise contribution smaller than
 2321 the signal charge from the smallest radiation event of interest. For gamma spec-
 2322 troscopy with CZT-like performance, $\rho > 10^9 \Omega \text{ cm}$ is typically required [56]. The
 2323 three-dimensional (3D) halide perovskites studied in this work achieve resistivities
 2324 in the range 10^9 – $10^{12} \Omega \text{ cm}$ in their highest-quality single-crystal forms, competitive
 2325 with or exceeding that of polycrystalline CZT. Second, the crystal must be thick
 2326 enough to stop a significant fraction of the incident radiation. The mean free path of
 2327 a 662-keV gamma ray (the primary emission of ^{137}Cs) in CsPbBr_3 is approximately
 2328 7 mm, meaning that crystals of order 5–10 mm are required for reasonable detec-
 2329 tion efficiency. Growing defect-free perovskite single crystals at these length scales
 2330 presents a significant fabrication challenge, discussed in Section 5.2.

2331 For fast neutron detection, the governing interaction is not photoelectric absorp-
 2332 tion but elastic proton recoil: a fast neutron of kinetic energy E_n collides with a
 2333 hydrogen nucleus (proton), transferring up to E_n of kinetic energy to the recoil pro-
 2334 ton in a head-on collision. The recoil proton then deposits its energy densely in
 2335 the crystal volume, generating a large number of electron-hole pairs. The detection
 2336 efficiency of a hydrogenous material is therefore proportional to its hydrogen atom
 2337 density n_H . Table 5.1 compares n_H for the key materials considered in this work.

Table 5.1: Hydrogen atom density comparison for fast neutron detector materials. MHyPbCl₃ achieves a hydrogen density comparable to organic scintillators despite being an inorganic-organic hybrid perovskite.

Material	Formula	Density (g cm ⁻³)	n_H (cm ⁻³)
Stilbene	C ₁₄ H ₁₂	1.16	5.67×10^{22}
Rubrene	C ₄₂ H ₂₈	1.27	3.62×10^{22}
EJ-276 plastic	C _n H _m	1.02	4.82×10^{22}
MHyPbCl ₃	CH ₆ N ₂ PbCl ₃	3.25	$\sim 4.0 \times 10^{22}$

5.2 Materials, Crystal Growth, and Device Fabrication

5.2.1 Perovskite Crystal Systems

Four distinct halide perovskite crystal chemistries were investigated as detector platforms. MAPbI₃ doped with hexagonal boron nitride (hBN) was used as an X-ray detector medium. The hBN inclusions were incorporated to improve grain boundary passivation and reduce the density of deep-level trap states that would otherwise act as recombination centres for radiation-generated carriers. The 3D bromide perovskites CsPbBr₃, MAPbBr₃, and FAPbBr₃ — where Cs, MA, and FA denote the inorganic caesium, methylammonium, and formamidinium A-site cations, respectively — were pursued as gamma-ray spectrometers on account of their wide bandgap (~ 2.3 eV), which provides the high dark resistivity essential for low-noise spectroscopy, their large effective atomic number driven by the Pb and Br sublattices, and their excellent carrier transport properties demonstrated in the photovoltaic literature [87, ?]. Finally, MHyPbCl₃ (methylhydrazinium lead trichloride), where MHy denotes the CH₃NH₂NH₂⁺ methylhydrazinium cation, was selected as a fast neutron converter on

the basis of its high hydrogen density, which places it on a par with conventional organic fast-neutron scintillators (Table 5.1), combined with the direct charge-collection advantage of a semiconductor [77].

5.2.2 Crystal Growth Procedures

MHyPbCl₃ Synthesis

Methylhydrazinium chloride (MHyCl) was first synthesised as the organic precursor salt. To obtain MHyCl, 2.63 mL (50 mmol) of methylhydrazine was diluted with 5 mL of anhydrous isopropanol (IPA), and 100 mL of 0.5 M HCl solution was added to the IPA mixture without stirring. After the combined solution was stored overnight in an inert atmosphere at room temperature, white plate crystals of MHyCl precipitated spontaneously. The crystals were filtered, washed with cold IPA, and dried under vacuum at room temperature, yielding 3.10 g (~38%).

Single crystals of MHyPbCl₃ were subsequently grown using an anti-solvent assisted crystallisation method. Two millimoles each of MHyCl and PbCl₂ were dissolved in a co-solvent mixture of 5 mL DMF and 1 mL DMSO. After stirring for 3 h, the solution was filtered through a 0.22- μ m syringe filter to remove undissolved particles, then transferred to a 20-mL vial. This vial was placed inside a 100-mL bottle containing 15 mL of toluene as the anti-solvent and sealed in a glovebox. After one week at room temperature, several clear hexagonal MHyPbCl₃ crystals formed. The crystals were harvested and dried under vacuum. Immediately prior to electrode deposition, the crystal surfaces were polished with polishing paper to remove the polycrystalline surface layer that accumulates during growth.

2375 **Bromide Perovskite Crystals**

2376 CsPbBr₃, MAPbBr₃, and FAPbBr₃ single crystals were grown externally and re-
2377 ceived by the research group for detector fabrication. Upon receipt, all crystals were
2378 inspected under optical magnification for internal cracks, inclusions, and colour uni-
2379 formity. Crystals judged to be free of macroscopic defects were then mechanically
2380 planarised using silicon carbide abrasive paper, followed by sequential mechanical
2381 polishing with diamond polishing paste down to a 10-nm grain size. The crystals
2382 were rinsed with a hexane bath between each polishing stage to remove residual abra-
2383 sive particles. The final polished faces were inspected for surface uniformity and
2384 flatness prior to contact deposition. One FAPbBr₃ crystal cracked during the polish-
2385 ing stage and was excluded from subsequent electrical testing; a bare CsPbBr₃ crystal
2386 sustained internal cracking from an accidental drop during handling, and while it re-
2387 mained electrically functional, the crack likely introduced additional carrier trapping
2388 sites and reduced the geometric solid-angle efficiency for spectroscopy measurements.

2389 **5.2.3 Contact Architecture and Deposition**

2390 Two distinct contact architectures were used, reflecting the different electronic
2391 requirements of the ohmic and rectifying contact geometries.

2392 **MHyPbCl₃ Ohmic Contact Device**

2393 For the MHyPbCl₃ fast neutron detector, an ohmic-ohmic contact configuration
2394 was adopted. On the lower face of the crystal, an electron-transport bilayer consisting
2395 of 30 nm of C₆₀ (fullerene) and 6 nm of bathocuproine (BCP) was deposited, followed
2396 by an 80-nm Cu top layer to complete the lower ohmic contact. The upper face of the
2397 crystal received a 50-nm Au contact deposited by thermal evaporation. The C₆₀/BCP

bilayer serves simultaneously as an electron-selective interlayer and as a buffer that reduces the density of interface states at the Cu-perovskite junction. Current-voltage characterisation confirmed ohmic, symmetric behaviour under both polarities of applied bias, consistent with the intended ohmic-ohmic device architecture.

Bromide Perovskite Schottky Devices

For the bromide perovskite gamma detectors, a Schottky (rectifying) contact architecture was pursued in order to create a depletion region that would both reduce the dark current and define the active detection volume of the device. A pixelated contact geometry was used, with a 2×2 array of 1.25-mm-diameter titanium pixel contacts surrounded by an integrated guard ring to intercept and divert surface leakage currents away from the individual pixel measurement channels. On the ohmic side of each crystal, a 50-nm gold layer was deposited by sputtering.

The Ti Schottky contacts were deposited using an AJA Orion RF/DC sputter deposition system. A shadow mask was drilled from an aluminium sheet to define the pixel and guard-ring geometry. The titanium was deposited at 450 W with a deposition rate of approximately $1.15 \text{ \AA s}^{-1}$, followed immediately by a 10-nm nickel passivating overlayer at 250 W (rate $\approx 2.1 \text{ \AA s}^{-1}$) to prevent atmospheric oxidation of the titanium surface upon venting. The gold ohmic contact was deposited on the opposite crystal face with a Denton sputter coater at a rate of $\sim 100 \text{ \AA per 90 s}$. Electrical connection was made to each pixel contact using copper wire leads bonded with silver epoxy, and the assembled device was mounted in a shielded enclosure for electrical characterisation.

2420 5.3 Electrical Characterisation

2421 5.3.1 Resistivity and Dark Current Stability

2422 Current-voltage (IV) characteristics for all devices were measured using a Keithley
2423 2612A SourceMeter. Sweep protocols ranged from the low-voltage regime (± 2 V at
2424 a logarithmic current resolution of 10^{-12} A to confirm ohmic behaviour and extract
2425 contact resistance) to high-voltage unipolar sweeps (0 to -100 V or 0 to -200 V) for
2426 resistivity extraction and leakage current characterisation under detector operating
2427 conditions. All sweeps were performed with the device in a light-tight enclosure to
2428 prevent photogenerated carrier contributions.

2429 The MHyPbCl₃ crystal yielded a resistivity of 4.43×10^{11} Ω cm, extracted from the
2430 slope of the ohmic low-field IV characteristic. This value places MHyPbCl₃ squarely
2431 in the range required for high-sensitivity radiation detectors and exceeds the resistiv-
2432 ity of commercial polycrystalline CZT by roughly two orders of magnitude, reflecting
2433 the low intrinsic carrier density achievable in high-quality perovskite single crystals.
2434 The CsPbBr₃ device exhibited a working leakage current in the range of 315–710 nA
2435 at 1000 V reverse bias. This current was not stable upon first application of bias:
2436 the leakage overshoot its quasi-steady-state value immediately after the voltage was
2437 stepped, then decayed gradually over a period of 45 minutes to one hour before ap-
2438 proaching a more slowly varying baseline. The overshooting behaviour is consistent
2439 with rapid redistribution of mobile halide vacancies upon application of the field,
2440 followed by slower re-equilibration of the resulting space-charge distribution. The
2441 FAPbBr₃ device demonstrated substantially lower leakage under high-voltage oper-
2442 ation, with a maximum leakage current of 5 ± 2.5 nA during continuous operation

2443 above 700 V, and showed no signs of electrical instability over more than five hours
2444 of sustained high-voltage testing.

2445 IV hysteresis measurements — sweeping the bias from forward to reverse and
2446 immediately back to forward at a scan rate of 20 mV s^{-1} — served as a sensitive diag-
2447 nostic for ion-migration-driven charge redistribution. The MHyPbCl₃ device exhib-
2448 ited a measurable sweep-versus-hold difference in the low-voltage regime, indicative
2449 of a polarisation effect driven by slow ionic relaxation. For the bromide perovskites,
2450 hysteresis was observed at all bias levels and was most pronounced at elevated bias,
2451 consistent with field-driven amplification of the ion-migration rate [?]. The mechanis-
2452 tic origins of this hysteresis and its consequences for detector operation are examined
2453 in detail in Section 5.7.

2454 5.3.2 Mobility-Lifetime Product

2455 The $\mu\tau$ product was extracted from photocurrent measurements under X-ray ex-
2456 citation as a function of applied bias, fitting the resulting data to Equation (5.1). For
2457 MHyPbCl₃, this procedure yielded $\mu\tau = 9.1 \times 10^{-3} \text{ cm}^2 \text{ V}^{-1}$, a value comparable to
2458 that reported for high-quality CsPbBr₃ single crystals in the literature [?] and ap-
2459 proximately an order of magnitude larger than typical values for polycrystalline CZT
2460 radiation detector grade material. The high $\mu\tau$ of MHyPbCl₃ indicates that carriers
2461 generated by neutron-induced proton recoil events can traverse centimetre-scale crys-
2462 tal thicknesses without significant trapping losses, provided the applied electric field
2463 is sufficiently large. Physically, the large $\mu\tau$ reflects the extended carrier diffusion
2464 lengths achievable in defect-lean single crystals grown by the anti-solvent method,

2465 where the slow precipitation kinetics produce crystals with lower intrinsic vacancy
 2466 concentrations than those formed by faster-nucleation techniques.

2467 **5.4 Polycrystalline X-Ray Detection**

2468 This section presents two polycrystalline perovskite detector systems developed
 2469 for X-ray sensing. The first is the MAPbI₃-hBN device fabricated and characterised in
 2470 this work, which demonstrated the role of atmospheric humidity in limiting detector
 2471 performance under applied bias. The second is the 2D perovskite (PEA₂MA₂Pb₃I₁₀)
 2472 detector with a fluorinated surface treatment, reported by Tsai et al. [?] with the
 2473 present author as co-investigator, which directly addressed the voltage-induced in-
 2474 stability problem through a hydrophobic 5F-PEAI barrier layer and achieved X-ray
 2475 sensitivities approaching 1000 $\mu\text{C} (\text{Gy}_{\text{air}} \text{cm}^2)^{-1}$.

2476 **5.4.1 MAPbI₃-hBN X-Ray Detector**

2477 **Experimental Setup and Dosimetry Calibration**

2478 X-ray characterisation was performed using a benchtop X-ray tube operated at
 2479 tube voltages of 30–50 kVp. The tube output was collimated to a conical beam with
 2480 an exit aperture of approximately 2 cm at the tube port. The primary detector under
 2481 X-ray evaluation in this subsection is the MAPbI₃-hBN device. The photocurrent
 2482 induced by X-ray exposure was measured as a function of applied bias using the
 2483 Keithley 2612A SourceMeter, with the X-ray beam switched on and off periodically
 2484 to distinguish the radiation-induced signal above the dark-current baseline. The X-
 2485 ray sensitivity S is defined as

$$S = \frac{\Delta I}{A \cdot \dot{D}}, \quad (5.2)$$

2486 where ΔI is the X-ray-induced change in current, A is the active contact area, and $\dot{D}$
 2487 is the X-ray dose rate incident on the detector. The dose rate was calibrated using a
 2488 RaySafe 452 dosimeter positioned in the beam. The RaySafe 452 contains 24 silicon
 2489 diodes and one Geiger-Müller tube distributed over a large active area, which intro-
 2490 duced a systematic calibration uncertainty when the detector active area was smaller
 2491 than the dosimeter sensing region. In particular, the conical beam geometry caused
 2492 the dose rate to vary across the dosimeter face, leading to an underestimation of the
 2493 dose rate incident on the small perovskite contact area. To address this, NanoDot
 2494 optically stimulated luminescence (OSL) dosimeters — which integrate dose over a
 2495 localised area comparable to the perovskite contact dimensions — were identified as
 2496 a more appropriate calibration tool for subsequent experiments.

2497 **X-Ray Sensitivity Results**

2498 Figure 5.1 shows the extracted X-ray sensitivity of the MAPbI₃-hBN device as a
 2499 function of applied bias voltage. At 5 V reverse bias the sensitivity was $0.107 \mu\text{C} (\text{Gy cm}^2)^{-1}$.
 2500 Increasing the bias to 10 V raised the sensitivity to a measured mean of $0.245 \mu\text{C} (\text{Gy cm}^2)^{-1}$
 2501 (two independent measurements yielding 0.245 and $0.252 \mu\text{C} (\text{Gy cm}^2)^{-1}$ at the same
 2502 bias, confirming reproducibility), and at 20 V the sensitivity reached $0.411 \mu\text{C} (\text{Gy cm}^2)^{-1}$.
 2503 The monotonic increase of sensitivity with applied bias is consistent with Equa-
 2504 tion (5.1): as the electric field increases, the carrier drift length grows, fewer carriers
 2505 are trapped before reaching the contacts, and the collected charge per unit deposited
 2506 energy increases correspondingly. The sub-linear growth of sensitivity with voltage
 2507 — a factor of ~ 3.8 increase in sensitivity for a factor of four increase in voltage —
 2508 suggests that the device has not yet entered the charge-collection saturation regime
 2509 at 20 V, and that further gains could be achieved at higher fields.

Table 5.2: X-ray sensitivity of the MAPbI₃-hBN detector as a function of applied bias. Sensitivity values are extracted from the ratio of the X-ray-induced current to the calibrated dose rate multiplied by the contact area (Equation 5.2).

Applied Bias (V)	Sensitivity ($\mu\text{C} (\text{Gy cm}^2)^{-1}$)
5	0.107
10	0.245
10	0.252
20	0.411

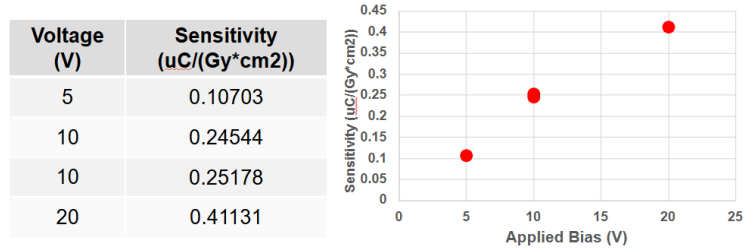


Figure 5.1: X-ray sensitivity of the MAPbI₃-hBN detector as a function of applied reverse bias. Sensitivity increases monotonically with field, consistent with improved charge collection efficiency described by the Hecht equation (Equation 5.1). Error bars represent the spread between repeated measurements at the same bias condition.

2510 A bi-polar X-ray response was observed at zero applied bias in some MAPbI₃-hBN
2511 device configurations, wherein a spontaneous photovoltaic-like current transient was
2512 generated by X-ray exposure without any external voltage source. This response is
2513 attributed to the asymmetric contact architecture, which creates a built-in potential
2514 difference between the hBN/MAPbI₃ and the Au contact interfaces. The built-in field
2515 sweeps radiation-generated carriers toward the respective contacts even in the absence
2516 of applied bias, producing a net photocurrent that switches polarity depending on the
2517 direction of the dominant carrier flow.

2518 **Vacuum Testing and Humidity Effects**

2519 The role of atmospheric humidity in X-ray detector performance was directly
2520 evaluated by repeating X-ray sensitivity measurements on the MAPbI₃-hBN device
2521 under vacuum conditions (base pressure $< 10^{-3}$ Torr) and comparing the results to
2522 measurements performed in ambient laboratory air. Figures ??-?? present the X-
2523 ray current transients at 5, 10, and 20 V applied bias under both conditions. In
2524 each case, the X-ray-induced current was larger and more stable under vacuum than
2525 under ambient atmosphere at matched bias, and the dark-current baseline was lower
2526 under vacuum. These improvements are consistent with the suppression of moisture-
2527 accelerated ion migration under vacuum. When water molecules are present at grain
2528 boundaries and crystal surfaces, they facilitate the diffusion of halide vacancies under
2529 the applied field, creating shunting pathways and broadening the internal electric
2530 field distribution in ways that reduce the effective field in the bulk and hence the
2531 charge collection efficiency. Removal of ambient moisture under vacuum disrupts this
2532 pathway and partially restores the intrinsic carrier transport properties of the crystal.

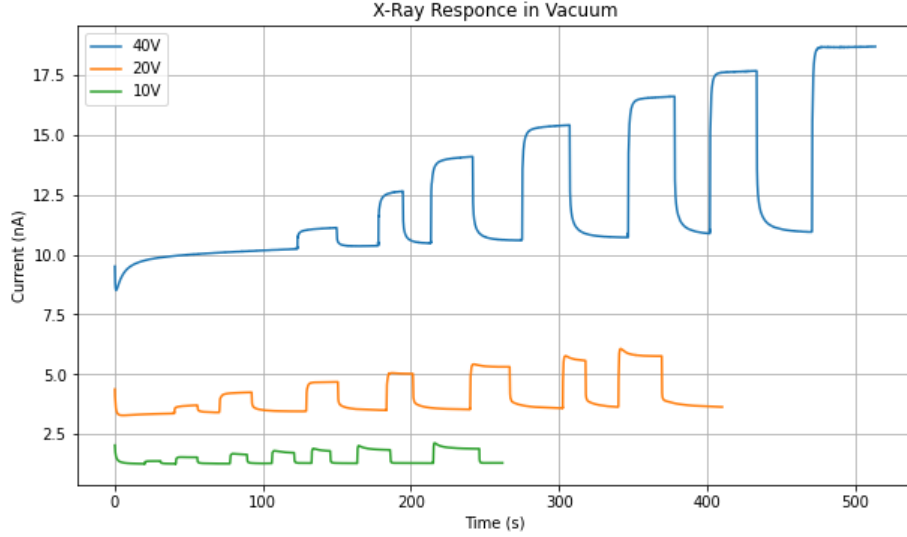


Figure 5.2: Comparison of X-ray current transients at 5, 10, and 20 V applied bias for the MAPbI₃-hBN device in ambient air (solid curves) and under vacuum (dashed curves). Vacuum operation consistently produces a larger and more stable radiation-induced current, attributed to suppression of moisture-accelerated ion migration.

2533 Despite the improvements afforded by vacuum operation, the MAPbI₃-hBN de-
 2534 vice ultimately failed after approximately five hours of continuous operation at 400 V
 2535 applied bias. Post-failure inspection showed that the device could no longer sustain
 2536 any significant reverse bias, consistent with the formation of a low-resistance shunting
 2537 pathway through the crystal bulk. This failure mode is attributed to the catastrophic
 2538 redistribution of ionic species under the large applied field: as mobile iodide vacan-
 2539 cies accumulate at the electrode interfaces, the local electric field at those interfaces
 2540 increases dramatically, eventually driving a decomposition of the perovskite lattice
 2541 into its constituent binary and ternary phases. Vacuum operation delays but does
 2542 not eliminate this failure mode.

2543 Charge Trapping and Sensitivity Degradation

2544 Beyond the acute device failure observed at 400 V, a more gradual form of sen-
2545 sitivity degradation was observed under continuous X-ray exposure at moderate bias
2546 levels, referred to here as charge trapping. During prolonged irradiation at -50 V,
2547 the X-ray-induced current was observed to decrease monotonically over the course of
2548 several minutes, even though the incident X-ray flux remained constant. A compara-
2549 ble decay was recorded at $+50$ V. A measurement of sensitivity performed after this
2550 trapping period showed a reduced value relative to the pre-trapping measurement at
2551 the same bias.

2552 The trapping phenomenon is attributed to the progressive filling of mid-gap trap
2553 states within the crystal. Radiation-generated carriers that become trapped at these
2554 sites are immobilised and unable to contribute to the measured current; additionally,
2555 the trapped charge builds an opposing space-charge field that reduces the net elec-
2556 tric field available for sweeping subsequently generated carriers to the contacts. This
2557 behaviour is distinct from, though mechanistically related to, ion migration: both
2558 processes reduce charge collection efficiency under sustained applied bias, but trap-
2559 ping occurs on a timescale of seconds to minutes governed by the carrier capture cross
2560 sections of the trap states, whereas ion migration proceeds on timescales of tens of
2561 minutes to hours governed by the diffusivity of the ionic species. The distinction has
2562 implications for mitigation: trap-state filling can in principle be reversed by briefly re-
2563 ducing the bias to zero (allowing trapped carriers to recombine or detrap thermally),
2564 whereas ionic redistribution is reversed only by allowing the crystal to equilibrate for
2565 extended periods, and may in extreme cases represent permanent structural damage
2566 to the crystal.

2567 Discussion

2568 The absolute sensitivity values measured for the MAPbI₃-hBN device — of order
2569 $10^{-1} \mu\text{C} (\text{Gy cm}^2)^{-1}$ at modest bias voltages — are substantially below those reported
2570 for optimised 2D perovskite thin-film detectors, as discussed in the following subsec-
2571 tion. The gap reflects the thick absorber geometry of the 3D crystal, which requires
2572 much larger absolute voltages to achieve equivalent electric fields compared to a 9- μm
2573 thin film. Nevertheless, the MAPbI₃-hBN results establish two critical findings: first,
2574 that the sensitivity follows the Hecht equation dependence on field, confirming that
2575 charge collection — not absorption — is the performance-limiting factor at these bias
2576 levels; and second, that vacuum operation demonstrably suppresses the moisture-
2577 accelerated degradation pathway, confirming the mechanistic link between humidity
2578 and voltage-induced instability that motivated the 5F-PEAI surface treatment strat-
2579 egy described next.

2580 5.4.2 2D Perovskite Detector with 5F-PEAI Surface Treat- 2581 ment

2582 The voltage-induced instability observed in the MAPbI₃-hBN device is not unique
2583 to 3D single crystals; it is a general challenge for all perovskite direct-conversion detec-
2584 tors that require sustained reverse bias. To address this instability in polycrystalline
2585 thin-film detectors, Tsai et al. developed a fluorinated surface treatment strategy that
2586 was reported in [?], with the present author contributing to device fabrication and
2587 X-ray characterisation. The key results of that work are summarised here to pro-
2588 vide context for the mitigation strategies discussed in Section 5.7 and to establish the

2589 state-of-the-art X-ray sensitivity achievable in perovskite thin-film detectors when the
2590 voltage instability problem is mitigated.

2591 **Device Structure and Material Characterisation**

2592 The 2D perovskite detector employed a p–i–n diode architecture, where the p-type
2593 contact was made with poly[bis(4-phenyl)(2,4,6-trimethylphenyl)amine] (PTAA) and
2594 the n-type contact was composed of a bilayer of fullerene (60 nm) and bathocuproine,
2595 BCP (10 nm). The detector was completed with a gold top contact without any
2596 additional encapsulation. The intrinsic perovskite layer targeted a $\text{PEA}_2\text{MA}_2\text{Pb}_3\text{I}_{10}$
2597 composition, where PEA denotes phenylethylammonium. Cross-sectional scanning
2598 electron microscopy (SEM) showed a continuous, compact film approximately 9 μm
2599 thick, with most of the film composed of large crystal grains extending through the
2600 film thickness and a relatively more disordered structure with smaller crystallites near
2601 the surface. Synchrotron grazing incidence wide-angle X-ray scattering (GIWAXS)
2602 confirmed high crystallinity, with strong discrete Bragg spots indicating preferred
2603 crystal orientations alongside weak ring-like Debye–Scherrer patterns from the smaller
2604 disordered crystallites near the surface.

2605 **Humidity-Dependent Instability**

2606 A systematic investigation of the relationship between ambient humidity and de-
2607 vice stability established that the severity of voltage-induced instability is directly
2608 correlated with the relative humidity (RH) of the operating environment. Devices
2609 were subjected to IV sweeps and bias-hold tests at controlled RH levels of 0%, 20%,
2610 and 40% in a sealed chamber.

At 0% RH (inert atmosphere), the devices exhibited no measurable I–V hysteresis across the tested voltage range and remained stable under sustained reverse bias to voltages beyond -15 V. As the RH was raised to 20% and then to 40%, the hysteresis in the forward-to-reverse sweep became progressively more pronounced, and the breakdown voltage decreased dramatically: devices at 40% RH failed irreversibly at bias voltages as low as -8 to -10 V, whereas their counterparts tested at 0% RH survived to voltages more than twice as large. Time-resolved dark current measurements at constant -5 V bias in 40% RH air showed that devices failed within two to three minutes, while the same devices at zero bias in 40% RH remained stable for over one hour. The combination of reverse bias and humidity was therefore identified as the critical driver of device failure, with moisture alone being insufficient to cause instantaneous breakdown.

5F-PEAI Surface Treatment and Results

To combat the moisture-accelerated breakdown, a fluorinated phenylethylammonium iodide (5F-PEAI) barrier layer was applied to the perovskite surface. The 5F-PEAI molecule replaces all five hydrogen atoms on the phenyl ring of PEAi with fluorine atoms, increasing the surface energy and producing a hydrophobic barrier. After incorporation with 5F-PEAI, the water contact angle of the perovskite surface increased from 66.1° to 76.0° (approximately 15%), confirming a higher surface energy.

Time evolution of the dark current at constant -5 V bias and RH = 40% showed striking differences between treated and untreated devices. The unprotected device

2633 exhibited large dark current fluctuations after two minutes, followed by device shunt-
2634 ing with sudden current jumps of three orders of magnitude. In contrast, the 5F-
2635 PEAI-protected device remained stable over 30 minutes of continuous biasing under
2636 the same conditions.

2637 The 5F-PEAI concentration was varied to optimise the protection. At 1 mg mL^{-1} ,
2638 the coating provided negligible improvement, possibly due to non-uniform coverage.
2639 At 2 mg mL^{-1} , the device survived up to -14 V with current fluctuations appear-
2640 ing only near the highest bias, approaching its limit at -16 V ($1.8 \text{ V } \mu\text{m}^{-1}$). At
2641 4 mg mL^{-1} , the device sustained up to -16 V without any sign of failure. Time-of-
2642 flight secondary ion mass spectroscopy (ToF-SIMS) fluorine mapping revealed that
2643 at the higher concentration, fluorine atoms penetrated below the surface and diffused
2644 through the grain boundaries, providing the comprehensive moisture protection re-
2645 sponsible for the enhanced breakdown threshold.

2646 Overlaying the I-V curves of all four devices before their breakdown thresholds
2647 revealed that the 2 mg mL^{-1} treatment reduced the dark current in the reverse bias
2648 regime while leaving forward injection unaffected, indicating surface passivation with-
2649 out interfering with carrier transport. The 4 mg mL^{-1} treatment produced a reduced
2650 forward injection current, indicating that a thicker 5F-PEAI layer introduces an en-
2651 ergy barrier at the interface that becomes more insulating.

2652 **X-Ray Detection Performance**

2653 Photodetection and X-ray characterisation were performed on the 5F-PEAI-treated
2654 devices. External quantum efficiency (EQE) measurements under visible-light illu-
2655 mination showed that all devices exhibited vanishing efficiency near zero bias due to
2656 their thick absorber volumes (approximately $8\text{--}9 \text{ } \mu\text{m}$), where a significant fraction of

carriers recombine before reaching the contacts. As the reverse bias increased from 0 to -5 V, EQEs rose rapidly from 3–6% to approximately 15%. At higher reverse bias (-10 V), the uncoated device’s EQE saturated toward 30%, while the 5F-PEAI-coated devices continued to grow, ultimately reaching 50%. The lower efficiency at low bias for the thicker 5F-PEAI coating is understood as a consequence of the charge extraction barrier formed by the insulating fluorinated layer at the interface.

X-ray sensitivity was measured using an Amptek mini-X-ray tube with an Ag target, producing characteristic radiation at 22.1 keV and a maximum bremsstrahlung energy of 50 keV. The X-ray tube was cycled on and off at increasing tube currents, tuning the dose rate from 2 to 70 $\mu\text{Gy}_{\text{air}} \text{s}^{-1}$. Sensitivity was extracted from the slope of the linear regression between the X-ray-induced photocurrent and the dose rate. At a bias of 4 V, the 5F-PEAI-treated device achieved a higher average sensitivity of 862.5 $\mu\text{C} (\text{Gy}_{\text{air}} \text{cm}^2)^{-1}$ compared to 726.2 $\mu\text{C} (\text{Gy}_{\text{air}} \text{cm}^2)^{-1}$ for the reference device. The maximum reported sensitivity at 4 V was 996.3 $\mu\text{C} (\text{Gy}_{\text{air}} \text{cm}^2)^{-1}$ for the 5F-PEAI device and 828.3 $\mu\text{C} (\text{Gy}_{\text{air}} \text{cm}^2)^{-1}$ for the reference, confirming the superior performance and stability of the treated device. The five electrode pixels on the 5F-PEAI device consistently reached 6–7 V with measurable sensitivities, whereas the reference device pixels could only operate at 5–6 V before exhibiting sharply increased noise.

Mechanistic Understanding

The mechanism by which 5F-PEAI suppresses the moisture-accelerated breakdown was elucidated through photoluminescence (PL) mapping and first-principles

quantum chemical simulations. PL peak-position maps acquired after voltage stress-
ing at -1 , -5 , and -7 V for 30 minutes showed that the unprotected device devel-
oped pronounced spatial inhomogeneity in its PL wavelength and intensity after -7 V
stress, with local hot spots of shifted emission indicative of ion-migration-driven phase
segregation. The PL peak distribution of the untreated device shifted toward lower
energy and broadened significantly, consistent with defect creation and the formation
of lower- n -numbered phases through field-driven phase segregation. In contrast, the
5F-PEAI-treated device maintained a sharp PL distribution centred near 735 nm even
after identical voltage stress.

DFT calculations of the water migration pathway through the organic layers re-
vealed a key mechanistic difference. On the PEA-covered surface, the activation
energy for water penetration was only approximately 5 meV — far below the thermal
energy at room temperature — meaning water molecules percolate through the or-
ganic capping layer essentially without barrier. Moreover, water became more stable
(by approximately 58 meV) in the vicinity of the iodide atoms on the inorganic Pb/I
surface, thermodynamically favouring moisture ingress. In contrast, the fluorinated
5F-PEA layer raised the activation barrier for water penetration to approximately
120 meV. The fluorinated molecules make the surface predominantly electronegative,
so approaching water molecules encounter strong electrostatic repulsion between oxy-
gen and fluorine atoms. This computational result directly explains why the 5F-PEAI
coating prevents the moisture-accelerated degradation observed in both the 2D de-
tector and, by extension, the 3D MAPbI₃-hBN device tested in this work.

2701 Discussion

2702 The absolute sensitivity values measured for the MAPbI₃-hBN device (of order
2703 $10^{-1} \mu\text{C} (\text{Gy cm}^2)^{-1}$) are three to four orders of magnitude below those demonstrated
2704 for the 5F-PEAI-treated 2D perovskite detector (approaching $10^3 \mu\text{C} (\text{Gy}_{\text{air}} \text{cm}^2)^{-1}$).
2705 This gap reflects fundamental differences in device geometry rather than intrinsic ma-
2706 terial quality: the 2D perovskite's $\sim 9\text{-}\mu\text{m}$ thin-film geometry charges and discharges
2707 rapidly under modest applied fields, whereas the thick 3D crystal requires much larger
2708 absolute voltages to achieve equivalent electric field strengths. The 2D detector also
2709 benefits from a built-in p-i-n junction that provides efficient carrier extraction even
2710 at low bias.

2711 The 5F-PEAI surface treatment demonstrated by Tsai et al. establishes a clear
2712 pathway for stabilising polycrystalline perovskite detectors against the moisture-
2713 accelerated breakdown that limited the MAPbI₃-hBN device in this work. The combi-
2714 nation of enhanced breakdown voltage (up to -16 V at 40% RH, compared to -10 V
2715 for the untreated device) and high X-ray sensitivity ($\sim 1000 \mu\text{C} (\text{Gy}_{\text{air}} \text{cm}^2)^{-1}$) makes
2716 the 5F-PEAI-treated detector competitive with the highest reported sensitivities for
2717 polycrystalline perovskite detectors of comparable absorber thickness. Extending this
2718 passivation strategy to the 3D single crystal systems used for gamma-ray and neutron
2719 detection in the following sections is identified as a high-priority direction for future
2720 work.

2721 5.5 Single-Crystal Gamma-Ray Detection

2722 This section presents the gamma-ray spectroscopy results obtained from the bro-
2723 mide perovskite single-crystal detectors. The three crystals — CsPbBr₃, FAPbBr₃,

2724 and MAPbBr₃ — share the same Schottky contact architecture and readout chain but
2725 produced markedly different outcomes, ranging from proof-of-concept spectroscopy
2726 (FAPbBr₃) to complete fabrication failure (MAPbBr₃). Each crystal system is dis-
2727 cussed in turn below.

2728 5.5.1 Experimental Setup and Acquisition Chain

2729 Gamma-ray spectroscopy experiments were performed using two isotopic sources:
2730 a ¹³⁷Cs source emitting 662-keV photons, and a ⁵⁷Co source emitting characteristic X-
2731 rays and gamma rays at 14, 122, and 136 keV. The measurement electronics consisted
2732 of a CAEN A1422 charge-sensitive preamplifier connected directly to the detector
2733 pixel contacts via copper wire leads, followed by an Ortec 926 multichannel analyser
2734 (MCA) shaping amplifier with a 10- μ s shaping time constant, and a CAEN N6724 16-
2735 bit analogue-to-digital converter (ADC) with 16,384 channels (16k ADC) for digital
2736 pulse height analysis. The 10- μ s shaping time was selected to be long relative to
2737 the expected carrier transit time across the millimetre-scale crystal, ensuring that
2738 the shaped output pulse height is proportional to the total collected charge (i.e.
2739 minimising ballistic deficit). Digitiser settings were: channel threshold at 15 channels,
2740 trigger hold-off of 1500 ns, input rise time of 1200 ns, trapezoid rise time of 8 μ s,
2741 trapezoid flat-top of 5 μ s, and energy fine gain of 10. High voltages were supplied
2742 by a regulated power supply; the pixel contact and guard ring contacts were biased
2743 independently to allow guard-ring isolation experiments to be performed.

2744 The experimental geometry for gamma spectroscopy placed the source at a fixed
2745 distance from the detector face, with the geometry defined by the active pixel area.
2746 Because the crystals used in this work were of order 2–3 mm in thickness and 5–8 mm

in lateral dimension, the geometric efficiency for 662-keV gamma rays was inherently low: the mean free path of a 662-keV photon in CsPbBr₃ ($Z_{\text{eff}} \approx 45$, $\rho \approx 4.5 \text{ g cm}^{-3}$) is approximately 7 mm, meaning that fewer than 50% of photons traversing the crystal along its thin dimension undergo an interaction.

5.5.2 CsPbBr₃

Characterisation

The CsPbBr₃ device exhibited a working leakage current in the range of 315–710 nA at 1000 V reverse bias, as detailed in Section 5.3.1. The leakage current was not stable upon first application of bias, showing a characteristic overshoot followed by a gradual decay over 45 minutes to one hour. This overshooting behaviour, consistent with redistribution of mobile halide vacancies upon application of the field, introduced additional electronic noise during the stabilisation period and limited the effective acquisition window for spectroscopy.

Results

The CsPbBr₃ detector was operated at 1000 V reverse bias for ¹³⁷Cs spectroscopy measurements. A comparison of count rates acquired with the source present versus background-only conditions yielded a source count rate of 17.45 counts s⁻¹ against a background of 9.11 counts s⁻¹, confirming a statistically significant response to the 662-keV source. However, no resolvable photopeak at the expected 662-keV channel position was observed in the accumulated pulse-height spectrum. The expected photopeak channel was estimated from a CZT energy calibration reference and the amplifier gain settings to fall at approximately channels 300–400 in the 16k ADC.

2769 Instead, the acquired spectrum displayed a broad continuum of counts at low chan-
 2770 nels consistent with incomplete charge collection and a spurious artefact near channel
 2771 16,382 (the maximum channel of the ADC), which was attributed to ADC saturation
 2772 or ringing from large current transients generated during ion migration events.

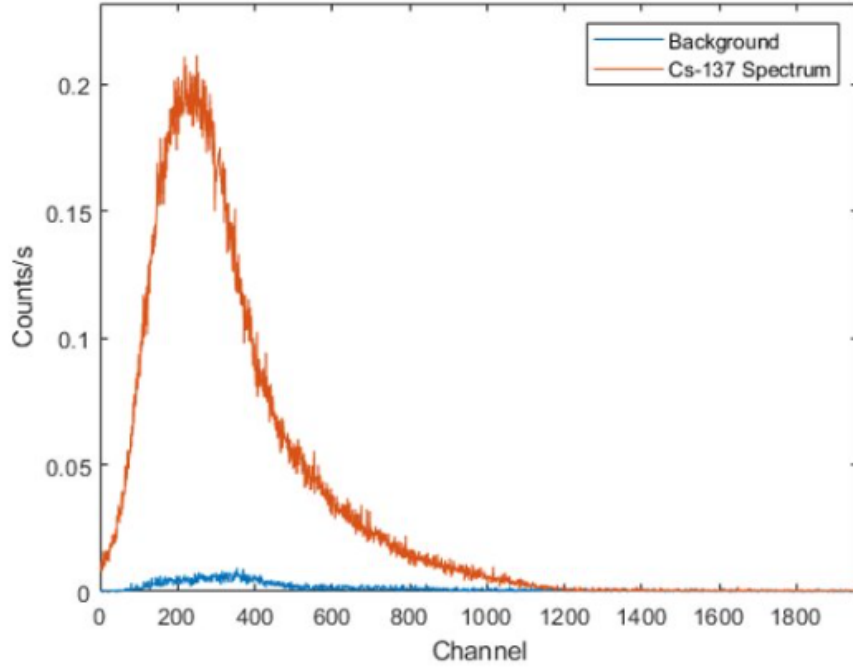


Figure 5.3: Pulse-height spectrum acquired from the CsPbBr₃ detector at 1000 V reverse bias with the ¹³⁷Cs source present (solid line) and background-only (dashed line). A clear elevation in total count rate is observed with the source, but no resolved photopeak is apparent due to the high electronic noise from the large device leakage current. The anomalous feature near channel 16,382 is attributed to ADC saturation from ion-migration-induced current spikes.

2773 A Geant4 Monte Carlo simulation of the CsPbBr₃ crystal geometry exposed to
 2774 ¹³⁷Cs (662-keV) gammas in a 4 π source configuration was performed to compute
 2775 the theoretical energy deposition spectrum. The simulated spectrum showed the

characteristic features expected for a small-volume Compton spectrometer: a Compton continuum, a Compton edge, and a full-energy peak whose intensity relative to the Compton continuum reflects the small crystal volume and the resulting high Compton-to-photoelectric ratio at 662 keV in a medium-Z material.

Discussion

The primary reason for the absence of a resolved photopeak is the high electronic noise contribution from the leakage current of the device. At 1000 V reverse bias, the CsPbBr₃ device drew a working leakage current of 315–710 nA during the period immediately following bias application. The root-mean-square (RMS) noise generated by a leakage current of this magnitude is far larger than the signal charge deposited by a single 662-keV gamma photon (approximately 0.4 pC for a 3-mm-thick CsPbBr₃ crystal with unity charge collection efficiency), rendering spectroscopic pulse resolution impossible during the transient stabilisation period. Even at quasi-steady-state, the residual leakage and its associated shot noise remained the dominant contribution to the equivalent noise charge of the preamplifier-detector system at the 10- μ s shaping time. Comparison of the simulated and measured spectra confirmed that the measured spectrum was dominated by electronic noise rather than any intrinsic limitation of the CsPbBr₃ detector material: the resolution-limiting factor was the large, unstable leakage current and its noise contribution rather than the charge transport properties of the crystal itself.

5.5.3 FAPbBr₃

Characterisation

The FAPbBr₃ single-crystal detector, with a crystal thickness of 2.35 mm and contact area of $5.2 \times 6.15 \text{ mm}^2$, demonstrated substantially better high-voltage stability than the CsPbBr₃ device. Electrical characterisation was first performed with IV sweeps from -20 to $+20 \text{ V}$ and from -200 to $+200 \text{ V}$ to assess the device leakage and confirm acceptable Schottky behaviour. A bias stability test at -200 V over a six-hour period showed that the leakage current reached a pseudo-steady-state with a maximum measured value of $5 \pm 2.5 \text{ nA}$ during high-voltage operation, approximately sixty times lower than the CsPbBr₃ leakage at comparable bias. The lower leakage in FAPbBr₃ at the same applied bias is attributable to the better Schottky barrier quality achieved with the Ti contact on this particular crystal, which created a wider depletion region and a steeper band-bending profile at the metal-semiconductor interface.

Results

Spectroscopic measurements with ^{137}Cs were performed at bias voltages from 700 V upward, as shown in Figure 5.4. At 700 V , the spectrum showed a broad peak feature centred near the expected channel position for the 662-keV photopeak, with a count rate of approximately $17.45 \text{ counts s}^{-1}$ above a background of $9.11 \text{ counts s}^{-1}$. At 1600 V bias, a simultaneous acquisition with both the ^{137}Cs (662-keV) and ^{57}Co (122-keV) sources produced a spectrum in which two distinct features could be identified at channel positions consistent with the expected photon energies when calibrated against a reference CZT detector. Figure 5.4b compares the ^{137}Cs and ^{57}Co spectra

2819 acquired at 1600 V. The count rates during acquisition were 9.11 counts s⁻¹ for back-
 2820 ground, 17.45 counts s⁻¹ for ¹³⁷Cs, and 26.60 counts s⁻¹ for ⁵⁷Co, reflecting the higher
 2821 specific activity of the ⁵⁷Co source used (100 μCi vs. 5 μCi for ¹³⁷Cs).

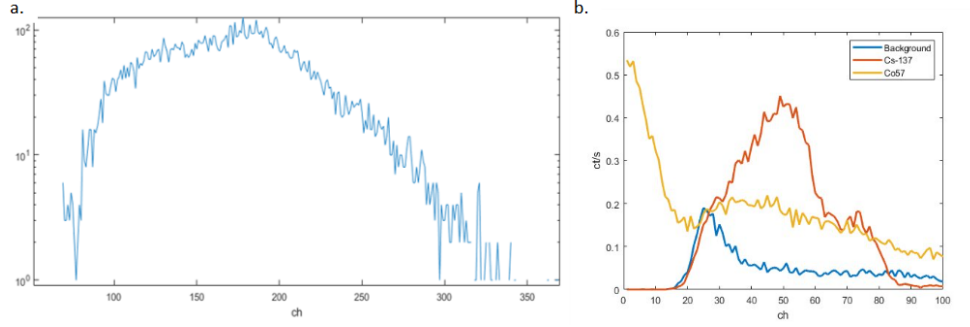


Figure 5.4: (a) Pulse-height spectrum from the FAPbBr₃ detector under ¹³⁷Cs irradiation at 700 V reverse bias, showing a broad peak feature near the expected photopeak channel. (b) Spectrum comparison at 1600 V with ¹³⁷Cs (5 μCi) and ⁵⁷Co (100 μCi). Acquisition count rates: background 9.11 cps, ¹³⁷Cs 17.45 cps, ⁵⁷Co 26.60 cps. The distinct features associated with each source's primary gamma energies can be resolved from the background continuum.

2822 Discussion

2823 The FAPbBr₃ results represent a significant improvement over the CsPbBr₃ device,
 2824 and the detection of spectral features attributable to both ¹³⁷Cs and ⁵⁷Co at 1600 V
 2825 constitutes a proof-of-concept demonstration of gamma spectroscopy in a solution-
 2826 grown halide perovskite crystal. However, the energy resolution remains far below
 2827 that required for isotope identification in a practical nuclear security application.
 2828 The resolution is limited by a combination of factors: residual leakage-current noise
 2829 even at the low operating leakage of 5 nA; incomplete charge collection arising from

2830 carrier trapping analogous to that described in Section 5.4.1; and ion migration-
2831 induced fluctuations in the device’s internal electric field on timescales comparable to
2832 the spectroscopic acquisition period, which cause the effective gain of the detector to
2833 drift slowly and broaden any spectral features.

2834 The superior performance of FAPbBr₃ relative to CsPbBr₃ under high-voltage
2835 operation merits comment. The FA⁺ cation, being larger than Cs⁺, creates a more
2836 tightly packed lattice with fewer pathways for halide vacancy migration, which may
2837 partly account for the lower leakage current and greater electrical stability observed
2838 over extended testing periods. This hypothesis is consistent with literature reports
2839 that FA-based perovskites exhibit reduced ion migration rates compared to Cs-based
2840 and MA-based compositions at comparable temperatures [?].

2841 **5.5.4 MAPbBr₃ Fabrication Challenges**

2842 The MAPbBr₃ crystal was processed through the same pixelated Schottky contact
2843 fabrication protocol as the other bromide systems, but failed to produce a functional
2844 gamma spectrometer in the configuration tested. The primary cause was the poor
2845 quality of the Ti Schottky contact achieved on this crystal. The Ti deposition showed
2846 weak adhesion to the polished MAPbBr₃ surface; upon venting the deposition cham-
2847 ber to atmosphere, the Ti layer was observed to partially delaminate, and subsequent
2848 IV characterisation showed near-ohmic rather than Schottky behaviour. This outcome
2849 is attributed to two compounding factors. First, the high sputter power used for Ti
2850 deposition (~ 450 W) produces fast-moving Ti adatoms with high kinetic energies
2851 that may damage the soft perovskite surface upon impact, creating a disordered in-
2852 terfacial layer rather than a well-defined Schottky junction. Literature on Ti Schottky

2853 contacts to halide perovskites suggests that lower sputter power and higher chamber
 2854 vacuum produce higher barrier heights and improved interface quality, a condition
 2855 not satisfied with the sputtering parameters used in this work [87]. Second, resid-
 2856 ual polishing compound at the crystal surface — despite the hexane wash protocol
 2857 — may have introduced an insulating contamination layer that prevented intimate
 2858 metal-semiconductor contact. As a consequence of the failed Schottky barrier, the
 2859 dark current in the MAPbBr₃ device was too large to permit spectroscopic measure-
 2860 ments; the maximum usable bias before the measurement was overwhelmed by noise
 2861 was 20 V, at which point the device had not yet entered a field regime where charge
 2862 collection efficiency was appreciable.

2863 **5.6 MHyPbCl₃ Fast Neutron Detection**

2864 **5.6.1 Motivation and Material Selection**

2865 Fast neutron detection poses a unique challenge because neutrons carry no charge
 2866 and therefore do not interact directly with the electron clouds that govern the re-
 2867 sponse of conventional radiation detectors. Instead, fast neutrons must be detected
 2868 indirectly by observing the secondary charged particles — predominantly recoil pro-
 2869 tons — produced in elastic scattering collisions with hydrogen nuclei ($^1\text{H}(n,n)^1\text{H}$).
 2870 The efficiency of this conversion process is proportional to the hydrogen number den-
 2871 sity n_H of the detector material, which for conventional hydrogenous semiconductors
 2872 is typically far too small to be competitive with hydrogen-rich organic scintillators
 2873 such as stilbene or EJ-276 plastic.

2874 MHyPbCl₃ was identified as a candidate fast neutron semiconductor detector be-
 2875 cause its methylhydrazinium cation ($\text{CH}_3\text{NH}_2\text{NH}_2^+$) contributes a substantial number

2876 of hydrogen atoms per formula unit, giving a calculated hydrogen density of approx-
 2877 imately $4.0 \times 10^{22} \text{ cm}^{-3}$ (Table 5.1), comparable to stilbene ($5.67 \times 10^{22} \text{ cm}^{-3}$) and
 2878 rubrene ($3.62 \times 10^{22} \text{ cm}^{-3}$) [77]. Crucially, MHyPbCl₃ also exhibits the high resistiv-
 2879 ity and large $\mu\tau$ product demonstrated in Section 5.3, making it suitable for direct
 2880 charge-collection detection of the recoil proton pulses generated by neutron scattering
 2881 events. In this mode of operation, an individual fast neutron interaction produces
 2882 a recoil proton of kinetic energy $E_p \leq E_n$ (the equality holding for a head-on col-
 2883 lision) that subsequently deposits its energy entirely within the crystal, generating
 2884 an electron-hole pair population proportional to E_p/W . The resulting charge pulse
 2885 is collected by the applied field and read out by a charge-sensitive preamplifier, en-
 2886 abling pulse-by-pulse energy analysis and — in principle — reconstruction of the fast
 2887 neutron energy spectrum from the measured recoil proton spectrum [?].

2888 An important complication for MHyPbCl₃ as a neutron detector is the presence of
 2889 lead in the crystal lattice. With an atomic number of $Z = 82$, lead is a highly efficient
 2890 photon absorber via the photoelectric effect at photon energies below its K-edge
 2891 (88 keV) and by Compton scattering at higher energies. In a mixed neutron-gamma
 2892 radiation field such as that produced by a research reactor, the detector therefore
 2893 responds to both neutrons and gammas simultaneously, making clean separation of
 2894 the two contributions a significant analytical challenge.

2895 **5.6.2 Beam Facility and Experimental Characterisation**

2896 Neutron beam testing was performed at the Fast Beam Facility (FBF) of the Ohio
 2897 State University Research Reactor (OSURR), a 500-kW_{th} pool-type research reactor.
 2898 The FBF beamport incorporates a 4-inch (10.16-cm) single-crystal bismuth gamma

2899 filter embedded in the beam collimator assembly, which preferentially attenuates the
 2900 prompt reactor gamma flux while transmitting fast neutrons. A downstream 8-inch
 2901 (20.32-cm) lead gamma shutter — consisting of a cylindrical drum filled with 1/16-
 2902 inch Pb shot — can be inserted into the beam path to provide an additional factor
 2903 of ~ 300 in gamma reduction at the cost of a factor of 58 in fast neutron attenuation.
 2904 An external 2-mm cadmium (Cd) foil positioned at the beam exit port can further
 2905 reduce the thermal neutron content of the beam; the Cd foil was characterised via
 2906 gold foil activation measurements per ASTM E262-17 to yield a Cadmium Ratio of
 2907 1.1 and a thermal neutron reduction factor of 110.

2908 The neutron beam energy spectrum at the FBF, determined by the SAND-II and
 2909 STAYSL spectrum unfolding techniques using activation of copper, gold, and nickel
 2910 wire dosimeters, is shown in Figure 5.5. The spectrum peaks in the fast neutron
 2911 regime, with a dominant fast neutron flux of approximately $4.3 \times 10^5 \text{ cm}^{-2} \text{ s}^{-1}$ when
 2912 the gamma shutter is closed and no Cd filter is in place. With the gamma shutter
 2913 open and no Cd filter, the fast neutron flux rises to $1.5 \times 10^7 \text{ cm}^{-2} \text{ s}^{-1}$, accompanied
 2914 by a gamma dose rate of 15.787 R hr^{-1} from the prompt fission gamma field. Insertion
 2915 of the lead gamma shutter reduces the gamma dose rate to $50.525 \text{ mR hr}^{-1}$ — a 300-
 2916 fold reduction — while attenuating the fast neutron flux by only a factor of 35 to
 2917 $4.3 \times 10^5 \text{ cm}^{-2} \text{ s}^{-1}$. Table 5.3 summarises the beam conditions used in this work.

Table 5.3: FBF beam conditions used for MHyPbCl₃ neutron detection experiments.
 All measurements at OSURR operating power.

Configuration	Fast Neutron Flux ($\text{cm}^{-2} \text{ s}^{-1}$)	Gamma Dose Rate
Shutter open, no Cd	1.5×10^7	15.787 R hr^{-1}
Shutter closed, no Cd	4.3×10^5	$50.525 \text{ mR hr}^{-1}$

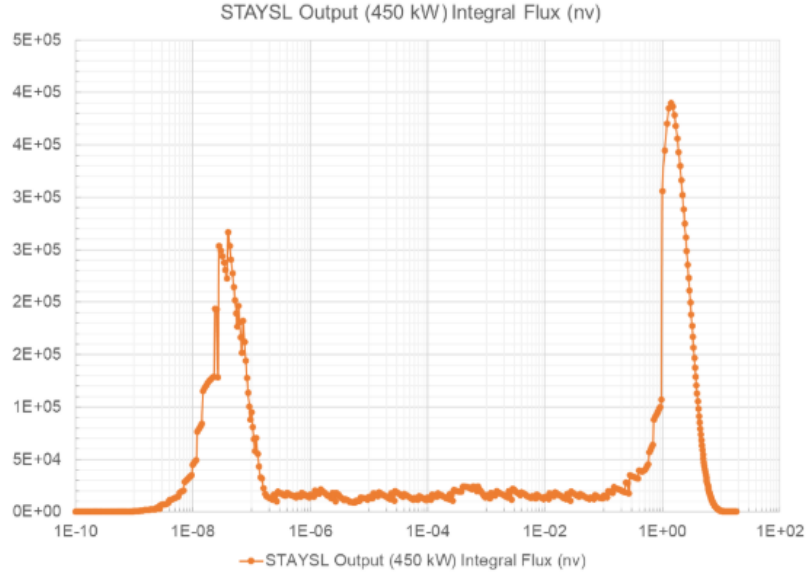


Figure 5.5: Neutron energy spectrum at the OSURR Fast Beam Facility determined by SAND-II unfolding of Cu, Au, and Ni activation wire measurements. The spectrum is dominated by fast neutrons with energies from 0.1 to 10 MeV.

2918 Data were acquired using a CAEN A1422 charge-sensitive preamplifier connected
 2919 directly to the MHyPbCl₃ detector, with the preamplifier output recorded on a Tek-
 2920 tronix MSO54 oscilloscope in waveform-capture mode. A Keithley 2612A SourceMe-
 2921 ter provided the detector bias voltage and simultaneously monitored the device leak-
 2922 age current. A cadmium foil chopper driven by a motor was positioned in the beam
 2923 path during selected measurements to modulate the thermal neutron flux periodi-
 2924 cally, enabling comparison of chopper-on and chopper-off waveform populations. The
 2925 raw waveform data were processed offline using a Python script that applied a Lo-
 2926 cally Weighted Scatterplot Smoothing (LOWESS) filter to suppress high-frequency
 2927 electronic noise while preserving the temporal profile of genuine radiation-induced

2928 pulses. Filtered waveforms were then passed through a pulse-height analysis (PHA)
 2929 algorithm to reconstruct the energy deposition spectrum.

2930 5.6.3 Results

2931 The MHyPbCl₃ detector was operated at reverse bias voltages of 80 V and 120 V
 2932 for the FBF measurements. Figure 5.6 shows representative waveforms acquired from
 2933 the preamplifier at 120 V reverse bias with the gamma shutter open (mixed neutron-
 2934 gamma field) and with the gamma shutter closed (fast neutron and residual gamma
 2935 only).

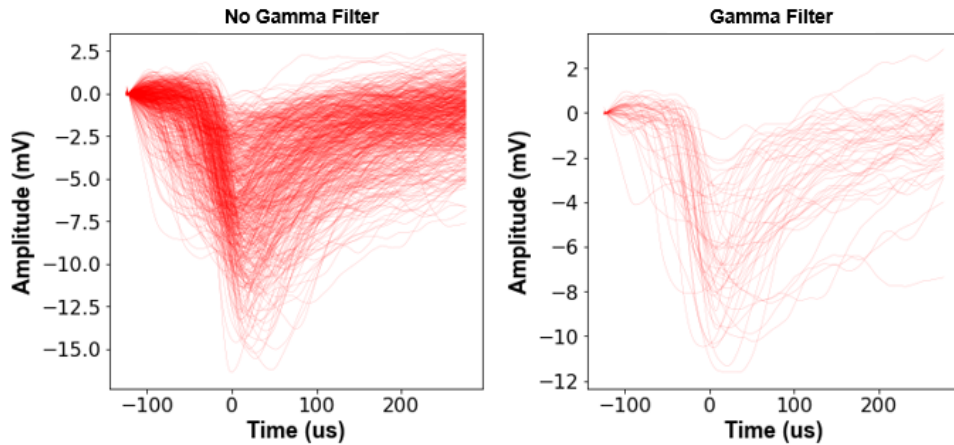


Figure 5.6: Representative preamplifier waveforms from the MHyPbCl₃ detector at 120 V reverse bias at the OSURR FBF. (a) Gamma shutter open (mixed field). (b) Gamma shutter closed (reduced gamma, fast neutron dominant). (c) Beam background acquired in the shutter-closed configuration with the reactor at power. Radiation-induced pulses are visible above the noise floor in all three configurations.

2936 At 120 V reverse bias with the gamma shutter open, 1090 waveforms containing
 2937 radiation-induced pulses were collected over the acquisition interval. After closing
 2938 the gamma shutter, 1001 pulses were recorded over a comparable interval; the beam

background (reactor at power, beam blocked entirely) yielded 394 pulses. The count rate with the shutter closed ($\sim 0.5 \text{ counts s}^{-1}$) was not meaningfully different from the shutter-open count rate at 120 V in the forward bias configuration (Figure ??), a result that is discussed further below.

At 80 V reverse bias, the total waveform counts with the gamma shutter open were 1040 (shutter open) and 507 (shutter closed) versus 252 for background. The ratio of source-to-background counts at 80 V is higher than at 120 V, suggesting that the lower field at 80 V produces a different carrier collection regime in which the relative contribution of high-charge-deposition events (which would include both neutron proton-recoil events and direct gamma interactions) is more clearly separated from the background noise floor.

Pulse height analysis of the filtered waveforms at 120 V reverse bias is shown in Figure 5.7. The resulting spectrum does not display a sharp Gaussian peak indicative of full-energy deposition from a mono-energetic source; rather, it exhibits a broad continuum of pulse heights extending from the noise threshold to the maximum digitised amplitude. This spectral shape is physically expected for fast neutron detection: recoil protons from elastic scattering of poly-energetic fast neutrons produce a broad energy deposition distribution, and the convolution of this distribution with the incomplete charge collection at the bias voltages used here further broadens the response. Nevertheless, the waveform population is clearly distinct from the background (reactor off) waveform population, confirming that the MHyPbCl_3 crystal is responding to radiation from the reactor beam.

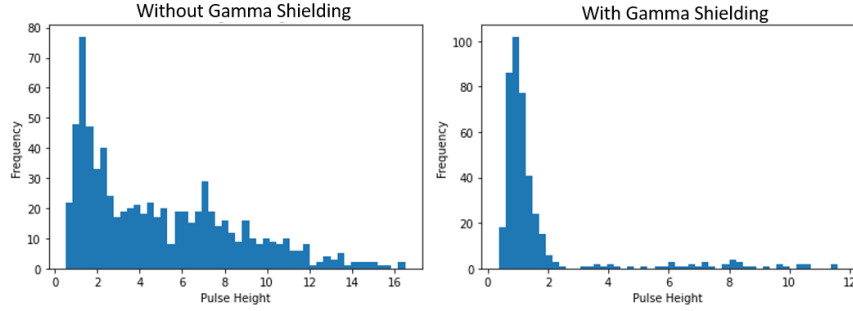


Figure 5.7: Pulse-height spectrum reconstructed from LOWESS-filtered waveforms from the MHyPbCl₃ detector at 120 V reverse bias. (a) No gamma filter applied (mixed field). (b) Gamma shutter closed. The broad continuum is consistent with the expected proton-recoil energy deposition spectrum from a poly-energetic fast neutron beam.

5.6.4 Discussion: Neutron-Gamma Discrimination and Detection Efficiency

The neutron-gamma discrimination challenge is illustrated by the comparison between shutter-open and shutter-closed waveform populations. When the lead gamma shutter is closed, the gamma dose rate is reduced by a factor of approximately 300 (from 15.787 R hr⁻¹ to 50.525 mR hr⁻¹), while the fast neutron flux decreases by only a factor of 35. If the detector were responding purely to neutrons, one would expect the pulse count rate with the shutter closed to be approximately $35/300 \approx 0.12$ times the shutter-open count rate. The observed shutter-open to shutter-closed count rate ratio at 120 V reverse bias (1090:1001, after background subtraction) does not show this level of suppression. The most likely interpretation is that the detector response at 120 V bias is dominated by direct gamma interactions (via Compton scattering and photoelectric absorption in the Pb sublattice) rather than neutron-induced proton recoil events. This result highlights the fundamental difficulty of neutron-gamma

discrimination in a Pb-containing semiconductor: the heavy Pb sublattice that makes the material mechanically robust and electrically well-behaved also makes it an efficient gamma absorber that overwhelms the neutron-induced signal in mixed radiation fields.

An upper bound estimate of the intrinsic detection efficiency for fast neutrons in the MHyPbCl₃ crystal can be computed from the hydrogen density, the crystal dimensions, and the elastic scattering cross section. At a neutron energy of 1 MeV, the ¹H(*n,n*)¹H elastic cross section is approximately 3 b [?]. For a crystal with an effective thickness of 5 mm normal to the beam and a hydrogen density of $4.0 \times 10^{22} \text{ cm}^{-3}$, the interaction probability per incident neutron is

$$P = 1 - \exp(-n_H \sigma_{\text{el}} t) \approx 1 - \exp(-4.0 \times 10^{22} \times 3 \times 10^{-24} \times 0.5) \approx 5.8\%, \quad (5.3)$$

where $t = 0.5 \text{ cm}$ is the path length. At the fast neutron flux of $4.3 \times 10^5 \text{ cm}^{-2} \text{ s}^{-1}$ (shutter closed) and the $\sim 0.3\text{-cm}^2$ detector area, this implies an expected proton-recoil interaction rate of ~ 750 events per second — far larger than the $0.5 \text{ counts s}^{-1}$ actually observed. The discrepancy is explained primarily by the oscilloscope waveform-write-rate bottleneck: the Tektronix MSO54 oscilloscope in single-shot waveform acquisition mode is capable of writing at a maximum rate that was the dominant constraint on the measurable count rate in this configuration. Increasing the data acquisition throughput — for example by switching to a hardware MCA or a fast digitiser in continuous acquisition mode — is the highest-priority hardware improvement identified for future neutron detection experiments.

2995 5.7 Ion Migration and Device Stability

2996 The experimental results presented in the preceding sections are pervaded by a
2997 single overarching materials-physics challenge: the migration of halide ions under
2998 the large applied electric fields required for efficient charge collection. Ion migration
2999 is not unique to the detector geometry; as noted in Chapter 4, it also affects the
3000 photovoltaic devices studied in that chapter under their built-in fields. However, the
3001 detector geometry exacerbates the problem in two important ways. First, the fields
3002 required for efficient charge collection ($\sim 1 \text{ V } \mu\text{m}^{-1}$ across millimetre-scale crystals)
3003 are several orders of magnitude larger than the built-in fields in a photovoltaic device.
3004 Second, the crystal must sustain these fields continuously for hours at a time during a
3005 spectroscopy acquisition, rather than the millisecond timescales over which a radiation
3006 monitoring current measurement is averaged. The combination of large field and long
3007 exposure duration creates an environment in which ion migration is not merely a
3008 nuisance but a device-lifetime-limiting degradation mechanism.

3009 The humidity-dependent instability mechanism was characterised in detail through
3010 the 2D perovskite work described in Section 5.4.2, where the correlation between rela-
3011 tive humidity and breakdown voltage was directly measured, and the 5F-PEAI surface
3012 treatment (Section 5.4.2) demonstrated an effective mitigation strategy for thin-film
3013 devices. The vacuum testing of the MAPbI₃-hBN device (Section 5.4.1) indepen-
3014 dently confirmed the same mechanistic link between moisture and voltage-induced
3015 instability in the 3D crystal geometry.

3016 5.7.1 Contact Degradation at Elevated Bias

3017 A second and distinct degradation mechanism was identified during testing of
3018 the FAPbBr₃ device: the chemical reaction of the titanium Schottky contact with
3019 mobile bromide ions generated and driven toward the contact interface under the large
3020 applied reverse bias. After a full day of electrical testing at the Ti1 contact of the
3021 FAPbBr₃ crystal, the contact surface was visually observed to have changed colour
3022 and the metallic titanium layer appeared to have partially receded into or reacted
3023 with the perovskite crystal surface. Elemental analysis of the relevant ionic species
3024 identifies the reaction as plausible: mobile bromide ions (charge -1) accumulating at
3025 the Ti contact (Ti in the $+3$ oxidation state in many common compounds) can form
3026 stable titanium bromide complexes (e.g. TiBr_4^{2-}), consuming the metal electrode. The
3027 relevant charge states of the species present — Br^- , Ti^{3+} , Pb^{2+} , and the organic FA^+
3028 cation — are all present at the interface under the reverse-bias condition that drives
3029 Br^- toward the titanium contact.

3030 This failure mode has profound implications for contact material selection in bro-
3031 mide perovskite radiation detectors. Metals that form stable halide compounds under
3032 the electrochemical conditions encountered at the reversed-biased contact — titanium
3033 being one such metal — are fundamentally incompatible with sustained high-field op-
3034 eration in halide perovskite detectors, regardless of their electronic Schottky barrier
3035 properties. Gold, by contrast, forms only kinetically stable halide compounds under
3036 mild conditions and showed no evidence of analogous chemical degradation at the
3037 ohmic contact in any device examined in this work. Alternative contact materials
3038 identified in the broader perovskite detector literature as candidates for improved
3039 chemical stability include chromium (Cr) and tin oxide (SnO_2), both of which were

3040 evaluated on the CsPbBr_3 crystal at the $\text{Cr}/\text{SnO}_2/\text{Cr}$ contact in preliminary ex-
3041 periments; however, these contacts were not carried through to full spectroscopic
3042 evaluation within the scope of this work.

3043 **5.7.2 Ion Migration Consequences for Spectroscopy**

3044 Beyond the acute failure modes described above, ion migration impairs spectro-
3045 scopic performance through three mechanisms that together set a practical upper
3046 bound on the achievable energy resolution in as-fabricated perovskite single crystal
3047 detectors.

3048 First, the redistribution of ions under the applied field creates a non-uniform
3049 internal electric field profile. In an ideal crystal with uniform space charge, the
3050 electric field is constant across the bulk between the contacts. As mobile anions
3051 accumulate at the anode and cations at the cathode, a positive space-charge layer
3052 forms near the cathode and a negative space-charge layer near the anode. This space
3053 charge creates an opposing electric field that adds to the applied field at one interface
3054 and subtracts from it at the other, producing a spatially non-uniform field profile in
3055 the bulk. Carriers generated at different depths in the crystal therefore experience
3056 different net fields and have different collection probabilities, broadening the pulse-
3057 height distribution for monoenergetic radiation events and degrading the achievable
3058 energy resolution [?].

3059 Second, the current overshoot observed upon initial application of bias — where
3060 the leakage current transiently rises above its quasi-steady-state value before decaying
3061 over tens of minutes — causes the internal field distribution to evolve continuously

during the acquisition. Since the peak position in a pulse-height spectrum is proportional to the instantaneous charge collection efficiency (which in turn depends on the internal field profile), the slow time evolution of the field distribution during ion migration causes the effective detector gain to drift, further broadening accumulated spectral peaks.

Third, on long timescales, the accumulated ionic charge at the interfaces depresses the net electric field in the bulk, reducing the drift velocity of radiation-generated carriers and increasing the probability of trapping before collection — the same trapping mechanism identified in Section 5.4.1 for X-ray sensitivity degradation. The combination of these three effects means that the energy resolution of a perovskite detector operated at high bias in ambient conditions will inevitably worsen over time, even if the initial resolution at the moment of bias application is acceptable.

5.7.3 Mitigation Strategies

Several mitigation strategies were evaluated or identified in the course of this work. Vacuum operation was demonstrated to extend the operational lifetime of the MAPbI₃-hBN device and to improve X-ray sensitivity at matched bias voltages (Section 5.4.1). However, vacuum operation is impractical for most field applications of radiation detectors, and it addresses only the humidity-acceleration pathway rather than the intrinsic ion-migration driving force provided by the applied electric field.

Surface passivation with 5F-PEAI, characterised in detail in Section 5.4.2, represents the most promising approach demonstrated in this work for thin-film device geometries. At a concentration of 4 mg mL⁻¹, 5F-PEAI penetrates the grain boundaries and raises the water-penetration activation energy from ~5 meV to ~120 meV,

3085 stabilising the device to -16 V in 40% RH air. Extending this passivation strategy
3086 to the 3D single crystal systems examined in this work is identified as a high-priority
3087 direction for future research.

3088 Contact material engineering is a second avenue for mitigating contact-specific
3089 degradation. The chemical reaction of Ti contacts with mobile Br^- ions — docu-
3090 mented here for the first time in the context of perovskite radiation detectors —
3091 motivates the replacement of Ti with contact metals that do not form stable halide
3092 compounds, such as Au (already used on the ohmic side), Pt, or Cr-based multi-
3093 layers. Independently, the use of carrier-selective interlayers (such as the C_{60} :BCP
3094 bilayer used in the MHyPbCl_3 device) both at the Schottky and ohmic contacts can
3095 reduce the effective recombination velocity at the interfaces and thereby extend the
3096 field range over which stable detector operation is achievable.

3097 A third mitigation approach is improved crystal quality — specifically, the reduc-
3098 tion of halide vacancy concentrations that provide the mobile ionic carriers. Growth
3099 conditions that minimise the stoichiometric excess of either the organic cation or the
3100 lead halide precursor, such as the slow anti-solvent crystallisation used for MHyPbCl_3 ,
3101 or the use of single-crystal growth methods under inert atmosphere, have been shown
3102 to produce lower intrinsic defect densities and improved electrical stability in the
3103 photovoltaic literature [87]. Applying these best-practice growth conditions system-
3104 atically to the detector crystal geometries required here is an important near-term
3105 research goal.

3106 5.8 Summary

3107 This chapter has described the development and characterisation of halide per-
3108 ovskite direct-conversion radiation detectors for three distinct radiation types: X-rays,
3109 gamma rays, and fast neutrons.

3110 For X-ray detection, two polycrystalline systems were presented. The MAPbI₃-
3111 hBN device demonstrated a monotonically increasing sensitivity with applied bias:
3112 0.107, 0.245, and 0.411 $\mu\text{C} (\text{Gy cm}^2)^{-1}$ at 5, 10, and 20 V, respectively, consistent with
3113 the Hecht equation prediction for improving charge collection at higher fields. Vacuum
3114 testing confirmed that the removal of ambient moisture enhanced both sensitivity and
3115 stability, directly attributing the performance degradation observed in ambient air to
3116 moisture-accelerated ion migration. The device failed irreversibly after approximately
3117 five hours at 400 V, setting an empirical upper bound on the sustainable bias for this
3118 material-contact combination in the absence of surface passivation. The 2D perovskite
3119 (PEA₂MA₂Pb₃I₁₀) detector with 5F-PEAI surface treatment demonstrated that this
3120 moisture-accelerated instability can be effectively mitigated: the fluorinated barrier
3121 layer raised the breakdown voltage from -10 V to -16 V at 40% RH and enabled X-
3122 ray sensitivities approaching $\sim 1000 \mu\text{C} (\text{Gy}_{\text{air}} \text{cm}^2)^{-1}$ at 4 V bias, among the highest
3123 reported for polycrystalline perovskite detectors of comparable thickness.

3124 For gamma-ray detection, three bromide perovskite single-crystal systems were
3125 evaluated. The CsPbBr₃ device produced a statistically significant count-rate re-
3126 sponse to a ¹³⁷Cs source (17.45 cps vs. 9.11 cps background) at 1000 V bias, but
3127 the high and unstable leakage current (315–710 nA) generated electronic noise that
3128 prevented resolution of the 662-keV photopeak. The FAPbBr₃ device, which achieved

3129 substantially lower operating leakage (5 ± 2.5 nA maximum at high bias) and sus-
3130 tained stable high-voltage operation for over five hours, produced spectral features
3131 attributable to both ^{137}Cs and ^{57}Co in a joint acquisition at 1600 V bias, representing
3132 a proof-of-concept demonstration of gamma spectroscopy in a solution-grown halide
3133 perovskite crystal. The MAPbBr_3 device failed to produce functional Schottky con-
3134 tacts under the deposition conditions used, due to high sputter power and surface
3135 contamination from polishing, motivating a revised contact deposition protocol at
3136 reduced sputter power.

3137 For fast neutron detection, MHyPbCl_3 crystals were characterised as the first
3138 organic-metal halide perovskite semiconductor evaluated in a reactor fast neutron
3139 beam. The material combines a high hydrogen density ($\sim 4.0 \times 10^{22} \text{ cm}^{-3}$, comparable
3140 to stilbene) with an exceptional resistivity of $4.43 \times 10^{11} \Omega \text{ cm}$ and a $\mu\tau$ product
3141 of $9.1 \times 10^{-3} \text{ cm}^2 \text{ V}^{-1}$. Waveform data acquired at the OSURR FBF confirmed
3142 radiation-induced pulse activity distinct from the beam background, and pulse-height
3143 analysis produced spectra consistent with the expected broad proton-recoil energy
3144 deposition distribution from a poly-energetic fast neutron beam. Neutron-gamma
3145 discrimination remains a challenge due to the large Pb content of MHyPbCl_3 , and the
3146 data acquisition bottleneck imposed by the oscilloscope waveform write rate prevented
3147 collection of sufficient statistics for quantitative efficiency analysis.

3148 Across all three detection modes, ion migration under applied electric fields emerged
3149 as the central performance-limiting mechanism. The correlation of device instability
3150 with ambient humidity — confirmed both by the vacuum tests on MAPbI_3 -hBN and
3151 by the systematic humidity-dependence study on 2D perovskite detectors — points

3152 to surface passivation, such as the 5F-PEAI treatment, as the most promising near-
3153 term strategy for extending the operational stability of perovskite radiation detectors
3154 toward the performance levels required for practical deployment.

Chapter 6: Thermal Radiation Sensing Concepts for Nuclear Systems

6.1 Introduction

In addition to the direct detection approaches explored in the preceding chapters, indirect methods of radiation sensing—those that infer radiation fields from their thermal signatures—represent an alternative pathway for monitoring nuclear environments. This chapter describes the investigation of two such concepts: the Optical Fiber-Based Gamma Thermometer (OFBGT) and the Optical Fiber-Based Neutron Dosimeter (OFBND). Both devices exploit the temperature differential that arises when ionizing radiation deposits energy in a thermally isolated material, and both rely on distributed fiber-optic temperature sensing to measure that differential.

The OFBGT was originally conceived and developed by Birri [11] at The Ohio State University. The present work extends that effort to a new sensor design intended for deployment in the Central Irradiation Facility (CIF) of the Ohio State University Research Reactor (OSURR), carried out in partnership with Luna Innovations [107]. The author’s contribution centered on executing and validating thermal simulations of the redesigned OFBGT using analytical models previously established by Birri,

3172 as well as extending those models to predict the performance of a proposed neutron-
3173 sensitive variant, the OFBND. The simulation methodology, calculational results, and
3174 implications for the experimental campaign are presented in the sections that follow.

3175 **6.2 Gamma Heating and the Gamma Thermometer Principle**

3176 When a material is placed in a mixed radiation field inside a reactor core, gamma
3177 rays interact with the material primarily through the photoelectric effect, Compton
3178 scattering, and pair production. These interactions transfer kinetic energy to sec-
3179 ondary electrons, which subsequently deposit that energy locally as heat through
3180 ionization and excitation of the lattice. The volumetric rate at which energy is de-
3181 posited, Q''' (W/m³), depends on the gamma-ray flux, the energy spectrum of the
3182 gamma-ray field, and the mass energy-absorption coefficient of the target material.

3183 A gamma thermometer exploits this heating effect by measuring the temperature
3184 difference, ΔT , that develops between a thermally insulated internal mass—referred
3185 to here as the thermal mass or insulating tube—and an outer metallic sheath that is
3186 in thermal contact with the reactor coolant. Because the insulating gas gap between
3187 the thermal mass and the sheath represents the dominant thermal resistance, the
3188 temperature within the thermal mass is elevated relative to the sheath. When the
3189 sensor is calibrated by applying a known electrical power to an internal heating wire, a
3190 relationship between the linear power density q' and ΔT is established. During reactor
3191 operation, the measured $\Delta T(z)$ at each axial position z can then be converted to the
3192 local $q'(z)$, and hence to the local gamma-ray dose rate [11, 12].

3193 **6.2.1 Optical Frequency Domain Reflectometry**

3194 The temperature sensing modality employed in both the OFBGT and OFBND is
3195 optical frequency domain reflectometry (OFDR). In OFDR, a tunable laser is swept
3196 in wavelength and coupled into a sensing fiber. The backscattered Rayleigh signal,
3197 or the reflected signal from fiber Bragg gratings (FBGs) inscribed along the fiber, is
3198 interfered with a reference signal to produce a spatial map of spectral shift along the
3199 fiber. A shift in the reflected spectrum at a given location is linearly related to the
3200 local temperature change. Phase-shifted, continuously written (PSC) type II FBGs,
3201 inscribed by ultrafast laser pulses, were chosen for this application because of their
3202 demonstrated radiation hardness and thermal stability at elevated temperatures [11].

3203 **6.3 Optical Fiber-Based Gamma Thermometer Design**

3204 **6.3.1 Cross-Sectional Geometry**

3205 A generalized cross-section of the OFBGT active sensing region is shown in Fig-
3206 ure 6.1. The central element is a repurposed four-hole thermocouple-wire electrical
3207 insulator tube composed of 80% mullite and 20% silica. Two of the four holes carry
3208 optical fibers that measure the temperature of the thermal mass; the remaining two
3209 holes carry a nichrome heating wire that forms a loop for calibration purposes. When
3210 a known current is passed through the heating wire, a known q' is generated, estab-
3211 lishing the calibration relationship between q' and ΔT .

3212 The insulating tube/thermal mass is surrounded by an argon gas gap at atmo-
3213 spheric pressure (760 mmHg). Argon was selected for its chemical inertness and well-
3214 characterized thermal conductivity ($\kappa_G = 22.6 \text{ mW}/(\text{m}\cdot\text{K})$ at room temperature).

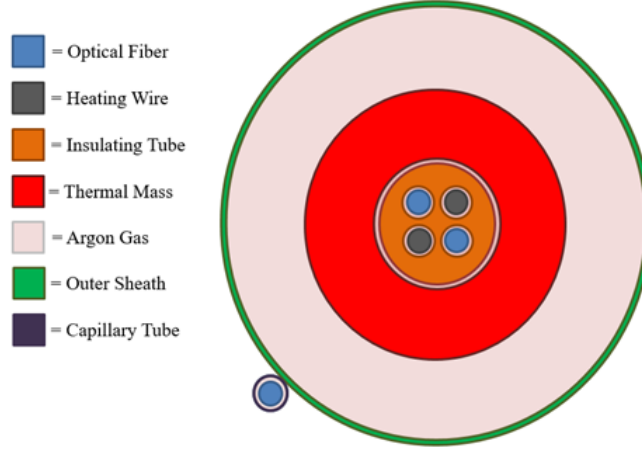


Figure 6.1: Cross-sectional view of the optical fiber-based gamma thermometer, showing the insulating tube/thermal mass, optical fibers, heating wire, argon gas gap, and nested outer sheath tubes. Adapted from the OFBGT Simulation Response Document [15], p. 2.

3215 Because of its low thermal conductivity, the gas gap constitutes the dominant ther-
 3216 mal resistance in the radial heat transfer path, making the temperature distribution
 3217 within the thermal mass nearly uniform.

3218 Two nested aluminum 6061 tubes form the outer sheath. The inner nested tube
 3219 has two grooves machined into its outer surface, each containing an optical fiber that
 3220 senses the sheath temperature. The temperature difference $\Delta T(z) = T_{\text{fiber,inner}}(z) -$
 3221 $T_{\text{fiber,outer}}(z)$ between the fibers in the thermal mass and those in the outer sheath is
 3222 the principal measurement quantity of the OFBGT.

3223 6.3.2 Axial Layout

3224 An overall view of the OFBGT design as configured for the OSURR is shown in
 3225 Figure 6.2. The sensor has three regions in the axial direction. The bottom-most
 3226 region is the active sensor region, where the thermal mass, the gas gap, and the outer

3227 sheath are present. The active sensor length was approximately 24 inches (61 cm),
 3228 constrained by the longest available continuous length of the mullite insulator. The
 3229 bottom of the OFBGT is welded shut to maintain the inert argon atmosphere. Glass
 3230 disc spacers keep the thermal mass centered within the sheath.

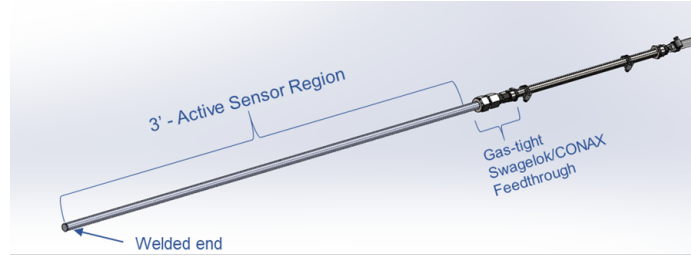


Figure 6.2: Overall view of the OFBGT design for the OSURR, showing the active sensor region, the gas-tight Swagelok/CONAX feedthrough, and the welded end. Adapted from the OFBGT Simulation Response Document [15], p. 4.

3231 6.4 Optical Fiber-Based Neutron Dosimeter Concept

3232 The OFBND extends the gamma thermometer concept to neutron fields by replac-
 3233 ing the mullite insulator with borosilicate glass. Borosilicate glass contains boron-10,
 3234 which has a large thermal-neutron absorption cross-section ($\sigma_B \approx 755$ barns for nat-
 3235 ural boron). When a thermal neutron is absorbed by ^{10}B , the resulting $^{10}\text{B}(\text{n},\alpha)^7\text{Li}$
 3236 reaction releases charged particles whose kinetic energy is deposited locally as heat:

$$\langle Q_\alpha^B + Q_{Li}^B \rangle = 3.75 \times 10^{-13} \text{ J/reaction} \quad (6.1)$$

3237 where the angle brackets denote averaging over the two branches of the reaction. The
 3238 volumetric energy deposition rate in the borosilicate thermal mass is then

$$Q''' = N_B \sigma_B \varphi \langle Q_\alpha^B + Q_{Li}^B \rangle \quad (6.2)$$

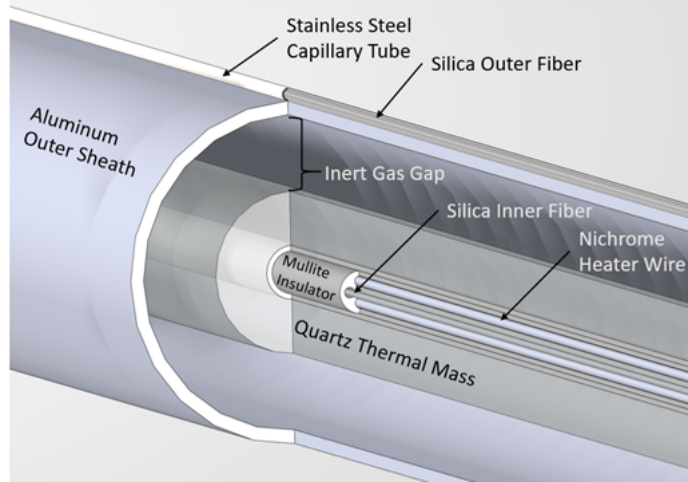


Figure 6.3: Internal view of the active sensor region of the OFBGT, showing the mullite insulator, nichrome heater wire, silica optical fibers, inert gas gap, and aluminum outer sheath. Adapted from the OFBGT Simulation Response Document [15], p. 4.

where N_B is the boron number density and φ is the thermal neutron flux.

The OFBND is geometrically similar to the OFBGT but uses a smaller-diameter thermal mass. Whereas the OFBGT employs a 1/8-inch (3.175 mm) OD mullite tube, the OFBND uses a 1/16-inch (1.5875 mm) OD borosilicate glass tube. The smaller diameter was driven by the availability of borosilicate capillary tubing. The neutron-sensitive thermal mass is to be deployed in the Auxiliary Irradiation Facility (AIF) of the OSURR, where the thermal neutron flux is $6.0 \times 10^{12} \text{ n cm}^{-2}\text{s}^{-1}$.

By placing a standard gamma-sensitive OFBGT and a neutron-sensitive OFBND at adjacent positions in the reactor, it becomes possible in principle to separate the neutron and gamma contributions to the local dose rate. The neutron-to-gamma heating ratio in the borosilicate glass thermal mass is approximately 4:1 at the thermal neutron fluxes encountered in the AIF, making the OFBND substantially more responsive to neutrons than to gamma rays [15].

3252 6.5 Simulation Methodology

3253 6.5.1 Analytical Thermal Model

3254 The thermal response of the OFBGT and OFBND was predicted using an ana-
3255 lytical model originally developed by Birri [12, 11]. The model treats the sensor as
3256 a series of concentric cylindrical regions—thermal mass, gas gap, and outer sheath—
3257 with azimuthal symmetry assumed. Deviations from azimuthal symmetry due to the
3258 heating wire and optical fiber holes are small and were neglected.

3259 Under the assumption that the linear power density q' is axially uniform and
3260 the outer sheath surface temperature $T(r_{So}, z)$ is axially constant, the temperature
3261 difference between the fiber location in the thermal mass and the outer sheath surface
3262 is

$$\Delta T(z) = T(r_f, z) - T(r_{So}, z) = q' R_{r_f}^{r_{So}} \quad (6.3)$$

3263 where $q' = Q''' \pi r_{TM}^2$ is the linear power density for the sensor, r_f is the radial position
3264 of the optical fiber within the thermal mass, r_{TM} is the outer radius of the thermal
3265 mass, and $R_{r_f}^{r_{So}}$ is the thermal resistance from the fiber location to the outer sheath
3266 surface:

$$R_{r_f}^{r_{So}} = \frac{1}{2\pi} \left[\frac{1 - \frac{r_f^2}{r_{TM}^2}}{2\kappa_{TM}} + \frac{\ln\left(\frac{r_{Si}}{r_{TM}}\right)}{\kappa_G} + \frac{\ln\left(\frac{r_{So}}{r_{Si}}\right)}{\kappa_S} \right] \quad (6.4)$$

3267 Here κ_{TM} , κ_G , and κ_S are the thermal conductivities of the thermal mass, gas gap,
3268 and sheath, respectively, and r_{Si} and r_{So} are the inner and outer radii of the outer
3269 sheath.

3270 This expression is exact for the case of axially constant q' and axially constant
 3271 outer sheath temperature. In a reactor, however, $q'(z)$ varies axially following the neu-
 3272 tron flux profile. To characterize the resulting error, a modulation transfer function
 3273 (MTF) formalism was employed.

3274 6.5.2 Modulation Transfer Function

3275 The concept of a modulation transfer function, borrowed from optical imaging
 3276 theory, was applied to the OFBGT by Palmer and Blue [75]. When the volumetric
 3277 energy deposition $q'''(z)$ varies sinusoidally with spatial frequency $k = 2\pi/\lambda$, axial
 3278 diffusion of thermal energy in the thermal mass, gas gap, and sheath causes the
 3279 measured $\Delta T(z)$ to be attenuated (modulated) relative to the true $q'''(z)$ distribution.
 3280 During calibration with a spatially uniform heat source ($k = 0$), this axial diffusion
 3281 does not occur. The normalized MTF quantifies the ratio of the sensor response at
 3282 spatial frequency k to the response at $k = 0$:

$$\text{MTF}_{\text{norm}}(k) = \frac{\text{MTF}(k)}{\text{MTF}(k = 0)} = \frac{R_{TM}(r_f)|_k}{R_{TM}(r_f)|_{k=0}} \quad (6.5)$$

3283 where $R_{TM}(r_f)|_k$ is a function of the modified Bessel functions I_0 and K_0 evaluated
 3284 at the boundaries of the three regions, determined by solving a 5×5 matrix equation
 3285 that enforces continuity of temperature and heat flux at the interfaces [12, 15].

3286 A value of MTF_{norm} close to unity indicates that the OFBGT faithfully reproduces
 3287 the axial variation in $q'''(z)$ at that spatial frequency. As k increases, the MTF
 3288 decreases, indicating a loss of axial resolution.

3289 6.5.3 Computational Implementation

3290 Two computational implementations of the analytical model were used:

3291 1. **Python model.** A point calculation based on Equations 6.3–6.4 was imple-
3292 mented in Python to compute ΔT at the axial location of maximum q' . This
3293 model assumed constant (room-temperature) thermal conductivities.

3294 2. **MATLAB model.** A more comprehensive model was implemented in MAT-
3295 LAB that solved the matrix equation for the temperature distribution $T(r, z)$
3296 throughout the entire sensor, iteratively updating the temperature-dependent
3297 thermal conductivities of the argon gas gap, the thermal mass, and the outer
3298 sheath at each axial and radial position until convergence was achieved. The
3299 MATLAB model also computed the normalized MTF as a function of spatial
3300 frequency k .

3301 The MATLAB code iterated on the thermal conductivities as follows: an initial
3302 temperature field was computed using room-temperature conductivities; the thermal
3303 conductivities were then re-evaluated at the local temperatures using tabulated data
3304 and the matrix equation was re-solved; this process was repeated until the maximum
3305 change in ΔT between successive iterations fell below 10^{-6} °C.

3306 6.6 Sensor Design Parameters

3307 The design parameters for the OFBGT and OFBND are summarized in Tables 6.1
3308 and 6.2, respectively. The irradiation conditions and resulting linear power densities
3309 are given in Table 6.3.

3310 The linear power density for the OFBGT was calculated from the absorbed dose
3311 rate for silicon in the CIF ($\dot{D} = 87$ Mrad/h), scaled to mullite using the ratio of
3312 mass energy-absorption coefficients evaluated at 1 MeV ($\mu_e^{\text{Mullite}}/\mu_e^{\text{Si}} = 1.296$), yielding
3313 $Q''' = 0.731$ W/cm³ and $q' = 5.77$ W/m [15]. The linear power density for the OFBND

Table 6.1: Part details and parameters for the OFBGT. Adapted from the OFBGT Simulation Response Document [15], p. 10.

Component	Material	Dimensions	Parameters
Outer sheath (inner tube)	Al 6061	5/16" OD, 0.243" ID	$r_{Si} = 3.09$ mm, $r_{So} = 3.9$ mm $\kappa_S = 177$ W/(m·K)
Outer sheath (outer tube)	Al 6061	3/8" OD, 0.319" ID	(Not used in calculation)
Insulating tube	80% Mullite, 20% Silica	1/8" OD, 0.02" ID	$r_f = 1.058$ mm $r_{TM} = 1.588$ mm $\kappa_{TM} = 5$ W/(m·K)
Gas gap	Argon, 760 mmHg	$t_G = 1.5$ mm	$\kappa_G = 22.6$ mW/(m·K)

Table 6.2: Part details and parameters for the OFBND. Adapted from the OFBGT Simulation Response Document [15], p. 11.

Component	Material	Dimensions	Parameters
Outer sheath (inner tube)	Al 6061	5/16" OD, 0.243" ID	$r_{Si} = 3.09$ mm, $r_{So} = 3.9$ mm $\kappa_S = 177$ W/(m·K)
Outer sheath (outer tube)	Al 6061	3/8" OD, 0.319" ID	(Not used in calculation)
Insulating tube	Borosilicate glass	0.062" OD, 0.015" ID	$r_f = 0.529$ mm $r_{TM} = 0.794$ mm $\kappa_{TM} = 1.15$ W/(m·K)
Gas gap	Argon, 760 mmHg	$t_G = 2.3$ mm	$\kappa_G = 22.6$ mW/(m·K)

Table 6.3: Irradiation facilities and calculated linear power densities for the OFBGT and OFBND, not accounting for the volume of the through-holes. Adapted from the OFBGT Simulation Response Document [15], p. 12.

Sensor	Material	Dimensions	Facility	q' (W/m)
OFBGT	Mullite	3.175 mm OD, 0.508 mm ID	CIF	5.77
OFBND	Borosilicate glass	1.5875 mm OD, 0.381 mm ID	AIF	16.6

3314 was calculated from the thermal neutron flux in the AIF ($\varphi = 6.0 \times 10^{12} \text{ n cm}^{-2}\text{s}^{-1}$)
 3315 using Equation 6.2, assuming 13% boron trioxide by weight and a borosilicate glass
 3316 density of 2.23 g/cm^3 , yielding $Q''' = 8.52 \text{ W/cm}^3$ and $q' = 16.6 \text{ W/m}$ [15]. The
 3317 detailed derivations of these quantities are provided in the simulation response doc-
 3318 ument [15].

3319 For both sensors, the radial position of the fiber within the thermal mass was
 3320 estimated as $r_f = \frac{2}{3}r_{TM}$, since exact dimensions of the through-hole locations were
 3321 not available from the vendor. The sensitivity of ΔT to this assumption is small
 3322 because the temperature distribution within the thermal mass is nearly uniform.

3323 6.7 Simulation Results

3324 6.7.1 Predicted Temperature Differential

3325 The results of the ΔT calculations from both the Python and MATLAB models
 3326 are summarized in Table 6.4.

Table 6.4: Comparison of the OFBGT and OFBND modeling results from the Python (constant thermal conductivity) and MATLAB (temperature-dependent thermal conductivity, iterative) models. Adapted from the OFBGT Simulation Response Document [15], p. 14.

Parameter	OFBGT	OFBND
Thermal mass material	Mullite	Borosilicate glass
Thermal mass radius (mm)	1.588	0.794
Gas gap thickness (mm)	1.5	2.3
κ_{TM} (W/(m·K))	5	1.15
q' (W/m)	5.77	16.6
ΔT from Python (°C)	27.1	159.4
ΔT from MATLAB (°C)	29.66	154.8
Relative error (%)	8.77	2.96

3327 The Python model predicted a maximum ΔT of 27.1 °C for the OFBGT and
3328 159.4 °C for the OFBND. The MATLAB model, which accounts for the tempera-
3329 ture dependence of the argon thermal conductivity, predicted values of 29.66 °C and
3330 154.8 °C, respectively. The relative errors between the two models were 8.77% for the
3331 OFBGT and 2.96% for the OFBND. The principal source of discrepancy is the treat-
3332 ment of the gas gap thermal conductivity: the Python model uses a single constant
3333 value, whereas the MATLAB model evaluates it locally at the temperature-dependent
3334 gas conductivity, which is significant because of the large temperature gradient across
3335 the gap.

3336 The substantially larger ΔT predicted for the OFBND relative to the OFBGT
3337 arises from two factors. First, the linear power density is approximately 2.9 times
3338 greater for the OFBND ($q' = 16.6$ W/m versus 5.77 W/m). Second, the gas gap is
3339 53% wider in the OFBND (2.3 mm versus 1.5 mm), increasing the thermal resistance.
3340 Although the borosilicate glass thermal mass has a lower thermal conductivity than
3341 mullite (1.15 versus 5 W/(m·K)), the contribution of the thermal mass resistance to
3342 the overall ΔT is small relative to the gas gap resistance.

3343 6.7.2 Temperature Distributions

3344 The temperature distributions computed by the MATLAB model, plotted as con-
3345 tour maps in the radial-axial (r - z) plane, are shown in Figure 6.4. The axial variation
3346 follows the sinusoidal flux profile of the reactor, with the maximum temperatures oc-
3347 ccurring near the axial center of the core. The radial temperature drop is concentrated
3348 in the gas gap region, consistent with the dominance of the gap thermal resistance.

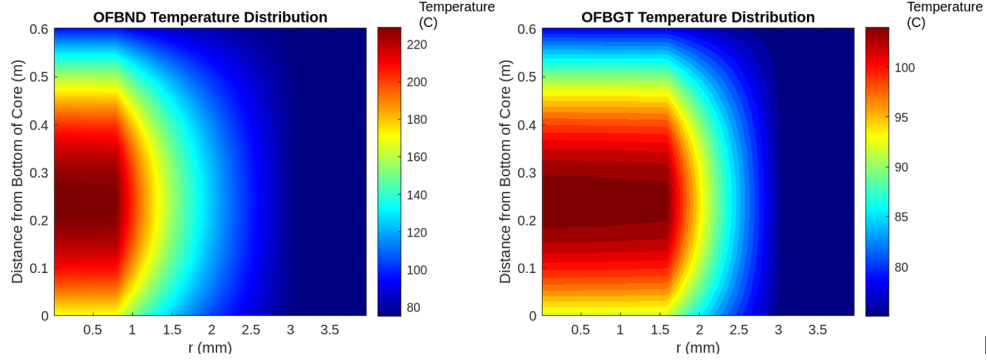


Figure 6.4: Temperature distributions plotted as contour maps in the radial (r) and axial (z) directions for the OFBND (left) and the OFBGT (right). The OFBND reaches significantly higher peak temperatures due to its larger linear power density. Adapted from the OFBGT Simulation Response Document [15], p. 15.

6.7.3 Modulation Transfer Function Results

The normalized MTF curves for both the OFBGT and OFBND are compared in Figure 6.5. The MTF quantifies the sensor's ability to faithfully reproduce axial variations in the energy deposition rate at different spatial frequencies. A value of unity corresponds to perfect fidelity; lower values indicate spatial smoothing.

The OFBND exhibited a higher MTF at all spatial frequencies than the OFBGT. This improved axial resolution is a consequence of the smaller cross-sectional area of the borosilicate glass thermal mass, which reduces the length scale over which axial thermal diffusion can smooth out spatial variations. The MTF curves for the new (Luna) OFBGT design were found to be nearly identical to those of the original Birri design, confirming that the design modifications did not degrade the sensor's spatial resolution characteristics.

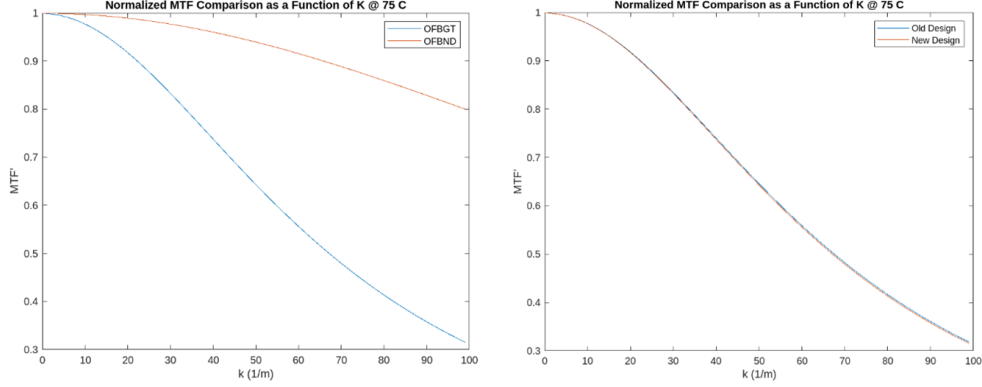


Figure 6.5: Left: comparison of the normalized MTF curves for the OFBGT and OFBND as a function of spatial frequency k . The OFBND exhibits a higher MTF at all spatial frequencies due to its smaller cross-sectional area and lower thermal mass conductivity. Right: comparison of the new (Luna) OFBGT design with the original (Birri) OFBGT design, showing nearly identical MTF performance. Adapted from the OFBGT Simulation Response Document [15], p. 16.

6.8 Experimental Testing and Discussion of the OFBGT

6.8.1 Reactor Testing Campaign

Following the simulation work, an OFBGT was fabricated by Luna Innovations and tested in the CIF of the OSURR during a seven-hour reactor test campaign [107]. The sensor was equipped with ten FBG sensor pairs (inner and outer fiber) distributed along the 24-inch active sensor length. The reactor was operated at multiple power levels (50%, 70%, 80%, and 90% of full power), and the temperature data from each FBG pair were recorded in real time.

6.8.2 Discrepancy Between Predicted and Measured ΔT

The simulations presented in Section 6.7 predicted a maximum ΔT of approximately 27–30 °C for the OFBGT when deployed in the CIF at full reactor power,

3372 which exceeded the estimated measurement threshold of approximately 25 °C needed
 3373 for reliable power inference. However, the experimental measurements yielded a sig-
 3374 nificantly lower ΔT , on the order of 17 °C at full power. This discrepancy rendered
 3375 the ΔT insufficient for reliable calibration and power measurement, and the test was
 3376 consequently deemed unsuccessful in meeting the original performance targets.

3377 The primary source of the discrepancy was identified as the neglect of radiative
 3378 heat transfer in the simulation models. Both the Python and MATLAB models con-
 3379 sidered only conductive heat transfer through the gas gap. However, at the operating
 3380 temperatures of the thermal mass (approaching 100 °C above ambient), radiative heat
 3381 transfer across the gas gap between the thermal mass surface and the inner surface of
 3382 the outer sheath provides a parallel pathway for heat dissipation that is not accounted
 3383 for by conduction alone. The radiative heat flux is governed by

$$q''_{\text{rad}} = \frac{\sigma(T_1^4 - T_2^4)}{\frac{1}{\varepsilon_1} + \frac{(1-\varepsilon_2)r_{TM}}{\varepsilon_2 r_{Si}}} \quad (6.6)$$

3384 where σ is the Stefan–Boltzmann constant, T_1 and T_2 are the absolute temperatures
 3385 of the thermal mass surface and the sheath inner surface, respectively, and ε_1 and ε_2
 3386 are the emissivities of these surfaces.

3387 The inclusion of this radiative pathway effectively reduces the thermal resistance
 3388 of the gas gap, lowering the equilibrium ΔT . The MATLAB code contained provisions
 3389 for incorporating radiative heat transfer (with $\varepsilon_1 = 0.76$ for the mullite surface and
 3390 $\varepsilon_2 = 0.05$ for the aluminum sheath), but this term was set to zero in the simulations
 3391 presented in the response document based on the recommendation to neglect it at the
 3392 time [15]. When radiative heat transfer was subsequently included in the model, the
 3393 predicted ΔT dropped from approximately 27 °C to approximately 17 °C, in much
 3394 closer agreement with the experimental observations.

3395 This result underscores the importance of including all relevant heat transfer
3396 mechanisms—conduction, convection (if applicable), and radiation—in the thermal
3397 modeling of gamma thermometers, particularly when the gas gap temperature differ-
3398 ential is large. The lesson is especially pertinent for the OFBND, which operates at
3399 still higher temperatures and for which the neglect of radiative heat transfer would
3400 introduce even greater modeling errors.

3401 6.9 Status of the OFBND

3402 The OFBND concept was investigated through simulation only during this re-
3403 search effort. The modeling results predict a substantially larger ΔT (on the order of
3404 155–159 °C without radiative heat transfer corrections) for the OFBND in the AIF
3405 than for the OFBGT in the CIF, suggesting that the neutron dosimeter concept is
3406 viable from a signal-strength perspective. The higher MTF of the OFBND further
3407 suggests superior axial resolution relative to the OFBGT.

3408 However, no OFBND prototype has yet been fabricated or tested experimentally.
3409 Several aspects of the OFBND design require experimental validation before the con-
3410 cept can be considered proven:

3411 1. **Thermal performance.** The simulations predict operating temperatures in
3412 excess of 200 °C in the thermal mass. At such temperatures, radiative heat
3413 transfer—which was shown to be the dominant source of modeling error for
3414 the OFBGT—will be even more significant. Updated simulations including
3415 radiative heat transfer are essential before a prototype is fabricated.

3416 2. **Material compatibility.** The borosilicate glass thermal mass must survive
3417 prolonged exposure to the combined thermal, gamma, and neutron environment

3418 of the AIF without degradation in mechanical integrity or thermal properties.
3419 The transmutation of boron-10 through the (n,α) reaction will deplete the boron
3420 content over time, reducing the sensor's neutron sensitivity and introducing
3421 helium gas into the glass matrix.

3422 **3. Calibration methodology.** The calibration of the OFBND with an internal
3423 heating wire is analogous to that of the OFBGT. However, the separation of
3424 the neutron and gamma contributions to the measured ΔT requires a paired
3425 measurement with a collocated, purely gamma-sensitive OFBGT. The practical
3426 feasibility of this approach remains to be demonstrated.

3427 **4. Signal-to-noise ratio.** While the predicted ΔT is large, the FBG measure-
3428 ment uncertainty (typically ± 0.1 °C) must be small relative to the minimum
3429 resolvable ΔT at the lowest flux levels of interest.

3430 The OFBND therefore remains a promising but unproven concept that requires
3431 experimental testing to validate the theoretical predictions presented here.

3432 **6.10 Summary**

3433 This chapter has presented the simulation and analysis work performed in sup-
3434 port of the OFBGT and OFBND sensor development. The analytical thermal model,
3435 originally developed by Birri, was applied to a redesigned OFBGT intended for de-
3436 ployment in the CIF of the OSURR and extended to a neutron-sensitive variant
3437 (OFBND) intended for the AIF.

3438 Two modeling approaches—a simplified Python model and an iterative MAT-
3439 LAB model with temperature-dependent thermal conductivities—yielded consistent

3440 predictions of ΔT for both sensors. The simulations predicted a maximum ΔT of
3441 approximately 27–30 °C for the OFBGT in the CIF and 155–159 °C for the OFBND
3442 in the AIF, without accounting for radiative heat transfer. The MTF analysis con-
3443 firmed that both sensors possess adequate axial resolution for the spatial frequencies
3444 expected in the OSURR flux profiles, with the OFBND exhibiting superior resolution
3445 due to its smaller thermal mass.

3446 Experimental testing of the OFBGT revealed that the measured ΔT was approx-
3447 imately 17 °C, falling short of the approximately 25 °C threshold required for reliable
3448 operation. The discrepancy was attributed to the omission of radiative heat transfer
3449 in the original simulations. When radiative heat transfer was included retrospectively,
3450 the model predictions agreed closely with the experimental observations.

3451 The OFBND has not yet been fabricated or tested. While the simulations indicate
3452 a favorable signal strength, experimental validation is required to confirm the ther-
3453 mal performance, material compatibility, and calibration methodology of the neutron
3454 dosimeter concept. The inclusion of radiative heat transfer in future OFBND simu-
3455 lations is essential, given the high operating temperatures predicted for this sensor.

Chapter 7: Digital Electronics Verification Methods for Radiation Environments

7.1 Introduction

Radiation environments impose two fundamentally distinct threats on digital electronics: displacement damage and total ionizing dose (TID) acting as cumulative material-degradation mechanisms, and single-event effects (SEEs) acting as stochastic, event-driven disturbances. Of these, single-event upsets (SEUs) are of particular concern for SRAM-based field-programmable gate arrays (FPGAs) because such devices store both user data and their own circuit definition in volatile static-RAM cells that are as vulnerable to ionizing particles as any other SRAM. Assessing and mitigating this vulnerability requires purpose-built verification instruments that can observe radiation-induced state changes in real time, with sufficient state-volume coverage to yield statistically meaningful upset rates, yet compact enough to deploy inside a beam facility with limited infrastructure.

This chapter describes the design, implementation, and experimental results of the Flip-Flop Shift Register (FFSR) test structure used to evaluate Xilinx Spartan-7 FPGAs under fast neutron irradiation at the Ohio State University Research Reactor (OSURR) Fast Neutron Beam Facility (FNBF). Section 7.2 reviews relevant radiation

3474 effects in FPGAs. Section 7.3 discusses verification challenges specific to FPGA-based
3475 systems in nuclear environments. Section 7.4 presents the FFSR RTL architecture
3476 and governing algorithms. Section 7.5 describes the test configuration and radiation
3477 environment. Section 7.6 presents the experimental results from two reactor runs
3478 and the intervening overnight monitoring period. Section 7.7 discusses the observed
3479 upset mechanisms and their implications, and Section 7.8 summarises the principal
3480 findings.

3481 7.2 Radiation Effects in FPGAs

3482 SRAM-based FPGAs are widely used in nuclear instrumentation and space-borne
3483 electronics because of their reconfigurability, high logic density, and short design
3484 cycles. However, their reliance on SRAM storage cells for both configuration memory
3485 and user state makes them susceptible to a range of radiation-induced failure modes.

3486 **Single-Event Upsets.** An SEU occurs when a charged particle deposits suffi-
3487 cient energy in an SRAM cell to flip its stored bit. In the user data fabric, an SEU
3488 manifests as a transient or persistent logical error in the circuit under test. In the
3489 configuration SRAM, the consequence is more severe: a bit flip can alter LUT con-
3490 tents, routing multiplexers, embedded constant strings, or timing-control registers,
3491 effectively changing the implemented circuit itself. Because the configuration SRAM
3492 is not typically protected by on-chip error correction in commercial devices, such
3493 upsets persist until the device is power-cycled or a scrubbing operation rewrites the
3494 affected cell.

3495 **Single-Event Latch-up.** High-energy particle strikes can trigger parasitic bipo-
3496 lar structures in bulk CMOS, creating a low-impedance path between supply and

3497 ground. The resulting latch-up condition draws large currents, halts device opera-
3498 tion, and can cause permanent damage if not cleared by a prompt power cycle. The
3499 probability of latch-up increases with particle flux and LET.

3500 **Total Ionizing Dose Effects.** Prolonged irradiation builds up trapped charge
3501 in gate oxides and field oxides, increasing leakage current, shifting threshold voltages,
3502 and ultimately degrading transistor drive strength. TID effects accumulate monoton-
3503 ically with fluence and manifest at the system level as a gradual increase in supply
3504 current drawn by the device.

3505 **Displacement Damage.** Fast neutrons displace lattice atoms through nuclear
3506 recoil, creating interstitial-vacancy pairs that act as recombination centres and carrier
3507 traps. In CMOS logic this mechanism is generally less significant than in minority-
3508 carrier devices (e.g., solar cells or bipolar transistors), but at high fast-neutron fluences
3509 it contributes to parametric degradation.

3510 7.3 FPGA Verification Challenges in Nuclear Systems

3511 Several practical constraints complicate the experimental characterisation of FPGA
3512 upset susceptibility in nuclear beam facilities.

3513 **State-volume coverage.** An SEU cross-section measurement requires observing
3514 a statistically significant number of events. The cross-section for a single memory
3515 cell is of order 10^{-14} to 10^{-12} cm², so at a flux of 10^5 – 10^7 n/cm²/s a single cell
3516 would require impractically long exposures to accumulate even one event on average.
3517 Maximising the monitored state population is therefore essential.

3518 **Observability without intrusion.** Reading every flip-flop off-chip individually
3519 during irradiation is infeasible for large state populations and introduces additional

3520 wiring susceptible to radiation and noise. A compact on-chip checker that compresses
3521 the health of the state array into a small digital flag is preferable.

3522 **Configuration vs. data upsets.** Any experimental design must be capable of
3523 distinguishing a user-data upset from a configuration memory upset. The two mech-
3524 anisms require different mitigation strategies: data-path redundancy (DMR/TMR)
3525 addresses the former, while periodic configuration scrubbing addresses the latter.

3526 **Telemetry reliability.** The UART or other communication channel used to
3527 export results is itself implemented in FPGA resources and therefore subject to the
3528 same configuration upsets. A robust transmitter design reduces the risk that the
3529 observability channel itself is corrupted during an experiment.

3530 7.4 FFSR RTL Architecture

3531 7.4.1 System-Level Overview

3532 The FFSR test structure is an FPGA-resident self-test instrument designed to
3533 address the observability and state-volume challenges described in Section 7.3. Its
3534 purpose is to instantiate a large, controllable population of sequential elements, ex-
3535 ercise those elements with known scan/capture behaviour, and report any deviation
3536 from the expected internal state in real time. The design contains four cooperating
3537 subsystems:

- 3538 1. A **top-level experiment controller** (`top_selftest_100mhz_dbg`) that gen-
3539 erates experiment timing, reset sequences, capture windows, and telemetry ca-
3540 dence.
- 3541 2. A **scan-farm fabric** (`ff_scanfarm / ff_scanfarm_sig`) that instantiates many
3542 shift-register chains in parallel, scaling the monitored state volume.

3543 3. A **signature checker** that computes an expected compact signature from the
3544 chain contents and compares it against the observed signature to produce live
3545 and latched error flags.

3546 4. A **UART telemetry path** that serialises concise status words for logging on
3547 a host computer, enabling offline correlation with beam current, fluence, and
3548 shutter state.

3549 Rather than streaming every monitored bit off chip, the design compresses the
3550 health of a large state space into a smaller set of mismatch indicators and coun-
3551 ters. This separation of upset *occurrence* from complete state readout enables larger
3552 monitored populations and simpler experimental integration while still supporting
3553 controlled injection tests and persistent error latching.

3554 7.4.2 Module Descriptions

3555 Table 7.1 summarises the primary RTL blocks.

3556 **top_selftest_100mhz_dbg**

3557 The `top_selftest_100mhz_dbg` module is the supervisory block of the FFSR
3558 experiment. It establishes the operating clock domain, applies reset, sequences the
3559 scan and capture phases, clears latched errors when requested, and routes status
3560 information into the UART telemetry path. In a typical implementation, operation
3561 is divided into repeated frames or epochs: the design allows the farm to evolve or
3562 hold state, then enables capture, then enables scan shifting, then examines mismatch
3563 outputs, and finally serialises the results.

Table 7.1: Summary of primary RTL blocks in the FFSR design.

Module	Primary Role	Key Parameters / Signals	Main Idea	Algorithmic
<code>top_selftest_100mhz_dbg</code>	Top-level experiment controller and telemetry wrapper	Clock, reset, scan enable, capture enable, <code>clear_errors</code> , UART handshake, status/counter outputs	Coordinates test phases; periodically packages status for serial transmission	
<code>ff_scanfarm</code>	Wrapper layer for the scan farm	<code>CHAINS</code> , <code>BITS_PER_CHAIN</code> , <code>INJECT_EN</code> , <code>INJ_TAP</code> , <code>error_live</code> , <code>error_any</code>	Preserves a stable top-level interface while instantiating the robust signature-based implementation	
<code>ff_scanfarm_sig</code> / scan-chain fabric	Core monitored state array	<code>clk_scan</code> , <code>scan_en</code> , <code>capture_en</code> , <code>scan_in/out</code> , signature bits, mismatch flags	Updates each chain per shift/capture rules; compares compact state signature against expected value	
<code>scan_chain</code> primitive	One lane of monitored sequential storage	Serial input/output, per-stage registers, optional injection tap	Implements shift-register recurrences; optional controlled bit-flip insertion for self-verification	
<code>prbs31</code>	Pseudo-random pattern source	31-bit state, enable/reset, PRBS output bit/word	Maximal-length LFSR using $x^{31} + x^{28} + 1$; $2^{31} - 1$ states before repetition	
<code>uart_tx</code> / <code>uart_tx_robust</code>	UART serialiser; robust variant adds framing protection	<code>tx_start</code> , <code>tx_data</code> , baud divider, <code>tx_busy</code> , <code>txd</code>	Converts parallel byte to start bit + 8 data bits + stop bit at programmable baud rate; robust variant prevents malformed frames	

3564 **ff_scanfarm and ff_scanfarm_sig**

3565 The **ff_scanfarm** module acts as a named interface to the monitored sequential
3566 fabric. In the wrapper variant used in this project, it preserves a stable external inter-
3567 face while instantiating the more robust internal implementation **ff_scanfarm_sig**.
3568 This decouples the project hierarchy from changes in the checker implementation.
3569 The key outputs are a real-time mismatch indicator, **error_live**, and a sticky fault
3570 memory, **error_any**.

3571 The scan-farm core is parameterised by the number of parallel chains C and the
3572 number of bits per chain B , giving a total monitored state volume of:

$$N_{\text{state}} = C \times B. \quad (7.1)$$

3573 **scan_chain Primitive**

3574 At the lowest level, the **scan_chain** logic implements the per-stage shift-register
3575 recurrence. A chain receives a serial input bit, shifts stored data through successive
3576 registers when **scan_en** is asserted, and presents the final stage at **scan_out**. An
3577 optional injection feature can flip a selected stage, allowing the checker path to be
3578 validated without radiation. Because the chain state is deterministic under com-
3579 manded scan and capture sequences, any uncommanded state change is detectable
3580 through signature mismatch.

3581 **7.4.3 Core Algorithms**

3582 **Shift-Register State Update**

3583 Let $q_{c,i}[k]$ denote the state of stage i in chain c at clock step k , where $i =$
3584 $0, 1, \dots, B-1$ and $c = 0, 1, \dots, C-1$. During scan mode the chain obeys the standard

3585 shift recurrence:

$$q_{c,0}[k+1] = u_c[k], \quad (7.2)$$

$$q_{c,i}[k+1] = q_{c,i-1}[k], \quad i > 0, \quad (7.3)$$

3586 where $u_c[k]$ is the applied serial scan input for chain c . In hold mode, $q_{c,i}[k+1] =$
 3587 $q_{c,i}[k]$.

3588 **Optional Upset Injection for Verification**

3589 To validate the detector path without radiation exposure, a controlled injection
 3590 can be applied at tap t_{inj} :

$$q_{c,t_{\text{inj}}}[k+1] = q_{c,t_{\text{inj}}}^{\text{nom}}[k+1] \oplus \text{inj}[k], \quad (7.4)$$

3591 where $\text{inj}[k]$ is asserted only during a commanded test window and q^{nom} denotes the
 3592 nominal update in the absence of injection.

3593 **Signature Computation and Mismatch Detection**

3594 A compact checker is formed by reducing chain contents into parity-like signatures.
 3595 The signature for chain c is:

$$s_c[k] = \bigoplus_{i=0}^{B-1} q_{c,i}[k]. \quad (7.5)$$

3596 The fabric-level signature vector is $\mathbf{S}[k] = [s_0[k], s_1[k], \dots, s_{C-1}[k]]$. The instantana-
 3597 neous mismatch flag is:

$$e_{\text{live}}[k] = \begin{cases} 1 & \text{if } \mathbf{S}_{\text{obs}}[k] \neq \mathbf{S}_{\text{exp}}[k], \\ 0 & \text{otherwise.} \end{cases} \quad (7.6)$$

3598 **Sticky Error Memory**

3599 A sticky fault flag preserves the occurrence of any prior mismatch even if the chain
3600 later returns to agreement:

$$e_{\text{any}}[k+1] = (e_{\text{any}}[k] \vee e_{\text{live}}[k]) \wedge \overline{\text{clear_errors}[k]}. \quad (7.7)$$

3601 This form is essential in beam testing because transient upsets are not lost between
3602 telemetry packets.

3603 **PRBS31 Recurrence**

3604 The `prbs31` module generates a pseudo-random binary sequence using a 31-bit
3605 linear feedback shift register with the primitive polynomial $p(z) = z^{31} + z^{28} + 1$. Using
3606 zero-based indexing with x_{30} as the most-significant stage, the feedback bit is:

$$f[k] = x_{30}[k] \oplus x_{27}[k], \quad (7.8)$$

3607 and the next state is:

$$\mathbf{x}[k+1] = \{x_{29}[k], x_{28}[k], \dots, x_0[k], f[k]\}. \quad (7.9)$$

3608 Provided the all-zero state is avoided, this recurrence cycles through $2^{31} - 1$ non-
3609 zero states before repeating, yielding strong transition density and good statistical
3610 coverage for stressing the sequential fabric.

3611 **UART Frame Formation**

3612 For an 8-bit telemetry byte $D = [d_0, d_1, \dots, d_7]$, the UART transmitter outputs:

$$\text{TX frame} = [0, d_0, d_1, \dots, d_7, 1], \quad (7.10)$$

3613 where 0 is the start bit and 1 is the stop bit. The nominal bit-period divider is:

$$N_{\text{baud}} = \left\lceil \frac{f_{\text{clk}}}{f_{\text{baud}}} \right\rceil, \quad (7.11)$$

3614 where $\lceil \cdot \rceil$ denotes rounding to the nearest integer. The `uart_tx_robust` variant
 3615 extends the baseline transmitter with stronger start-request handling, safer state
 3616 transitions, and additional protection against malformed back-to-back frames under
 3617 transient control behaviour.

3618 7.5 Test Configuration

3619 7.5.1 Devices Under Test

3620 Five Digilent CMOD S7 boards, each loaded with the FFSR dual modular redun-
 3621 dancy (DMR) firmware on a Xilinx Spartan-7 FPGA, were monitored simultaneously.
 3622 Each device was connected to the host data acquisition PC via a dedicated USB-to-
 3623 serial COM port at 115 200 baud. The DMR comparator within the FFSR design
 3624 monitored two redundant data paths and incremented a 32-bit upset counter when-
 3625 ever a mismatch was detected. The counter value was reported over UART every
 3626 5 seconds. A non-zero counter output indicated detection of one or more SEUs in the
 3627 user data path. Table 7.2 summarises the device configuration and test timeline.

Table 7.2: CMOD S7 device configuration and test timeline.

Device	COM Port	Baseline Start	Reactor Start	Reactor End
FFSR1	COM8	21:21, Jan 26	11:42, Jan 27	16:34, Jan 27
FFSR2	COM7	22:41, Jan 26	11:42, Jan 27	16:34, Jan 27
FFSR3	COM6	22:41, Jan 26	11:42, Jan 27	16:34, Jan 27
FFSR4	COM9	22:41, Jan 26	11:42, Jan 27	16:34, Jan 27
FFSR5	COM4	22:41, Jan 26	11:42, Jan 27	16:34, Jan 27

3628 7.5.2 Radiation Environment

3629 The test was conducted at the OSURR Fast Neutron Beam Facility over two days
3630 using distinct beam configurations that produced substantially different neutron flux
3631 and fluence levels.

3632 **Day 1 (gamma shutter closed, January 27).** The reactor operated at approx-
3633 imately 300 kW (steady-state), reached within ~ 10 minutes of start-up and sustained
3634 for approximately 4.7 hours. The fast neutron flux (steady-state) was approximately
3635 2×10^5 n/cm²/s; the thermal component was approximately one order of magnitude
3636 lower. Total accumulated fast neutron fluence was approximately $6\text{--}7 \times 10^9$ n/cm².
3637 The fast and total fluence curves were nearly coincident throughout, confirming fast-
3638 neutron dominance.

3639 **Day 2 (gamma shutter open, January 28).** Reactor power remained at
3640 approximately 300 kW, but the open-shutter configuration produced a fast neutron
3641 flux of approximately 1.5×10^7 n/cm²/s—roughly two orders of magnitude higher
3642 than Day 1. The thermal flux was approximately 6×10^6 n/cm²/s (within a factor
3643 of 2–3 of the fast flux). Total accumulated fast neutron fluence was approximately
3644 $8\text{--}9 \times 10^{11}$ n/cm², also approximately two orders of magnitude greater than Day 1.

3645 Figures 7.1–7.3 show the neutron fluence, flux, and power profiles for the Day 1
3646 run; Figures 7.4–7.6 show the corresponding plots for the Day 2 run. Figures 7.7–7.9
3647 provide overlay comparisons of total fluence, flux, and reactor power between the two
3648 runs.

3649 The Day 1 beam configuration corresponds to the gamma shutter *closed* condition
3650 at the OSURR Fast Neutron Beam Facility. The FNBF beamline incorporates a

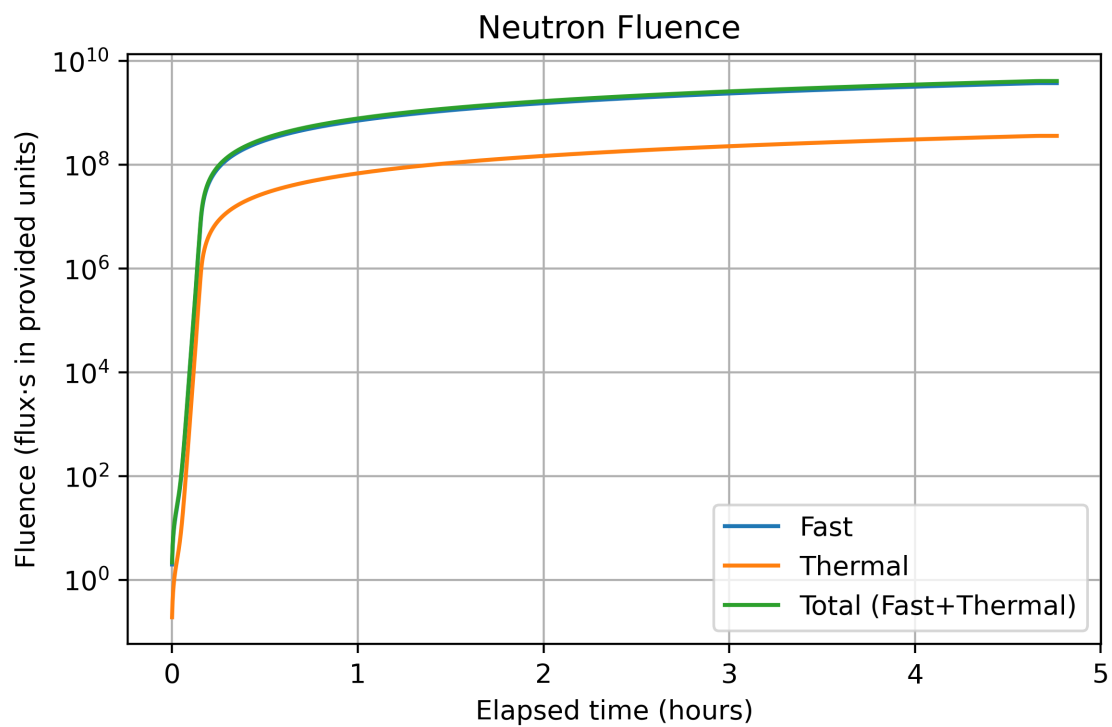


Figure 7.1: Neutron fluence (fast, thermal, and total) accumulated during the Day 1 (gamma shutter closed) reactor run at the OSURR FNBF. The fast and total curves are nearly coincident, confirming fast-neutron dominance at this beam configuration.

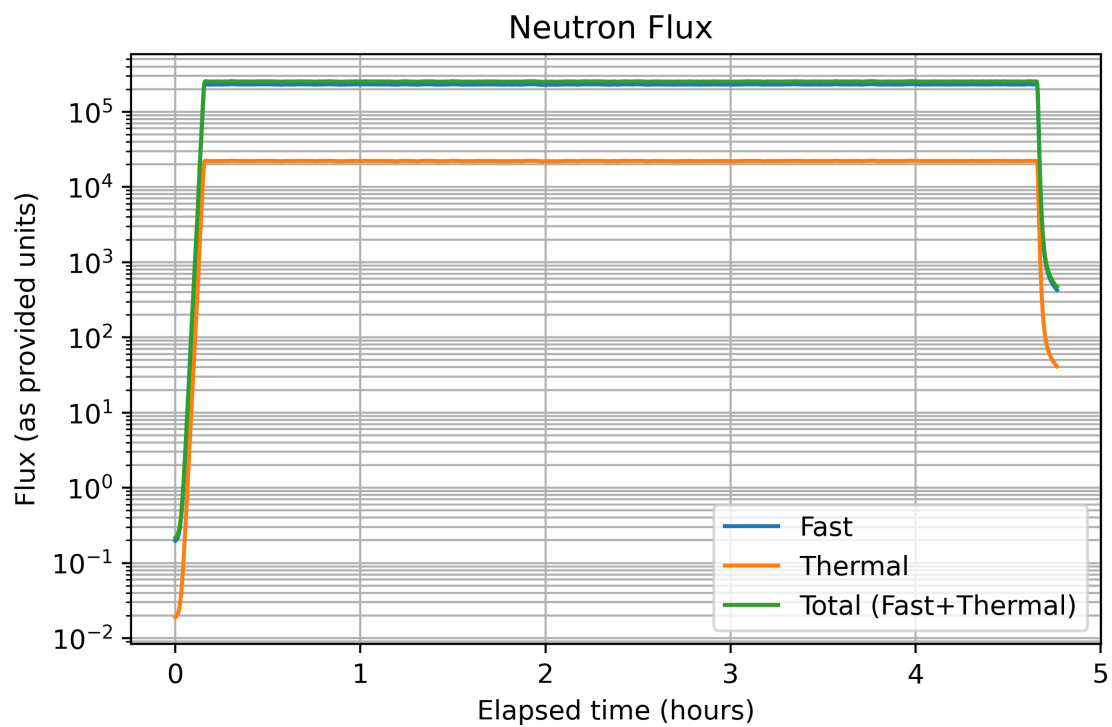


Figure 7.2: Neutron flux (fast, thermal, and total) during the Day 1 (gamma shutter closed) run. Fast flux reached a steady-state value of approximately 2×10^5 n/cm²/s; the thermal component was approximately one order of magnitude lower.

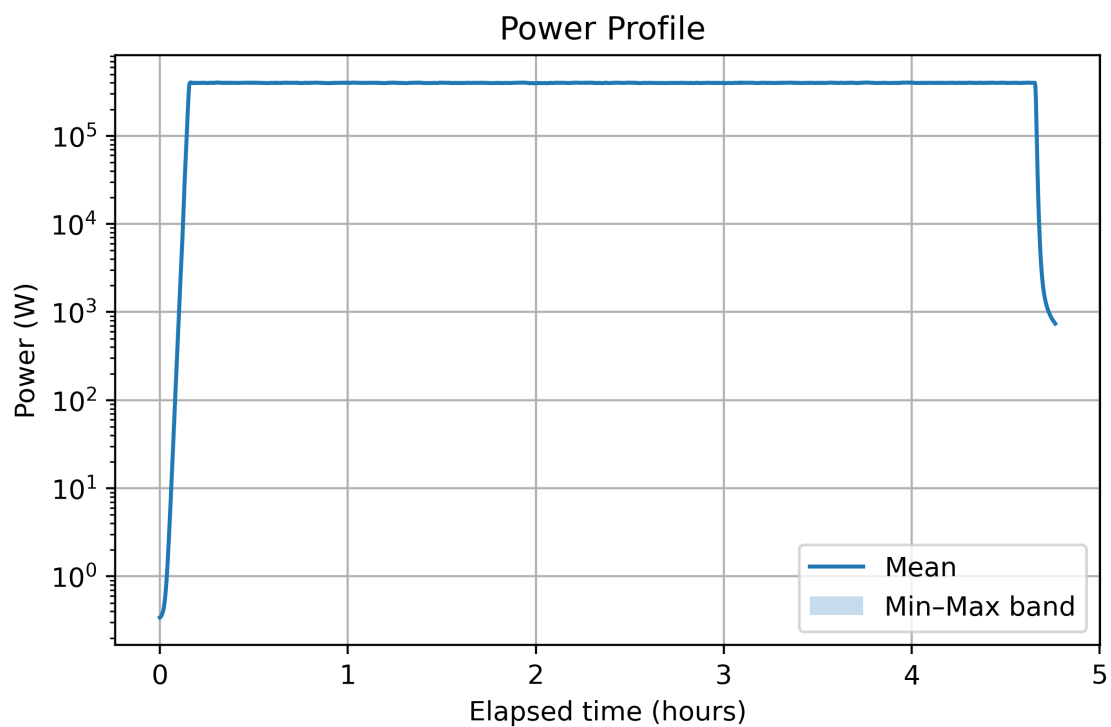


Figure 7.3: Reactor power profile during the Day 1 run. Full power (~ 300 kW) was reached within approximately 10 minutes of start-up and sustained for approximately 4.7 hours before controlled shutdown.

3651 4-inch (10.16 cm) single-crystal bismuth block near the reactor core as a passive in-
 3652 line gamma filter, followed by an 8-inch (20.32 cm) lead gamma shutter embedded
 3653 in the reactor's concrete shielding wall [?]. When the lead shutter is closed, the
 3654 gamma-ray content of the beam is reduced by a factor of 311 relative to the fully
 3655 open configuration, while fast neutrons are simultaneously attenuated by a factor
 3656 of 58. Under these conditions, the steady-state fast neutron flux ($E_n > 0.5$ eV)
 3657 at the beam collimator exit is approximately 2.6×10^5 n/cm²/s, while the thermal
 3658 neutron flux ($E_n < 0.5$ eV) is approximately 2.5×10^4 n/cm²/s—roughly one order of
 3659 magnitude lower than the fast component. The residual gamma dose rate with the
 3660 shutter closed is approximately 50 mR/h. The neutron energy spectrum of the FNBF
 3661 beam, characterised using the SAND-II and STAYSL unfolding techniques with Cu,
 3662 Au, and Ni metal wire neutron activation, spans thermal energies through the fast
 3663 fission spectrum, but the fast and epi-thermal components dominate the integrated
 3664 flux [?]. As a result, the fast and total neutron fluence curves are nearly coincident in
 3665 Figure 7.1, with the thermal contribution representing a minor fraction throughout
 3666 the irradiation period.

3667 The Day 2 beam configuration corresponds to the gamma shutter *open* condition.
 3668 With the lead shutter removed from the beam path, the fast neutron flux at the
 3669 collimator exit increases by a factor of approximately 58, reaching a steady-state
 3670 value of approximately 1.5×10^7 n/cm²/s [?]. The thermal neutron flux simultaneously
 3671 rises to approximately 8.3×10^6 n/cm²/s—now within approximately a factor of two
 3672 of the fast component, a substantially narrower fast-to-thermal ratio than the one
 3673 order of magnitude separation observed in the shutter-closed configuration. This
 3674 narrowing occurs because the lead shutter attenuates neutrons across the full energy

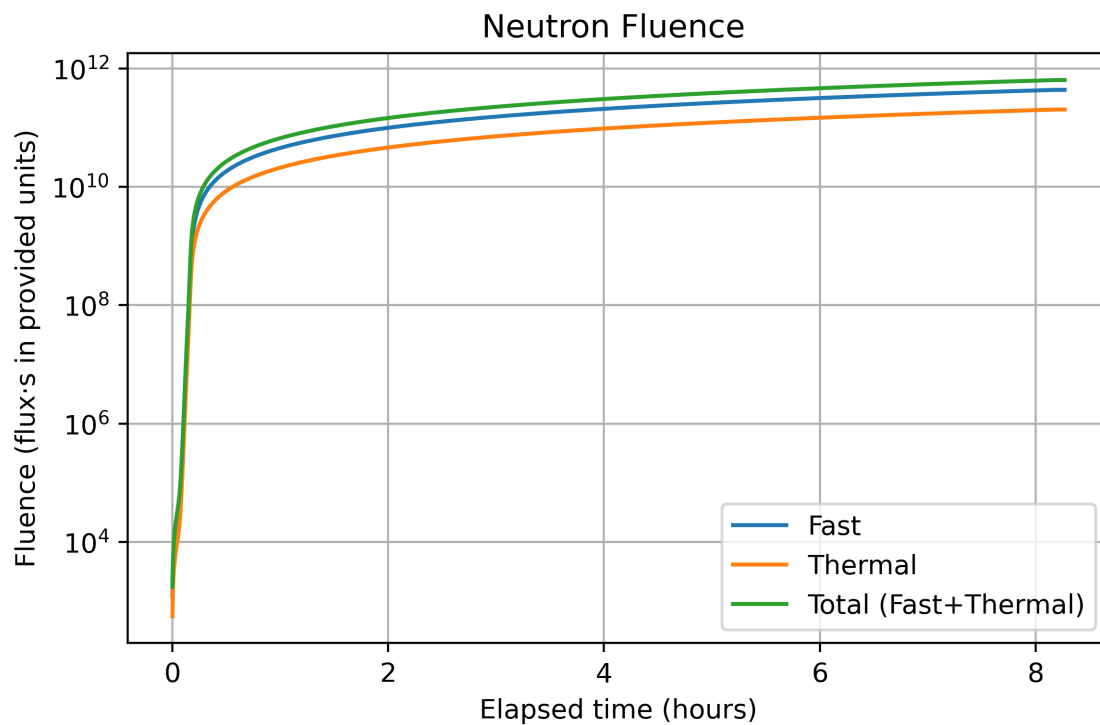


Figure 7.4: Neutron fluence (fast, thermal, and total) accumulated during the Day 2 (gamma shutter open) reactor run. Total fast fluence reached approximately $8\text{--}9 \times 10^{11}$ n/cm², roughly two orders of magnitude greater than Day 1.

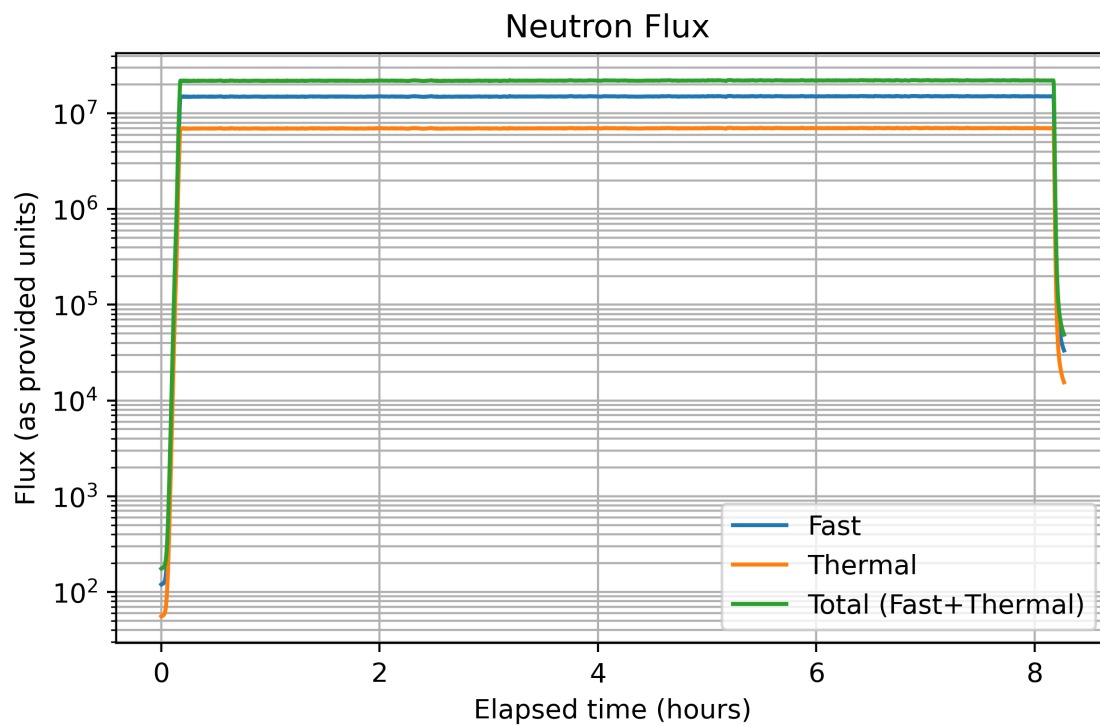


Figure 7.5: Neutron flux during the Day 2 (gamma shutter open) run. Fast flux reached approximately 1.5×10^7 n/cm²/s and thermal flux approximately 6×10^6 n/cm²/s, narrowing the fast-to-thermal ratio compared with Day 1.

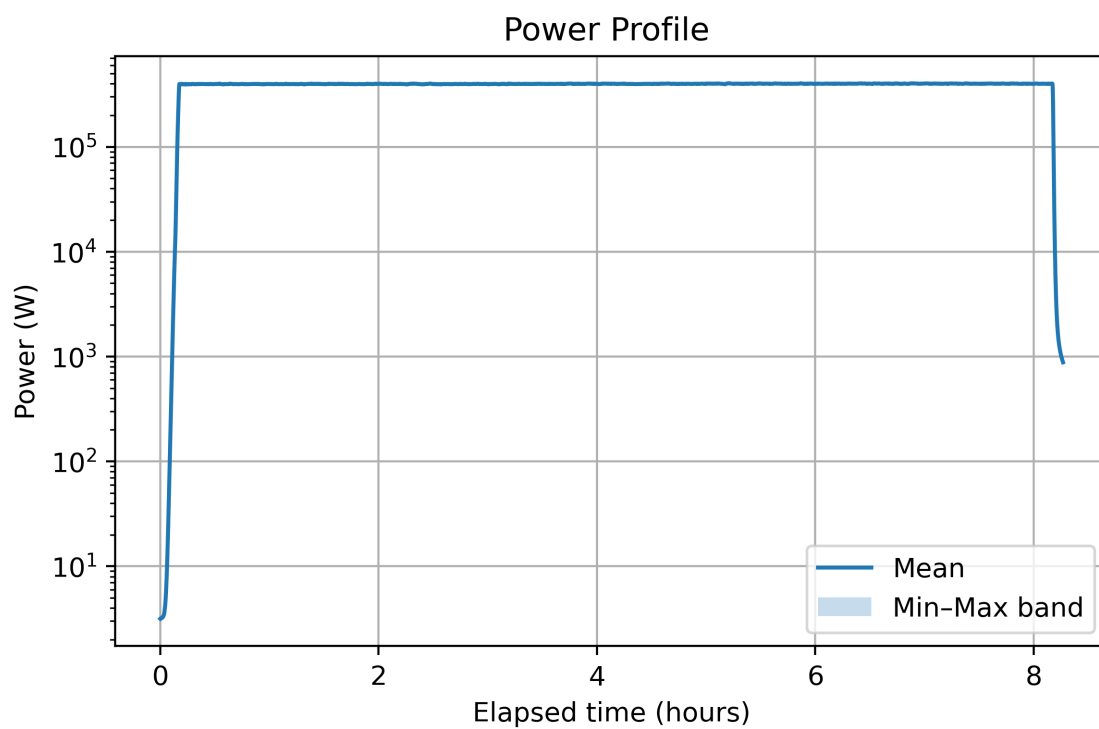


Figure 7.6: Reactor power profile during the Day 2 run. Full power was sustained for approximately 8.5 hours before shutdown.

spectrum; when it is removed, the beam carries a larger thermal contribution that is otherwise suppressed by the lead. The elevated thermal flux in the open-shutter configuration is relevant for interpreting FPGA upset behaviour because both fast and thermal neutrons can induce nuclear recoil events in silicon, though fast neutrons dominate displacement damage due to their higher energy transfer per elastic scatter interaction. Taken together, the Day 2 exposure accumulated a total fast neutron fluence of approximately $8\text{--}9 \times 10^{11}$ n/cm² over the 8.5-hour run—roughly two orders of magnitude greater than Day 1—and it is this step change in fluence that drives the qualitative difference in observed device behaviour between the two runs.

The overlay figures below place both beam configurations on common axes, making the shutter-state-driven difference in delivered fluence and flux directly apparent and confirming that the reactor power profile was consistent across the two days.

7.6 Experimental Results

7.6.1 Baseline Test (January 26, Out of Beam)

Prior to transport to the reactor facility, all five devices were operated at ambient background radiation levels to establish a baseline and to confirm correct data acquisition system operation. FFSR1 began logging at 21:21 on January 26; FFSR2–5 began at 22:41. All devices reported CNT:00000000 continuously throughout the baseline period, confirming zero false positives from the test setup and demonstrating that the DMR detection logic was functioning correctly.

Logging on all five devices terminated simultaneously at approximately 22:59 due to a Windows Update reboot on the acquisition PC. This was not a device failure. After the reboot, logging was not restarted until the devices were at the reactor

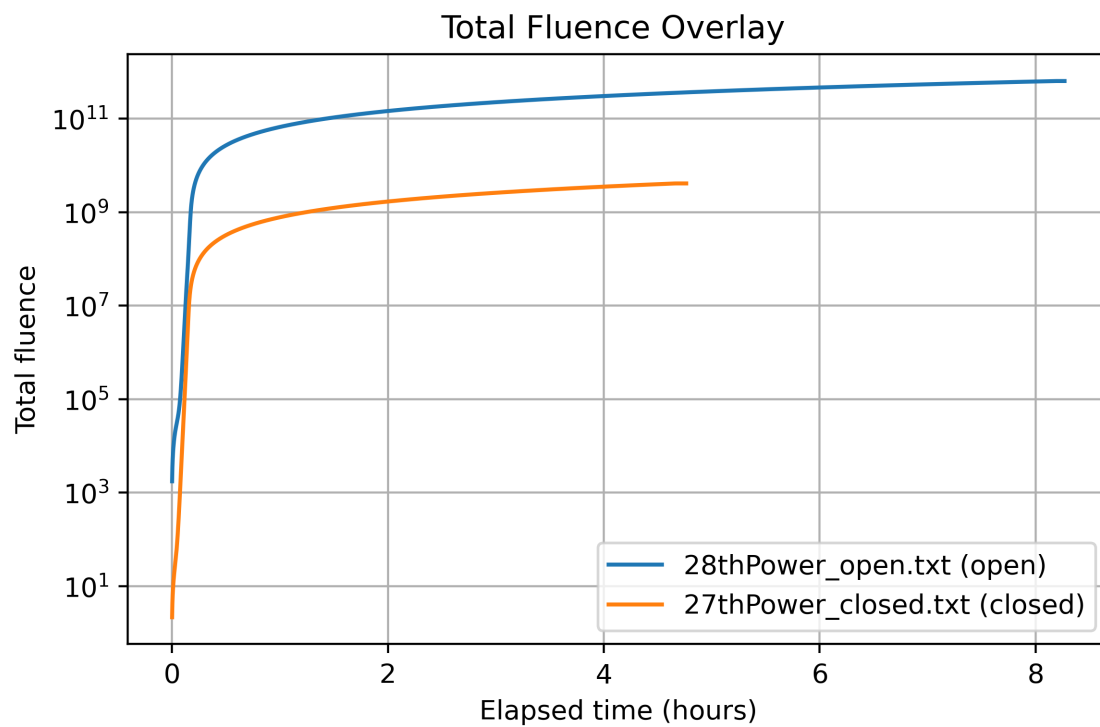


Figure 7.7: Overlay of total neutron fluence (fast + thermal) for the Day 1 (shutter closed) and Day 2 (shutter open) runs, illustrating the approximately two-orders-of-magnitude difference in accumulated fluence between the two configurations.

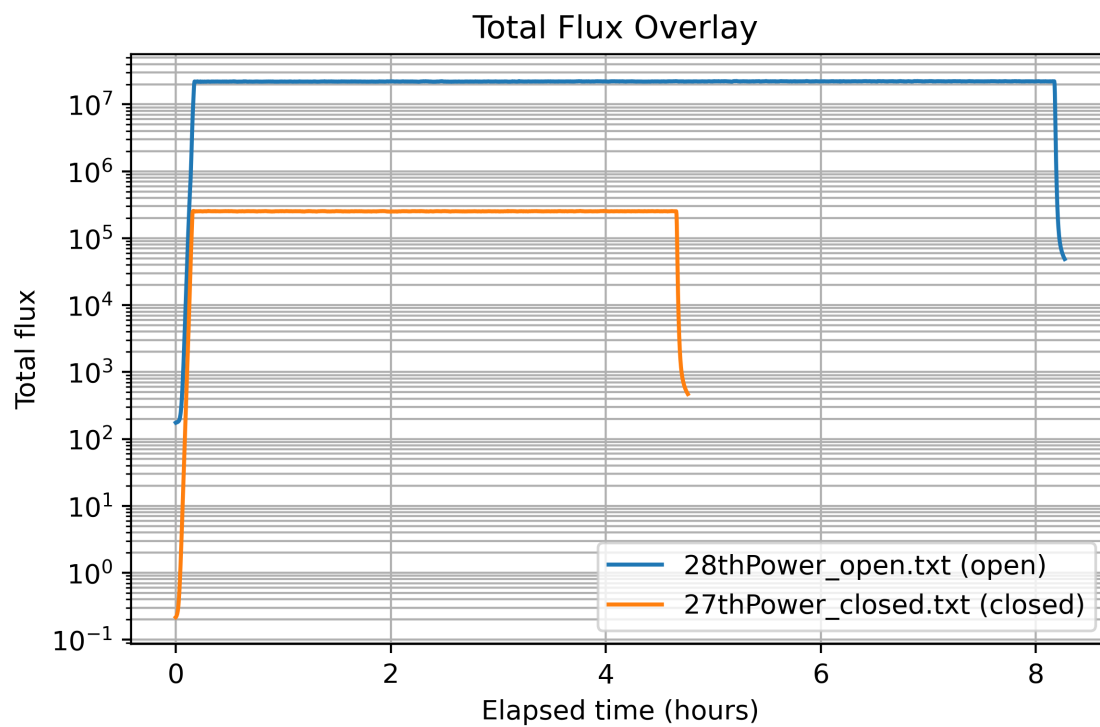


Figure 7.8: Overlay of total neutron flux for both runs. The Day 2 (open shutter) flux exceeded the Day 1 (closed shutter) flux by approximately two orders of magnitude throughout the steady-state period.

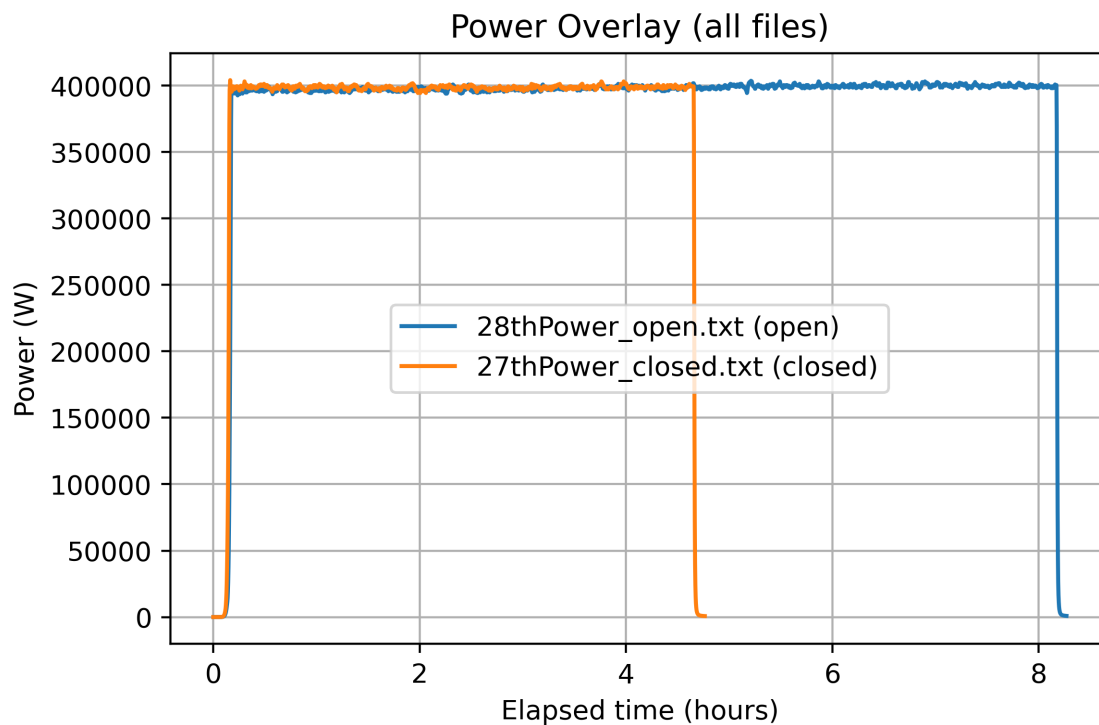


Figure 7.9: Reactor power overlay for both runs on a linear scale. Both days operated at the same nominal reactor power ($\sim 300\text{--}400$ kW); the large difference in neutron flux between the two days arose entirely from the gamma shutter configuration, not from a change in reactor power.

3698 facility the following morning. Future tests should disable automatic OS updates on
 3699 the acquisition PC for the duration of the experiment.

3700 **7.6.2 Day 1 Reactor Run (January 27)**

3701 Logging resumed at the reactor facility at approximately 11:42 on January 27 for
 3702 all five devices. The reactor reached full power within the first 10–15 minutes. All
 3703 devices monitored continuously through reactor shutdown at approximately 16:34,
 3704 providing approximately five hours of reactor run data.

3705 **Upset Count Results**

3706 Zero SEUs were detected on any of the five devices throughout the entire Day 1
 3707 reactor run at a total fast neutron fluence of approximately $6\text{--}7 \times 10^9$ n/cm². All
 3708 devices reported a consistent CNT:00000000 output for the full irradiation duration.
 3709 Table 7.3 summarises the Day 1 results.

Table 7.3: Day 1 reactor run upset detection results. *FFSR1 count includes 250 CNT-format and $\approx 3\,252$ CNZ-format readings (see Section 7.6.2).

Device	COM	Start	End	Readings	SEUs
FFSR1	COM8	11:42	16:34	$\approx 3\,502^*$	0
FFSR2	COM7	11:42	16:34	3 491	0
FFSR3	COM6	11:42	16:34	3 496	0
FFSR4	COM9	11:42	16:34	3 497	0
FFSR5	COM4	11:42	16:34	3 496	0

3710 **PSU Current Monitoring—Day 1**

3711 Power supply current was monitored for the CMOD board array throughout the
3712 Day 1 run. As shown in Figure 7.10, the current profile exhibits a power-on inrush
3713 transient from approximately 0.175 A, rising rapidly to a steady-state value of approx-
3714 imately 0.283–0.286 A within the first ~ 200 seconds as the boards initialised. The
3715 current then remained essentially flat for the entire $\sim 17\,500$ -second run (~ 4.9 hours).

3716 One small transient spike to approximately 0.289 A is visible at $\sim 12\,500$ s, consis-
3717 tent with a brief radiation-induced transient event; the current returned immediately
3718 to the baseline level with no permanent change. Critically, no monotonic TID-induced
3719 current increase is evident across the Day 1 run. The flat current profile is consistent
3720 with the Day 1 fluence level being insufficient to produce measurable TID-related
3721 leakage current degradation in the CMOS circuitry, providing a useful baseline for
3722 comparison with the Day 2 profile (Section 7.6.4).

3723 **FFSR1 UART Format Change—Candidate Configuration Memory SEU**

3724 At approximately 12:03 on January 27, the UART output format on FFSR1
3725 changed from CNT:00000000 to CNZ0000. The label changed from CNT to CNZ, the
3726 colon delimiter disappeared, and the counter field width reduced from 8 hex digits
3727 to 4. This format change persisted for the remainder of the run; the upset count
3728 remained zero throughout in both formats; and FFSR2–5 showed no similar change.

3729 This behaviour is consistent with a single-event upset in the FPGA configuration
3730 memory (SRAM) rather than in the user data path. The Spartan-7 is an SRAM-
3731 based FPGA, meaning the design implementation—LUT contents, routing resources,
3732 and embedded string constants—is stored in volatile SRAM configuration bits as

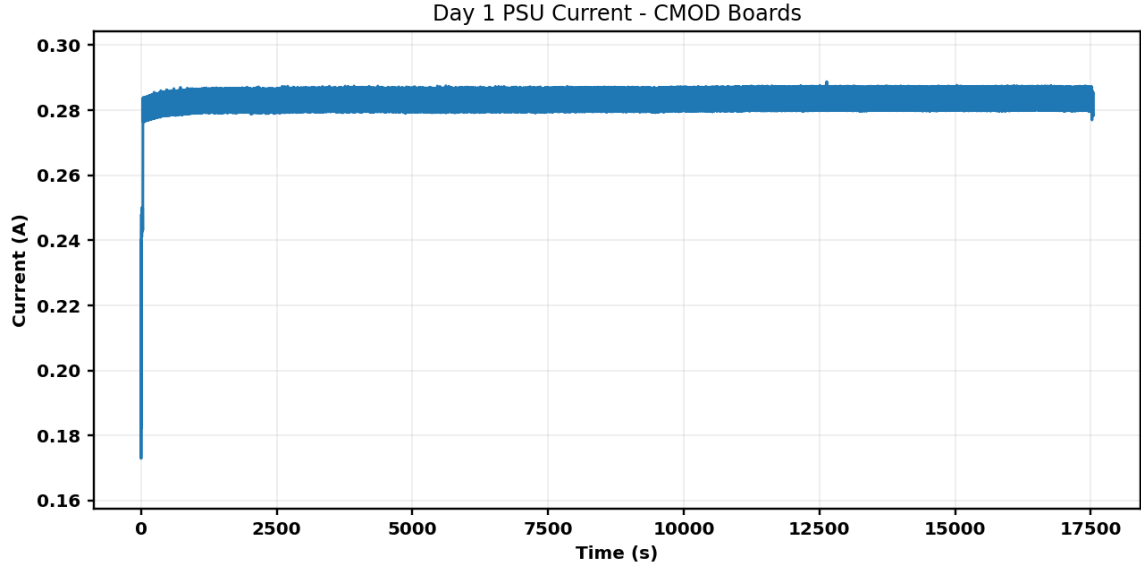


Figure 7.10: PSU current for the CMOD board array during the Day 1 (gamma shutter closed) reactor run. Current stabilises near 0.284 A after initial power-up and remains flat throughout, indicating no measurable TID-induced leakage accumulation at the Day 1 fluence level ($\sim 6\text{--}7 \times 10^9 \text{ n/cm}^2$).

3733 susceptible to neutron-induced upsets as any other SRAM. A bit flip in one or more
 3734 configuration cells controlling the UART formatting logic could plausibly alter the
 3735 output label and field width in exactly the structured manner observed.

3736 This event is significant for two reasons. First, the DMR logic correctly protected
 3737 the user flip-flop data path and reported zero upsets throughout, but it does not pro-
 3738 tect the configuration memory; the observation highlights that data-path redundancy
 3739 alone is insufficient for SRAM-based FPGAs in radiation environments. Second, the
 3740 single affected device (out of five) is consistent with the stochastic nature of SEUs at
 3741 the Day 1 fluence level.

3742 7.6.3 Overnight Monitoring (January 27–28)

3743 Following the Day 1 reactor run, all five devices were left running overnight out of
 3744 beam at the reactor facility. Logging ran from approximately 16:42 on January 27 to
 3745 approximately 07:13 on January 28, providing roughly 14.5 hours of out-of-beam mon-
 3746 itoring at ambient background levels. All five devices reported zero upsets throughout
 3747 the overnight period, as expected. Table 7.4 summarises the results.

Table 7.4: Overnight (out-of-beam) monitoring results.

Device	COM	Log Start	Log End	Readings	SEUs
FFSR1	COM8	16:48, Jan 27	07:13, Jan 28	13 981	0
FFSR2	COM7	16:48, Jan 27	07:13, Jan 28	10 389	0
FFSR3	COM6	16:48, Jan 27	07:13, Jan 28	10 354	0
FFSR4	COM9	16:48, Jan 27	07:13, Jan 28	10 353	0
FFSR5	COM4	16:48, Jan 27	07:13, Jan 28	10 353	0

3748 Notably, FFSR1 continued to output in the CNZ0000 format throughout the en-
 3749 tire overnight period, including after three reconnect attempts (16:42, 16:45, 16:48).
 3750 For an SRAM-based FPGA, configuration memory is volatile—loaded from flash at
 3751 power-up and held in SRAM during operation. A bit flip in the configuration SRAM
 3752 will persist indefinitely until either the device is power-cycled or a scrubbing oper-
 3753 ation overwrites the affected cell. The persistence of the CNZ format through log-
 3754 ging restarts, which do not cause a power cycle, but the expectation that it would
 3755 be cleared by a full power cycle, is consistent with a configuration SRAM upset.
 3756 This observation reinforces the importance of distinguishing between logging-session

3757 restarts and full device power cycles when interpreting FPGA behaviour in radiation
3758 test environments.

3759 Figure 7.11 shows the PSU current during the overnight period. Current was
3760 stable near 0.2835 A with occasional small transient spikes, confirming normal device
3761 operation at ambient background levels and providing a clean pre-Day-2 baseline.

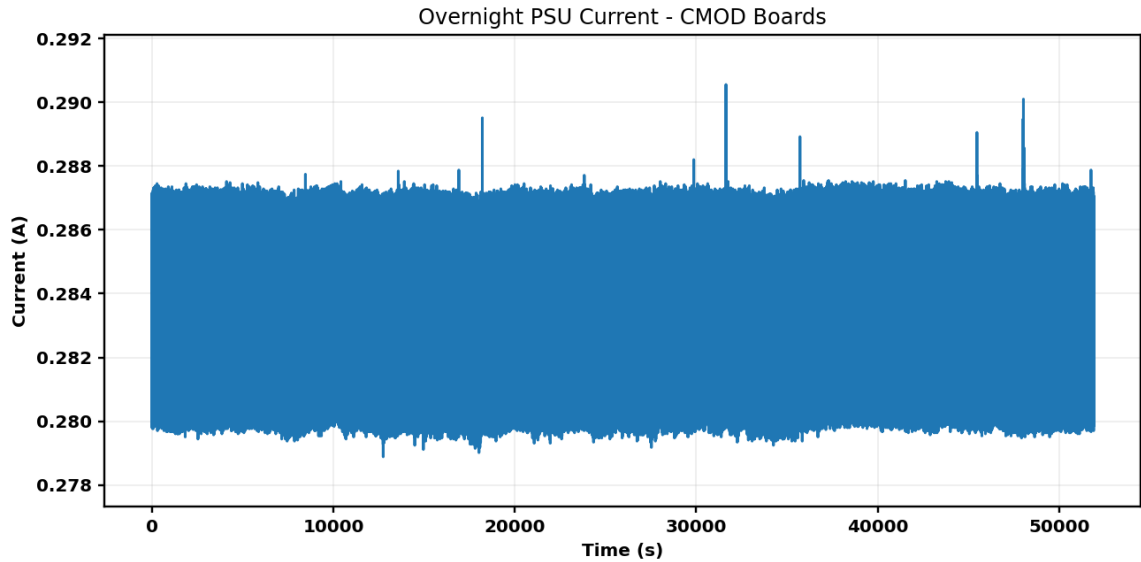


Figure 7.11: PSU current for the CMOD board array during the overnight (out-of-beam) monitoring period (January 27–28). Current is stable near 0.284 A with occasional small transient spikes, confirming normal device operation at ambient background radiation levels.

3762 7.6.4 Day 2 Reactor Run (January 28)

3763 The Day 2 reactor run was conducted with the gamma shutter open, producing
3764 a fast neutron flux approximately two orders of magnitude higher than Day 1. All

3765 five devices began logging at 07:21 and the reactor reached full power within approx-
 3766 imately 20 minutes. The reactor ran for approximately 8.5 hours before controlled
 3767 shutdown. Table 7.5 summarises the Day 2 results.

Table 7.5: Day 2 (gamma shutter open) upset detection results.

Device	COM	Log Start	Log End	Readings	SEUs / Notes
FFSR1	COM8	07:21, Jan 28	15:54, Jan 28	715	CNZ0800 detected (Sec. 7.6.4)
FFSR2	COM7	07:21, Jan 28	14:44, Jan 28	5 298	0
FFSR3	COM6	07:21, Jan 28	08:10, Jan 28	580	0—early dropout (~49 min)
FFSR4	COM9	07:21, Jan 28	07:57, Jan 28	431	0—early dropout (~36 min)
FFSR5	COM4	07:21, Jan 28	15:55, Jan 28	6 144	0

3768 **FFSR3 and FFSR4 Early Dropouts**

3769 FFSR4 ceased transmitting at approximately 07:57 (~36 minutes into the run)
 3770 and FFSR3 at approximately 08:10 (~49 minutes in). Both devices stopped cleanly
 3771 with no error messages; data simply ended mid-run and neither device recovered.
 3772 Two plausible mechanisms are consistent with these observations:

- 3773 • **Radiation-induced latch-up.** A high-energy neutron interaction could trig-
 3774 ger a latch-up condition in the FPGA I/O or core circuitry, causing the device
 3775 to lock up and halt UART output. Latch-up typically requires a power cycle to
 3776 recover and can cause permanent damage if not cleared promptly.
- 3777 • **Critical configuration memory upset.** A severe or multi-bit upset in a
 3778 critical portion of the configuration memory (e.g., clock management or I/O
 3779 control logic) could halt device operation entirely, even without a true latch-up.

3780 The PSU current data (Figure 7.12) shows step increases at approximately 2 800 s
3781 and 5 000 s into the run. FFSR4 stopped at approximately 2 160 s and FFSR3 at
3782 approximately 2 940 s, placing both dropouts near the step increase at $\sim 2\,800$ s. An
3783 abrupt change in aggregate board current at this time is consistent with a latch-up
3784 event altering one or both devices' current draw. These early dropouts represent
3785 the most severe functional failures observed during the two-day campaign and are
3786 consistent with the two-orders-of-magnitude increase in fast neutron flux on Day 2.

3787 **PSU Current Monitoring—Day 2**

3788 Figures 7.12 and 7.13 show the CMOD board aggregate PSU current during the
3789 Day 2 run. Key observations are:

- 3790 • **Part 1 ($\sim 07:21\text{--}14:15$):** Current increased steadily from approximately 0.28 A
3791 at run start to approximately 0.315 A by the end of the logging window—an
3792 increase of approximately 12.5%. This is in stark contrast to the flat Day 1 pro-
3793 file and is consistent with TID effects accumulating in the FPGA and support
3794 circuitry, increasing leakage current monotonically under the much higher Day 2
3795 fluence. Step increases near 2 800 s and 5 000 s correspond to the approximate
3796 dropout times of FFSR4 and FFSR3 respectively.
- 3797 • **Part 2 ($\sim 14:28\text{--}16:00$):** Current was approximately 0.315 A at the start of
3798 this window with a continued slow increase to ~ 0.319 A. Brief sharp transients
3799 at approximately 4 300 s and 5 200 s into this window are consistent with tran-
3800 sient radiation events affecting the boards.

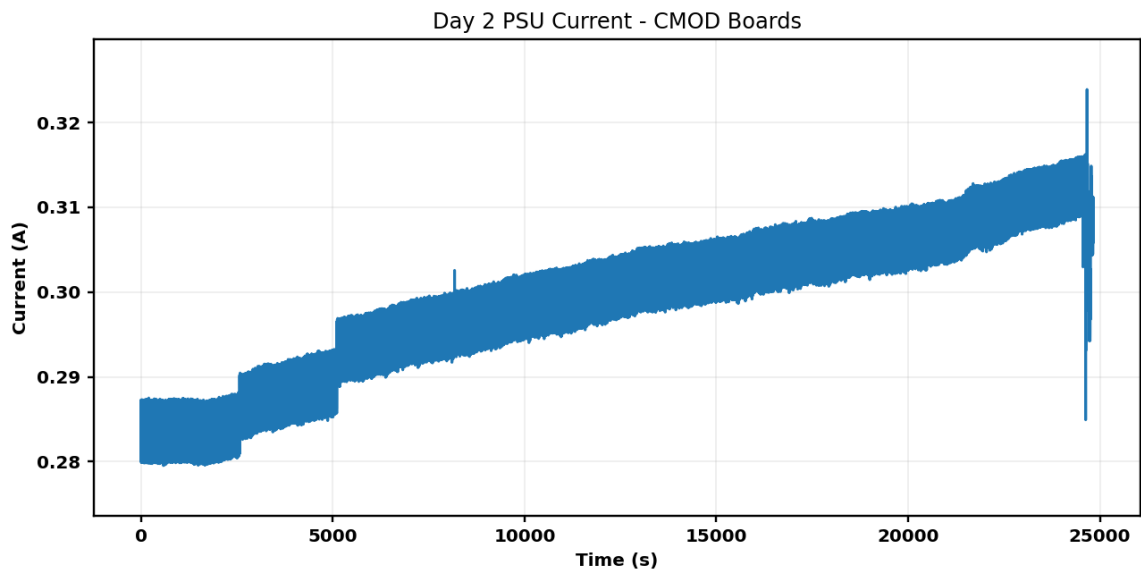


Figure 7.12: PSU current for the CMOD board array during the first part of the Day 2 (gamma shutter open) reactor run. A steady monotonic increase from approximately 0.28 A to 0.315 A ($\sim 12.5\%$) is visible over the ~ 7 -hour logging window, consistent with TID-induced leakage current accumulation. Step increases near 2800 s and 5000 s correspond to the approximate times of the FFSR4 and FFSR3 dropouts.

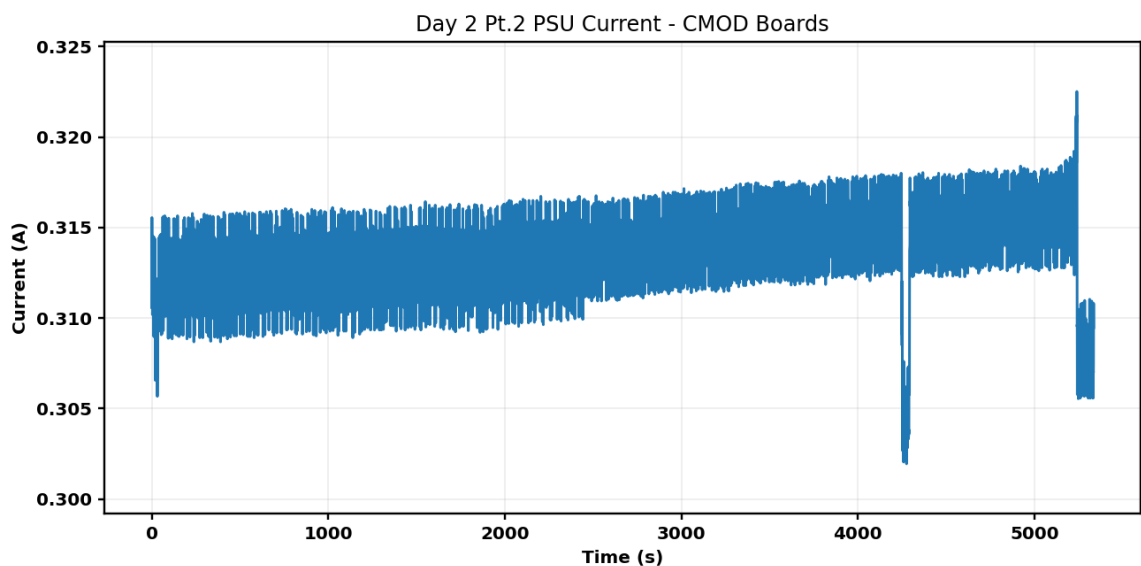


Figure 7.13: PSU current for the CMOD board array during the second part of the Day 2 run. Current continues near 0.315 A with a slow upward trend; brief sharp transients are consistent with transient radiation-induced events.

3801 **FFSR1 End-of-Run Burst Event and Confirmed DMR Upset**

3802 The most significant finding of the entire test campaign occurred on FFSR1 at
3803 approximately 15:54:44 on January 28. Under normal operation, FFSR1 transmitted
3804 one CNZ output line every ~ 5 seconds. At 15:54:44, the device abruptly began trans-
3805 mitting at an extremely high rate—approximately 400 readings within a 2-second win-
3806 dow, roughly $500\times$ the normal output rate. Within this burst the following anomalies
3807 were observed:

- 3808 • Multiple corrupted character substitutions: cNZ0000 (lowercase ‘c’), CLZ0000
3809 (‘L’ for ‘N’), KNZ0000 (‘K’ for ‘C’), CN[0000 (‘[’ for ‘Z’).
- 3810 • A truncated output: CNZ0 (partial transmission).
- 3811 • A non-zero upset count: CNZ0800 (hexadecimal 0x0800 = 2048 decimal).

3812 The value CNZ0800 is the *first and only non-zero DMR upset count recorded across*
3813 *the entire two-day test campaign*, representing a confirmed detection of one or more
3814 single-event upsets in the FFSR user data path by the DMR comparator logic.

3815 The burst event itself—the $\sim 500\times$ increase in UART transmission rate—is strongly
3816 indicative of a second configuration memory SEU on FFSR1, this time affecting the
3817 UART baud rate generation logic. The baud rate divider is implemented in FPGA
3818 configuration resources; a bit flip in the timing register can reduce the inter-character
3819 delay to near zero, producing exactly the rapid-fire transmission pattern observed.
3820 The coincidence of the baud rate upset, the character corruption events, and the
3821 non-zero DMR count within a 2-second window is consistent with a high-rate SEU
3822 cluster—multiple configuration and data memory upsets occurring in rapid succession
3823 near the end of the reactor run when cumulative fluence was at its maximum.

3824 7.7 Discussion

3825 7.7.1 SEU Sensitivity of the DMR Design

3826 The absence of any DMR-detected upsets across all five devices at the Day 1 fast
3827 neutron fluence of approximately $6\text{--}7 \times 10^9 \text{ n/cm}^2$ indicates either that the DMR
3828 design successfully masked data path upsets that would otherwise be detectable,
3829 or that the Spartan-7 shift-register fabric did not experience a sufficient number of
3830 upset events to register a DMR mismatch at this fluence level. To distinguish between
3831 these interpretations, a companion test without DMR protection (e.g., a bare shift-
3832 register comparison) would be needed to establish the intrinsic SEU cross-section of
3833 the Spartan-7 fabric under this beam condition.

3834 On Day 2, the first confirmed DMR data path SEU (counter value 2048) was
3835 observed on FFSR1 at the highest fluence level of the campaign ($\sim 8\text{--}9 \times 10^{11} \text{ n/cm}^2$
3836 total fast fluence). The two-orders-of-magnitude increase in flux and fluence from
3837 Day 1 to Day 2 is qualitatively consistent with the transition from zero detectable
3838 data-path upsets on Day 1 to device failures, latch-up events, and confirmed DMR
3839 detection on Day 2.

3840 7.7.2 Configuration Memory Vulnerability

3841 The FFSR1 observations across both days provide clear evidence of configuration
3842 memory SEUs manifesting as persistent, structured changes in device behaviour. The
3843 Day 1 UART label change (Section 7.6.2) and the Day 2 burst event (Section 7.6.4)
3844 both represent configuration memory upsets whose effects persisted until the de-
3845 vice was power-cycled, consistent with the volatile SRAM nature of the Spartan-7
3846 configuration store. These observations highlight that SRAM-based FPGAs have

3847 two distinct radiation vulnerability mechanisms: upsets in the user data registers
3848 (addressed by DMR/TMR) and upsets in the configuration SRAM (addressed by
3849 periodic scrubbing). Radiation-hardened designs intended for extended exposure en-
3850 vironments must address both.

3851 **7.7.3 TID Effects in PSU Current Data**

3852 The PSU current monitoring data provides a consistent and clear picture of TID
3853 accumulation across the two days. The Day 1 current profile was flat throughout
3854 (~ 0.284 A), indicating negligible TID-induced leakage at the Day 1 fluence level. In
3855 contrast, the Day 2 current profile exhibited a steady monotonic increase of approxi-
3856 mately 12.5% over the approximately 7-hour run. This is consistent with TID effects
3857 in the FPGA and support circuitry at the much higher Day 2 fluence. The step
3858 increases in the Day 2 current profile near 2800 s and 5000 s provide independent
3859 corroborating evidence for latch-up events at the times of the FFSR4 and FFSR3
3860 dropouts.

3861 **7.7.4 Implications for FPGA Radiation Hardening**

3862 Taken together, the results of this two-day test campaign demonstrate several
3863 conclusions relevant to the use of SRAM-based FPGAs in fast neutron environments.

3864 First, at moderate fluence levels ($\sim 10^9$ n/cm²), configuration memory SEUs can
3865 occur before data-path SEUs are detectable at the DMR counter level, suggesting
3866 that configuration scrubbing should be considered even in designs that rely on DMR
3867 or TMR for data-path protection.

3868 Second, at higher fluence levels ($\sim 10^{11}$ n/cm² and above), latch-up, multiple si-
3869 multaneous SEUs, and TID-induced leakage all become significant concerns that re-
3870 quire active mitigation strategies including latch-up detection and recovery circuitry,
3871 TID-hardened component selection, and TID-aware power budget margins.

3872 Third, the FFSR architecture proved effective as a radiation test instrument.
3873 The DMR comparator correctly identified the single confirmed data-path upset event
3874 while maintaining consistent zero-count output throughout normal operation, and the
3875 UART telemetry path provided a reliable real-time window into device state for the
3876 duration of both reactor runs.

3877 7.8 Summary

3878 This chapter presented the design and experimental results of the FFSR DMR
3879 irradiation test conducted at the OSURR Fast Neutron Beam Facility on January 26–
3880 28, 2026. The principal findings are as follows.

3881 **Day 1 (gamma shutter closed, $\sim 6\text{--}7 \times 10^9$ n/cm² fast fluence).** Zero DMR-
3882 detected SEUs across all five devices during approximately five hours of irradiation.
3883 FFSR1 exhibited a UART output format change consistent with a configuration mem-
3884 ory SEU while the DMR upset counter remained zero. No measurable TID effects
3885 were observed in the PSU current data.

3886 **Overnight (out of beam, January 27–28).** Zero upsets across all five devices
3887 during approximately 14.5 hours of ambient monitoring. The FFSR1 CNZ format
3888 persisted through all logging restarts, confirming a persistent SRAM hardware state
3889 change rather than a software artefact.

3890 **Day 2 (gamma shutter open, $\sim 8\text{--}9 \times 10^{11}$ n/cm² fast fluence).** FFSR3 and
3891 FFSR4 experienced early functional dropouts at approximately 36 and 49 minutes
3892 respectively, consistent with radiation-induced latch-up supported by the PSU current
3893 data. FFSR2 and FFSR5 completed the full run with zero DMR upsets. FFSR1
3894 experienced a severe multi-SEU cluster event near the end of the run, producing
3895 the first and only confirmed non-zero DMR upset count of the campaign (CNZ0800,
3896 0x0800 = 2048) along with a second configuration memory SEU affecting the UART
3897 baud rate. PSU current monitoring showed a sustained $\sim 12.5\%$ increase over the
3898 Day 2 run, consistent with TID-induced leakage current accumulation.

3899 Overall, these results demonstrate that SRAM-based FPGAs operating in this
3900 fast neutron environment are susceptible to configuration memory SEUs at moder-
3901 ate fluence levels, and to data-path SEUs, latch-up, and TID degradation at higher
3902 fluence levels. The FFSR DMR architecture successfully fulfilled its design objective
3903 of detecting data-path upsets while operating reliably throughout the test campaign,
3904 and highlights the need for configuration memory scrubbing and latch-up recovery
3905 circuitry for SRAM-based FPGAs intended for extended operation in radiation envi-
3906 ronments.

Chapter 8: Methods and Algorithms of the Fixed-Point Thermal Control Benchmark

8.1 Introduction

The preceding chapter described experiments conducted with the Free-Form Spatial Reference (FFSR) benchmark to evaluate FPGA behaviour under irradiation. The present chapter introduces a complementary benchmark designed for the same class of irradiation campaign, but with a richer computational workload and built-in self-comparison capability. Whereas the FFSR design exercised basic reconfiguration and readback paths, the fixed-point thermal control benchmark couples closed-loop feedback control, emulated physical plant dynamics, nonlinear actuator mapping, and dual-modular redundancy (DMR) within a single synthesizable Verilog design. Because the two benchmarks share the same neutron irradiation facility and beam configuration, the experimental infrastructure described in Chapter ?? applies directly here, and the present chapter focuses exclusively on the design methods, algorithms, and implementation choices that distinguish this benchmark.

3922 The benchmark is implemented entirely in synthesizable, fixed-point Verilog and is
3923 intended to operate continuously during an irradiation session, producing a 100 Hz se-
3924 rial telemetry stream that records the internal state of the control loop. Any radiation-
3925 induced perturbation that alters arithmetic results, corrupts stored state, or disrupts
3926 the routing of signals within the FPGA is therefore observable in real time through
3927 the telemetry output or, for subtle divergences, through the pipelined mismatch flag
3928 generated by the DMR wrapper. This combination of control significance and fault-
3929 detection capability makes the benchmark especially well suited to in-situ FPGA
3930 irradiation studies.

3931 The remainder of this chapter is organized as follows. Section 8.2 describes the
3932 top-level architecture and design rationale. Section 8.3 defines the fixed-point number
3933 representation and sampling structure. Sections 8.4–8.8 present the individual algo-
3934 rithmic subsystems in mathematical and implementation detail. Section 8.9 covers
3935 the redundancy, fault detection, and telemetry mechanisms. Section 8.10 maps each
3936 RTL module to its mathematical role. Section 8.11 describes the neutron irradiation
3937 experiment setup. Section 8.12 presents comparative observations from two teleme-
3938 try runs and discusses the failure mechanism. Section 8.13 discusses relevance to
3939 the radiation-effects objectives of this dissertation, and Section 8.14 concludes with
3940 a chapter summary.

3941 **8.2 System-Level Methodology**

3942 At the highest level, the benchmark is organized as a sampled-data loop exe-
3943 cuted inside a single FPGA clock domain running at 100 MHz. A one-cycle sam-
3944 ple strobe at 1 kHz acts as the logical update event for the control algorithm, the

Table 8.1: Implementation snapshot—key parameters of the fixed-point thermal control benchmark as configured in the present irradiation experiments.

Parameter	Value / Description
Clock frequency	100 MHz top-level system clock
Control/plant sample rate	1 kHz global tick generated in the stimulus block
Arithmetic format	Predominantly signed Q1.15; Q1.30 internal accumulation in the thermal plant
Redundancy	Two duplicated control-plant chains compared inside the DMR wrapper with CSV-style UART telemetry
Button control	20 ms debounce; 3 s hold to stop; short press-and-release issues <code>restart_pulse</code> and resumes run mode
UART output decimation	<code>decim = 10</code> , giving one CSV record every 10 ms (100 Hz) while running
DMR compare tolerance	<code>dmr_tol_q15 = 0x0000</code> (exact comparison)

3945 plant emulation, and telemetry capture. In the current top-level implementation
3946 (`top_pid_thermal_dmr.v`), this strobe is generated in `u_stim.v` and forwarded as
3947 the global tick only while the run-enable signal is asserted, so the dynamic state of
3948 the benchmark freezes when the experiment is stopped. The top-level integration ad-
3949 ditionally includes a dedicated power-on-reset counter, button-driven run and restart
3950 control, a millisecond time base that is reset by the restart pulse, the DMR-protected
3951 control-plant core, and the UART output path.

3952 The design is organized around five functional layers: (1) input stimulus genera-
3953 tion, (2) the closed-loop control algorithm, (3) actuator and nonlinear valve mapping,
3954 (4) the two-node thermal plant and measurement model, and (5) redundancy and
3955 telemetry infrastructure. Each layer is implemented in one or more dedicated RTL
3956 modules, and the inter-module interfaces use fixed-point Q1.15 values throughout.

3957 This methodology has several advantages for radiation-effects experimentation.
 3958 First, the state evolution is deterministic and repeatable, which allows observed
 3959 anomalies to be attributed more confidently to hardware perturbations rather than
 3960 software timing jitter. Second, the use of bounded fixed-point states and saturating
 3961 arithmetic prevents uncontrolled numeric growth during upset conditions. Third, the
 3962 benchmark simultaneously exercises arithmetic pipelines, state registers, lookup ta-
 3963 bles, comparators, counters, and serial communication logic—all categories of FPGA
 3964 resource that can respond differently to configuration upsets or transient single-event
 3965 effects. Finally, the explicit run gating, exact DMR comparison, and determinis-
 3966 tic restart behaviour make it well suited to repeatable irradiation campaigns where
 3967 reproducible initial conditions are important.

3968 **8.3 Numerical Representation and Sampling Structure**

3969 Nearly all signals in the design are encoded in signed Q1.15 format. In this
 3970 representation, a 16-bit signed integer x_{q15} represents the real quantity x through the
 3971 scaling relation

$$x \approx \frac{x_{q15}}{2^{15}}. \quad (8.1)$$

3972 This choice provides compact arithmetic, deterministic synthesis, and direct use
 3973 of FPGA DSP-slice resources while maintaining sufficient precision for the normal-
 3974 ized thermal benchmark, where all physical quantities are expressed as dimensionless
 3975 fractions of their respective full-scale values.

3976 The reusable multiplier pipeline in `q15_mul_pipe.v` implements rounded fixed-
 3977 point multiplication. For Q1.15 operands a and b , the full-precision product is formed

3978 in Q1.30, rounded by adding one-half least-significant bit of the target scale, and then
 3979 shifted right by 15 bits:

$$p_{q15} = \text{round}\left(\frac{a_{q15} \cdot b_{q15}}{2^{15}}\right). \quad (8.2)$$

3980 Simple additions and subtractions that could exceed the numeric range are per-
 3981 formed with saturation rather than wraparound. This is especially important in a
 3982 closed-loop design, because two's-complement overflow produces nonphysical values
 3983 that are indistinguishable in the binary record from genuine radiation-induced up-
 3984 sets. The thermal plant goes one step further and stores its internal accumulator
 3985 state in signed Q1.30 format, thereby reducing deadband and truncation error in the
 3986 forward-Euler integration even though block inputs and outputs remain Q1.15.

3987 Sampling is derived from the 100 MHz FPGA clock by a divider in `u_stim.v`.
 3988 One global tick is generated every 100,000 system-clock cycles, corresponding to a
 3989 sample period $T_s = 1$ ms. Because `tick_1k` is taken directly from the stimulus-valid
 3990 pulse, the controller, plant, DMR wrapper, and UART logger all advance only when
 3991 the benchmark is in run mode. The control chain is pipelined over several 100 MHz
 3992 cycles so that the algorithmic latency from tick assertion to updated measured output
 3993 is approximately 70 ns—negligible relative to the 1 ms control period.

3994 **8.4 Input Stimulus and Run/Restart Control**

3995 **8.4.1 Button-Controlled Run Logic**

3996 Human interaction with the benchmark during an irradiation session is handled
 3997 by `btn_run_ctrl.v`. The pushbutton signal is synchronized through a two-flip-flop
 3998 chain to eliminate metastability, then debounced by requiring a stable level for a

3999 programmable window of 20 ms. A short press-and-release sequence issues a one-cycle
4000 restart pulse and enables run mode; a long press of at least 3 s latches the design
4001 into a stopped state. This approach prevents mechanical bounce from spuriously
4002 retriggering the experiment and provides an operator-friendly means to restart the
4003 benchmark between irradiation intervals without requiring FPGA reconfiguration.

4004 8.4.2 Stimulus Generator

4005 The block `u_stim.v` generates the closed-loop reference $r[k]$ and the disturbance
4006 (thermal power input) term $Q[k]$. The pattern repeats every 600 s and is indexed
4007 by a millisecond counter t_{ms} that wraps after 600 000 updates. The reference target
4008 follows a piecewise-constant schedule, while the disturbance is applied as a direct step
4009 profile:

$$r_{\text{tgt}}(t) = \begin{cases} 0.55, & 0 \leq t < 30 \text{ s} \\ 0.60, & 30 \leq t < 250 \text{ s} \\ 0.57, & 250 \leq t < 400 \text{ s} \\ 0.61, & 400 \leq t < 530 \text{ s} \\ 0.55, & 530 \leq t < 600 \text{ s} \end{cases} \quad (8.3)$$

$$Q(t) = \begin{cases} 0.0850, & 0 \leq t < 300 \text{ s} \\ 0.0935, & 300 \leq t < 600 \text{ s} \end{cases} \quad (8.4)$$

4010 The reference actually delivered to the controller is slew-rate limited to prevent
4011 discontinuous command changes. It is updated every 10 ms by a bounded increment
4012 Δr . The slew law implemented in RTL is

$$r[k+1] = \text{sat}\left\{r[k] + \text{clip}(r_{\text{tgt}}[k] - r[k], -\Delta r, +\Delta r)\right\}, \quad (8.5)$$

4013 with $\Delta r = 5/32768 \approx 1.53 \times 10^{-4}$. This produces a finite ramp toward each
4014 new target, yielding a more informative transient response and avoiding step-like

4015 commands that would otherwise cause the controller output to rail immediately upon
4016 each reference change. Table 8.2 summarizes the piecewise schedule.

Table 8.2: Piecewise input schedule implemented by the stimulus generator. Each interval exercises a different combination of reference and disturbance conditions.

Interval	r_{tgt}	Q	Purpose in experiment
0–30 s	0.55	0.0850	Initial settling and baseline establishment
30–250 s	0.60	0.0850	Positive reference step and steady-state recovery
250–300 s	0.57	0.0850	Small reference decrease before disturbance change
300–400 s	0.57	0.0935	Disturbance increase at constant reference
400–530 s	0.61	0.0935	Combined reference increase under higher disturbance
530–600 s	0.55	0.0935	Return toward baseline before pattern repeats

4017 8.5 Controller Algorithm: Reverse-Acting I-PD with Anti- 4018 Windup

4019 The controller is implemented in `ipd_ctrl_q15.v` as a reverse-acting I-PD struc-
4020 ture. In this formulation the proportional and derivative actions act on the measured
4021 plant output rather than on the tracking error directly, which prevents actuator kicks
4022 when the reference changes abruptly. Only the integral state is updated from the
4023 control error. The sampled error and measured increment are

$$e[k] = r[k] - y_m[k], \quad (8.6)$$

$$\Delta y[k] = y_m[k] - y_m[k-1]. \quad (8.7)$$

4024 The derivative channel is low-pass filtered to suppress finite-difference noise. Using
 4025 smoothing coefficient $\alpha \in [0, 1]$, the filtered derivative state is updated as

$$\Delta y_f[k] = \alpha \Delta y_f[k-1] + (1 - \alpha) \Delta y[k]. \quad (8.8)$$

4026 The pre-saturation control command combines the proportional contribution on
 4027 measurement, the derivative contribution on filtered measurement change, and the
 4028 integral accumulator:

$$u_{\text{pre}}[k] = K_p y_m[k] + K_d \Delta y_f[k] + I_{\text{int}}[k]. \quad (8.9)$$

4029 The physical command is bounded to the allowable actuator range:

$$u[k] = \text{sat}(u_{\text{pre}}[k], U_{\text{min}}, U_{\text{max}}). \quad (8.10)$$

4030 To prevent integrator windup during saturation—a condition of particular im-
 4031 portance in radiation testing where prolonged actuator saturation following an upset
 4032 could mask recovery dynamics—the RTL includes a back-calculation anti-windup
 4033 correction. The integral state update is

$$I_{\text{int}}[k+1] = I_{\text{int}}[k] - K_i e[k] + K_t (u[k] - u_{\text{pre}}[k]). \quad (8.11)$$

4034 The term $K_t(u - u_{\text{pre}})$ drives the integral state back toward consistency with
 4035 the saturated output, preventing prolonged recovery times after rail contact. In the

4036 current top-level configuration $K_p \approx 1.0$, $K_i \approx 5.49 \times 10^{-4}$, $K_d = 0$, $\alpha \approx 0.90$,
4037 $U_{\min} \approx 0.020$, $U_{\max} \approx 0.980$, and $K_t \approx 0.025$. The derivative gain is set to zero in
4038 the present benchmark configuration, so the instantiated RTL retains the full I-PD
4039 architecture but operates effectively as a PI controller.

4040 **8.6 Actuator Dynamics and Nonlinear Valve Mapping**

4041 **8.6.1 First-Order Actuator with Slew Limiting**

4042 The valve actuator is implemented in `valve_act_q15.v` as a first-order lag from
4043 the demanded command $u[k]$ to the realized valve position $v[k]$. The unconstrained
4044 update law is

$$v_{\text{pred}}[k] = a_v v[k-1] + (1 - a_v) u[k], \quad (8.12)$$

4045 where a_v approximates e^{-T_s/τ_v} . In the present configuration $a_v = 0.9502$. An
4046 explicit slew-rate limit and a hard clamp to $[0, 1]$ are also applied. With $\Delta v_{\max}$
4047 denoting the maximum admissible change per sample:

$$v[k] = \text{clamp}_{[0,1]} \left\{ \text{clip}(v_{\text{pred}}[k], v[k-1] - \Delta v_{\max}, v[k-1] + \Delta v_{\max}) \right\}. \quad (8.13)$$

4048 **8.6.2 Equal-Percentage Valve Characteristic**

4049 The relationship between valve position and normalized flow is implemented with a
4050 lookup table in `valve_map_case.v`. Rather than evaluate a transcendental mapping
4051 at run time, the RTL indexes a 256-entry case-ROM using the upper bits of the
4052 Q15 valve position. The table was generated offline to match a normalized equal-
4053 percentage characteristic with gain parameter $g = 2.0$ and exponent $n = 0.6$:

$$\varphi(v) = \frac{g^v - 1}{g - 1}, \quad 0 \leq v \leq 1. \quad (8.14)$$

4054 A second exported table stores a nonlinear power of this flow quantity:

$$\varphi_{\text{pow}}(v) = [\varphi(v)]^n. \quad (8.15)$$

4055 The LUT-based approach preserves a nonlinear actuator-to-flow relation while
 4056 remaining fully deterministic, synthesizable, and timing friendly. From a radiation-
 4057 testing perspective, the LUT content resides in FPGA configuration memory and is
 4058 therefore vulnerable to single-event upsets that alter the stored coefficients, making
 4059 the valve map a useful target for future fault-injection studies.

4060 8.6.3 Coefficient Builder

4061 The block `coeff_builder.v` converts the valve-map outputs into flow-dependent
 4062 plant coefficients. Floor values ε and ε_{pow} prevent the effective mass-flow and heat-
 4063 transfer terms from collapsing to zero:

$$\dot{m}_{\text{rel}}[k] = \max(\varphi[k], \varepsilon), \quad (8.16)$$

$$\dot{m}_{\text{rel,pow}}[k] = \max(\varphi_{\text{pow}}[k], \varepsilon_{\text{pow}}), \quad (8.17)$$

$$\text{mcp}[k] = \text{mcp}_0 + k_m \dot{m}_{\text{rel}}[k], \quad (8.18)$$

$$\text{hA}[k] = \text{hA}_0 + k_h \dot{m}_{\text{rel,pow}}[k]. \quad (8.19)$$

Both the coolant heat-capacity flow rate `mcp` and the effective thermal conductance `hA` therefore vary with valve opening, introducing a state-dependent nonlinearity into the plant model and exercising the coefficient-computation path of the FPGA on every sample tick.

8.7 Two-Node Thermal Plant

8.7.1 Two-Node Forward-Euler Integration

The thermal plant is implemented in `plant_two_node_q15.v` as a two-node lumped-parameter model consisting of a fuel node temperature T_f and a coolant node temperature T_c . The state equations are discretized with forward-Euler integration, with T_s/C_f and T_s/C_c supplied as precomputed coefficients:

$$T_f[k+1] = T_f[k] + \frac{T_s}{C_f} \left(Q[k] - hA[k](T_f[k] - T_c[k]) \right), \quad (8.20)$$

$$T_c[k+1] = T_c[k] + \frac{T_s}{C_c} \left(hA[k](T_f[k] - T_c[k]) - mcp[k](T_c[k] - T_{in}[k]) \right). \quad (8.21)$$

Several implementation choices are noteworthy from a radiation-testing perspective. First, internal state is stored in Q1.30 accumulators, reducing numeric deadband and making the integration less sensitive to quantization errors introduced by bit-flip upsets. Second, the plant snapshots its inputs on the update tick and executes a four-stage arithmetic pipeline before committing the new temperatures, ensuring that a partially completed pipeline state cannot produce a corrupted but plausible output. Third, an optional clamp constrains the final temperatures to $[0, 1)$, ensuring the emulated plant remains bounded even under prolonged actuator saturation following an upset.

4083 8.7.2 Outlet Transport Lag

4084 The measured outlet temperature is not taken directly as the coolant-node temper-
4085 ature. Instead, `transport_lag_q15.v` implements an additional first-order mixing
4086 lag:

$$y[k] = a_p y[k-1] + (1 - a_p) T_c[k], \quad (8.22)$$

4087 where $a_p = 0.99786$ approximates e^{-T_s/τ_p} . This added lag introduces a second
4088 persistent state register whose vulnerability to upsets is distinct from the plant accu-
4089 mulator states.

4090 8.8 Quantized Measurement Path

4091 The module `quantizer_fifo.v` models analog-to-digital conversion and optional
4092 transport latency. For an N -bit ADC, the input y is clamped to $[0, 1]$, optionally
4093 dithered, and quantized:

$$c[k] = \text{round}(y[k] \cdot (2^N - 1)), \quad (8.23)$$

$$y_m[k] = c[k] \cdot 2^{15-N}. \quad (8.24)$$

4094 In the present configuration, $N = 15$, the FIFO latency is zero, and dither is
4095 disabled. The measurement path therefore behaves almost transparently at present,
4096 but the block remains methodologically important as a clean insertion point for future
4097 studies involving reduced measurement resolution, explicit sample delay, or dither-
4098 assisted quantization noise shaping.

4099 **8.9 Redundancy, Fault Detection, and Telemetry**

4100 **8.9.1 Dual-Modular Redundancy**

4101 The module `dmr_wrap_ctrlplant.v` instantiates two identical copies of the com-
4102 plete control-plant chain. Chain A drives the live outputs; Chain B serves as the
4103 redundant checker. Selected outputs from both copies are snapshotted on the sam-
4104 ple tick and then compared signal-by-signal with a programmable tolerance. The
4105 mismatch condition for each monitored signal s is

$$m_s[k] = \begin{cases} 1, & \text{if } |s_A[k] - s_B[k]| > \text{tol} \\ 0, & \text{otherwise.} \end{cases} \quad (8.25)$$

4106 The present implementation checks six signals: u , v , T_f , T_c , y , and y_m . The six
4107 per-signal mismatch bits are first registered, then OR-reduced in a second pipeline
4108 stage to ease timing closure at 100 MHz. The result drives a sticky status bit and
4109 an event counter that increments on each rising edge of the mismatch condition.
4110 With `dmr_tol_q15` = 0, the comparison is exact. This arrangement detects silent
4111 divergence between redundant implementations without requiring invasive internal
4112 probes.

4113 **8.9.2 Saturation Activity Monitor**

4114 The project also includes the health-monitor block `sat_monitor.v`, which ob-
4115 serves whether the controller output is at or near either actuator rail. The saturation
4116 flag is

$$s[k] = \begin{cases} 1, & \text{if } u[k] \geq U_{\max} - \delta \text{ or } u[k] \leq U_{\min} + \delta \\ 0, & \text{otherwise,} \end{cases} \quad (8.26)$$

4117 and an exponentially weighted moving average (EMA) of saturation activity is
4118 computed as

$$\text{EMA}[k] = \text{EMA}[k-1] - 2^{-d} \text{EMA}[k-1] + 2^{-d} s[k]. \quad (8.27)$$

4119 Although `sat_monitor.v` is not instantiated in the current top-level path—the
4120 logger receives the direct saturation flag and the `sat_ema_q15` input is tied to zero—
4121 the block formalizes how saturation activity could be compressed into a single bounded
4122 diagnostic state in future experiments where sustained actuator saturation following
4123 a large upset is an expected outcome.

4124 8.9.3 UART Logging Method

4125 The benchmark exports its internal state through `uart_logger.v` and `uart_tx_simple.v`.
4126 The logger snapshots the principal state variables on the 1 kHz tick, applies a deci-
4127 mation factor of ten, and serializes the snapshots into comma-separated hexadecimal
4128 records at 115,200 baud. One record is therefore emitted every 10 ms, corresponding
4129 to a 100 Hz logging rate. The field list includes time stamp, reference, disturbance, in-
4130 let temperature, pre- and post-measurement outlet temperatures, control command,
4131 valve position, node temperatures, tracking error, controller saturation flag, the tied-
4132 off saturation EMA field, and the DMR mismatch status. This record structure
4133 provides a complete state history of the closed-loop system and is the primary data
4134 source for the radiation-effects analysis presented in Chapter ??.

4135 8.10 Timing and Module-to-Method Mapping

4136 The complete chain is pipelined across the fast FPGA clock. The controller and
4137 actuator sample on the primary tick; the coefficient path aligns the nonlinear valve
4138 map with the plant update; the two-node plant commits new temperatures after
4139 its four-stage internal arithmetic sequence; and the transport-lag and measurement
4140 blocks subsequently publish the final measured output. The latency from tick asser-
4141 tion to updated y_m is seven 100 MHz clock cycles (≈ 70 ns), negligible relative to the
4142 1 ms control period. Table 8.3 maps each RTL module to the mathematical method
4143 it implements.

4144 Table 8.4 lists the complete Q1.15-valued parameter set extracted from the current
4145 top-level RTL.

4146 8.11 Neutron Irradiation Experiment Setup

4147 The neutron irradiation experiment used to evaluate the fixed-point thermal con-
4148 trol benchmark was conducted at the same facility and under the same beam con-
4149 figuration as the FFSR experiments described in Chapter ?? . A full description of
4150 the Ohio State University Research Reactor (OSURR) fast neutron beam facility,
4151 including its collimation geometry, shutter arrangements, and beam characterization,
4152 is provided in Chapter ?? and is not repeated here. Table 8.5 summarizes the key
4153 beam parameters for reference.

4154 The FPGA board was positioned at the beam exit in the same geometry used for
4155 the FFSR experiments. The benchmark was programmed onto the device and the
4156 run/restart control was exercised prior to beam exposure to verify nominal teleme-
4157 try output. The UART telemetry stream was logged continuously throughout each

4158 irradiation interval. The DMR mismatch counter and sticky flag fields in the teleme-
4159 try record served as the primary real-time indicators of radiation-induced functional
4160 divergence. A comparative analysis of two telemetry runs—one exhibiting radiation-
4161 induced degradation and one serving as a nominal baseline—is presented in Sec-
4162 tion 8.12. Additional statistical analyses and cross-run comparisons are provided in
4163 Chapter ??.

4164 8.12 Experimental Observations and Discussion

4165 Two UART telemetry acquisitions from the irradiation campaign were selected
4166 for detailed comparison. Run 1 (beginning at 07:21:56 on Day 2) was acquired after
4167 approximately 19 hours of cumulative device operation—4 hours of irradiation on
4168 Day 1 plus 15 hours of overnight standby without reactor flux. Run 2 (beginning at
4169 11:42:24) was acquired under nominal conditions and serves as the reference baseline.
4170 Both runs executed the same deterministic ROM stimulus cycle. Time alignment
4171 was performed by detecting the first departure of the setpoint signal r from its initial
4172 plateau value; all times reported below are relative to this alignment edge.

4173 8.12.1 ROM Stimulus Integrity

4174 The ROM-sourced inputs Q (heat input) and T_{in} (inlet temperature) are identical
4175 between runs to within floating-point representation limits ($\Delta Q_{\text{max}} = 7.09 \times 10^{-13}$,
4176 $\Delta T_{\text{in}} = 0$), confirming that the BRAM cells storing these stimulus signals were not
4177 corrupted during or after irradiation.

4178 The setpoint r follows the identical cycling pattern in both runs for the first ap-
4179 proximately 10 000 s. After this point, Run 1 exhibits corrupted r values dropping

4180 to the range 0.50–0.54. This ROM corruption is temporally distinct from and sub-
 4181 sequent to the plant-state failure described below, and is consistent with a separate
 4182 configuration memory upset affecting the `rom_stim` address or data routing at a later
 4183 time. Figure 8.1 shows the setpoint overlay for both runs.

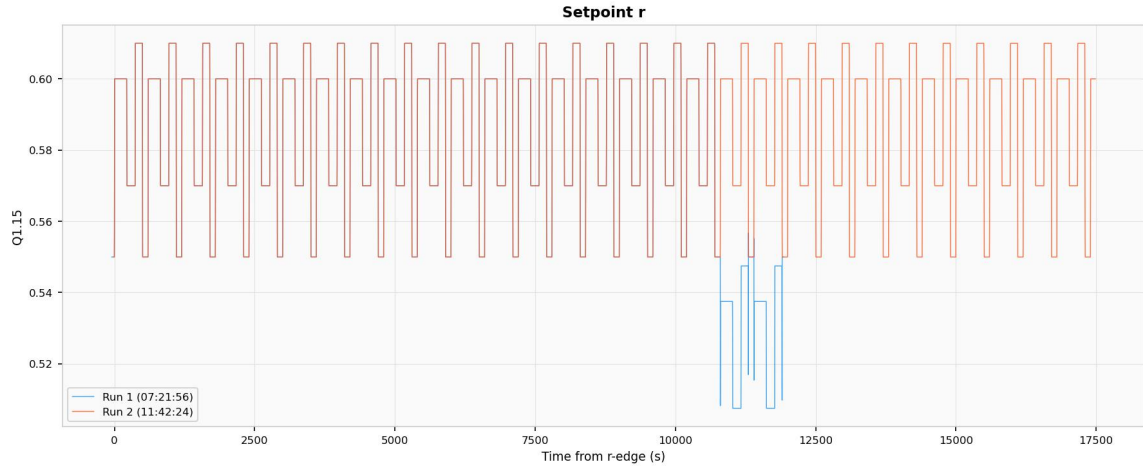


Figure 8.1: Setpoint r overlay. Run 1 (blue) and Run 2 (red) track identically for the first $\sim 10,000$ s. Run 1 exhibits corrupted setpoint values after $t = 10,000$ s, temporally distinct from the plant-state decay onset at $t \approx 2,500$ s.

4184 8.12.2 Plant-State Decay

4185 The primary failure signature in Run 1 is a smooth, monotonic decay of both plant-
 4186 model node temperatures (T_f and T_c), the outlet temperature y , and the measured
 4187 feedback signal y_m toward zero. The decay onset occurs at approximately $t = 2,500$ s
 4188 and reaches the zero clamp by approximately $t = 5,000$ s, a transition duration of
 4189 roughly 2 500 s. The decay profile is approximately exponential with no discontinuities

4190 or step changes. Run 2 maintains stable periodic operation throughout the same
 4191 interval. Figures 8.2–8.4 show the overlay plots for T_f , T_c , and y .

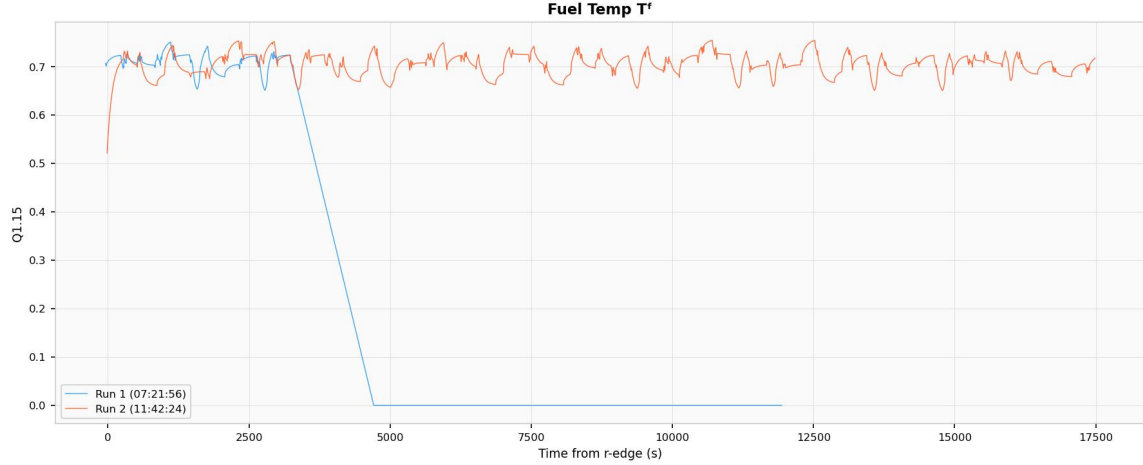


Figure 8.2: Fuel temperature T_f overlay. Run 1 (blue) begins a smooth decay near $t = 2,500$ s, reaching the zero clamp by $t \approx 5,000$ s. Run 2 (red) maintains stable periodic operation.

4192 The fuel temperature difference between runs reaches a sustained magnitude of
 4193 approximately -0.72 in Q1.15 units, the coolant temperature difference reaches ap-
 4194 proximately -0.62 , and the outlet difference reaches approximately -0.61 . These are
 4195 near-full-scale departures relative to the Q1.15 representable range of $[-1, +1]$. The
 4196 difference traces are shown in Figures 8.5–8.6.

4197 As the plant temperatures decay, the error signal ($e = r - y_m$) grows from near-
 4198 zero to approximately 0.57 – 0.60 (Figure 8.7), the controller output u saturates at
 4199 its limits, and the valve position v collapses to near-zero (Figure 8.8)—all expected
 4200 consequences of the plant state approaching zero while the setpoint remains at its
 4201 nominal value.

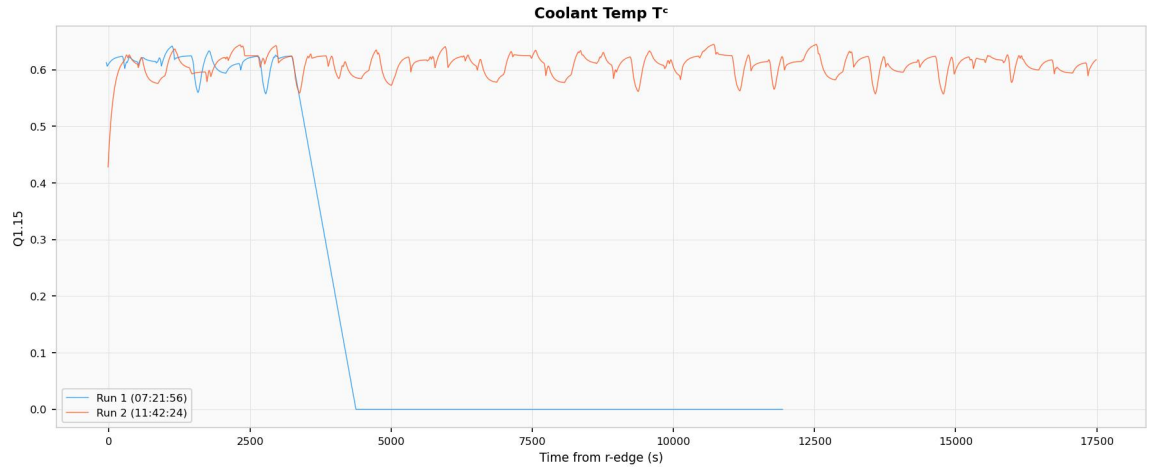


Figure 8.3: Coolant temperature T_c overlay. Same decay morphology as T_f with a slight lag, consistent with the thermal coupling between the two plant nodes.

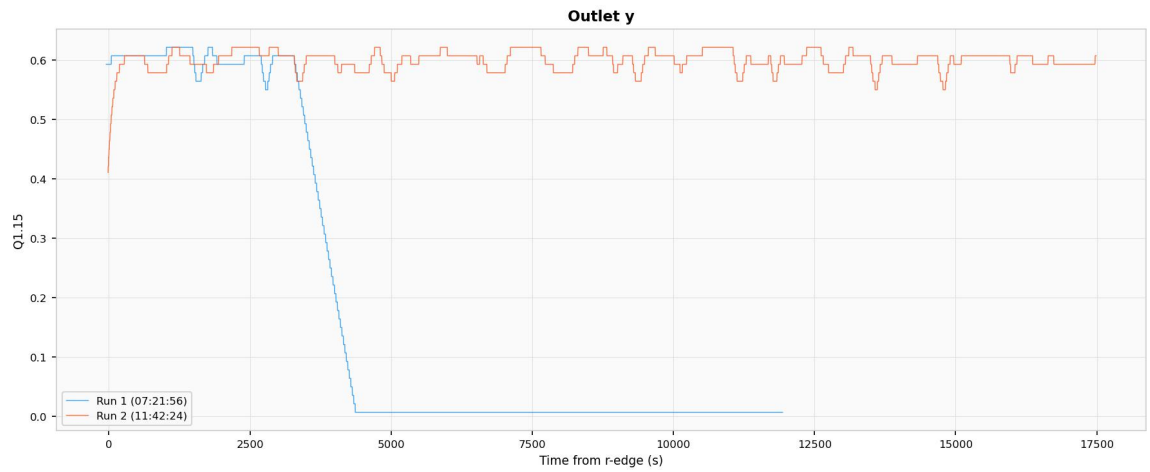


Figure 8.4: Outlet temperature y overlay. The outlet tracks the coolant temperature decay through the first-order transport lag.

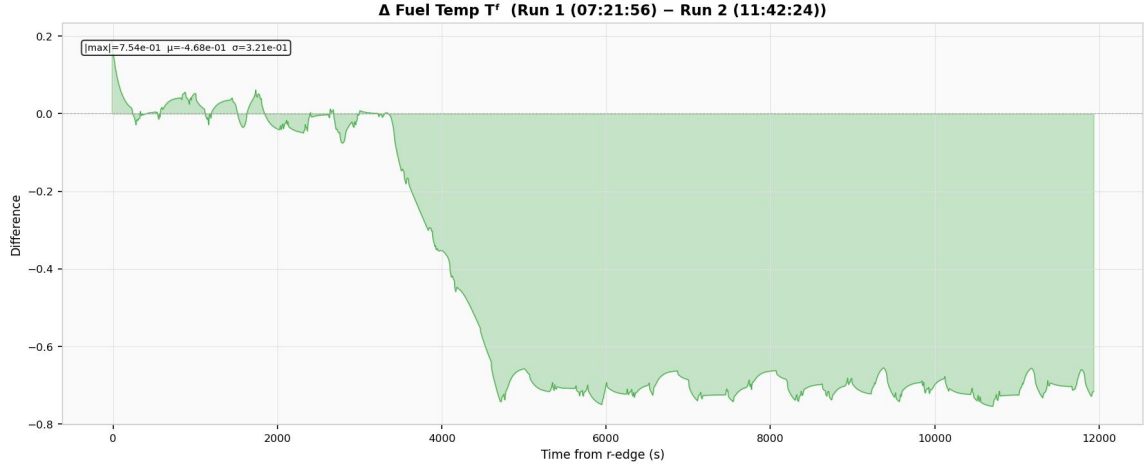


Figure 8.5: Fuel temperature difference (Run 1 – Run 2). The smooth onset near $t = 2,000$ s saturates at approximately -0.72 . The difference statistics are $|\max| = 7.54 \times 10^{-1}$, $\mu = -4.68 \times 10^{-1}$, $\sigma = 3.21 \times 10^{-1}$.

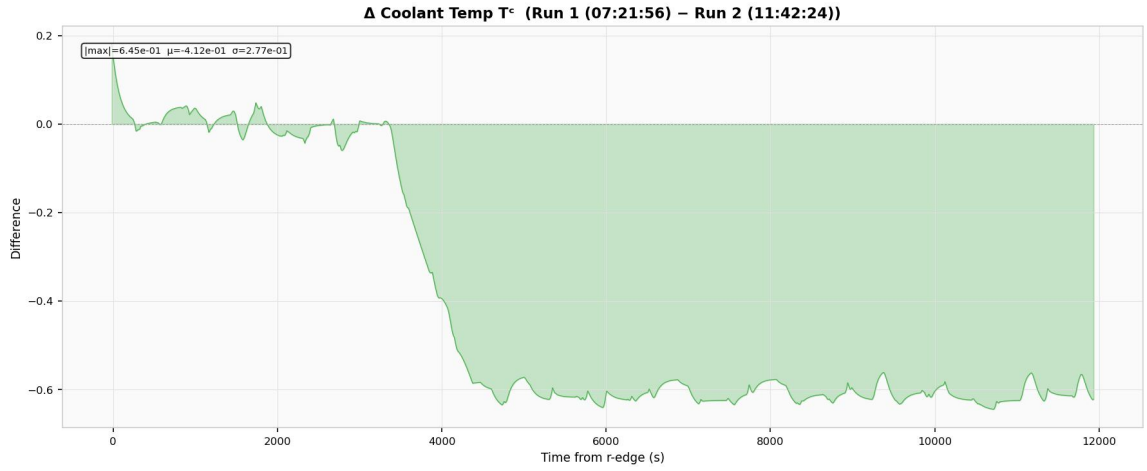


Figure 8.6: Coolant temperature difference (Run 1 – Run 2). Same morphology as the fuel temperature difference, saturating at approximately -0.62 .

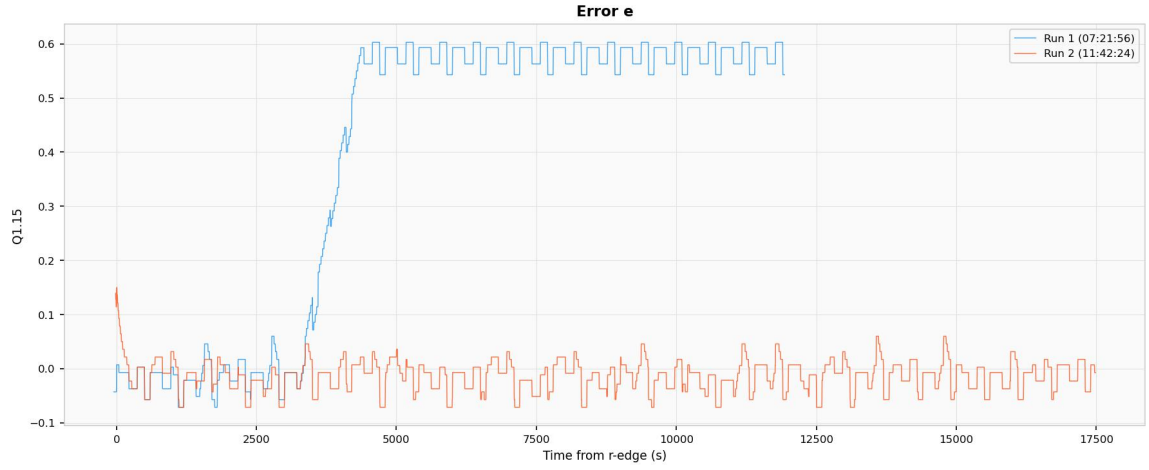


Figure 8.7: Error signal $e = r - y_m$ overlay. Run 1 error grows monotonically as plant temperatures collapse, reaching $e \approx 0.60$ (the setpoint value) once y_m reaches zero.

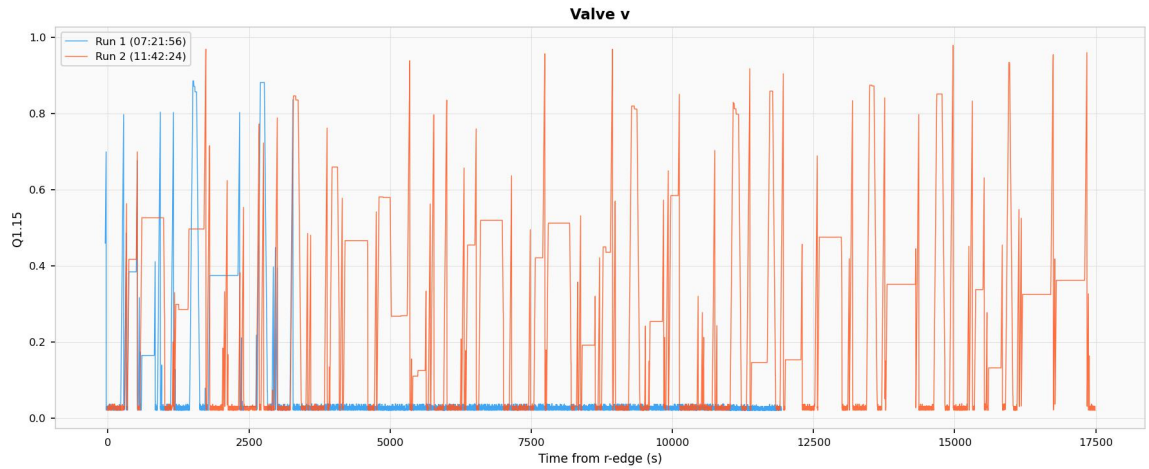


Figure 8.8: Valve position v overlay. Run 1 (blue) valve collapses to near-zero after the plant-state failure. Run 2 (red) exhibits the expected periodic actuation pattern driven by the ROM stimulus cycle.

4202 **8.12.3 DMR Response**

4203 No DMR mismatch events were reported during the plant-state decay. The
4204 `dmr_event_count` field remained at zero throughout the entire Run 1 acquisition.
4205 Both redundant controller–plant chains produced identical outputs at every compared
4206 sample tick, despite the anomalous decay of all plant-model state variables.

4207 **8.12.4 Discussion of Failure Mechanism**

4208 **Exclusion of Firmware-Induced Numerical Drift**

4209 To determine whether the observed decay could arise from numerical accumulation
4210 effects intrinsic to the Q1.15 fixed-point arithmetic, a bit-exact simulation of the
4211 complete closed-loop system was executed for 48 simulated hours. The simulation
4212 reproduced the identical arithmetic operations, rounding rules, and pipeline structure
4213 implemented in the Verilog RTL, including the stage-2 read-before-write dependency
4214 in the `plant_two_node_q15` module and the unsaturated integrator update in the
4215 controller. The simulation produced zero drift in all state variables: the system
4216 reaches a fixed point at which every Q1.15 multiply-accumulate operation rounds to
4217 zero incremental change per tick.

4218 Additionally, the decay morphology is inconsistent with integer overflow or wrap-
4219 around. An overflow of a 16-bit Q1.15 register would produce an instantaneous po-
4220 larity reversal (e.g., +0.99 wrapping to -1.0), not a smooth 2 500-second exponential
4221 decay. Firmware-induced numerical drift is therefore excluded as a causal mechanism.

4222 **Exclusion of BRAM Data Corruption**

4223 A single-event upset in a BRAM cell—for example, a corrupted entry in the
4224 `phi_q15` or `pow_q15` lookup table, or a corrupted ROM stimulus value—would af-
4225 fect only one of the two DMR chains, because BRAM instances are physically sep-
4226 arate per chain. The resulting output divergence would be detected by the DMR
4227 comparators. No such detection occurred, excluding single-chain BRAM corruption
4228 as the failure mechanism. This exclusion is further supported by the zero inter-run
4229 difference observed for the shared-BRAM stimulus signals Q and T_{in} .

4230 **Consistency with Configuration Memory SEU**

4231 The failure characteristics are collectively consistent with a single-event upset in
4232 the FPGA configuration SRAM (CRAM). CRAM bits define the logic function of
4233 every lookup table (LUT), the routing of every signal interconnect, and the configu-
4234 ration of every DSP48 slice. A CRAM SEU does not alter stored data; it alters the
4235 *function computed* by the logic fabric. Table 8.6 summarises the mapping between
4236 the observed failure characteristics and the CRAM SEU mechanism.

4237 The most probable specific upset sites, given the observed decay morphology, are:

- 4238 1. **Routing or computation of the parameter `Ts_over_Cc`.** This parameter
4239 has a raw Q1.15 value of 1 (0x0001). A single-bit flip to 0x0000 would com-
4240 pletely disable coolant temperature integration; a flip to 0x8001 would invert
4241 the integration direction, causing T_c to decay.
- 4242 2. **A high-order bit in the `hA` or `mcp` coefficient path.** These coefficients
4243 multiply temperature differences in the two-node model. A single-bit corruption

4244 in a high-order bit would introduce a large systematic error in the heat balance,
4245 biasing the forward-Euler integration toward decay.

4246 **3. A LUT truth-table bit in the behavioural mul_q15 function.** The plant
4247 uses behavioural function calls (not the pipelined DSP48 multiplier) for its
4248 internal multiplies. A corrupted LUT in this multiply chain would bias every
4249 product computation that traverses the affected input pattern.

4250 The persistence of the failure from Day 1 irradiation through the overnight standby
4251 period and into Day 2 Run 1 is consistent with the volatile SRAM nature of the
4252 Spartan-7 configuration store: CRAM upsets persist until the device is reconfigured
4253 or the affected configuration frame is scrubbed. This mirrors the persistent UART
4254 format change observed on FFSR1 (Chapter ??, Section 7.7.2), and together these
4255 two independent observations—one from a shift-register test structure, one from a
4256 closed-loop control system—establish that configuration memory SEUs are a primary
4257 concern for SRAM-based FPGAs at the fluence levels encountered in this campaign.

4258 **Implications for DMR Effectiveness**

4259 The complete functional collapse of the thermal control plant with zero DMR
4260 detection demonstrates a fundamental limitation of dual modular redundancy when
4261 both chains share common configuration memory. DMR protects against BRAM data
4262 upsets (which are physically instantiated per chain) but is transparent to CRAM logic
4263 and routing upsets (which affect both chains identically). Effective mitigation for
4264 this class of upset requires either configuration memory scrubbing via ICAP/PCAP
4265 readback, triple modular redundancy (TMR) with physical placement constraints to

4266 ensure distinct configuration frames per voter group, or periodic functional self-testing
4267 against known reference trajectories.

4268 **8.13 Relevance to Radiation-Effects Objectives**

4269 From a dissertation perspective, the value of this benchmark is not only that it
4270 performs closed-loop control, but that it does so in a form that is highly amenable to
4271 radiation-effects study. The design simultaneously exercises arithmetic pipelines, per-
4272 sistent state registers, lookup tables, comparators, counters, and serial communication
4273 logic—all categories of FPGA resource that can respond differently to configuration
4274 upsets or transient single-event effects. At the same time, the benchmark remains
4275 analytically tractable: each module corresponds to a clear mathematical operation,
4276 and the overall system response can be interpreted in terms of familiar control and
4277 thermal-system theory.

4278 The DMR wrapper transforms the benchmark from a simple controller into a self-
4279 checking experiment. Under nominal conditions the two chains evolve identically,
4280 so any discrepancy is attributable to an external perturbation. A radiation-induced
4281 upset that perturbs logic, corrupts a stored state, or disrupts routing within the
4282 FPGA may therefore be detected through inter-chain divergence even if the output
4283 trajectory has not yet become visibly unstable. However, as the experimental re-
4284 sults of Section 8.12 demonstrate, this detection capability has an important blind
4285 spot: CRAM upsets affecting logic or routing shared by both chains produce iden-
4286 tical corruption in both outputs, rendering the DMR comparator ineffective against
4287 this failure class. This finding refines the initial design expectation and motivates

4288 the complementary use of configuration scrubbing or TMR with physical separation
4289 constraints.

4290 The combination of a meaningful closed-loop workload, bounded fixed-point arith-
4291 metic, deterministic restart capability, and pipelined DMR comparison makes this
4292 benchmark a well-instrumented platform that complements the FFSR approach. Where
4293 the FFSR experiments focused on configuration integrity and readback verification,
4294 the fixed-point thermal control benchmark focuses on arithmetic correctness and
4295 control-loop functional continuity under irradiation. Together, the two benchmarks
4296 provide a multi-faceted evaluation of FPGA behaviour in a neutron radiation envi-
4297 ronment relevant to nuclear instrumentation and monitoring applications.

4298 **8.14 Chapter Summary**

4299 This chapter presented the methods and algorithms used to construct the fixed-
4300 point thermal control benchmark. The benchmark operates as a 1 kHz sampled-
4301 data loop on a 100 MHz FPGA clock and integrates a reverse-acting I-PD controller
4302 with anti-windup (Equations 8.6–8.11), first-order actuator dynamics with slew lim-
4303 iting (Equations 8.12–8.13), a nonlinear equal-percentage valve map implemented as
4304 a 256-entry lookup table (Equations 8.14–8.19), a two-node forward-Euler thermal
4305 plant with Q1.30 internal accumulators (Equations 8.20–8.21), an outlet transport
4306 lag (Equation 8.22), a configurable quantized measurement path (Equations 8.23–
4307 8.24), and a dual-modular redundancy wrapper that compares six key signals on
4308 every sample tick (Equation 8.25). Run gating, deterministic restart, and 100 Hz
4309 UART telemetry complete the design.

4310 The neutron irradiation experiment was conducted at the OSURR fast neutron
4311 beam facility under the same beam configuration used for the FFSR experiments
4312 in Chapter ???. Comparative analysis of two telemetry acquisitions—one exhibiting
4313 radiation-induced degradation (Run 1, acquired after 19 hours of cumulative operation
4314 including Day 1 irradiation) and one serving as a nominal baseline (Run 2)—revealed
4315 a smooth exponential decay of all plant state variables to zero over approximately
4316 2 500 s, with zero DMR mismatch detection throughout. Bit-exact 48-hour closed-
4317 loop simulation confirmed that the decay cannot arise from firmware numerical drift.
4318 The failure signature is consistent with a configuration memory (CRAM) SEU affect-
4319 ing shared arithmetic or routing logic, and demonstrates a fundamental limitation of
4320 DMR when both redundant chains share common FPGA configuration memory.

4321 By encoding the entire benchmark in synthesizable Verilog and tying each module
4322 to explicit mathematical relations, the design provides both a technically meaningful
4323 control workload and a well-instrumented, self-checking platform for the radiation-
4324 effects evaluation presented in the following chapter.

Table 8.3: Mapping between RTL modules and the mathematical methods they implement.

RTL Module	Implemented Method	Primary Mathematical Role
u_stim.v	Reference and disturbance scheduling	Piecewise input generation; slew-limited command shaping
ipd_ctrl_q15.v	Control law	I-PD feedback, derivative filtering, anti-windup
valve_act_q15.v	Actuator dynamics	First-order lag, rate limit, hard clamp
valve_map_case.v	Nonlinear static map	Valve position to normalized flow via 256-entry LUT
coeff_builder.v	Flow-dependent coefficients	mcp and hA generation from mapped flow
plant_two_node_q15.v	Thermal dynamics	Forward-Euler update of T_f and T_c
transport_lag_q15.v	Output lag	First-order outlet mixing/transport delay
quantizer_fifo.v	Measurement model	Quantization, optional dither, optional sample latency
dmr_wrap_ctrlplant.v	Fault detection	Redundant execution and signal-level comparison
sat_monitor.v	Optional health monitoring (inactive)	Rail detection and EMA of saturation activity
uart_logger.v	Telemetry	Decimated state snapshot and serialized CSV output
btn_run_ctrl.v	Run/restart supervision	Debounce, hold-time stop latch, short-press restart
q15_mul_pipe.v	Arithmetic primitive	Rounded pipelined Q1.15 multiplication
ctrl_plant_chain.v	Latency-aligned subsystem composition	Orders controller, actuator, LUT, coefficients, plant, lag, and measurement updates
top_pid_thermal_dmr.v	System integration	Binds parameters, gates run state, connects DMR and UART
uart_tx_simple.v	Serial transport	Bytewise UART at 115 200 baud

Table 8.4: Current top-level benchmark parameters extracted from the RTL implementation.

Parameter	Q15 Hex	Decimal	Module	Interpretation
K_p	0x7FFF	≈ 1.000	Controller	Proportional gain on measured output
K_i	0x0012	5.49×10^{-4}	Controller	Integral gain per sample
K_d	0x0000	0	Controller	Derivative gain (disabled)
α	0x7333	0.9000	Controller	Derivative low-pass coefficient
$U_{\min}/U_{\max}$	0x0290/0x7D70	0.020/0.980	Controller	Command saturation limits
K_t	0x0333	0.0250	Controller	Back-calculation anti-windup gain
a_v	0x79A0	0.9502	Actuator	Valve first-order pole
$\Delta v_{\max}$	0x0147	0.00998	Actuator	Maximum valve change per sample
mcp_0/k_m	0x3000/0x3000	0.375/0.375	Coeff. builder	Baseline and slope for mcp
hA_0/k_h	0x6666/0x7333	0.800/0.900	Coeff. builder	Baseline and slope for h
T_s/C_f	0x000D	3.97×10^{-4}	Plant	Fuel-node Euler step coefficient
T_s/C_c	0x0001	3.05×10^{-5}	Plant	Coolant-node Euler step coefficient
a_p	0x7FBA	0.99786	Transport lag	Outlet first-order lag pole
T_{in}	0x3333	0.4000	Plant	Normalized inlet temperature
$\varepsilon/\varepsilon_{\text{pow}}$	0x0008/0x0001	$2.44 \times 10^{-4}/3.05 \times 10^{-5}$	Coeff. builder	Lower bounds on mapped flow terms
dmr_tol	0x0000	0	DMR wrapper	Exact comparison threshold
$R_{\text{reset}}/Q_{\text{reset}}$	0x4666/0x0AE1	0.550/0.085	Stimulus gen.	Values loaded at POR/restart

Table 8.5: Summary of neutron beam facility parameters used in the fixed-point thermal control benchmark experiment. All values are identical to those reported in the FFSR irradiation campaign (Chapter ??).

Parameter	Value / Description	Notes
Facility	OSURR fast neutron beam	
Reactor power	450 kW _{th} (full power)	
Fast neutron flux (shutter open, no Cd)	$\approx 1.5 \times 10^7 \text{ cm}^{-2}\text{s}^{-1}$	Used for irradiation runs
Thermal neutron flux (shutter open, no Cd)	$\approx 8 \times 10^6 \text{ cm}^{-2}\text{s}^{-1}$	Suppressed with Cd filter
Fast neutron flux (shutter open, Cd filter)	$\approx 1.5 \times 10^7 \text{ cm}^{-2}\text{s}^{-1}$	Thermal component suppressed
Scattered flux (shutter closed)	$\approx 4.3 \times 10^5 \text{ cm}^{-2}\text{s}^{-1}$	Background reference
Beam configuration used	Shutter open, Cd filter installed, N chopper open	Predominantly fast neutron field

Table 8.6: Observed failure characteristics of the PID thermal control benchmark and their consistency with a CRAM SEU mechanism.

Observation	Consistency with CRAM SEU
DMR silent (no mismatches detected)	CRAM bits are shared between both DMR chains. The DMR architecture instantiates two chains from the same synthesised netlist, placed and routed in the same fabric. A CRAM upset on shared parameter routing (e.g., K_p , $h_{A,0}$, T_s/C_c) or on shared arithmetic logic (e.g., a LUT in the coefficient builder) delivers the same corrupted result to both chains. The DMR comparator sees matching but incorrect outputs.
Smooth exponential decay (not a step or spike)	A CRAM bit flip alters a LUT truth table for one specific input combination, or reroutes one signal net. This introduces a small systematic bias in the plant integration that accumulates over many sample ticks, producing a gradual drift rather than an instantaneous jump.
Temperatures decay to exactly zero (clamped)	The <code>plant_two_node_q15</code> module applies <code>clamp01_q15</code> , which clamps outputs to <code>[0, 0x7FFF]</code> . A negative integration bias drives temperatures through zero, where the clamp arrests further descent.
Onset at $t \approx 2,500$ s, not at $t = 0$	The CRAM upset likely occurred during Day 1 irradiation (hours prior to Run 1). The corrupted logic path may only activate for specific input combinations encountered during particular phases of the ROM stimulus cycle, producing no visible effect during other phases. The 2 500 s delay represents the time for the corrupted integration to accumulate past the noise floor.
ROM stimulus r corrupts later ($t > 10,000$ s)	Consistent with a second, independent CRAM upset affecting ROM address or data routing. Multiple CRAM upsets are expected after hours of cumulative irradiation.

Chapter 9: Conclusions and Future Work

9.1 Summary of Research Contributions

This dissertation has investigated radiation-responsive materials, detector devices, alternative sensing concepts, and radiation-tolerant digital electronics for advanced radiation detection and monitoring. The research was organized around three complementary objectives: (1) characterizing the radiation response of halide perovskite materials across multiple compositions and radiation types, (2) developing and demonstrating perovskite-based detection devices for specific nuclear monitoring applications, and (3) implementing and validating radiation-tolerant FPGA-based digital instrumentation systems for operation in nuclear environments. A fourth, ancillary objective addressed thermal radiation sensing concepts as an alternative pathway for nuclear environment monitoring. The following sections consolidate the principal findings, discuss their implications for nuclear monitoring systems, and identify directions for future research.

4339 9.2 Key Findings

4340 9.2.1 Perovskite Solar Cells as Current-Mode Radiation Mon- 4341 itors

4342 The work presented in Chapter 4 demonstrated that photovoltaic-architecture
4343 $\text{Cs}_{0.1}\text{FA}_{0.9}\text{PbI}_3$ perovskite solar cells can function as sensitive current-mode monitors
4344 for ionizing radiation spanning X-ray, gamma, and potentially neutron fields. Under
4345 ^{137}Cs gamma irradiation at $44.16 \text{ rad hr}^{-1}$, an induced current density of 2.19 nA cm^{-2}
4346 was measured, and a distinct step response was confirmed upon insertion and with-
4347 drawal of the source. Extended irradiation with ^{60}Co at $355.61 \text{ rad hr}^{-1}$ yielded
4348 3.56 nA cm^{-2} , confirming that the current-mode detection approach scales across a
4349 dose-rate range spanning nearly an order of magnitude. A linear degradation rate of
4350 approximately 0.74 nA rad^{-1} was observed during six-day continuous ^{137}Cs irradiation,
4351 attributable to cumulative defect creation and glass darkening in the substrate.

4352 The device bandwidth was found to be limited by the measurement electronics
4353 rather than the intrinsic perovskite photoresponse, with rise times of 33–84 ns
4354 depending on amplifier configuration. Under X-ray irradiation, the device produced
4355 measurable short-circuit current proportional to beam intensity, although a characteristic
4356 slow current decline was observed consistent with ion-migration field screening.
4357 Neutron response at the OSURR Fast Beam Facility was statistically inconclusive for
4358 a single module, and the signal-to-noise analysis indicated that scaling the detector
4359 array area by a factor of ten to twenty would be necessary to resolve the neutron-
4360 induced current above the combined gamma and electronic noise backgrounds.

4361 9.2.2 Perovskite Semiconductor Radiation Detectors

4362 Chapter 5 extended the investigation to direct-conversion semiconductor radiation
4363 detectors spanning three radiation types.

4364 For X-ray detection, the MAPbI₃-hBN polycrystalline device exhibited a mono-
4365 tonically increasing sensitivity with applied bias, reaching $0.411 \mu\text{C} (\text{Gy cm}^2)^{-1}$ at
4366 20 V, consistent with Hecht equation predictions for improving charge collection at
4367 higher fields. Vacuum testing confirmed that the removal of ambient moisture en-
4368 hanced both sensitivity and stability, directly linking performance degradation to
4369 moisture-accelerated ion migration. The 2D perovskite (PEA₂MA₂Pb₃I₁₀) detector
4370 with 5F-PEAI surface treatment demonstrated that this moisture-accelerated insta-
4371 bility can be effectively mitigated: the fluorinated barrier layer raised the breakdown
4372 voltage from -10 V to -16 V at 40% relative humidity and enabled X-ray sensitivi-
4373 ties approaching $\sim 1000 \mu\text{C} (\text{Gy}_{\text{air}} \text{cm}^2)^{-1}$ at 4 V bias, among the highest reported for
4374 polycrystalline perovskite detectors of comparable thickness.

4375 For gamma-ray detection, three bromide perovskite single-crystal systems were
4376 evaluated. The CsPbBr₃ device produced a statistically significant count-rate re-
4377 sponse to a ¹³⁷Cs source (17.45 cps vs. 9.11 cps background) at 1000 V bias, but
4378 high and unstable leakage current (315–710 nA) prevented resolution of the 662-
4379 keV photopeak. The FAPbBr₃ device achieved substantially lower operating leakage
4380 ($5 \pm 2.5 \text{ nA}$ maximum), sustained stable high-voltage operation for over five hours,
4381 and produced spectral features attributable to both ¹³⁷Cs and ⁵⁷Co in a joint ac-
4382 quisition at 1600 V bias—representing a proof-of-concept demonstration of gamma
4383 spectroscopy in a solution-grown halide perovskite crystal. The MAPbBr₃ device

4384 failed to produce functional Schottky contacts under the deposition conditions used,
4385 motivating a revised contact deposition protocol at reduced sputter power.

4386 For fast neutron detection, MHyPbCl₃ crystals were characterised as the first
4387 organic–metal halide perovskite semiconductor evaluated in a reactor fast neutron
4388 beam. The material combines a high hydrogen density ($\sim 4.0 \times 10^{22} \text{ cm}^{-3}$, compa-
4389 rable to stilbene) with an exceptional resistivity of $4.43 \times 10^{11} \Omega \text{ cm}$ and a mobility-
4390 lifetime product of $9.1 \times 10^{-3} \text{ cm}^2 \text{ V}^{-1}$. Waveform data acquired at the OSURR
4391 Fast Beam Facility confirmed radiation-induced pulse activity distinct from the beam
4392 background, and pulse-height analysis produced spectra consistent with the expected
4393 broad proton-recoil energy deposition distribution from a poly-energetic fast neutron
4394 beam. Neutron–gamma discrimination remains a challenge due to the large Pb con-
4395 tent, and data acquisition limitations prevented collection of sufficient statistics for
4396 quantitative efficiency analysis.

4397 Across all three detection modes, ion migration under applied electric fields emerged
4398 as the central performance-limiting mechanism. The correlation of device instabil-
4399 ity with ambient humidity—confirmed by vacuum tests on MAPbI₃-hBN and by the
4400 systematic humidity-dependence study on 2D perovskite detectors—identifies surface
4401 passivation, such as the 5F-PEAI treatment, as the most promising near-term strat-
4402 egy for extending the operational stability of perovskite radiation detectors.

4403 9.2.3 Thermal Radiation Sensing Concepts

4404 Chapter 6 presented the simulation and analysis work performed in support of the
4405 Optical Fiber-Based Gamma Thermometer (OFBGT) and the Optical Fiber-Based
4406 Neutron Dosimeter (OFBND). Two modeling approaches—a simplified Python model

4407 and an iterative MATLAB model with temperature-dependent thermal conductivities—
4408 yielded consistent predictions, forecasting a maximum temperature differential of ap-
4409 proximately 27–30 °C for the OFBGT in the CIF and 155–159 °C for the OFBND in
4410 the AIF of the OSURR.

4411 Experimental testing of the OFBGT revealed a measured temperature differential
4412 of approximately 17 °C, falling short of the approximately 25 °C threshold required for
4413 reliable operation. The discrepancy was attributed to the omission of radiative heat
4414 transfer in the original simulations; when radiative heat transfer was included retro-
4415 spectively, the model predictions agreed closely with the experimental observations.
4416 The Modulation Transfer Function analysis confirmed that both sensors possess ade-
4417 quate axial resolution for the spatial frequencies expected in the OSURR flux profiles.
4418 The OFBND has not yet been fabricated or tested, but the simulation results indicate
4419 a favorable signal strength contingent on the inclusion of radiative heat transfer in
4420 future design iterations.

4421 **9.2.4 Digital Electronics Verification for Radiation Environ-** 4422 **ments**

4423 Chapters 7 and 8 addressed the design and experimental validation of two comple-
4424 mentary FPGA-based test structures under fast neutron irradiation at the OSURR
4425 Fast Neutron Beam Facility.

4426 The Flip-Flop Shift Register (FFSR) DMR test structure (Chapter 7) was de-
4427 ployed across five Spartan-7 FPGAs during a two-day irradiation campaign. During
4428 Day 1 ($\sim 6\text{--}7 \times 10^9$ n/cm² fast fluence, gamma shutter closed), zero DMR-detected
4429 SEUs were observed across all five devices, although one device (FFSR1) exhibited a
4430 UART output format change consistent with a configuration memory SEU that the

4431 DMR counter was unable to detect. During Day 2 ($\sim 8\text{--}9 \times 10^{11}$ n/cm², gamma shutter
4432 open), two devices experienced early functional dropouts consistent with radiation-
4433 induced latch-up, one device experienced a severe multi-SEU cluster event producing
4434 2,048 detected upsets together with a second configuration memory SEU affecting the
4435 UART baud rate, and PSU current monitoring showed a sustained $\sim 12.5\%$ increase
4436 consistent with TID-induced leakage current accumulation.

4437 The fixed-point thermal control benchmark (Chapter 8) provided a complemen-
4438 tary assessment by exercising a closed-loop I-PD controller, forward-Euler thermal
4439 plant emulation, nonlinear valve mapping, and DMR comparison within a single syn-
4440 thesizable Verilog design. Comparative analysis of two telemetry acquisitions revealed
4441 a smooth exponential decay of all plant state variables to zero over approximately
4442 2,500 s, with zero DMR mismatch detection throughout. Bit-exact simulation con-
4443 firmed that the decay could not arise from firmware numerical drift. The failure
4444 signature was consistent with a configuration memory (CRAM) SEU affecting arith-
4445 metic or routing logic shared by both DMR chains, demonstrating a fundamental
4446 limitation of dual modular redundancy when both redundant chains share common
4447 FPGA configuration memory.

4448 Together, these two independent observations—one from a shift-register test struc-
4449 ture and one from a closed-loop control system—establish that configuration memory
4450 SEUs are a primary concern for SRAM-based FPGAs at the fluence levels encoun-
4451 tered in nuclear instrumentation environments, and that effective mitigation requires
4452 configuration memory scrubbing, triple modular redundancy with physical placement
4453 constraints, or periodic functional self-testing against known reference trajectories.

4454 9.3 Implications for Nuclear Monitoring Systems

4455 The findings of this dissertation carry several implications for the design and
4456 deployment of integrated radiation monitoring systems in nuclear environments.

4457 First, halide perovskite materials have been shown to produce measurable elec-
4458 trical responses to X-rays, gamma rays, and fast neutrons across a range of device
4459 architectures. The current-mode photovoltaic approach and the direct-conversion
4460 semiconductor approach represent complementary detection philosophies—the for-
4461 mer offering large-area, zero-bias sensing suitable for integrated dose monitoring and
4462 prompt detonation forensics, and the latter offering the possibility of pulse-by-pulse
4463 energy spectroscopy at the cost of applied high voltage and the attendant ion mi-
4464 gration challenges. The demonstration of gamma spectroscopy in a solution-grown
4465 FAPbBr_3 crystal and of fast neutron sensitivity in MHyPbCl_3 establishes that per-
4466 ovskite materials can address multiple radiation types within a common materials
4467 platform, a capability not available from conventional compound semiconductors
4468 without resorting to separate detector materials for each radiation species.

4469 Second, the thermal sensing concepts explored for the OFBGT and OFBND rep-
4470 resent an alternative, non-electronic sensing pathway that avoids the radiation sus-
4471 ceptibility of semiconductor detectors entirely. Although the experimental results
4472 highlight the importance of accounting for radiative heat transfer in the design, the
4473 fiber-optic sensing modality is inherently radiation-hard and offers distributed mea-
4474 surement capability along the entire sensor length, making it well suited for spatially
4475 resolved monitoring of flux profiles inside reactor cores.

4476 Third, the FPGA verification results provide quantitative evidence that SRAM-
4477 based FPGAs are susceptible to configuration memory upsets at moderate fast neu-
4478 tron fluences and to latch-up and TID degradation at higher fluence levels. The
4479 inability of DMR alone to detect configuration memory upsets that affect both re-
4480 dundant chains identically underscores the necessity of a defense-in-depth strategy
4481 combining TMR with physical separation, configuration scrubbing via ICAP/PCAP
4482 readback, and periodic functional self-testing for any FPGA-based instrumentation
4483 deployed in a sustained radiation environment.

4484 Fourth, the hierarchy investigated in this dissertation—from radiation-responsive
4485 materials through detector devices, alternative sensing approaches, and digital elec-
4486 tronics instrumentation—reflects the functional layers that must be addressed in any
4487 complete radiation monitoring system. The results demonstrate that no single tech-
4488 nology addresses all requirements: perovskite detectors offer material sensitivity but
4489 face ion migration stability challenges; thermal sensors offer radiation hardness but
4490 sacrifice time resolution and spectroscopic capability; and FPGA-based electronics
4491 offer reconfigurability but require dedicated radiation mitigation strategies. An in-
4492 tegrated monitoring architecture would therefore draw on each technology for the
4493 function to which it is best suited, with perovskite or conventional semiconductor
4494 detectors providing the primary sensing element, thermal sensors providing comple-
4495 mentary dose verification, and radiation-tolerant FPGAs providing the digital signal
4496 processing and data acquisition backbone.

4497 **9.4 Future Research Directions**

4498 The work presented in this dissertation opens several avenues for continued inves-
4499 tigation.

4500 **9.4.1 Perovskite Materials and Devices**

4501 The central limitation identified across all perovskite detector systems was ion
4502 migration under applied electric fields, exacerbated by ambient humidity. Future
4503 work should pursue the optimization of surface passivation strategies, building on the
4504 5F-PEAI treatment demonstrated in Chapter 5, with the goal of enabling sustained
4505 high-bias operation under ambient conditions. Encapsulation methodologies that
4506 maintain the low-humidity performance observed in vacuum testing, without requiring
4507 a permanent vacuum environment, would represent a significant step toward practical
4508 deployment.

4509 For gamma spectroscopy, the FAPbBr₃ proof-of-concept motivates the growth of
4510 larger single crystals with improved surface preparation and contact deposition. The
4511 revised low-power sputtering protocol identified as necessary for MAPbBr₃ should be
4512 evaluated systematically across all bromide compositions. Achieving spectroscopic en-
4513 ergy resolution comparable to that of established CZT detectors remains a long-term
4514 goal that will require simultaneous advances in crystal quality, contact metallurgy,
4515 and electronic noise reduction.

4516 For fast neutron detection, the MHyPbCl₃ results warrant follow-up experiments
4517 with a dedicated digital pulse-processing system to replace the oscilloscope-based
4518 waveform acquisition that limited the statistical power of the present measurements.

4519 Neutron–gamma discrimination techniques, including pulse-shape discrimination al-
4520 gorithms, should be investigated to assess whether the organic cation’s hydrogen
4521 content can be exploited for fast neutron detection while rejecting the gamma back-
4522 ground inherent to the lead-containing lattice.

4523 For the current-mode photovoltaic approach, the signal-to-noise analysis of Chap-
4524 ter 4 identified array area scaling as the path to resolving neutron-induced currents.
4525 Fabrication and testing of multi-module perovskite solar cell arrays, with parallel
4526 development of low-noise current amplification electronics, would directly test this
4527 prediction and move the concept toward the large-area, real-time monitoring appli-
4528 cations envisioned for post-detonation forensics.

4529 **9.4.2 Thermal Radiation Sensing**

4530 The OFBGT results highlight the critical importance of including radiative heat
4531 transfer in design-stage simulations. Future iterations of the sensor should incor-
4532 porate radiative corrections from the outset and explore design modifications—such
4533 as reduced gas-gap dimensions or alternative fill gases—to increase the temperature
4534 differential above the operational threshold. The OFBND concept remains to be val-
4535 idated experimentally, and fabrication and in-reactor testing of the borosilicate glass
4536 thermal mass should be prioritized as a near-term objective.

4537 **9.4.3 FPGA-Based Instrumentation**

4538 The DMR blind spot identified in this work—configuration memory upsets that
4539 corrupt both redundant chains identically—motivates the development and irradi-
4540 ation testing of a TMR implementation with explicit physical placement constraints
4541 ensuring that each voter group occupies distinct configuration frames. Integration of

4542 ICAP-based configuration scrubbing with the FFSR or thermal control benchmark
4543 would allow quantitative evaluation of scrubbing effectiveness against the CRAM up-
4544 set rates measured in this campaign. Additionally, the implementation and testing
4545 of automated latch-up detection and power-cycling recovery circuitry is needed to
4546 address the functional dropouts observed during the high-fluence Day 2 irradiation.
4547 The fixed-point thermal control benchmark, with its deterministic state evolution
4548 and 100 Hz telemetry, is particularly well suited as a test vehicle for evaluating these
4549 mitigation strategies in future irradiation campaigns.

4550 **9.4.4 System Integration**

4551 Looking further ahead, the integration of perovskite radiation sensors with radiation-
4552 tolerant FPGA-based readout electronics into a single prototype monitoring system
4553 would represent a natural convergence of the materials, device, and electronics work
4554 streams pursued in this dissertation. Such a prototype would provide a platform
4555 for evaluating system-level performance metrics—including detection sensitivity, en-
4556 ergy resolution, long-term stability, and radiation tolerance of the complete signal
4557 chain—under realistic nuclear environment conditions.

Appendix A: The Data on Cows

4559 This is the data that was used to produce the table in Table ??. In 1990, 57
4560 people had cows. In 1991, 80 people had cows. In 1992, 199 people had cows.

4561 **Appendix B: Important commands defined in osudissert96**

4562 The following is a list of all commands available in osudissert96.

```
4563 \author{First Middle Last}
4564 \title{The title of the thesis}
4565 \authordegrees{degree1, degree2}
4566 \unit{Department of Whatever The Name Is}
4567 \degree{Doctor of Philosophy}
4568 \committee{Dissertation}
4569 \advisorname{Name of advisor}   % Possible usage "Prof. Big Dude"
4570 \member{Name of committee member}
4571
4572 \thesis   % this makes it a thesis rather than a dissertation,
4573           % similar to the documentclass option [masters] or [ms].
4574
4575 \maketitle
4576
4577 \disscopyright           % or \blankpage
4578
4579 \begin{abstract}
4580 \end{abstract}
4581
4582 \begin{externalabstract}
4583 \end{externalabstract}
4584
4585
4586 \dedication{This is dedicated to \ldots}
4587
4588 \begin{acknowledgements}
4589 \end{acknowledgements}
4590
4591 \begin{vita}
```

```

4592 \dateitem{Important Date}{ Why its important}
4593
4594 \begin{publist}
4595 \researchpubs
4596 \pubitem{Bibliography item (from BibTeX?)}
4597
4598 \instructpubs
4599 \pubitem{Bibliography item (from BibTeX?)}
4600 \end{publist}
4601
4602 \begin{fieldsstudy}
4603 \majorfield*          % which uses \unit above
4604 % \majorfield{Your Major Field}
4605 \begin{studieslist}
4606 \studyitem{Topic 1}{Prof.\ 1}
4607 \studyitem{Topic 2}{Prof.\ 2}
4608 \studyitem{Topic 3}{Prof.\ 3}
4609 \end{studieslist}
4610 %% Note:  If there were only one field of study, the following list
4611 %%         would best be done using the following command:
4612 %% \onestudy{Only Topic}{Only Professor}
4613 \end{fieldsstudy}
4614
4615 \end{vita}

```


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